CHAPTER

ATA 31

INDICATING AND RECORDING SYSTEM

THIS MANUAL IS INTENDED FOR TRAINING PURPOSE ONLY

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CONTENTS

CONTENTS 3
INDICATING AND RECORDING SYSTEMS, GENERAL 4
ENGINE COCKPIT INTERFACE UNIT 7
INDEPENDENT INSTRUMENTS 12
RECORDERS 21
SIGNAL CONDITIONING UNIT 37
EXTENDED−STORAGE QUICK ACCESS RECORDER SYSTEM (EQAR) ..... **Error! Bookmark not defined.**
MICRO QUICK ACCESS RECORDER SYSTEM (MQAR) ..... **Error! Bookmark not defined.**
CENTRAL COMPUTER 39
CENTRAL WARNING SYSTEM 101
CAUTION AND WARNING LIGHTS SYSTEM 103
TAKE−OFF WARNING SYSTEM 125
ELECTRONIC INSTRUMENTS SYSTEM 128

INDICATING AND RECORDING SYSTEMS, GENERAL

Introduction

The indicating and recording system has different sub–systems to do the functions that follows:

- Show time
- Records aircraft parameter data
- Receive data from sensors and avionics systems and supply it to other systems
- Make warning tones
- Show caution and warning lights
- Show advisory lights
- Take–off warning calculation
- Show navigation, engine, and system parameters.

General Description

Refer Figure - INDICATING AND RECORDING BLOCK DIAGRAM

The indicating and recording system has the sub–systems that follow:

- Clocks
- Flight Data Recorder System (FDR)
- Central Computer
- Central Warning System
- Electronic Instruments System.

Clocks

The electronic clock has a quartz time base that supplies a continuous display of Greenwich Mean Time (GMT) or Local Time (LOC). The electronic clock can also be set to show the Elapsed Time (ET), the date or set to the chronometer function (CHR).

Flight Data Recorder System (FDR)

The Solid State Flight Data Recorder (SSFDR) records aircraft parameter data and stores it in crash–protected memory for future retrieval purposes.

Central Computer

The central computer has a Flight Data Processing System (FDPS) to receive data from sensors and avionics systems and supply it to other systems. The Flight Data Processing System (FDPS) also supplies warning tones that alert the crew to specific events or system failures.

Central Warning System

The central warning system is divided into the two parts that follow:

- Caution and Warning Lights
- Take–off Warning.

The caution and warning light system is divided into the two parts that follow:

- Caution and warning lights
- Advisory lights.

Caution and Warning Lights: The caution and warning lights system shows system malfunctions and other conditions that require a corrective action.

Advisory Lights: The advisory lights show malfunctions and other conditions that require a corrective action and safe and normal system operation.

The take−off warning system supplies an aural warning when a take−off is attempted with the aircraft not in the correct take−off configuration.

Electronic Instruments System

The Electronic Instrument System (EIS) shows navigation, engine, and system parameters. It interfaces with other systems to calculate, make, and show their images. The Electronic Instrument System (EIS) also monitors is calculations to prevent misleading information from being shown.

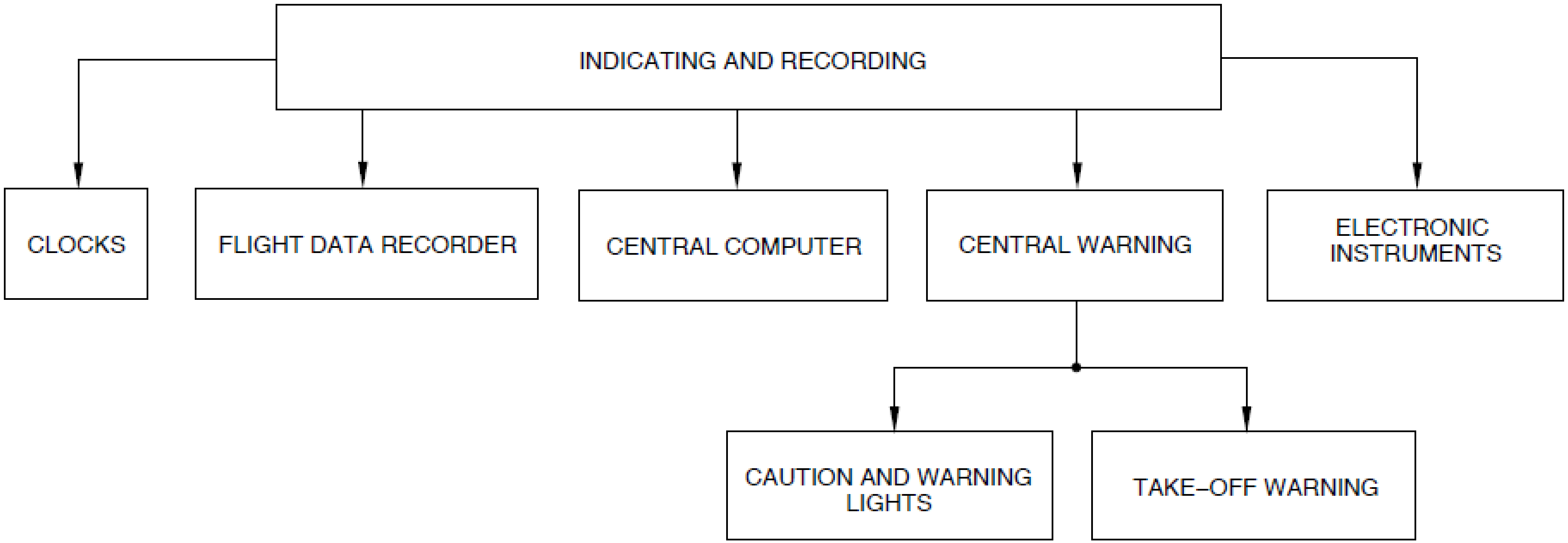


Figure - INDICATING AND RECORDING BLOCK DIAGRAM

ENGINE COCKPIT INTERFACE UNIT

General

Refer Figure - ENGINE COCKPIT INTERFACE UNIT

The engine cockpit interface unit (ECIU) transfers data between the cockpit and both full authority digital engine controllers (FADECs).

General Description

Refer Figure - ENGINE COCKPIT INTERFACE UNIT (ECIU) LOCATION

The ECIU is located in the flight compartment, on the lower shelf of the left circuit breaker console. It contains a single main circuit card assembly (CCA) that consists of two redundant channels (A and B).

The ECIU receives 24 inputs per channel from the cockpit that are transmitted to the FADEC and 16 inputs per channel from the engines that are transmitted to the cockpit.

Detailed Description

Refer Figure - ENGINE COCKPIT INTERFACE UNIT – SYSTEM BLOCK DIAGRAM

Inputs to the ECIU from the cockpit are hardware discretes and supply either a ground or an open indication. The ECIU monitors these discrete inputs and processes the information into ARINC data. The ECIU then transmits this data across four independent buses to each channel of both FADECs. The cockpit discrete inputs are shown below in the table.

TABLE 1 DISCRETE INPUTS

SOURCE	DISCRETE
Engine Control Panel	Maximum climb select* Maximum climb select* Maximum power select Power derate rest* Power derate decrement* 850 RPM approach select
Maintenance Panel	Maintenance test select * Rigging trim select*
Ignition Panel	Engine 1 manual ignition select Engine 1 ignition on/off Engine 2 manual ignition select Engine 2 ignition on/off
ECS Panel	ECS bleed 1 on/off ECS bleed 2 on/off ECS maximum select ECS minimum select
Spares	Eight spares available (two momentary)
* Momentary switches	

The ECIU can supply 16 discrete outputs per channel as either a ground or open indication. A 1.0–ampere discrete output drives the oil ejector solenoid while a 0.5–ampere output drives the other systems. The ECIU monitors the ARINC data and sets these discretes when the appropriate bits are set. The discrete outputs are shown in Table below.

breaker panel.

TABLE 2 DISCRETE OUTPUTS	
DESTINATION	**DISCRETE**
Lamps – O Red channel A and B	Engine 1 low oil pressure Engine 2 low oil pressure Engine 1 low fuel pressure Engine 2 low fuel pressure Engine 1 fuel filter bypass Engine 2 fuel filter bypass
Relay Drivers – simplex, engine 1 channel A only, engine 2 channel B only	Engine 1 oil cooler ejector Engine 2 oil cooler ejector Engine 1 oil cooler intermediate flap position Engine 2 oil cooler intermediate flap position Engine 1 oil cooler maximum flap position Engine 2 oil cooler maximum flap position
Spares	Four spares available

T1.8 information is also input to the ECIU from each engine FADEC channel (four per aircraft) across the ARINC 429 data buses. The ECIU processes the T1.8 signal and supplies this information on the four ARINC data buses.

The dual channel ECIU operates on 28 Vdc electrical power through 3–ampere circuit breakers. Channel A receives power from the left essential bus and channel B receives power from the right essential bus. The circuit breaker for channel A is located at position H5 on the left DC circuit breaker panel. The circuit breaker for channel B is located at position H5 on the right DC circuit

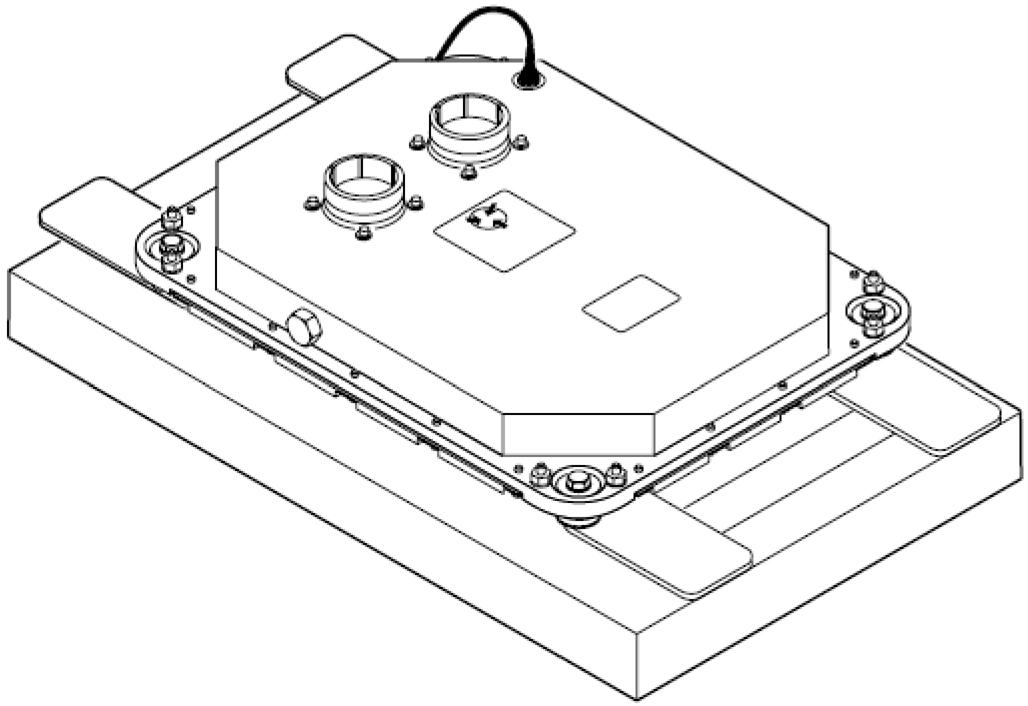


Figure - ENGINE COCKPIT INTERFACE UNIT

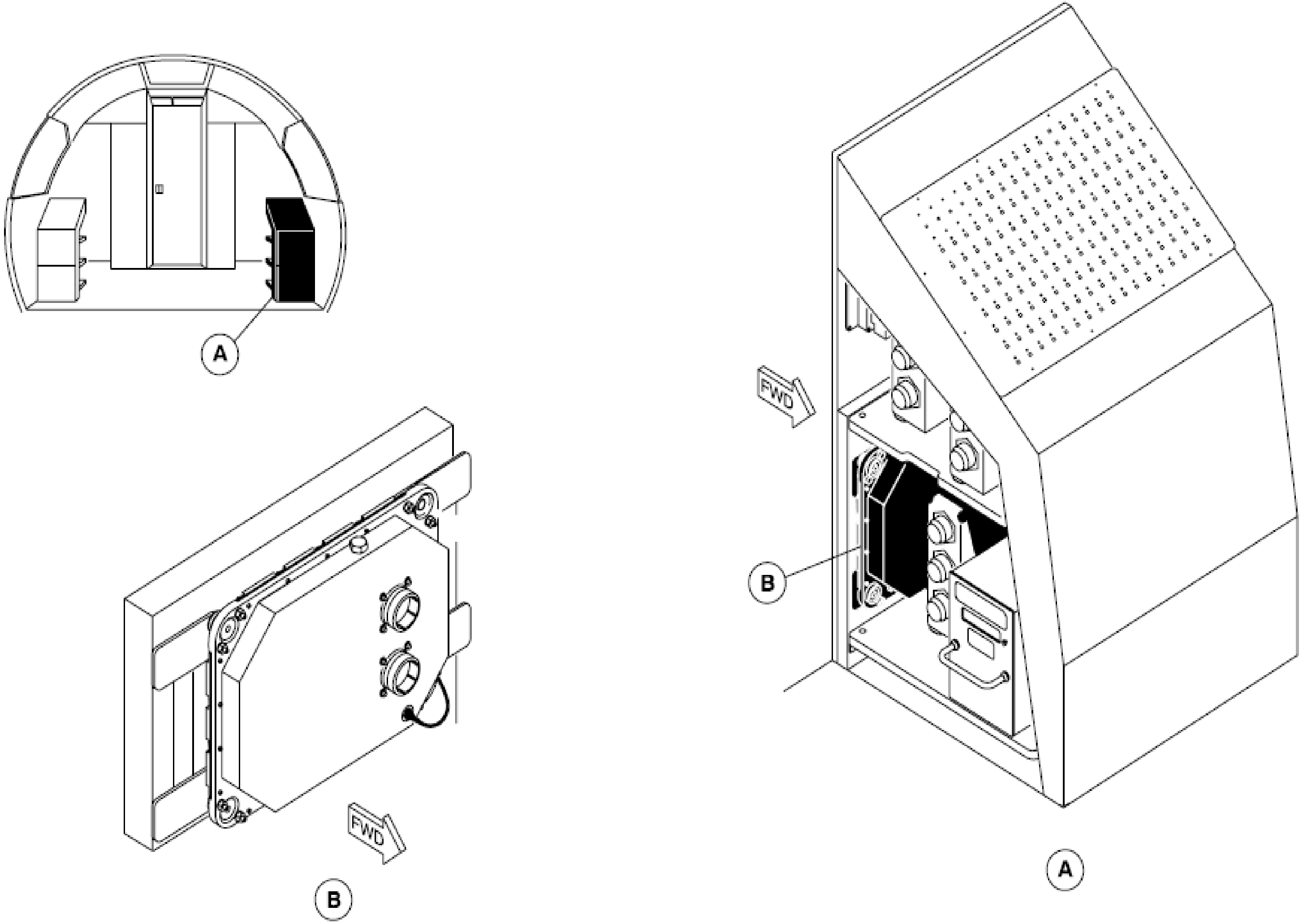


Figure - ENGINE COCKPIT INTERFACE UNIT (ECIU) LOCATION

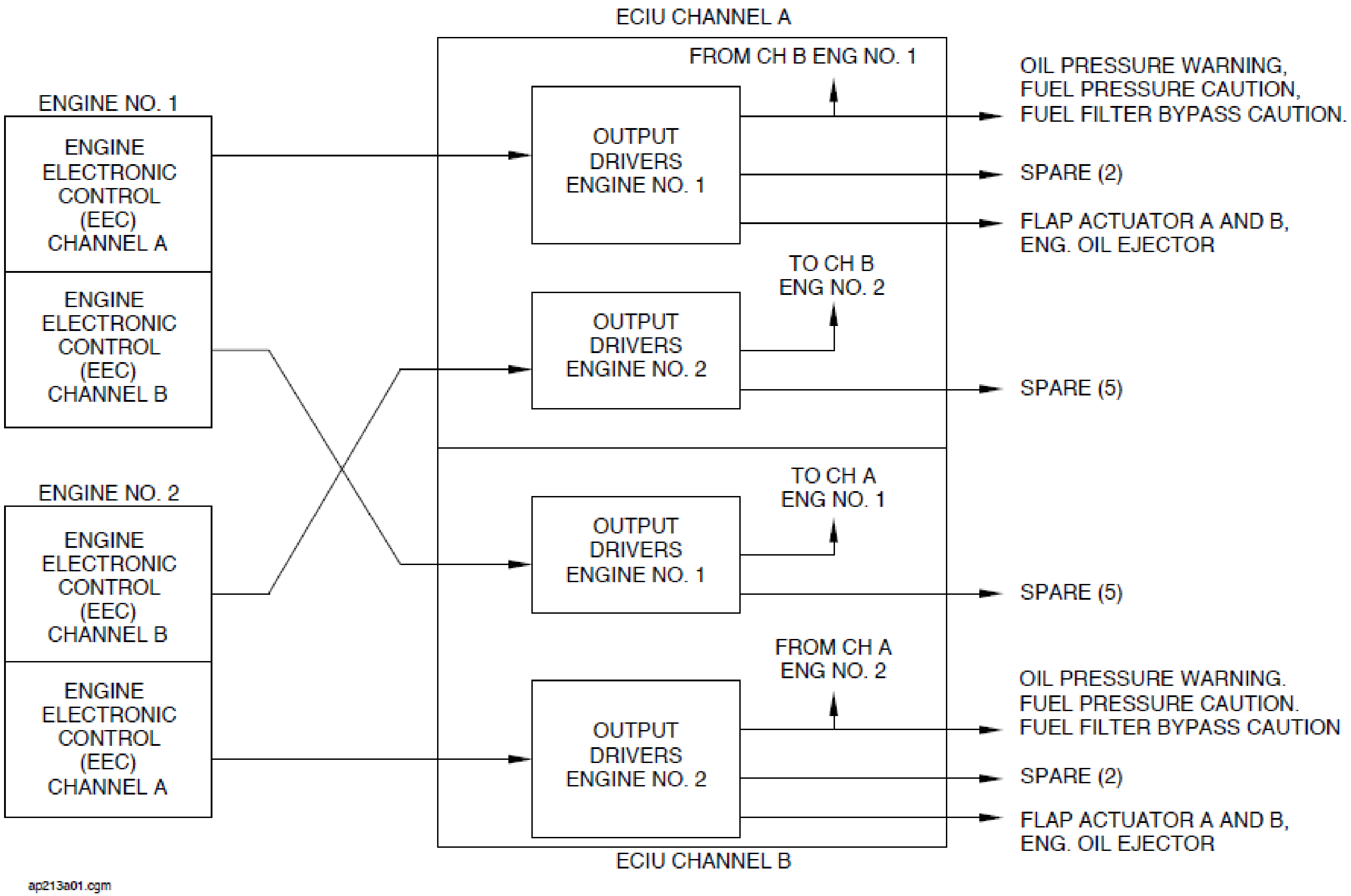


Figure - ENGINE COCKPIT INTERFACE UNIT – SYSTEM BLOCK DIAGRAM

INDEPENDENT INSTRUMENTS

General Description

Refer Figure - CLOCK SYSTEM BLOCK DIAGRAM

The electronic clock supplies the following functions that follow:

- Continuous digital display of GMT or LOC
- Digital display of ET
- Display of the chronometer (CHR) function. In this mode, the display of seconds is analog (sweep−hand) and that of minutes is digital.
- Display of date and year, when requested.

There are two clocks located in the flight compartment one on each side of the glare shield. The pilots set the type of time−based information to be shown on the display.

Detailed Description

Refer Figure - ELECTRONIC CLOCK

Each clock is independently operated from the switches located on the bezel.

The function selector positions are labeled DATE, LOC, GMT and SET. The selector is pushed to exit the SET mode.

When the function selector is placed in the SET position, the Elapsed Time (ET) button is pushed to cycle through the modes that follow:

- GMT minutes (displayed immediately when SET is selected)
- GMT hours
- LOC minutes
- LOC hours
- Days
- Months
- Years (default on power−up is 90).

At each momentary activation of the ET switch, the applicable area of the display flashes and the data is then entered using the CHR button.

The Elapsed Time (ET) switch located in the lower left area of the clock face gives 3−state and 2−state sequences dependent on the Weight on Wheels (WOW) status of the aircraft.

On the ground:

- First activation: Display of Elapsed Time
- Second Activation: Elapsed Time is reset to zero
- Third activation: Display of Chronometer minutes

Airborne:

- First Activation: Display of Elapsed Time
- Second Activation: Display of Chronometer minutes.

The Chronometer function switch (CHR) located in the top right corner of the clock face supplies the three states, in order, that follow:

First activation: START	Temporary removal of the Elapsed Time hour display Return to zero Chronometer minute count start Chronometer sweep−hand start
Second activation: STOP	Maintains the display of the current indication
Third activation: RESET	Sweep−hand returns to zero Elapsed Time display returns

When primary electrical power is removed, the time base is maintained by the aircraft battery bus, all displays are blanked, and the sweep−hand, if active, stops. Current parameters continue to increment with the exception of the Chronometer and Elapsed Time functions. When primary power is restored, the upper LCD display shows the original function data and the lower display indicates 00 00. The Chronometer sweep−hand returns to zero and can be re−enabled if set to start from zero.

The ARINC 429 output bus is not active during standby power operation.

Greenwich Mean Time (GMT) is shown in the top 4−digit readout area of the clock face from 00:00 to 23:59 minutes when the function selector is in the GMT position. A single dot is displayed above the GMT legend.

Local time (from 00:00 to 23:59) is shown in the same location as GMT when the selector switch is in the LOC position. A single dot appears above the LOC legend, to give an alternative means of distinguishing local time from GMT, additional to switch position.

When the selector switch is set to DATE, the day and month are shown in the top 4−digit area of the clock face. The two left digits (01 to 12) identify the month and the two right digits identifies the day (01 to 31) and the year (0 to 99). As the day and year occupy the same area, the display alternates each second between the two parameters. To aid interpretation while displaying the year, the left digits are blank. Leap years are programmed into clock operation.

Elapsed Time (ET) is indicated from 0 to 99:59 in the lower digital display area of the clock face and gives an indication of aircraft flight time. The mode is automatically enabled by a discrete Weight−Off−Wheels discrete input from the Proximity Switch Electronics Unit (PSEU) when the aircraft becomes airborne and can only be reset during a Weight−On−Wheels (WOW) condition. A colon separates the hours and minutes.

Minutes are indicated from 0 to 59 by the two right digits in the lower display area of the clock face with the left digits blanked. Seconds are shown against the round dial of the clock face by a sweep−hand activated by a stepper motor.

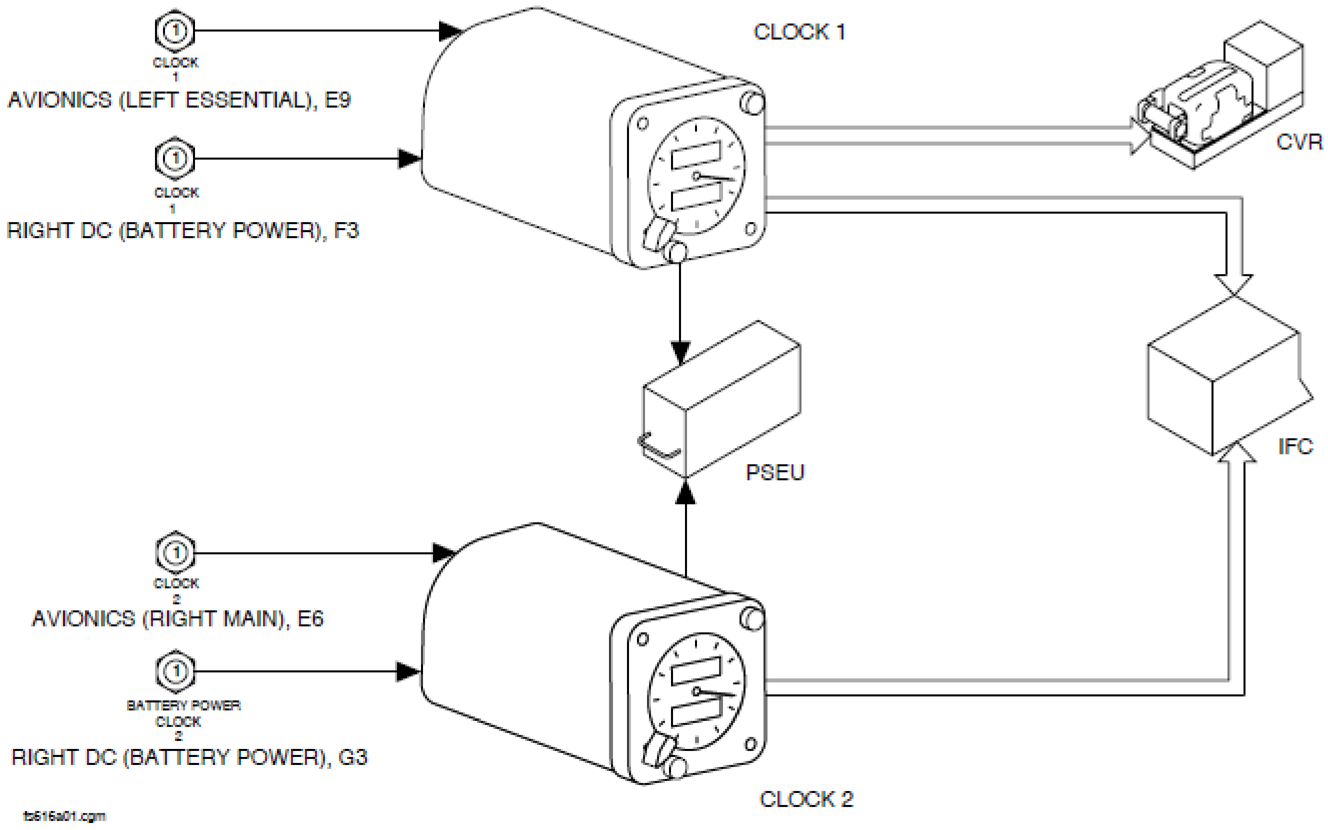
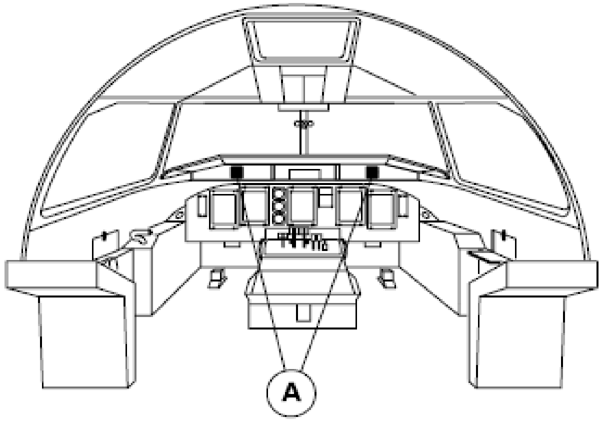


Figure - CLOCK SYSTEM BLOCK DIAGRAM



LEGEND

1. Function selector switch.
2. Elapsed time switch.
3. Chronometer function switch.

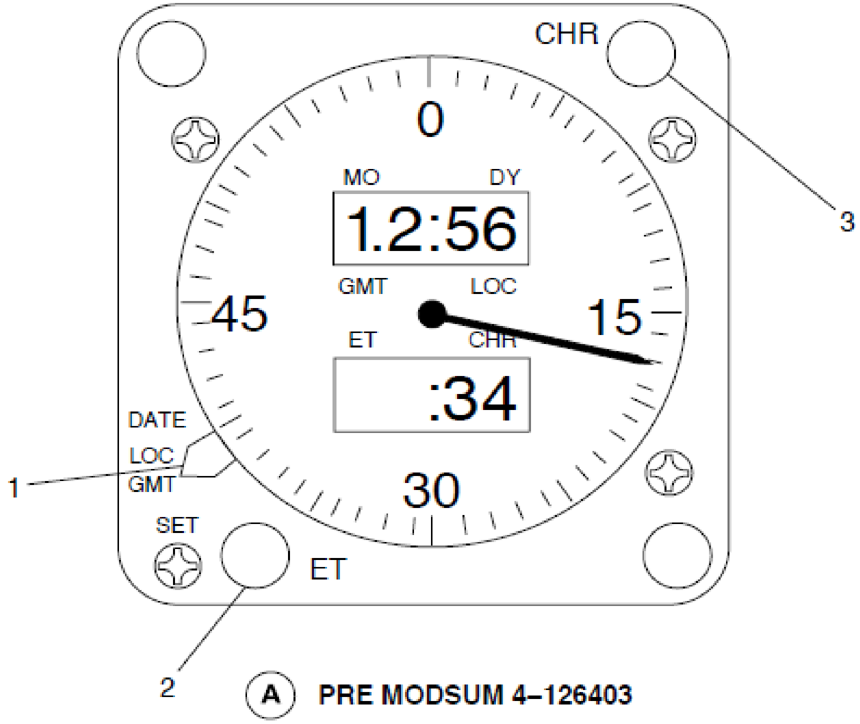


Figure - ELECTRONIC CLOCK

Refer Figure - Electronic Clock (Post ModSum 4-126403)

The clock has the indications that follow:

- UTC
- CHR
- ET.

UTC: The UTC time is displayed from 0 to 23 hours, 59 minutes and 59 seconds on the six digit LCD (upper).

The display can be:

- Synchronized with GPS if GPS is valid (GPS flag is shown)
- Internal time if GPS is invalid (INT flag is shown)
- Date, if this mode is selected (DT flag shown) (date format is mm/dd/yy)
- Local time, if this mode is selected (LT flag is shown).

The colon between the hours digits and the minutes digits is lit when the clock is on.

CHR: The chronometric time (minutes and seconds) is shown on the lower four−digit LCD from 0 to 99 hours, 59 seconds. The colon between the minutes and seconds is lit when the chronometer is on. The display is blanked when reset.

ET: Elapsed time is displayed from 0 to 99 hours, 59 minutes on the lower four−digit LCD. The elapsed time indication starts automatically when the aircraft is weight off wheels. The colon is lit when the ET indicator is on. Reset of the display is possible only when the aircraft is weight on wheels.

Mode: The MODE pushbutton switch allows selection of the various modes in turn:

- GPS: In GPS mode the clock is synchronized with the GPS, if the GPS is valid. Time and date are updated.
- INT: In the INT mode the clock runs in internal mode and uses its own time base if GPS is invalid. The default is to run in GPS mode.
- LT: In LT mode the local time is shown and the clock uses its internal time base.
- DT: In this mode the date is shown and the clock uses the GPS time or its own internal time base.

At start−up the clock automatically starts in the INT mode. If the GPS is valid the clock is automatically updated and switched in the GPS mode.

The time can be set with the SET function which is made available by pressing the MODE pushbutton for a minimum of two seconds. In this function the UTC minute digits flash (the INT flag is on). Use the ET RST pushbutton to increase the minute indication; use the ET SEL pushbutton to decrease the minute indication. While in this function, each push of the MODE button will cycle through:

- The hour indication (the INT flag is on)
- The year indication (the DATE flag is on)
- The month indication (the DATE flag is on)
- The day indication (the DATE flag is on)
- The LT minute indication (the LT flag is on)
- The LT hour indication (the LT flag is on).

For each of these indications, use the ET RST pushbutton to increase the value and the ET SEL pushbutton to decrease the value.

CHR: This function operates with the CHR pushbutton switch which, with each push, cycles through start (colon is lit), stop (colon is blanked) and reset (display is blanked). If the display was occupied by the ET function before CHR is selected, the indicator flag toggles between ET and CHR.

ET SEL and ET RST: The elapsed time function is operated with these two pushbutton switches. If the display was occupied by the CHR function before ET SEL is selected, the indicator flag will toggle between CHR and ET. A push of the ET RST button blanks the ET display (only when the aircraft is weight on wheels).

The left essential and battery power busses supply 28 Vdc electrical power through two 1 ampere circuit breakers to the CLOCK 1.

The right main and battery power busses supply 28 Vdc electrical power through two 1 ampere circuit breakers to the CLOCK 2.

Primary power is supplied from the left essential and right main 28 Vdc busses for CLOCK1 and CLOCK2 with a keep−alive voltage automatically supplied by the battery power bus when the primary electrical power is removed. The clocks are also supplied with variable 5 Vdc from the aircraft dimming bus to 4 display backlighting bulbs.

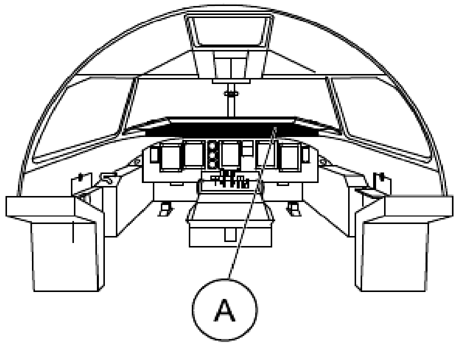
Each clock operates independently. The CLOCK1 is interfaced directly with the Cockpit Voice Recorder and both clocks are interfaced with the Flight Data Recorder (FDR) through the Flight Data Processing System (FDPS). The FDR normally records time from the CLOCK1 but will switch to CLOCK2 if the CLOCK1 malfunctions. Real time is recorded on both the Cockpit Voice Recorder (CVR) and Flight Data Recorder (FDR) to establish synchronization between the two recording systems.

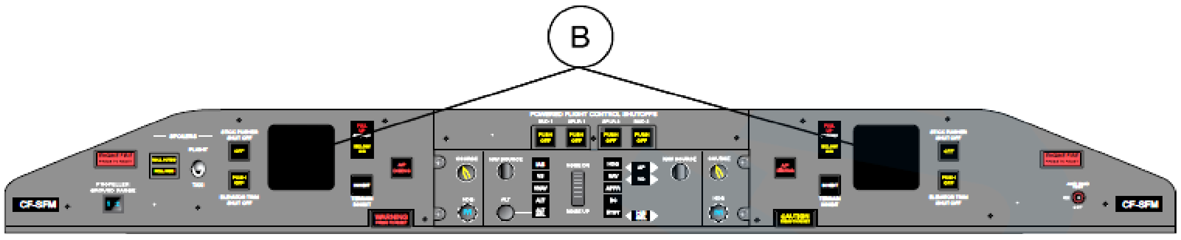
The date is also supplied to the Flight Data Processing System (FDPS) located in the Integrated Flight Cabinet (IFC1) to show when systems malfunctioned for events in Central Diagnostics System (CDS) BITE reports.

The clocks send data to the CVR and FDPS through an ARINC 429 data bus.

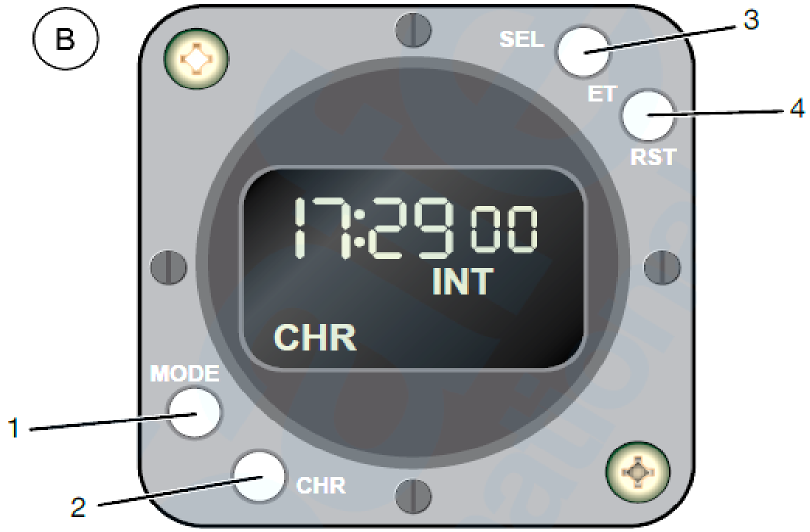
The clock installations interface with the aircraft systems that follow:

- IOP1, located in IFC 1, for input to the CDS and FDR
- PSEU for Weight−Off−Wheels discrete (to activate the Elapsed Time function)
- CVR for recording of real time and synchronization with the FDR.





GLARESHIELD PANEL



LEGEND

1. Mode pushbutton switch
2. Chronometer function switch
3. ET SEL pushbutton switch
4. ET RST pushbutton switch

Figure - Electronic Clock (Post ModSum 4-126403)

Clock, Electronic

Refer Figure - ELECTRONIC CLOCK LOCATOR

Two clocks are installed on the left and right sides of the glare shield panel assembly.

On aircraft without ModSum 4−126403 and 4−126434 incorporated, the clocks have dichroic LCDs that show white digits against a grey background.

On aircraft with ModSum 4−126403 or 4−126434 incorporated, the clocks have one six−digit and one−four digit LCDs that show white digits against a black background.

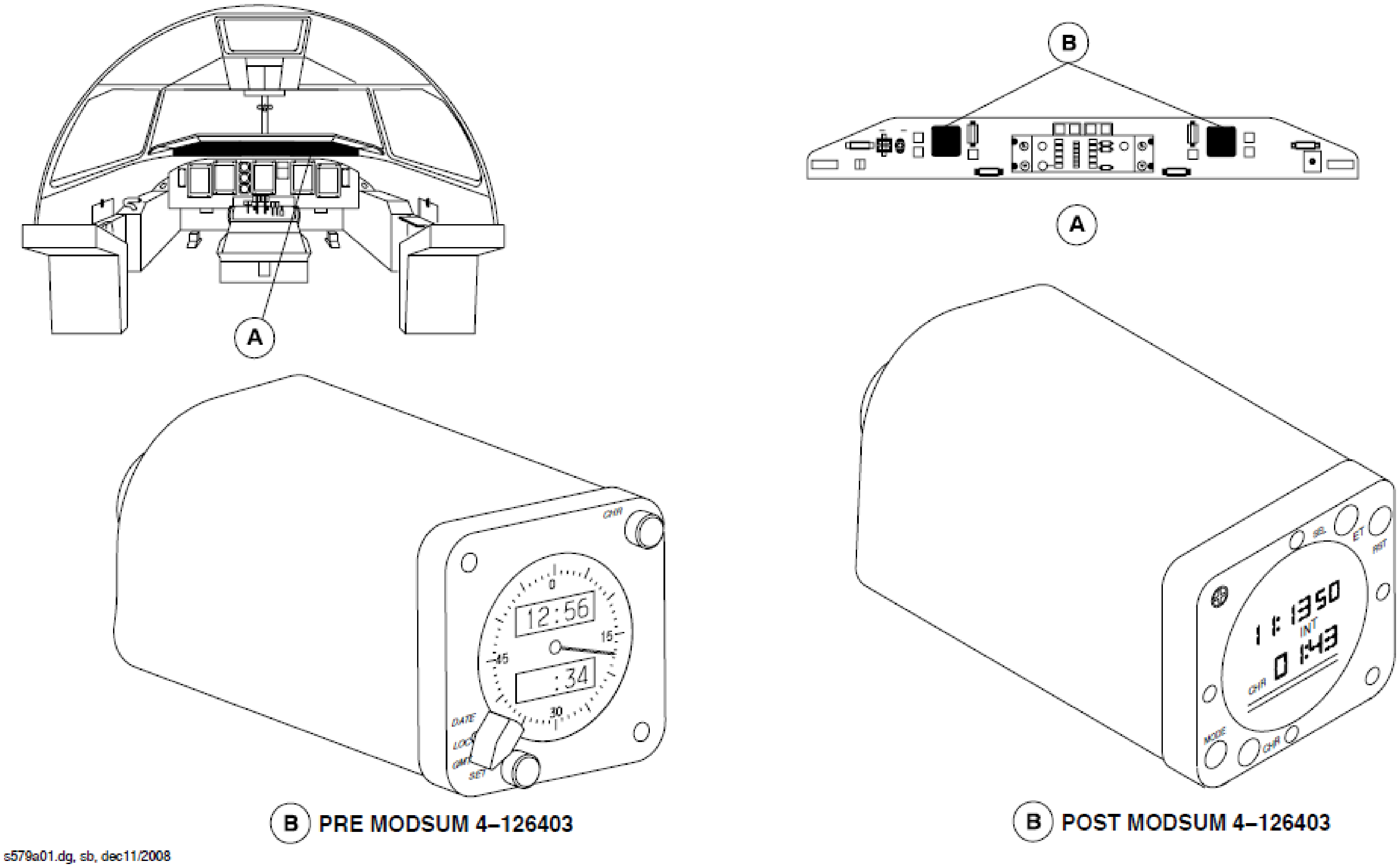


Figure - ELECTRONIC CLOCK LOCATOR

RECORDERS

Introduction

The Solid State Flight Data Recorder (SSFDR) records aircraft parameter data and stores it in crash−protected memory for future retrieval purposes.

General Description

Figure - FLIGHT DATA RECORDER BLOCK DIAGRAM
Figure - Flight Data Recorder (Universal) Block Diagram

The Solid State Flight Data Recorder (SSFDR) is a crash survivable unit that can record up to 25 hours of aircraft parameters and clock data.

The Solid State Flight Data Recorder (SSFDR) receives from the Integrated Flight Cabinet (IFC 1), the aircraft parameters that follow:

- Flight path
- Speed
- Attitude
- Engine power
- Configuration
- Operation.

An inertia switch removes power to the SSFDR and Solid−State Cockpit Voice Recorder (SSCVR) when exposed to a high acceleration. Pushing a mechanical reset button on the switch assembly manually resets the switch following activation.

On aircraft with ModSum 4Q126473 incorporated, two inertia switches are there, one each, for both the SSFDR and SSCVR is installed, which removes power to the SSFDR and SSCVR when exposed to a high acceleration.

Pushing a mechanical reset button on the switch assembly manually resets the switch following activation.

The Flight Data Recorder System (FDR) has the components that follow:

- Recorder, Flight Data
- Tray, Mounting
- Switch, Inertia
- Switch, Test
- Switch, Anti-collision.

An Underwater Locator Beacon (ULB) (also known as Underwater Locating Device (ULD)) is attached to the Solid State Flight Data Recorder (SSFDR) handle. When immersed in either fresh water or saltwater, the ULB transmits an acoustic signal to help locate the SSFDR.

Detailed Description

The flight data recorder system has SSFDR and an acceleration inertia switch. The system interfaces with:

- Flight Data Processing System (FDPS)
- Test circuit
- Input power interlock arrangement
- Caution and warning panel.

The ULB is attached to the front of the SSFDR to easily access it for servicing and to read its battery expiratory date. It may also be used as a carrying handle.

The ULB is automatically activated when immersed in either fresh water or salt water at depths from 0.5 to 20,000 feet (0.15 to 6096 meters).

When the ULB is activated, it will transmits an acoustic signal of 37.5 ±1 kHz frequency for a duration of 90 days or more. The ULB is self−powered by a battery with a typical service life of seven years from the date of manufacture. A label that shows the battery expiry date is attached to the ULB.

The SSFDR system receives mandatory and non−mandatory parameters and discrete from aircraft systems at 128 words per second through an ARINC 717 data bus. The unit records up to 25 hours of data in a crash−survivable memory module for subsequent retrieval and analysis. Ground−Based Equipment (GBE) and an RS422 interface are used to download the recorded data.

On aircraft with ModSum 4Q126473 OR SB84−31−82 incorporated, the SSFDR system receives mandatory and non−mandatory parameters and discrete from aircraft systems at 512 words per second through an ARINC 717 data bus. Ground−Based Equipment (GBE) and an Ethernet interface are used to download the recorded data.

The GBE performs the functions that follow:

- Enables the real−time monitoring of aircraft parameters and discrete
- Shows the internal status of the SSFDR
- Downloads data from the Crash Survivable Memory Unit (CSMU)
- Does a functional test of the SSFDR.

The GBE interface connector (with protective cover) is installed in the front panel of the SSFDR and is used for download, test and maintenance functions without removing the LRU from the aircraft.

The SSFDR operates in the modes that follow:

- Power off mode
- Initialization mode
- Record mode
- Continuous monitor mode
- Test mode
- Download mode
- Power down mode.

Power Off: The SSFDR operates when power is removed for more than 200 milliseconds. No functions are available. Applying power exits the power off mode.

Initialization: The SSFDR operates in the initialization mode immediately upon receiving power. It initializes Built−In−Test (BIT) within 500 millisecond (On aircraft with ModSum 4Q126473 OR SB84−31−82 incorporated the initialization is completed within 250 millisecond). It determines the SSFDR status.

If the BIT fails, the external BIT light on the front panel of the SSFDR comes on. The SSFDR exits the initialization mode of operation if it cannot record data due to a critical failure. It transmits a signal to the continuous monitor mode to stop it from recording, and the FLT DATA RECORDER annunciator on the Caution Warning Panel comes on. The Central Diagnostics System (CDS) receives and records an equivalent maintenance output through the Integrated Flight Cabinet (IFC 1).

Record Mode: Recording data begins immediately following the initialization and self−test mode. The SSFDR stops recording if it does not receive data for more than five seconds, causing the FLT DATA RECORDER caution light to come on.

The SSFDR stays in the record mode until it is connected to the Ground Based Equipment (GBE). The type of GBE selects what mode the SSFDR will enter next.

The SSFDR transmits the received data back to the Input/output Processor (IOP1). This makes sure that the SSFDR is properly receiving the input data. If the status discrete senses a failure, the "loopback" data is not present. The FLT DATA RECORDER caution light on the caution and warning panel will come on.

The conditions that follow exit the SSFDR from the record mode:

- Removing power
- Detection of the download mode
- Command from the IOP.

Monitor Mode: The monitor mode supplies aircraft installation diagnostics and troubleshooting.

The SSFDR continuously does the following:

- Background tests on system hardware
- Monitors data inputs.

In the monitor mode, the SSFDR records normal flight data in parallel with the data bus to the GBE. The GBE commands the SSFDR to operate in the monitor mode and to output a "data frame", once per second, that shows the status of the recorder, and the record data, 8 times per second.

Test Mode: The test mode supplies "Return to Service" testing of the SSFDR. The external Automated Test Unit (ATU) commands the SSFDR to operate in the test mode. The normal record mode stops when the unit is operating in the test mode. The ATU controls the exit from the test mode.

Down Load Mode: The SSFDR starts the down load mode when initiated by the external Down Load Unit (DLU).

Power down Mode: The SSFDR power supply maintains internal voltages to continue normal operation during a power interruption lasting less than 200 milliseconds. For power interruptions greater than 200 milliseconds:

- The data in the RAM is lost
- The initialization mode starts.

The SSFDR is electrically powered by 28 VDC through an interlock system. The SSFDR starts to record when any of the conditions are met that follow:

- Red anti-collision lights are switched ON
- White anti-collision lights are switched ON
- Both engines are operating, engine−operating oil pressure is sensed from both engines
- Weight−Off−Wheels is sensed
- Test switch is set to GND TEST.

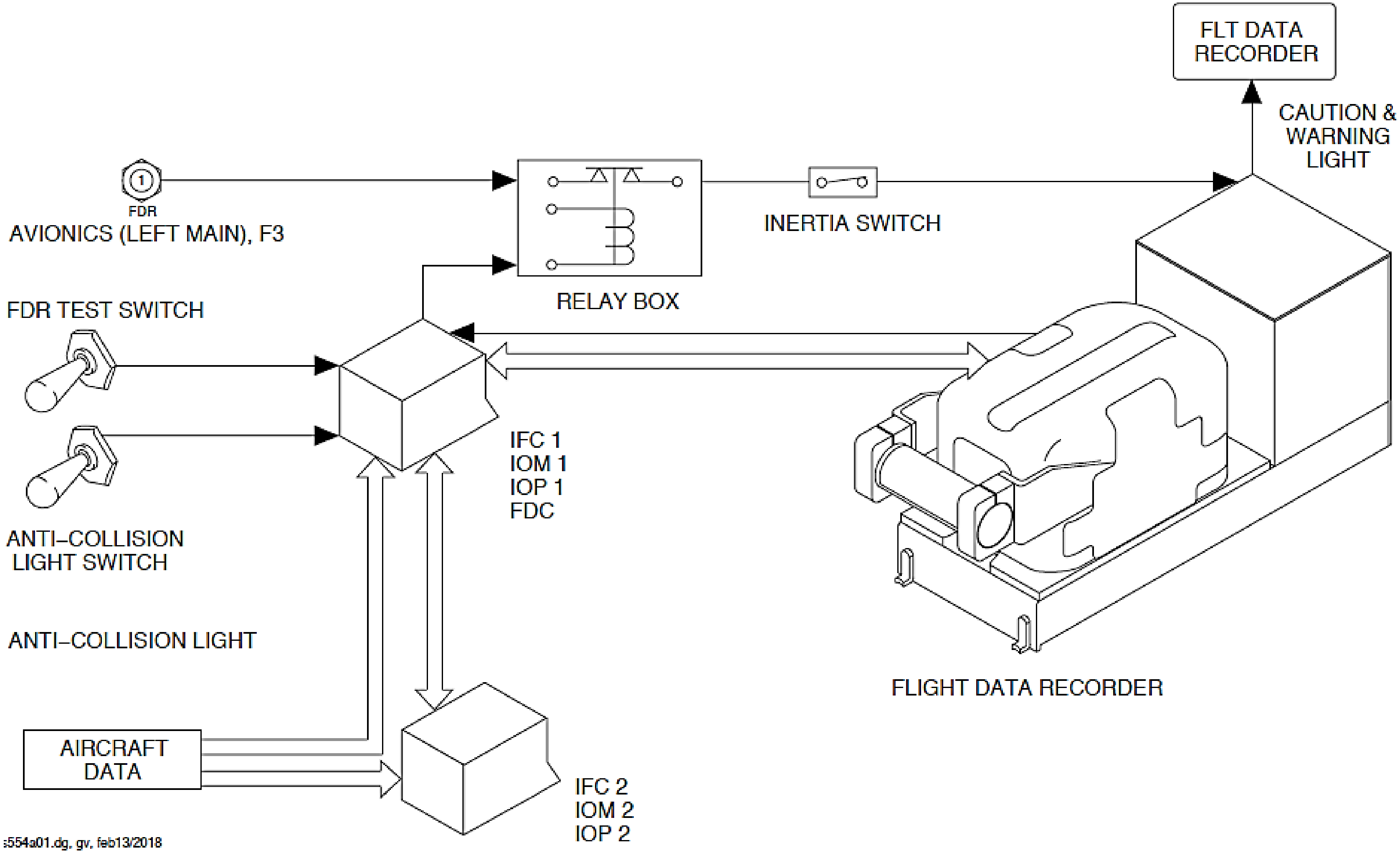


Figure - FLIGHT DATA RECORDER BLOCK DIAGRAM

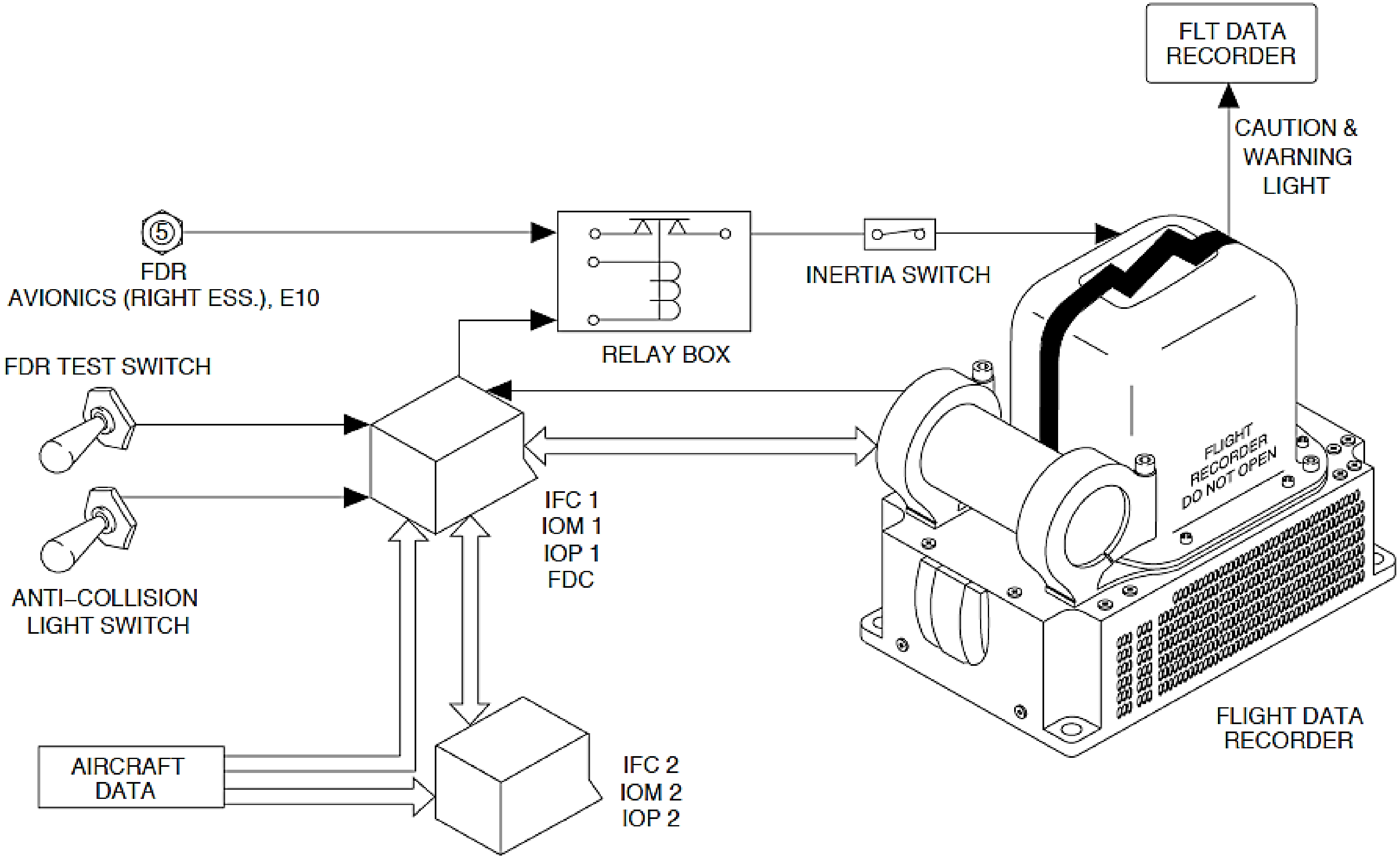


Figure - Flight Data Recorder (Universal) Block Diagram

Refer Figure - Flight Data Test Switch

The FLIGHT DATA RCDR toggle switch on the Flight Data Recorder Panel is set to the GND TEST position to test the SSFDR while the aircraft is on the ground.

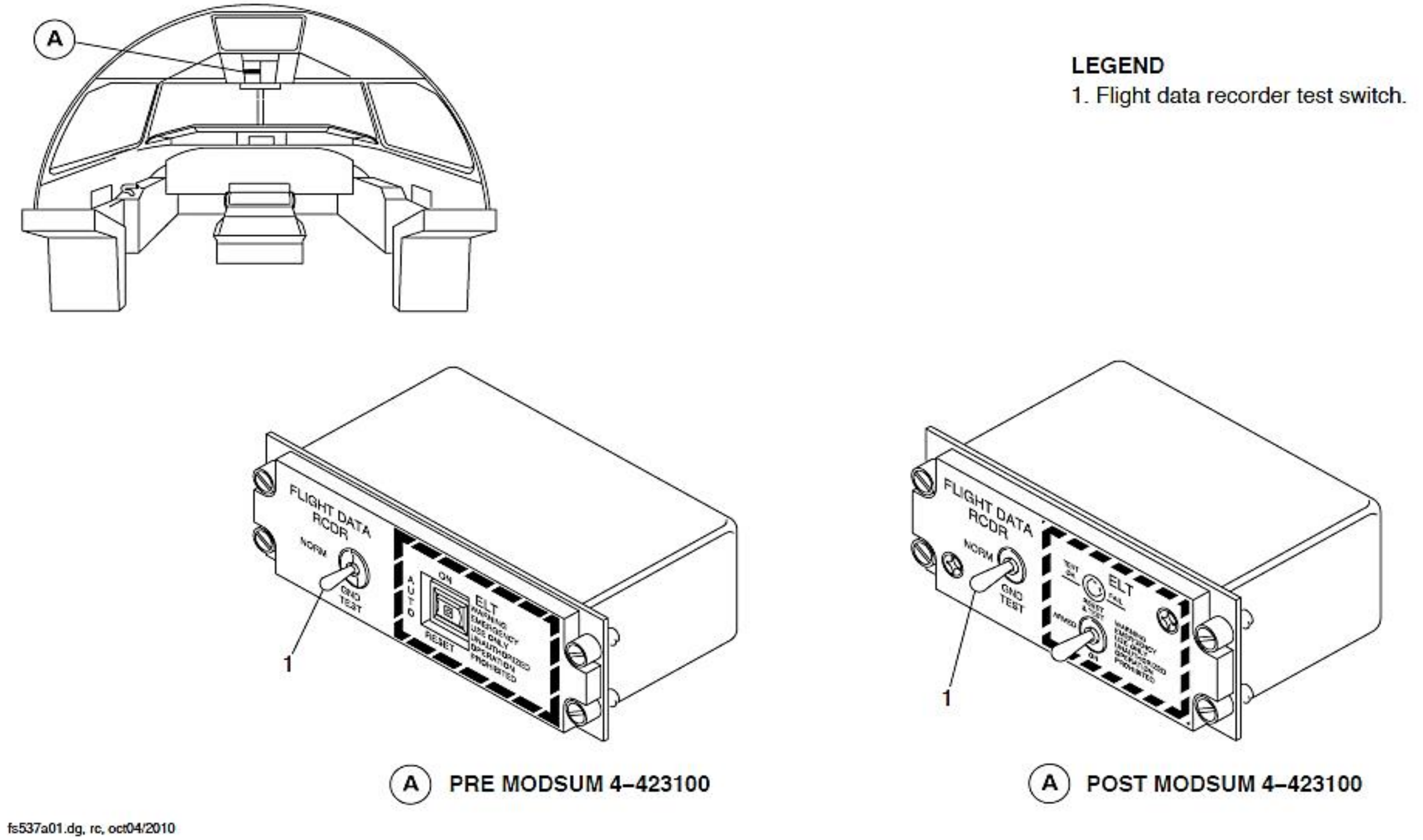


Figure - Flight Data Test Switch

Refer Figure - FDR CAUTION INDICATION

If the SSFDR BITE senses satisfactory system operation, the FLT DATA RECORDER indication in the caution and warning panel goes out.

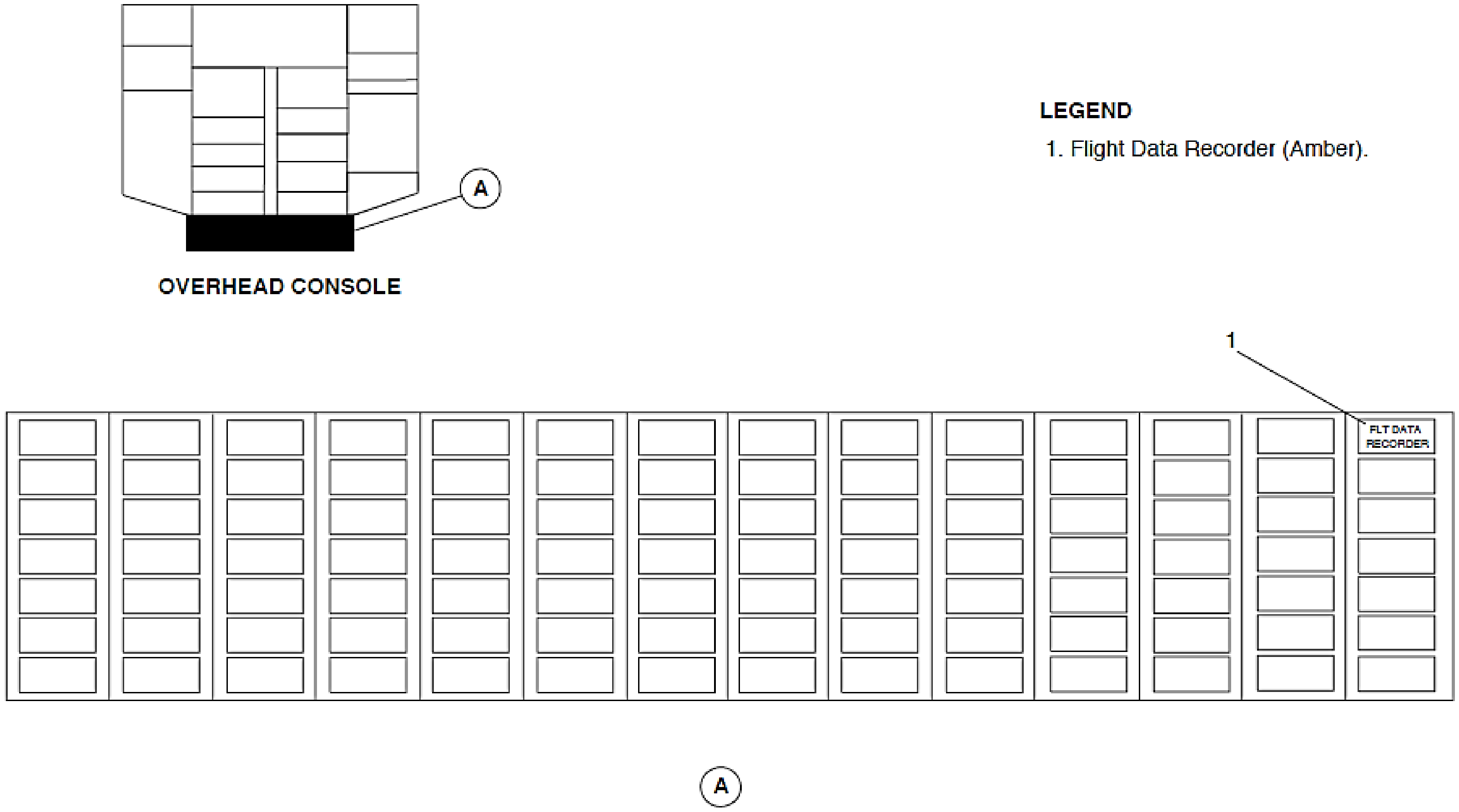


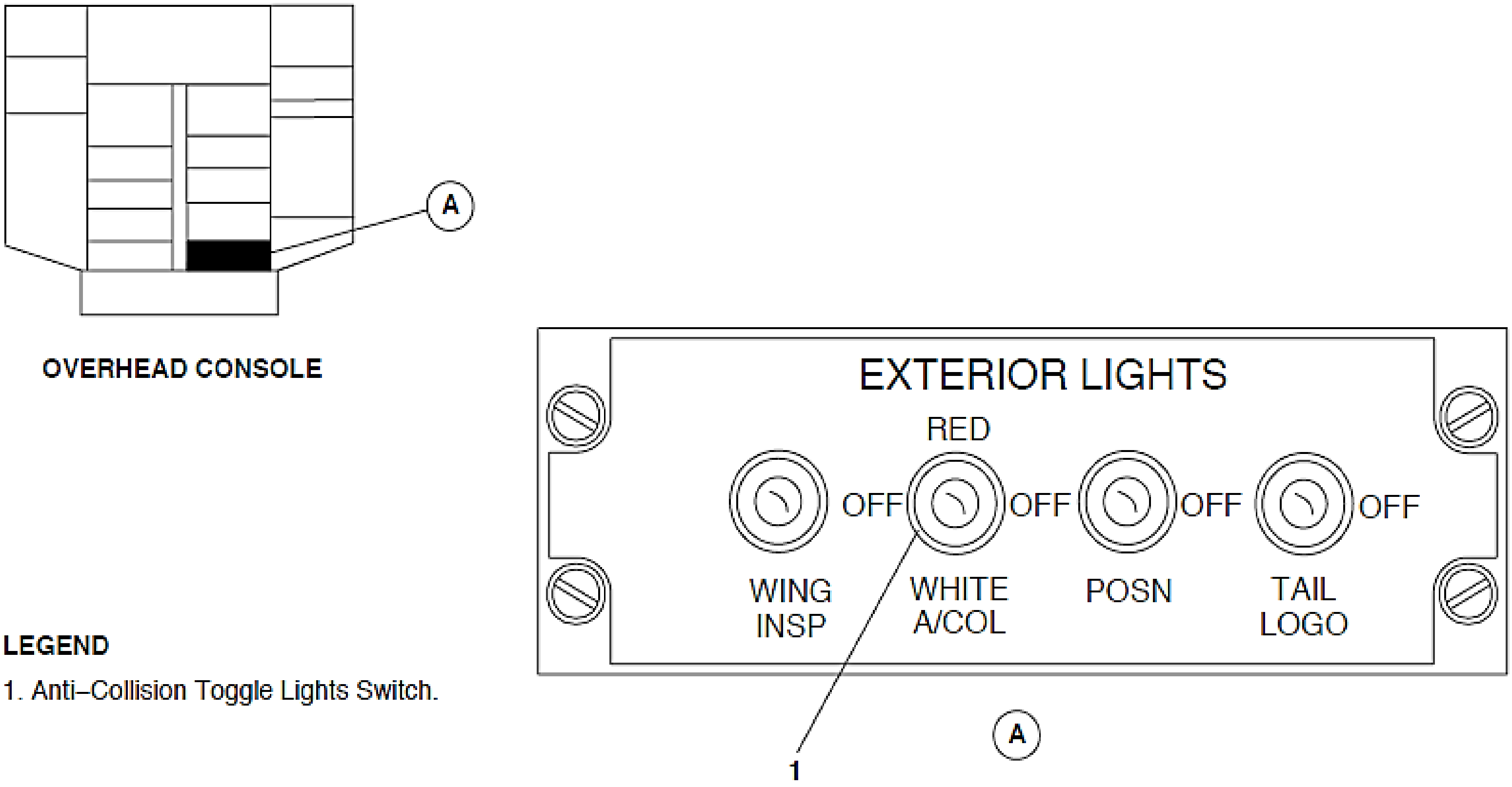
Figure - FDR CAUTION INDICATION

Refer Figure - ANTI−COLLISION LIGHT SWITCH

The RED−OFF−WHITE A/COL toggle switch on the Exterior Lights Panel is set to the WHITE or RED position to power the SSFDR.

The SSFDR system interfaces with the aircraft systems/busses that follow:

- Left Main 28 VDC bus through 1 A circuit breaker F3
- On aircraft with ModSum 4Q126473 OR SB84−31−82 incorporated, Right Essential 28 VDC bus through 5 A circuit breaker E10.
- Avionics circuit breaker panel
- IOM1 (located in IFC1) for input to the Centralized Diagnostic System (CDS) (maintenance flag)
- IOP1 (located in IFC1) for input ARINC 573/717 data and loop−back data
- Overhead anti-collision lights (red and white) switch for power interlock
- Proximity Sensor Electronic Unit (PSEU) for WOW input (part of power interlock)
- Overhead test switch
- Caution panel (status flag)
- Inertia switch (removes power for accelerations in excess of 5.5 g).



LEGEND

1. Anti–Collision Toggle Lights Switch.

Figure - ANTI−COLLISION LIGHT SWITCH

Recorder, Flight Data

Refer Figure - FLIGHT DATA RECORDER LOCATOR
Refer Figure - Flight Data Recorder (Universal) Locator

The SSFDR has the components that follow:

- Circuit Card Assemblies (CCAs)
- Controller
- Memory
- Power supply regulator
- Power supply filter.

The SSFDR chassis is painted a bright international orange and is marked with the black letters "FLIGHT RECORDER DO NOT OPEN" and "ENREGISTREUR DE VOL NE PAS OUVRIR".

Reflective tape is attached to the SSFDR external surface to further help recover it.

The SSFDR is designed to operate in extreme environmental conditions with no forced air cooling. When installed, air can flow around the unit and supply cooling.

The SSFDR is installed in the rear fuselage of the aircraft between station points X942.57 and X958.23 at Z134.46 and to the left of aircraft centerline.

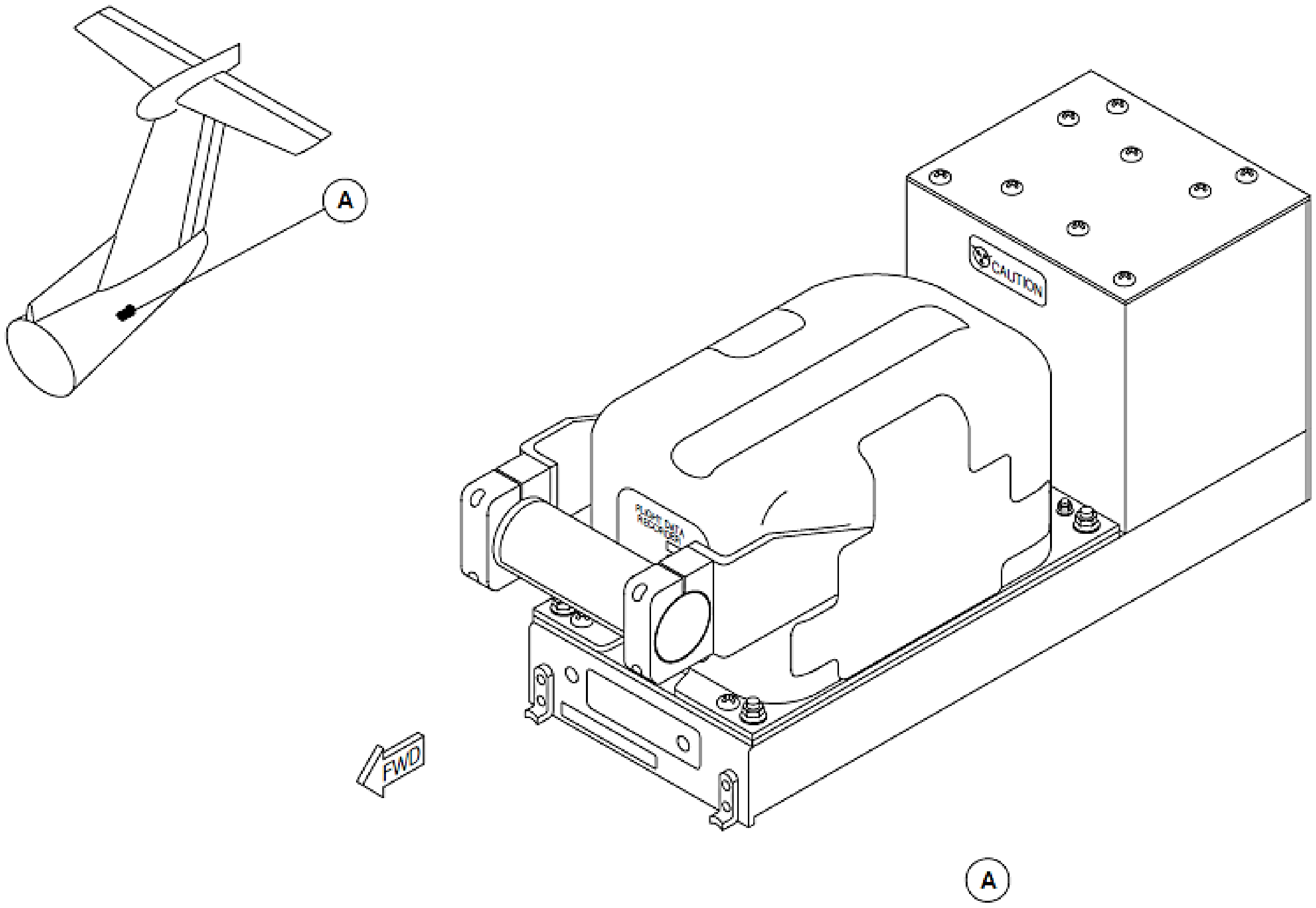


Figure - FLIGHT DATA RECORDER LOCATOR

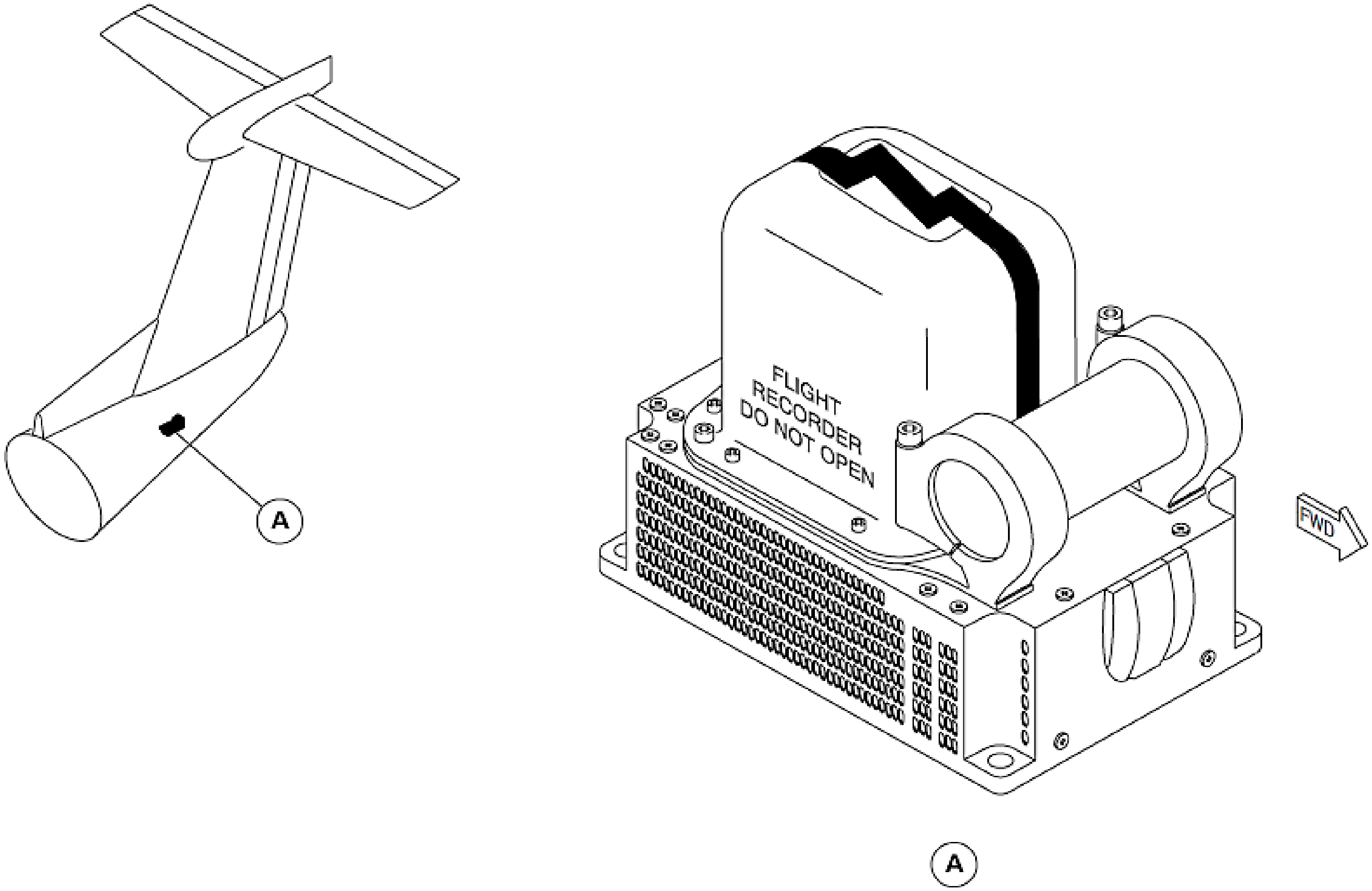


Figure - Flight Data Recorder (Universal) Locator

Tray, Mounting

The SSFDR is installed in a ½ ATR short mounting tray that is electrically bonded to the airframe by a bonding strap.

On aircraft with ModSum 4Q126473 OR SB84−31−82 incorporated, the SSFDR tray is directly mounted on to the aircraft structure with help of four bolts, which provides bonding to the chassis ground.

The SSFDR mounting tray is located in the tail of the aircraft at stations X958.23 and Z134.43.

Switch, Inertia

Refer Figure - Flight Data Recorder Inertia Switch Locator

The inertia switch supplies the SSFDR system with the capability to remove power from the system if an acceleration of greater than 5.5 g is applied to the switch as a consequence of high impact landing. This makes sure that SSFDR recorded data prior to the impact is stored and cannot be overwritten.

The normally−closed inertia switch has an impact activated double−pole latching switch; one pole of which is connected in series with the 28 VDC aircraft supply to the SSFDR. (The second pole is connected to the SSCVR). The switch latches to an open state when activated.

On aircraft with ModSum 4Q126473 incorporated, two inertia switches are there, one each, for both the SSFDR and SSCVR is installed. The existing inertia switch provides 28 VDC power to the SSFDR and the new inertia switch, installed above the SSFDR inertia switch, provides the 28 VDC power to the SSCVR.

A manually operated RESET BUTTON, located on the side of the inertia switch is used to re−arm the switch.

The current rating of each pole of the switch is 5 A at 28 VDC.

NOTE

The maximum DC current of the SSFDR is less than 500 mA.

The inertia switch connector is installed on the side panel of the SSFDR.

The unit is installed at 45 degrees with the reset switch pointing towards ground and the front of the aircraft to sense upward and downward acceleration. It is attached to a bracket located below the cabin floor at station point X195.8, Y0, Z88.9.

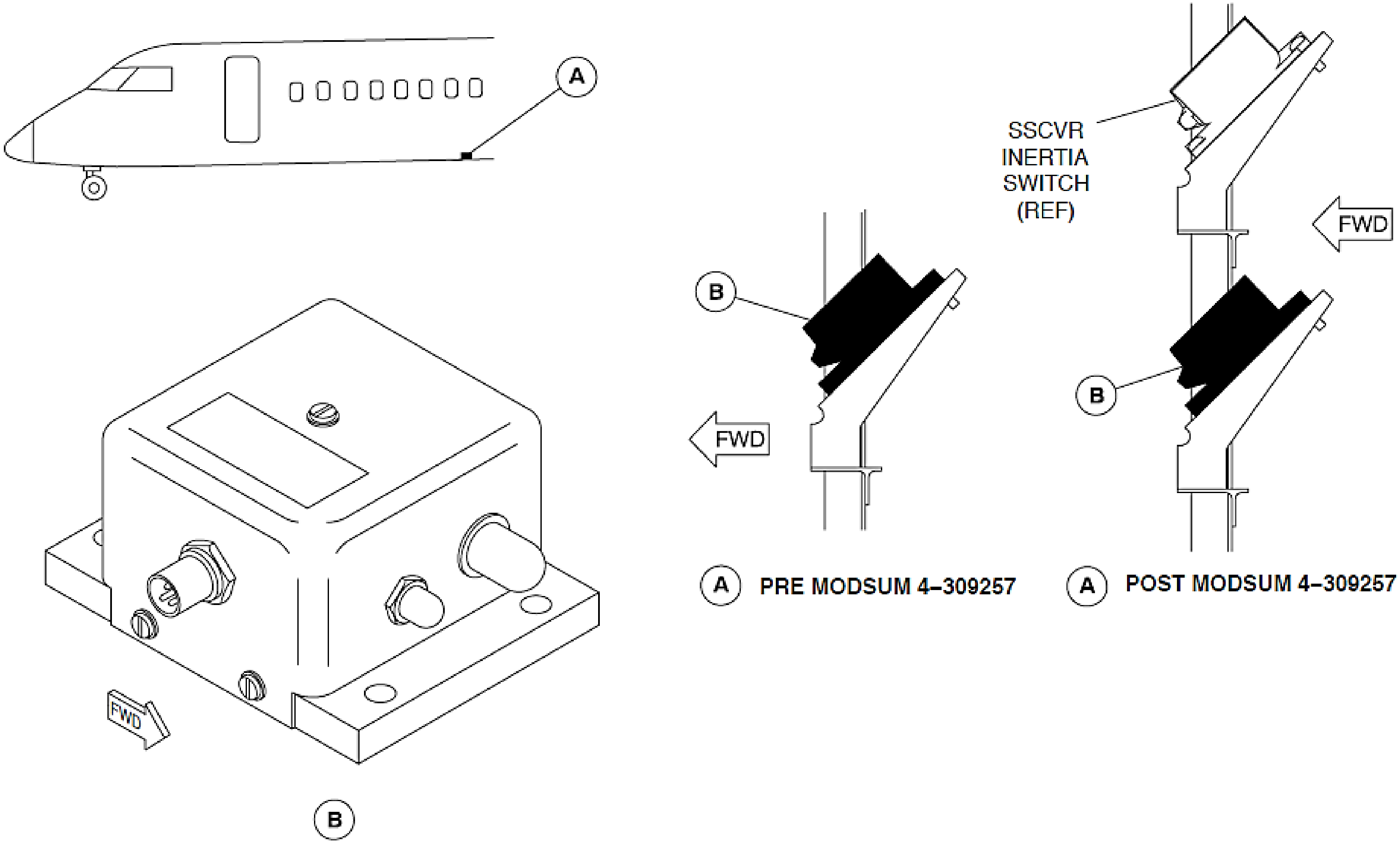


Figure - Flight Data Recorder Inertia Switch Locator

Switch, Test

Refer Figure - Flight Data Recorder Test Switch

When the FLIGHT DATA RCDR switch is set to the GND TEST position, the normal "on ground" power removal circuits are bypassed. The pre−flight GND TEST switch located in the flight compartment on the overhead panel, supplies power to the SSFDR for the duration that the switch is held. If the SSFDR BITE indicates satisfactory system operation, the caution panel FLT DATA RECORDER light goes out.

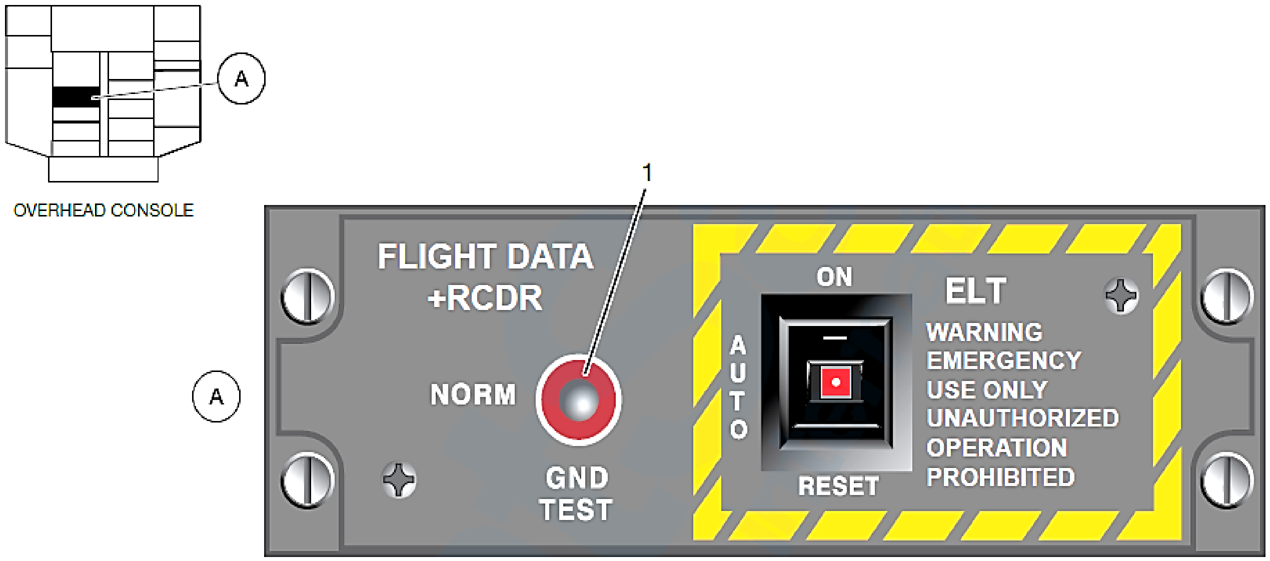
The NORM−GND TEST toggle switch is located in the flight compartment on the overhead FLIGHT DATA RCDR panel.

Switch, Anti−Collision

Refer Figure - ANTI−COLLISION LIGHT SWITCH

When the anti-collision lights switch is set to the RED or WHITE position, the normal "on ground" power removal circuits are bypassed. The anti-collision lights switch is located in the flight compartment on the overhead panel. If the SSFDR BITE indicates satisfactory system operation, the caution panel FLT DATA RECORDER light goes out.

The RED−OFF−WHITE A/COL switch is located in the flight compartment on the overhead EXTERIOR LIGHTS panel.



LEGEND

1. Flight Data Recorder Test Switch.

Figure - Flight Data Recorder Test Switch

SIGNAL CONDITIONING UNIT

Introduction

***Refer Figure 31- 18 Signal Conditioning* Unit**

The discrete, force, position and pressure sensors measure the control column inputs and also the affecter response. The aircraft system sensors convert the control column inputs into electrical signals and transmit them to Flight Data Recorder Signal Conditioning Unit (FSCU). The FSCU supplies excitation, demodulation and filtering signal for the sensing devices. The FSCU also supplies ARINC 429 data for the Flight Data Recorder (FDR).

General Description

Refer Figure - Signal Conditioning Unit Block Diagram

The FSCU operates in the states that follow:

- On
- Off

The FSCU interfaces with the aircraft 28 VDC power supply. The power supply converts the input into the conditioned supply and reference voltages.

The FSCU interfaces with the 24 discrete inputs and 4 discrete outputs.

The FSCU receives data from at least five ARINC 429 receivers at 100 kbps.

The FSCU supplies analog inputs to interface with the devices that follow:

- Potentiometers
- Linear Variable Differential Transformers (LVDTs)
- Pressure Sensors

The FSCU performs the tests to detect the critical faults that follow:

- Power supply fault
- A/D converter fault
- SRAM fault
- Program memory fault
- EEPROM checksum fault
- Watchdog fault

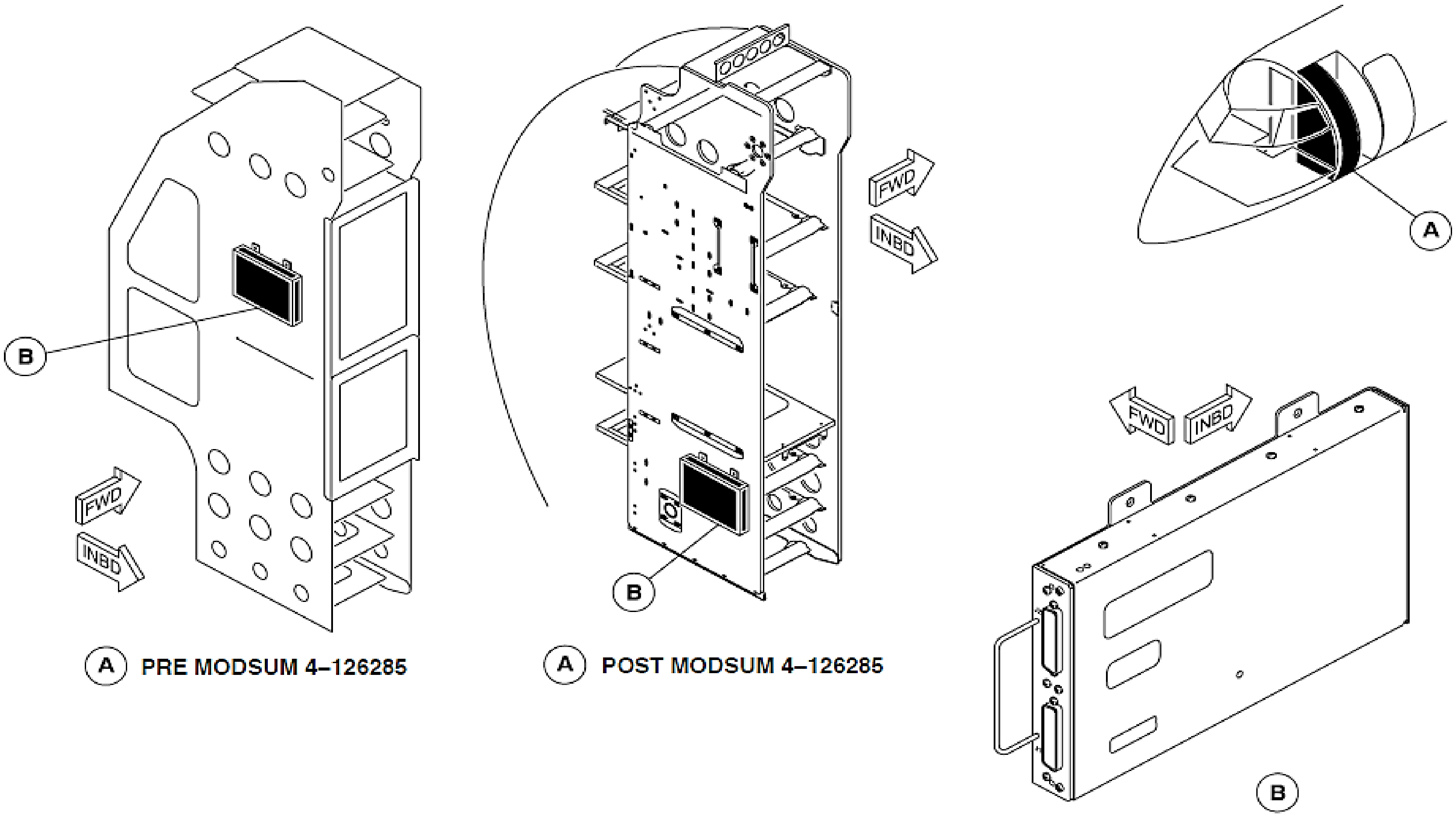


Figure - Signal Conditioning Unit Block Diagram

CENTRAL COMPUTER

Introduction

The central computer has a Flight Data Processing System (FDPS) to receive data from sensors and avionics systems and supply it to other systems. The Flight Data Processing System (FDPS) also causes different warning tones to sound if an important system has malfunctioned or if the aircraft is in a dangerous condition.

General Description

Refer Figure - FLIGHT DATA PROCESS SYSTEM BLOCK DIAGRAM
Refer Figure - FLIGHT DATA PROCESS SYSTEM BLOCK DIAGRAM, POWER

The two Flight Data Processing System (FDPS1, FDPS2) do the functions that follow:

- Receives and calculates aircraft and avionics parameters for the Flight Data Recorder (FDR)
- Receives data from different avionics and aircraft systems and routes their parameters to another systems through a single output
- Receives analogue, discrete, and digital data from other systems and changes it to ARINC 429 format
- Mismatch message calculations
- Monitors the Electronic Instrument System (EIS)
- Message generation
- Aircraft Configuration Management (ACM)
- Supplies Warning Tones (WTG 1, WTG 2) and manages its priority
- Maintenance tests
- Software teleloading.

Each Flight Data Processing System (FDPS 1, FDPS 2) has the modules that follow:

- Integrated Flight Cabinet
- Input/output Processor
- Input/output Module
- Prime Power Supply Module
- Aircraft Configuration Module.

The modules are located in two Integrated Flight Cabinets (IFC 1, IFC 2) installed in the Avionics rack.

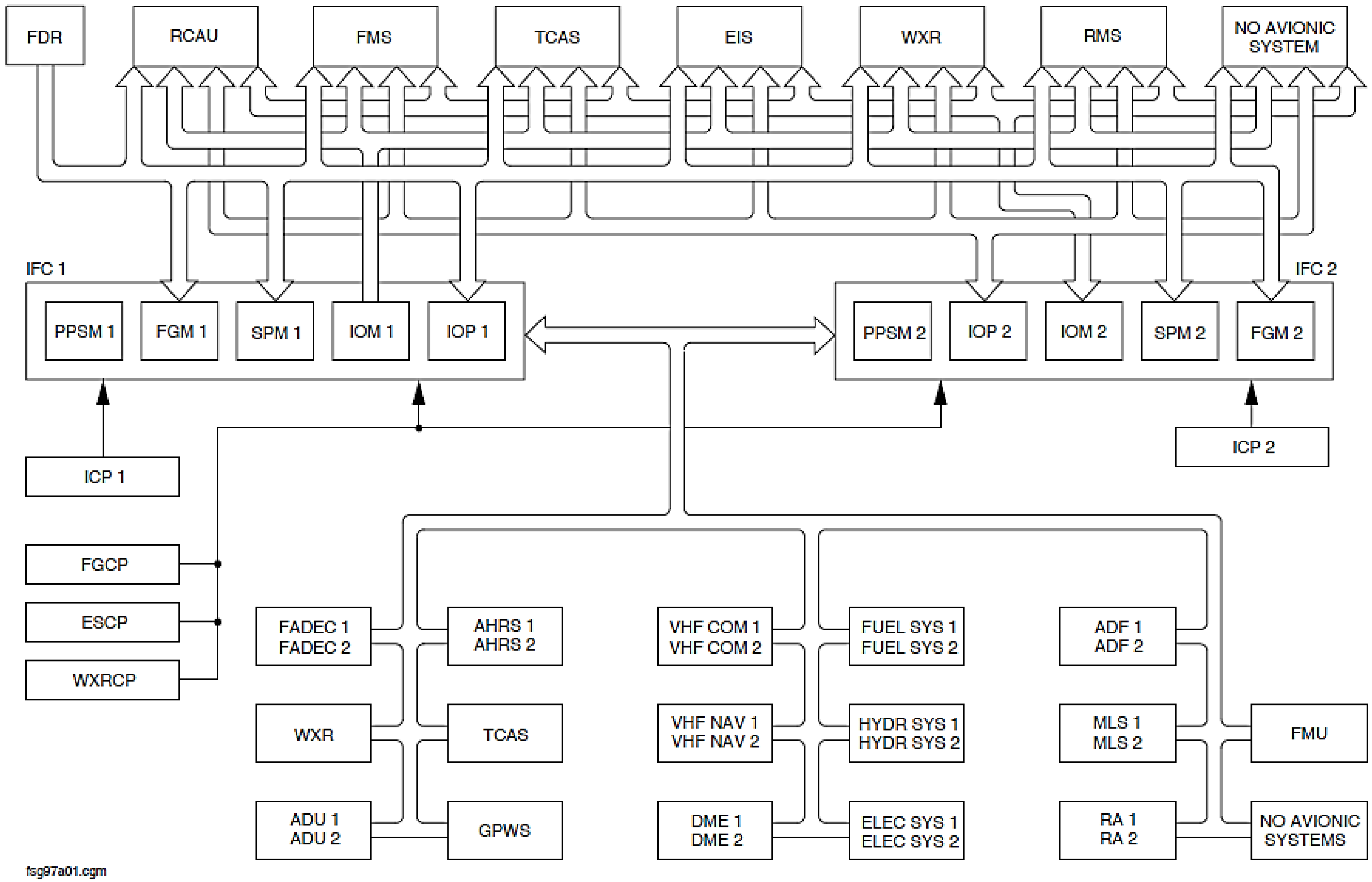


Figure - FLIGHT DATA PROCESS SYSTEM BLOCK DIAGRAM

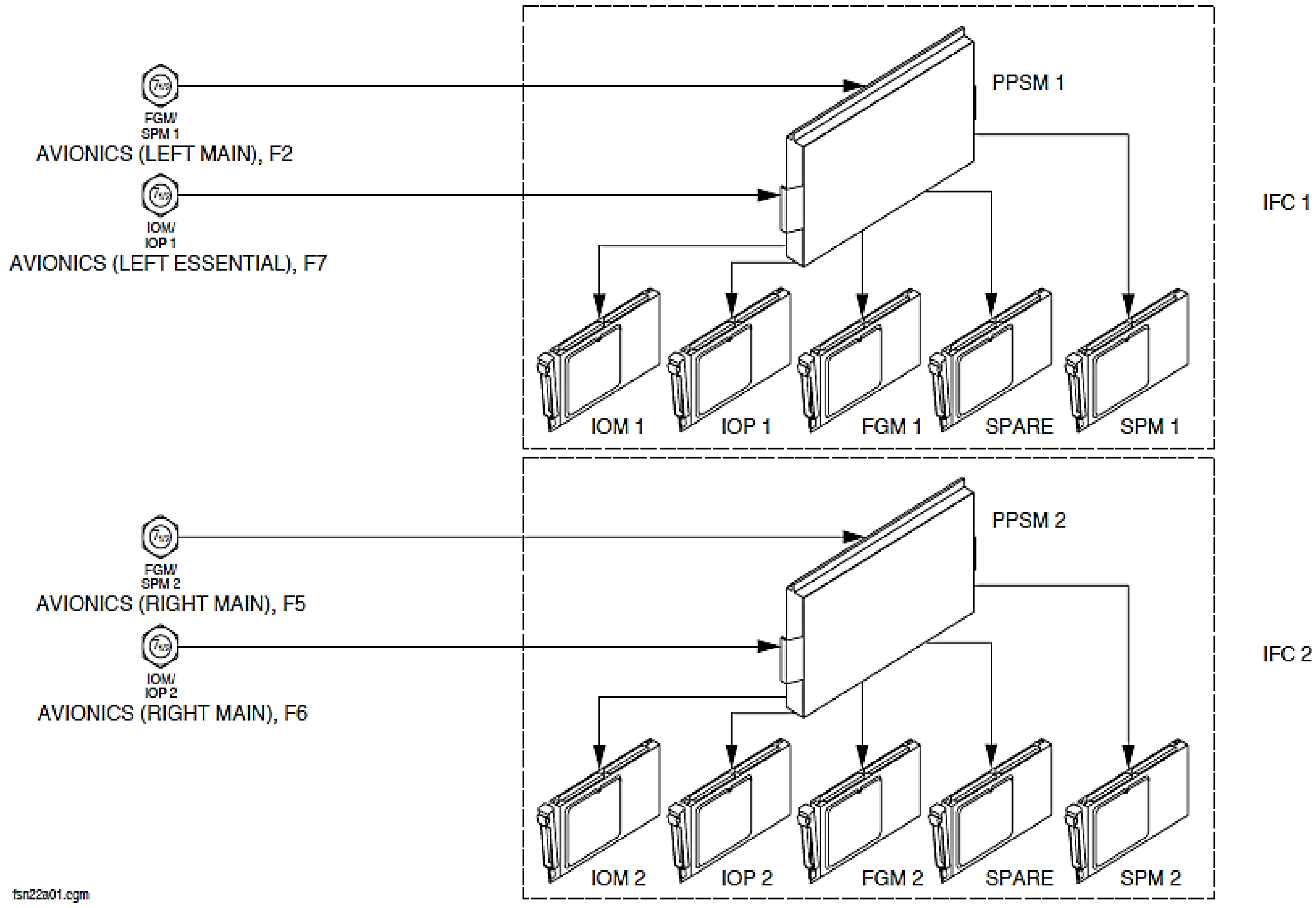


Figure - FLIGHT DATA PROCESS SYSTEM BLOCK DIAGRAM, POWER

Detailed Description

The Flight Data Processing System (FDPS 1, FDPS 2) functions in the modes that follow:

- Initialization (INIT)
- Power−On Self-Test (POST)
- Operational (OPER)
- Maintenance
- Teleloading.

Initialization (INIT) State: The FDPS functions in the initialization mode after an electrical power interruption. It initializes the hardware and software when it is on the ground.

If the FDPS does not receive a maintenance or teleloading request, it will then function in the Power−On Self-Test (POST) mode.

Power−On Self-Test (POST): The FDPS does a (POST) to make sure the hardware operates correctly before starting the operational mode.

The FDPS does a POST when the aircraft is on the ground and there is a long electrical power interruption that continues for more than 200 milliseconds.

The FDPS tests the interfaces that follow:

- Input/output Processor (IOP 1, IOP 2)
- Input/output Module (IOM 1, IOM 2)
- Ground Proximity Warning System Converter.

The result of the POST is stored in the Built In Test Equipment (BITE) and sent to the Central Diagnostic System (CDS).

The POST checks the parameters that follows:

- Program memory
- Data memory
- Watch dog
- Power supplies Hardware (HW) monitoring
- Module Validity HW and Software (SW) logic
- Memory partitioning HW monitoring
- Time partitioning HW monitoring.

If a parameter malfunctions, it will causes the FDPS to malfunction.

The POST also checks the parameters that follow:

- ARINC 429 inputs and outputs
- ARINC 422 inputs and outputs
- ARINC 717 inputs and outputs
- Discrete inputs and outputs
- IO1, IO2 board validity
- Analogue inputs and outputs
- Discrete inputs and outputs.

They do not cause a FDPS malfunction.

The POST sequence continues for 25 seconds.

Operational State: The FDPS does the functions that follow:

- Flight Data Concentrator (FDC)
- Data Hub Concentrator (DHC)
- Data Control (DCO)
- Mismatch calculations
- Electronic Instrument System (EIS) essential monitoring
- Advisory message generation
- Aircraft configuration management
- Warning Tone Generator (WTG 1, WTG 2).

Flight Data Concentrator (FDC): The FDC is located in IOP1. It receives data in different formats from the avionics and other aircraft systems and supplies digital data to the Solid State Flight Data Recorder (SSFDR) through an ARINC 573/717 bus.

Data Hub Concentrator (DHC): The FDPS receives noncritical data from avionics and other aircraft systems and supplies their parameters to other systems through a single output.

The FDPS supplies the concentrated parameters through ARINC 429 data buses to the avionics systems that follow:

- Electronic Instruments System (EIS)
- Flight Guidance Module (FGM 1, FGM 2)
- Stall Protection Module (SPM 1, SPM 2)
- Audio and Radio Control Display Units (ARCDU 1, ARCDU 2)
- Traffic Collision Avoidance System (TCAS)
- Ground Proximity Warning System (GPWS)
- Weather Radar (WXR)
- Flight Management System (FMS 1, FMS 2).

Two specific external non–avionics General Purpose Data Buses (GPDB 1, GPDB 2) are used to transmit data to systems that are not part of the avionics suite.

Data Control (DCO): The Data Control (DCO) makes calculations for the parameters before concentration.

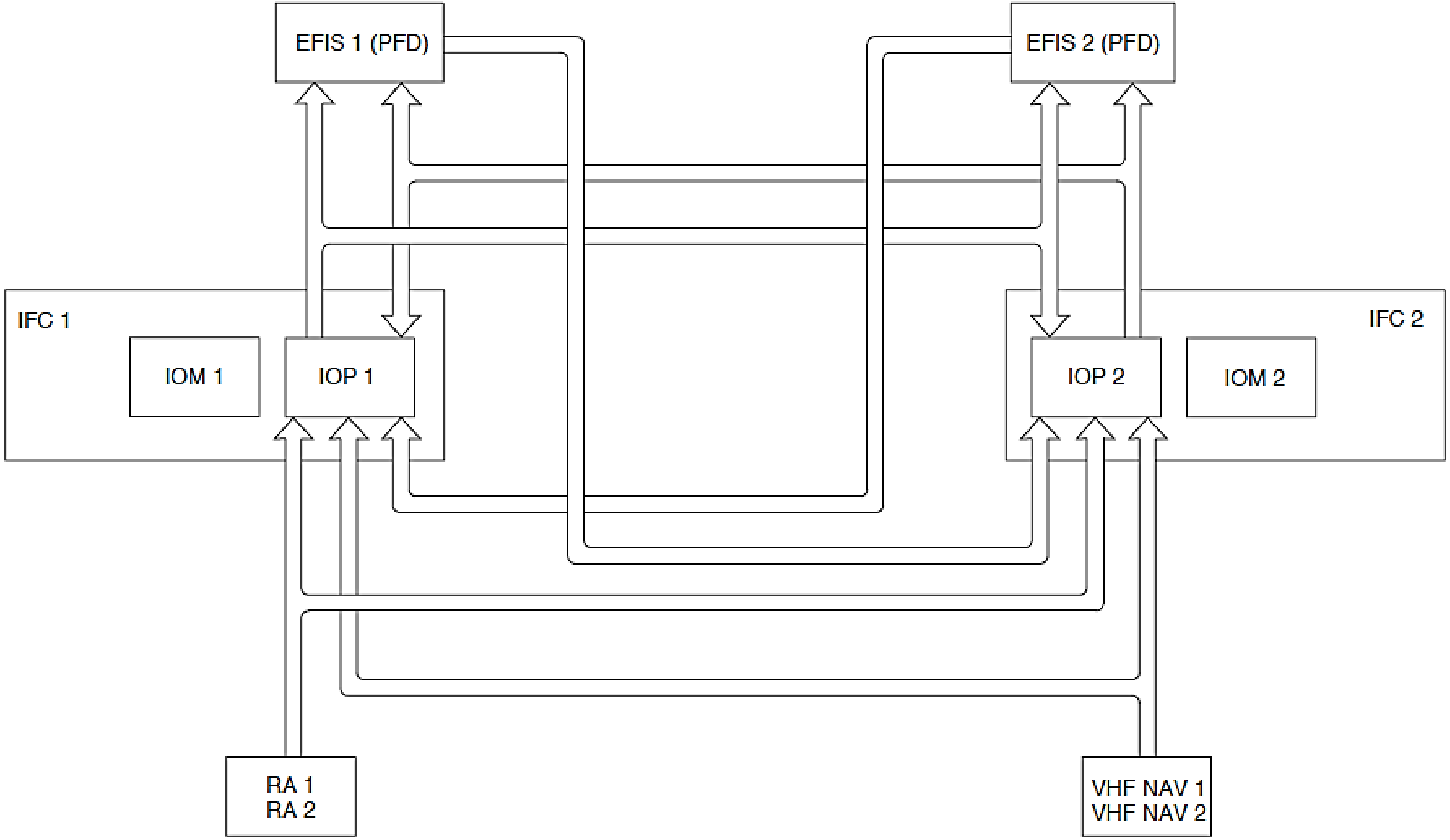


Figure - EIS ESSENTIAL MONITORING

Refer Figure - EIS ESSENTIAL MONITORING

Mismatch Calculations: The FDPS monitors different parameters. If an IOP senses a difference between its related value and the same the parameters it received from the other IOP, a mismatch message is shown by the Flight Mode Annunciators (FMA) in the opposite Primary Flight Displays (PFD).

The Flight Data Processing System (FDPS) Mismatch Messages that follow:

- PITCH MISMATCH
- HEADING MISMATCH
- IAS MISMATCH
- ALT MISMATCH
- RAD ALT MISMATCH
- GS MISMATCH
- LOC MISMATCH.

The calculations are done only when the parameters are valid. For localizer and glideslope mismatch messages, the two navigation receivers outputs must be valid and tuned to the same frequency.

Electronic Instrument System (EIS) essential monitoring: The FDPS does essential monitoring for the systems that follow:

- Radio Altimeter
- Localizer deviation
- Glideslope deviation.

The IOPs monitors the difference between a parameter received directly and same parameter received from the opposite PFD and IOP. When the difference is more than the predetermined value, the FDPS causes to the PFD to show a malfunction condition for that parameter.

Refer Figure - FDPS MESSAGE GENERATION

The essential monitoring parameter thresholds are shown in the table that follows:

Parameter	Difference
AGL altitude, less than 1000 ft	20 ft
LOC deviation, between 50 and 1000 ft AGL	2/3 of Expanded LOC Scale
G/S deviation, between 50 and 1000 ft AGL	1/6 of G/S Scale

Message Generation: The Message Generation System has the modes that follow:

- Advisory
- Flight Mode Annunciator (FMA).

Advisory: The advisory messages are shown in white letters on the Engine Display (ED). Each message has a location and is shown while the condition is present. There are two kinds of advisory messages as follows:

- First Family
- Second Family.

First Family: The first family messages relate to an applicable aircraft status, where crew awareness is required and an action may be necessary. These messages are located near its related indication.

Second Family: Second family messages relate to minor malfunctions and are located at the bottom of the ED. The messages relate to passive malfunctions do not affect a continued safe flight.

The FDPS causes the ED to show the advisory messages that follow:

- [BALANCE]
- INCR REF SPEEDS
- ICE DETECTED
- IFC messages
- Display messages.

NOTE

Note: The advisory messages are shown in order of decreasing importance. The most important IFC message is shown at the bottom left part of the ED and the most important display message is shown at the bottom right part of the ED.

[BALANCE] Message: The BALANCE message comes into view flashing for five seconds then stays on steady when a fuel imbalance is sensed by the left or right fuel gauging computers.

INCR REF SPEEDS Message: The INCR REF SPEEDS message comes into view when the Stall Protection System (SPS) stall sensing is changed for icing conditions. The FDPS sends a discrete data signal to Electronic Instruments System (EIS) from one or the other SPM.

ICE DETECTED Message: The ICE DETECTED message comes into view when one ice detector probe or the other senses ice accumulation.

IFC Messages: The most important IFC message is shown at the bottom left part of the ED. It shows the messages that follow:

- IOP1 FAIL, IOP2 FAIL, IOPS FAIL
- IOP BAD CONF
- IOM1 FAIL, IOM2 FAIL, IOMS FAIL
- WTG1 FAIL, WTG2 FAIL, WTGS FAIL
- GPWSC I/F FAIL
- WOW/IOP1 FAIL, WOW/IOP2 FAIL, WOW/IOPS FAIL
- IFC ACM1 FAIL, IFC ACM2 FAIL, IFC ACMS FAIL
- RA1 FAIL, RA2 FAIL, RAS FAIL.

IOP1 FAIL, IOP2 FAIL, IOPS FAIL: The IOP FAIL messages are shown when the IOP1 or IOP2 or the interface to the ED has malfunctioned.

IOP BAD CONF: The IOP BAD CONF message is shown when a bad aircraft configuration condition is sensed by one IOP or the other. The message comes into view when the aircraft is on the ground.

IOM1 FAIL, IOM2 FAIL, IOMS FAIL: The IOM FAIL messages are shown when IOM malfunctions are sensed.

WTG1 FAIL, WTG2 FAIL, WTGS FAIL: The WTG FAIL messages are shown when the WTG malfunctions. The message is shown when the aircraft is on the ground and engines are not running.

GPWSC I/F FAIL: The GPWSC I/F FAIL messages are shown when the IFC1 makes the GPWS inoperative when it malfunctions. The message is when the malfunction is sensed.

WOW/IOP1 FAIL, WOW/IOP2 FAIL, WOW/IOPS FAIL: The WOW/IOP FAIL messages are shown when the IOP senses a difference between the main and nose WOW signals from the PSEU. The IOP will not be able to do a POST after a power interruption. The message is shown when the aircraft is on the ground and engines are not running.

IFC ACM1 FAIL, IFC ACM2 FAIL, IFC ACMS FAIL: The IFC ACM FAIL messages are shown when a ACM malfunction is detected. A single ACM malfunction has no effect on the aircraft. When the two ACM malfunction, a bad DU BAD CONF messages is shown.

RA1 FAIL, RA2 FAIL, RAS FAIL: The RA FAIL message is shown when a dual Radar Altimeter system is installed and the RA malfunction is detected for more than ten seconds by the EIS ED. The message is displayed any time the malfunction is detected.

Display Messages: The most important display message is shown at the bottom right part of the ED. The FANS FAIL message is the only FDPS related display message indication. It comes into view when 2 or more avionics cooling fans do not operate. The FANS FAIL display message also comes into view when 2 or more avionics cooling fans do not operate and they are not inhibited by their related thermal switch while the aircraft is on the ground. The temperature switch 1 supplies data to IOP1 and temperature switch 2 supplies data to IOP2.

An IFC or display advisory message also causes the AVIONICS caution light to come on 2 minutes after the aircraft is on the ground and the air speed is less than 50 kts.

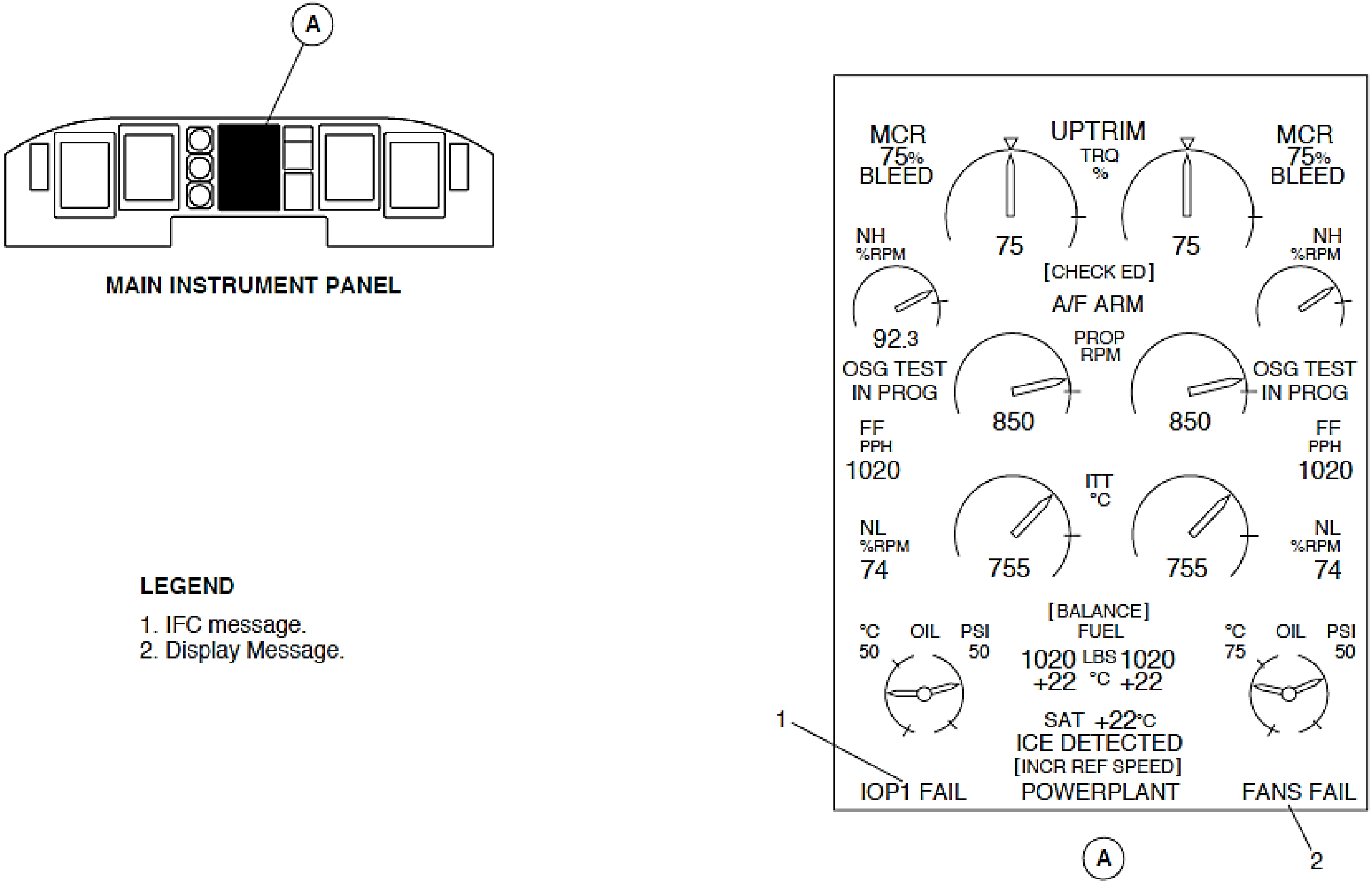


Figure - FDPS MESSAGE GENERATION

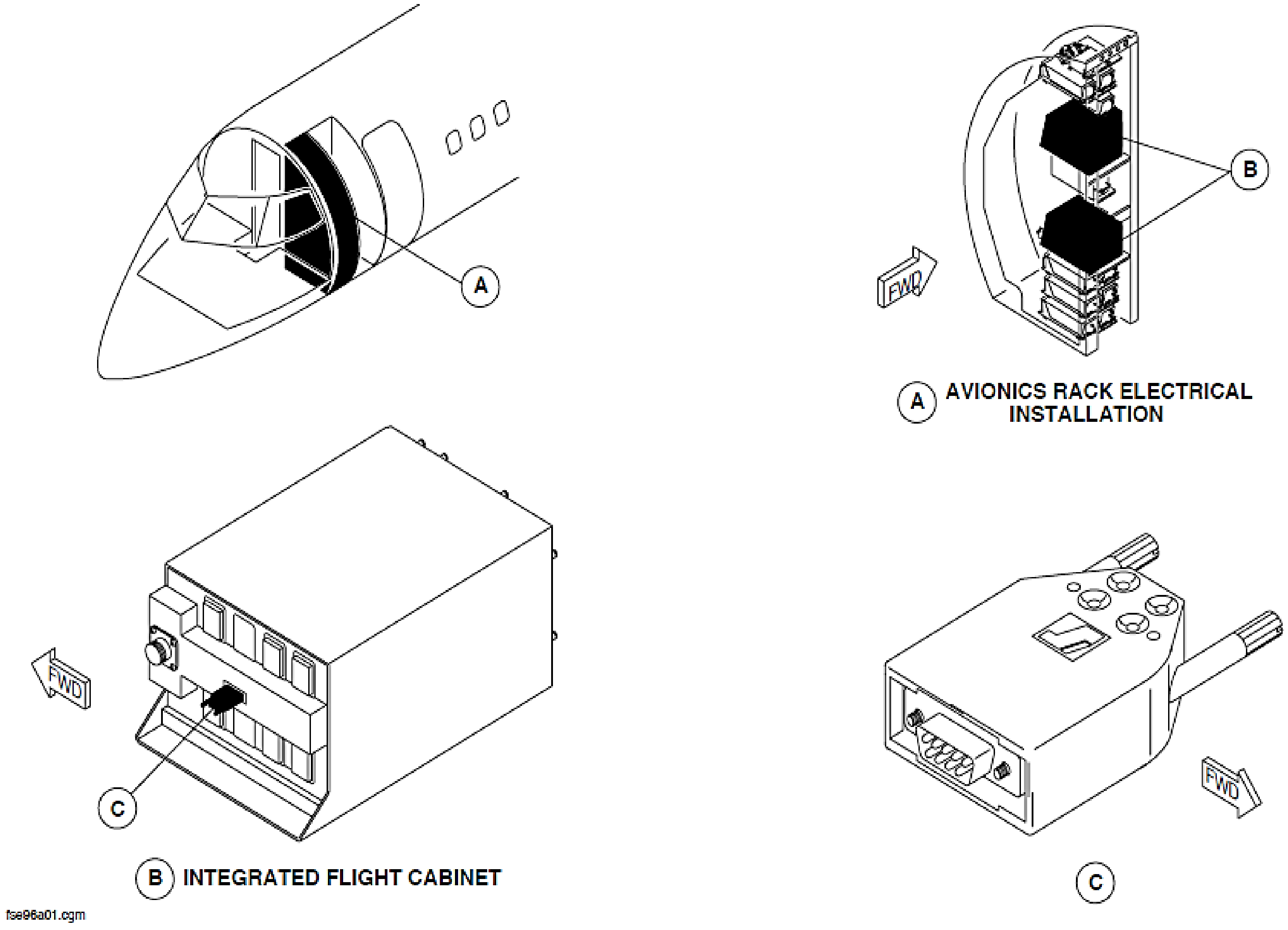


Figure - FDPS AIRCRAFT CONFIGURATION MODULES

Refer Figure - FDPS AIRCRAFT CONFIGURATION MODULES

Aircraft Configuration Management: The FDPS uses data that is stored in the Aircraft Configuration Modules (ACM 1, ACM 2). The ACMs are attached to the back−panel of its related IFC1. Each IOP supplies the configuration data to the systems that follow:

- Electronic Instrument System (EIS)
- Stall Protection System (SPS), Stall Protection Modules (SPM 1, SPM 2)
- Auto Flight Control System (AFCS), Flight Guidance Modules (FGM 1, FGM 2)
- Audio and Radio Control Display Units (ARCDU 1, ARCDU 2)

Each system monitors the configuration data. If a malfunction is sensed, it is stored in the Built in Test Equipment (BITE) and sent to the Central Diagnostics System (CDS) and the ED shows an IFC message.

The Aircraft Configuration Modules (ACM 1, ACM 2) are programmed using the Portable Multipurpose Access Terminal (PMAT).

Refer Figure - FDPS WARNING TONE GENERATORS (WTG1, WTG2)

Warning Tone Generators (WTG 1, WTG 2): The Warning Tone Generators (WTG 1, WTG 2) give different tones to tell the pilots of dangerous conditions or system malfunctions.

The tone that sounds depends on its priority. The most important tone always sound as follows:

PRIORITY	TONE	DESCRIPTION	SIGNAL TYPE	LEVEL, VOLTS
1	GPWS	Voice	N/A	N/A
2	TCAS, RA	Voice	N/A	N/A
3	Fire	Continuous chime until manually cancelled	A	4.8
4	Incorrect take off configuration	1000 Hz intermittent tone until cause of condition is removed	C	3.5
5	Autopilot disengage	250 Hz Intermittent tone for 1.5 seconds (manual) or until manually cancelled (automatic)	F	3.8
6	Pitch trim in motion	Continuous clicking until cause of condition is removed	D	4.2
7	Overspeed	1000 Hz intermittent tone until speed decreased	C	3.1

PRIORITY	TONE	DESCRIPTION	SIGNAL TYPE	LEVEL, VOLTS
8	Incorrect landing gear configuration	Continuous 800 Hz tone until cause of condition is removed	B	1.8
9	Altitude alert	2900 Hz tone for 1 seconds	B	2.9
10	Beta lockout warning	Continuous 1900 Hz to 2900 tone until cause of condition is removed	G	5.2
11	Warning annunciation	3 1000 Hz chimes	E	3.8
12	Caution annunciation	1 1000 Hz chime	E	5.0
13	TCAS, TA	Voice	N/A	N/A
14	SELCAL	1200 Hz tone for 3 seconds	N/A	2.7

Note

The WTGs generate the aural tones and control the priority of the voice sound from the GPWS and TCAS.

The WTG is not an independent system. It is part of the Input/output Modules' (IOM1, IOM2) function. One WTG supplies a warning tone to the Remote Control Audio Unit (RCAU) and other monitors it. The RCAU amplifies the signal and sends the tone to the flight compartment speakers and the pilots headsets.

GPWS: The GPWS is a system that makes its own synthesized voice sounds and connects directly to the Audio Integrating System (AIS).

When the WTG senses a GPWS audio on condition, it inhibits the other tones.

The voice sounds from the GPWS is inhibited if a GPWS malfunction is sensed while the aircraft is airborne. The GPWS audio on signal is monitored to prevent an inhibit of a different WTG tone caused by a partial GPWS malfunction. If the GPWS audio on signal stays on for more than 60 seconds, the GPWS's priority status is ignored. A GPWS and a TCAS or WTG tone is allowed to be heard at the same time.

The GPWS malfunction condition and that the TCAS and WTG continues to function is easily identified by the pilots. To be able to sense this malfunction condition, the FDPS does not inhibit the GPWS during the WTG test (ADC1 or ADC2 TEST toggle switch selection).

TCAS: The TCAS is a system that makes its own synthesized voice sounds and connects directly to the Audio Integrating System (AIS). It has two types of alerts with two different priority levels that follows:

- Resolution advisories
- Traffic advisories.

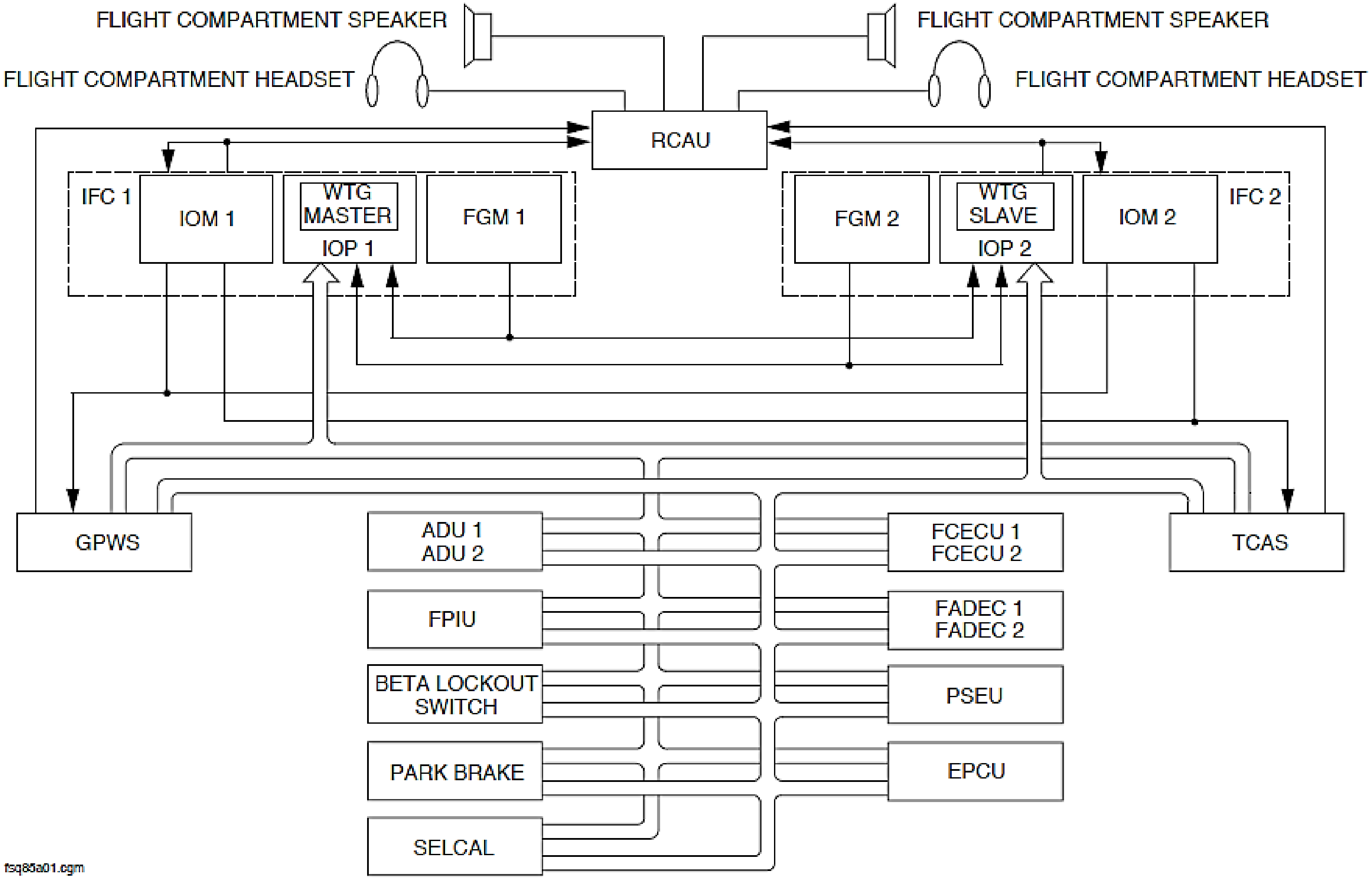


Figure - FDPS WARNING TONE GENERATORS (WTG1, WTG2)

The TCAS malfunction condition and that the GPWS and WTG continues to function is easily identified by the pilots. To be able to sense this malfunction condition, the FDPS does not inhibit the TCAS during the WTG test (ADC1 or ADC2 TEST toggle switch selection).

Engine Fire: The WTG inhibits the engine fire aural warnings when a more important tone sounds.

A continuous chime sounds when the WTG senses a fire bell discrete signal from the fire detection system.

Figure - FDPS Incorrect Take−Off Configuration Warning

Incorrect Take−Off Warning: The WTG inhibits the incorrect take−off warning indication when more a more important tone sounds.

Autopilot Disengagement: The WTG inhibits the autopilot disengagement tones when a more important tone sounds.

The WTG sounds an intermittent tone for 1.5 seconds for a manual AFCS disengagement. It sounds a 4000 Hz Intermittent tone when the Auto Flight Control System (AFCS) automatically disconnects until it is cancelled.

The two Flight Guidance Modules (FGM1, FGM2) supply Autopilot (AP) engagement/disengagement data through data buses to the WTGs.

The autopilot disengagement tones sound if one FGM or the other supplies a disengagement signal to the WTG.

Pitch Trim in Motion: The WTG inhibits the pitch trim in motion tones when a more important tone sounds.

The WTG sounds a clicking tone when the pitch trim is in motion for more than 3 seconds.

The two Flight Control Electronic Control Units (FC ECU1, FC ECU2) supply pitch trim in motion data through data buses to the WTGs.

The clicking tone sounds if one FCS ECU or the other supplies a pitch trim in motion signal to the WTG.

Overspeed Warning: The WTG inhibits the overspeed warning tones when a more important tone sounds.

The WTG sounds an intermittent 1000 Hz tone when the aircraft's speed is more than Maximum Velocity in Operation (VMO).

The two Air Data Units (ADU1, ADU2) supply overspeed data through data buses to the WTGs.

The overspeed tone sounds if one ADU or the other supplies an overspeed signal to the WTG.

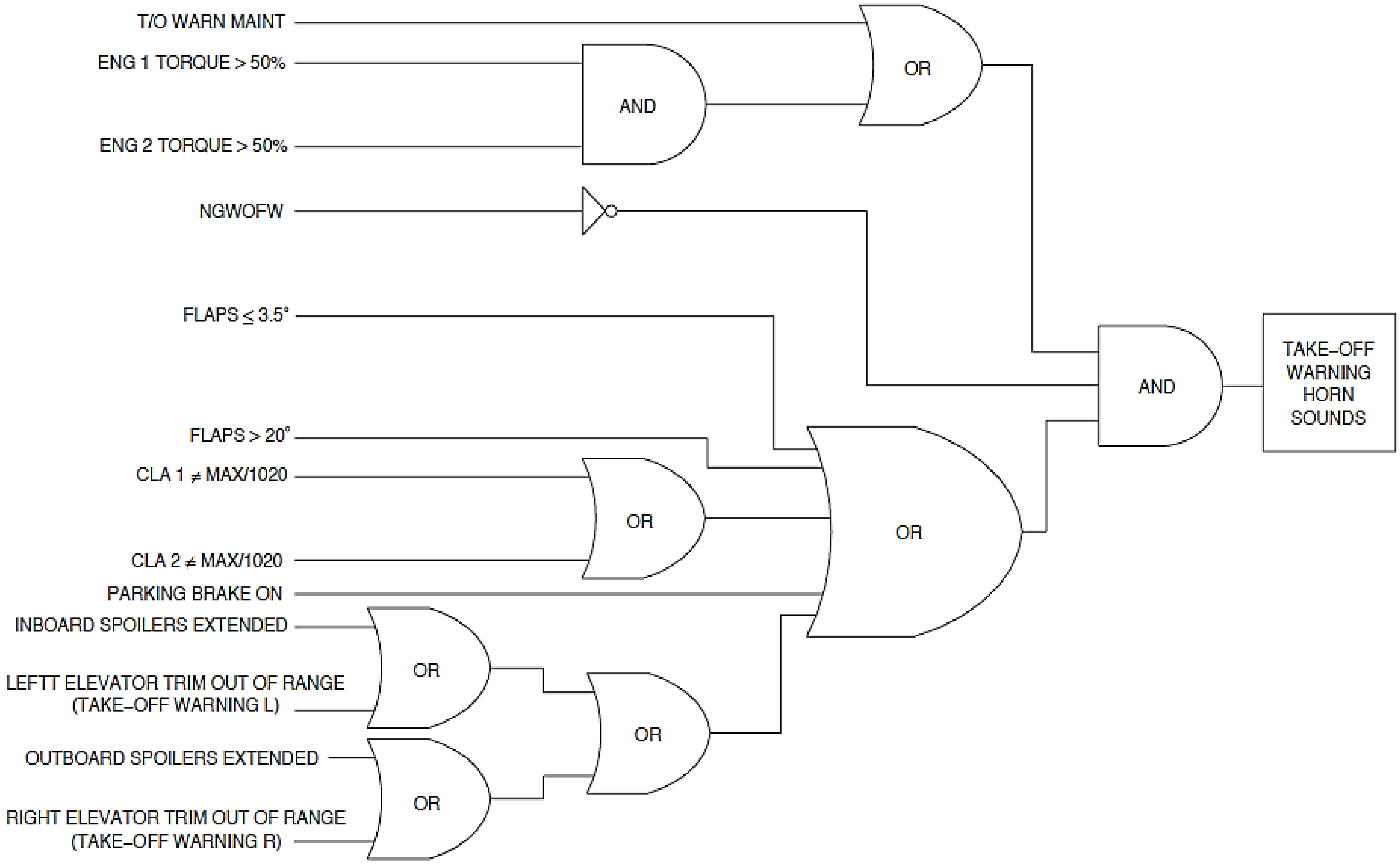


Figure - FDPS Incorrect Take–Off Configuration Warning

Refer Figure - FDPS INCORRECT LANDING GEAR CONFIGURATION

Incorrect Landing Gear Configuration: The WTG inhibits the incorrect landing gear configuration tones when a more important tone sounds.

The WTG sounds a continuous 800 Hz tone when the landing gear is not set to a safe for landing configuration.

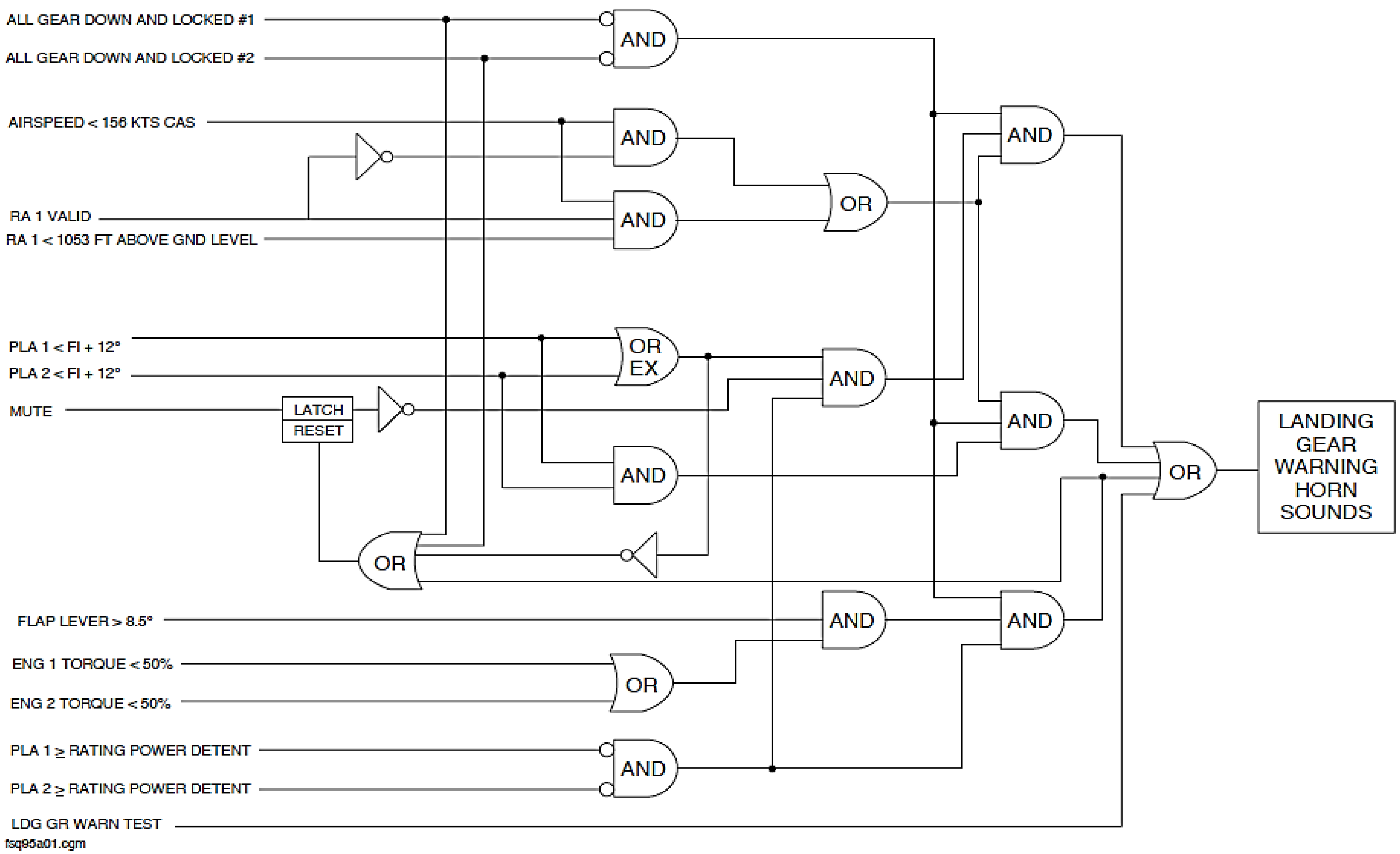


Figure - FDPS INCORRECT LANDING GEAR CONFIGURATION

Refer Figure - FDPS ALTITUDE ALERT

Altitude Alert: The WTG inhibits the altitude alert tones when a more important tone sounds.

The WTG sounds a continuous 2900 Hz tone for 1 second when the aircraft enters an area that is less than 1000 ft. (305 m) above or below a set altitude.

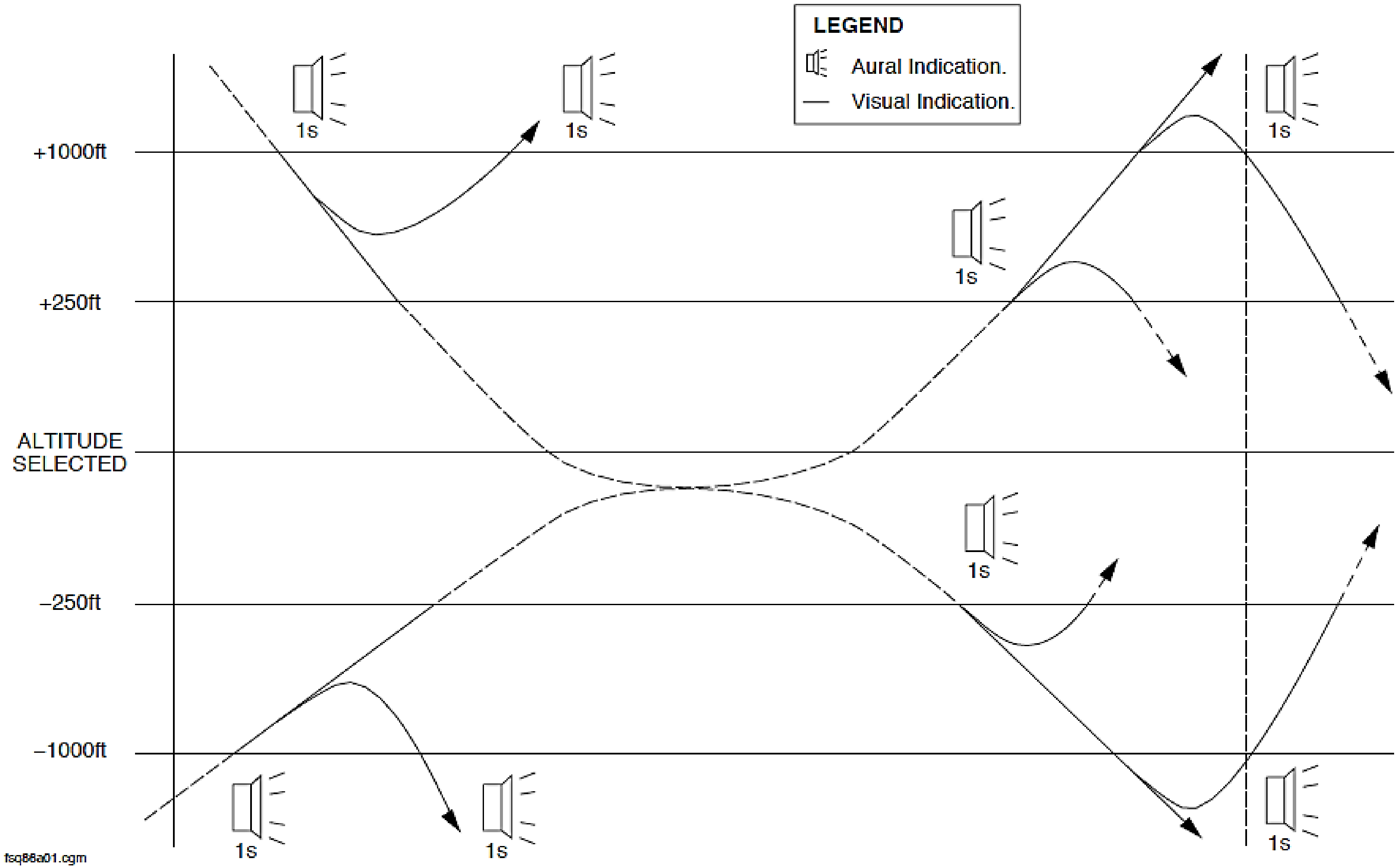


Figure - FDPS ALTITUDE ALERT

Refer Figure - FDPS BETA LOCKOUT WARNING

Beta lockout warning: The WTG inhibits the beta lockout alerts when a more important tone sounds.

The Warning Tone Generator (WTG 1, WTG 2) sounds a continuous 800 Hz tone when the Power Lever Angle (PLA) is set below the IDLE position while in flight. To give a beta lockout indication, the Flight Data Processing System (FDPS) senses the parameters that follow:

- Beta lockout switch position
- Power lever angle
- Main landing gear WOW.

Master Warning: The WTG inhibits the master warning tones when a more important tone sounds. The WTG sounds three chimes when one or the other red master warning light comes on.

Master Caution: The WTG inhibits the master caution tones when a more important tone sounds. The WTG sounds a single chime when one or the other amber master caution light comes on.

SELCAL: The WTG inhibits the SELCAL tone when a more important tone sounds. A continuous 1200 Hz tone sound for three seconds when the Selective Calling (SELCAL) system senses an incoming call.

There Warning Tone Generators (WTG 1, WTG 2). The WTG1 sounds the applicable tone when necessary while WTG2 functions in the standby mode. The WTG2 functions only when it senses that WTG1 has malfunctioned.

The WTG receives inputs from other systems to make it operate.

Each WTG monitors its output. If the WTGs calculate different tones, the system will use the calculations from WTG 1. Its calculations are more important than WTG 2.

The WTGs does the monitoring that follows:

- Software
- Hardware.

Software Monitor: Each Warning Tone Generator (WTG 1, WTG 2) uses their related inputs to calculate the tone logic. The tone calculations are compared and the WTG 1 supplies the applicable tone.

Hardware Monitor: Each Input/Output Module (IOM 1, IOM 2) has two Input/Output boards (IO1, IO2). The WTG 1 sends an analogue tone to the Remote Control Audio Unit (RCAU) through IO2. It routes the same signal to IO1 to make sure that the output tone is correct. If there is a difference, the WTG 1 stops functioning and makes itself invalid. Then, the WTG 2 functions when required.

The Warning Tone Generator (WTG 1, WTG 2) will remain off until the next power interruption when it is latched off because of a malfunction. The Power−On Self-Test (POST) will determine its validity.

A Warning Tone Generator (WTG 1, WTG 2) malfunction is stored in the Built in Test Equipment (BITE) and sent to the Central Diagnostic System. The Engine Display (ED) advisory message location will show a WTG FAIL message when a WTG malfunctions.

FDPS Abnormal Modes: Most aircraft and avionics system supplies data to other systems through the two FDPS's. Each aircraft or avionics system uses data from its related FDPS.

If its related FDPS malfunctions, the systems will automatically receive data from the other FDPS.

The Flight Data Recorder (FDR) receives data through the Input/Output Processor (IOP 1) from the two FDPS's.

If FDPS1 malfunctions, the Flight Data Recorder (FDR) function also malfunctions. If FDPS 2 malfunctions, the FDR will not record its parameters.

The data is supplied only to the FDPS1 from the systems that follow:

- Hydraulic quantity 1
- Hydraulic quantity 3
- Fuel inlet temperature 1
- Parking brake pressure
- Main Oil Pressure 1
- Ground Proximity Warning System Converter.

If FDPS1 malfunctions, its parameters are shown as dashes on the Engine Display (ED).

The data is supplied only to the FDPS2 from the systems that follow:

- Hydraulic quantity 2
- Fuel inlet temperature 2
- Traffic Collision Avoidance System (TCAS).

If FDPS2 malfunctions, its parameters are shown as dashes on the Engine Display (ED).

If one FDPS or the other malfunctions, the Engine Display (ED) will also show an IFC message in the advisory message area.

Maintenance Mode: The Built in Test Equipment (BITE) uses the Central Diagnosis System (CDS) to give the condition of the component. It stores faults in a Non-Volatile Memory (NVM) for reporting to line and shop maintenance.

The Built in Test Equipment (BITE) allows aircraft maintenance personnel to:

- Do fault isolation and return to service testing after completing maintenance actions
- Access failure reports from last or previous flight legs
- Get the avionics status report
- Get the part number of a given part

The Built in Test Equipment (BITE) modes monitors the condition of the component as follows:

- Power−On Self-Test (POST)
- Continuous Monitoring.

Power−On Self-Test (POST): The Power−On Self-Test (POST) checks the condition of the component at Power−Up or after a long power interruption.

Continuous Monitoring: The Continuous Monitoring checks the status of the component in flight. It records faults in a Non-Volatile Memory (NVM) for later troubleshooting using the Central Diagnosis System (CDS). A long power interruption causes the FDPS to start a Power−On Self-Test (POST) again.

Teleloading: The CDS is a communication connection between the FDPS and a Portable Multipurpose Access Terminal (PMAT). The PMAT connects to a Personal Computer (PC) to download a new software version when a software upgrade is necessary.

The CDS functions in the teleloading mode when the conditions that follow is correct:

- The Calibrated Air Speed (CAS) is less than 50 kts for more than 10 seconds
- The aircraft is on the ground
- The CDS GND MAINT toggle switch on maintenance panel is set
- The MAINT Key on the ARCDU is pushed.

The left essential bus supplies electrical power through a 10 A circuit breaker and the Prime Power Supply Module (PPSM1) to the Input/Output Processor (IOP1) and Input/Output Module (IOM1). The circuit breaker is located in position F7 on the avionics circuit breaker panel. The left main bus supplies electrical power through a 7.5 A circuit breaker and the PPSM1 to the Stall Protection Module (SPM1). The circuit breaker is located in position F2 on the avionics circuit breaker panel.

The right main bus supplies electrical power through a 10 A circuit breaker and the Prime Power Supply Module (PPSM 2) to the Input/Output Processor (IOP2) and Input/Output Module (IOM2). The circuit breaker is located in position F6 on the avionics circuit breaker panel. The right main bus supplies electrical power through a 7.5 A circuit breaker and the PPSM2 to the SPM2. The circuit breaker is located in position F5 on the avionics circuit breaker panel.

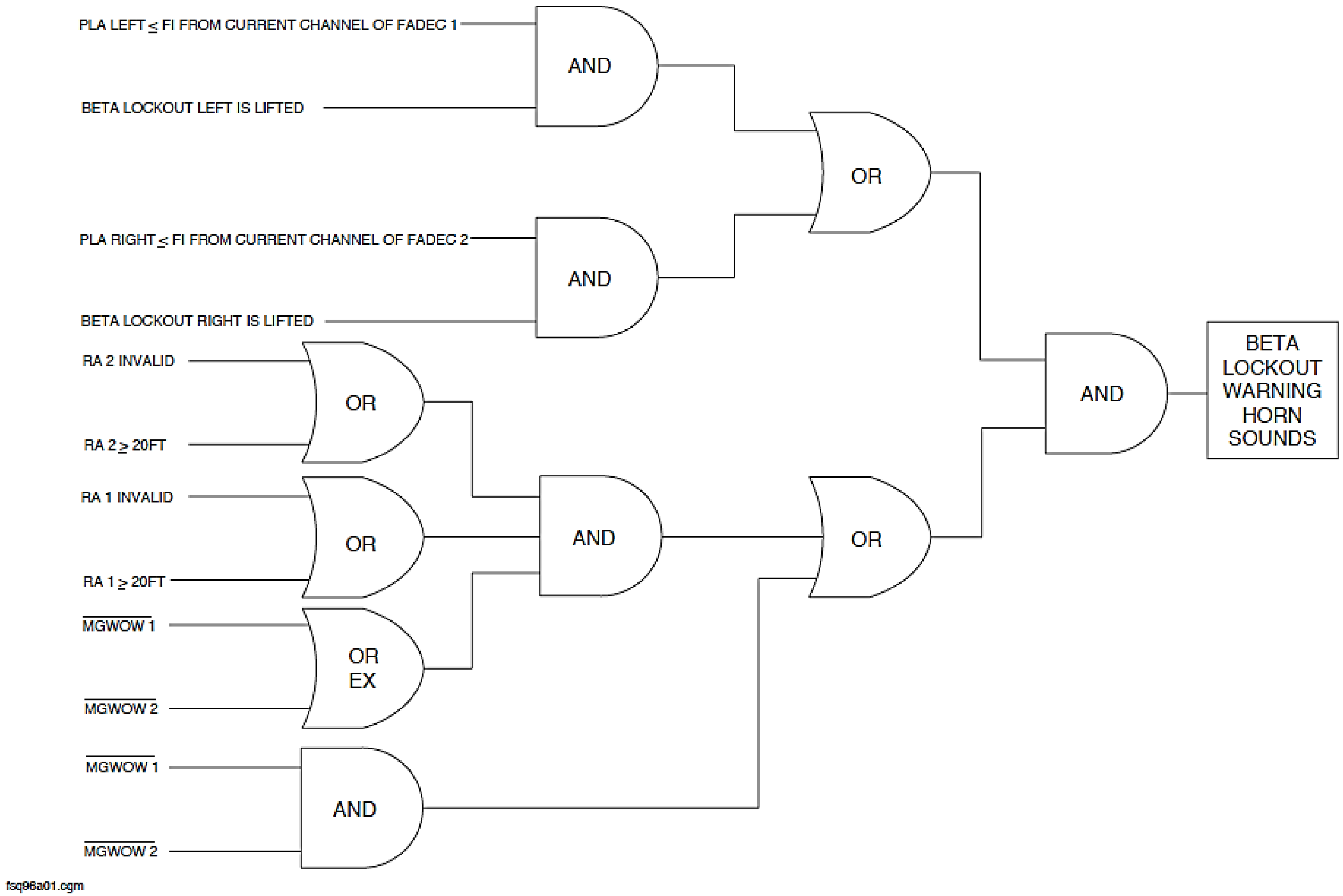


Figure - FDPS BETA LOCKOUT WARNING

Refer Figure - FDPS ARCDU CALCULATION AND CONCENTRATION

The FDPS receives data from the DME, ADF and MLS (option) systems and supplies it to the ARCDUs. It supplies the data as shown in the table that follows:

Parameter	Source
DME frequency	DME1, DME2
ADF frequencies	ADF1, ADF2
MLS channel	MLS1, MLS2
MLS selected azimuth angle	MLS1, MLS2
MLS maximum selected glidepath angle	MLS1, MLS2

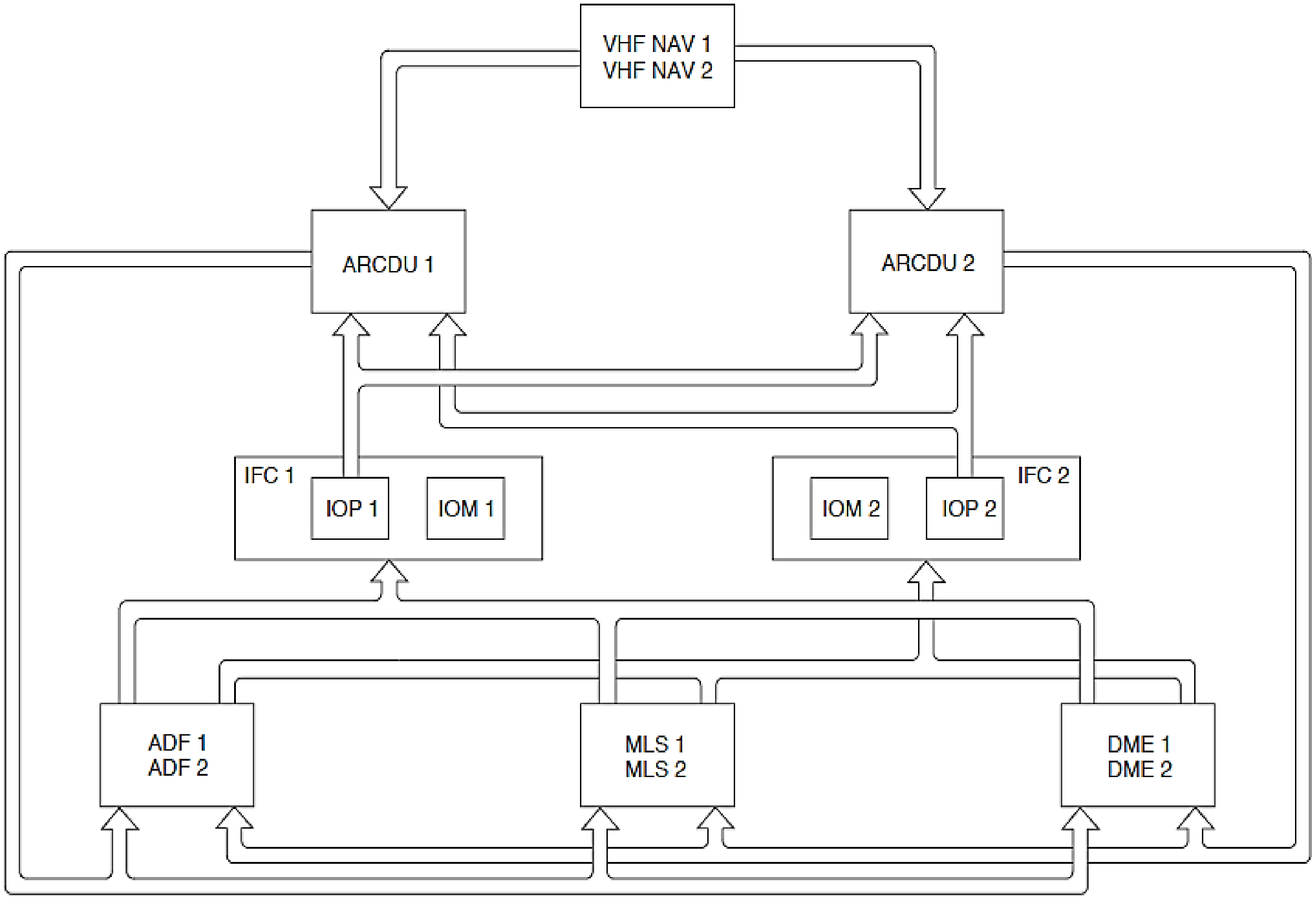


Figure - FDPS ARCDU CALCULATION AND CONCENTRATION

Refer Figure - FDPS EIS AND OPPOSITE IOP CONCENTRATION

The FDPS receives data from different non-important systems and supplies it to the EIS and the opposite IOP. It supplies the data as shown in the table that follows:

Parameter	Source
Selected Heading	FGCP
Selected Course	FGCP
Navigation source	FGCP
Selected DH	ICP1, ICP2
Speed Bugs	ICP1, ICP2
Rudder position	FCSECU1, FCSECU2
Elevator position	FCSECU1, FCSECU2
Spoiler position	FCSECU1, FCSECU2
MLS channel	MLS1, MLS2
MLS AZ/EL deviation	MLS1, MLS2
Radio Altitude	RA1, RA2
Fuel Quantity	FGC / FQC
Fuel discretes	FGC / FQC
Fuel Temperature	Fuel Temperature Sensor
Fuel Flow	From Fuel Flow Sensors
VOR/ILS frequencies	VHFNAV1, VHFNAV2
LOC/GS deviation	VHFNAV1, VHFNAV2
VOR bearing	VHFNAV1, VHFNAV2
Hydraulics Pressure	Hydraulics Sensor
Hydraulics Quantity	Hydraulics Sensor
Parking brake hydraulic pressure	Hydraulics Sensor
Air/Ground status	PSEU

Doors Status	PSEU
Main Oil Pressure	FADEC1, FADEC2
Engine torques	FADEC1, FADEC2
Flap position	FPIU
Static Air Temperature	ADU1, ADU2
ED Brightness	ESCP
System page status	ESCP
Reversion status	ESCP
APU generator volt	EPGDS
AC Bus Voltage	EPGDS
AC Bus Load	EPGDS
Generator Load	EPGDS
TRU Load	EPGDS
Secondary Bus Voltage	EPGDS
Battery Load	EPGDS
Essential Bus Voltage	EPGDS

Main Bus Voltage	EPGDS
Battery Temperature	EPGDS
External Power Discrete	EPGDS
frequency	DME1, DME2
Distance	DME1, DME2
Time To Station	DME1, DME2
Ground Speed	DME1, DME2
Display Control	TCAS
Vertical Resolution	TCAS
Output Sense Level and Reply Data	TCAS
Intruder range, altitude, bearing	TCAS
Failure status	TCAS
frequency	ADF1, ADF2
Bearing	ADF1, ADF2

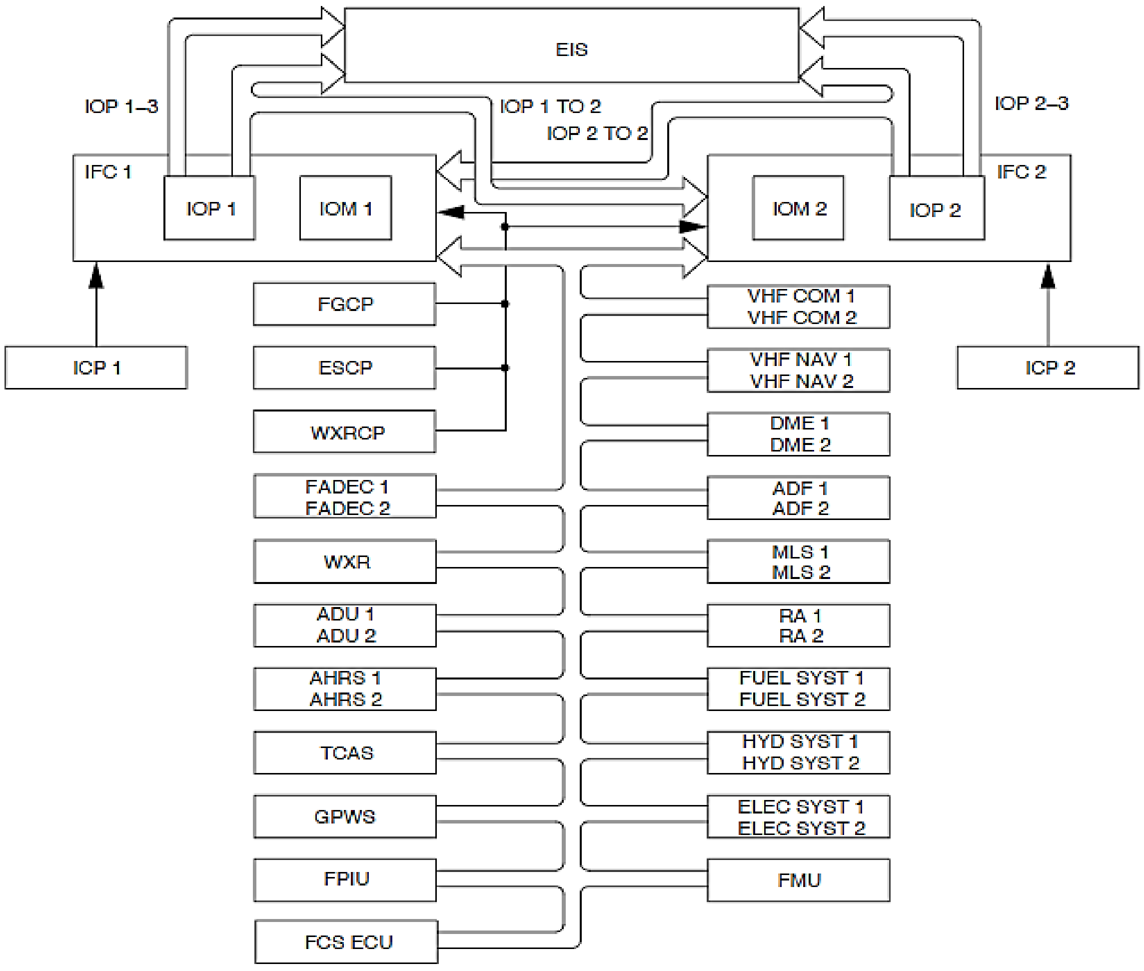


Figure - FDPS EIS AND OPPOSITE IOP CONCENTRATION

Refer Figure - FDPS TCAS AND WXR CALCULATION AND CONCENTRATION

The FDPS receives data from the Attitude and Heading Reference System (AHRS) and the EFIS Control Panels through the EIS and supplies it to the weather radar system. It supplies the data as shown in the table that follows:

Parameter	Source
Roll	AHRS1, AHRS2
Pitch	AHRS1, AHRS2
Magnetic Heading	AHRS1, AHRS2
WXR Range	ESCP

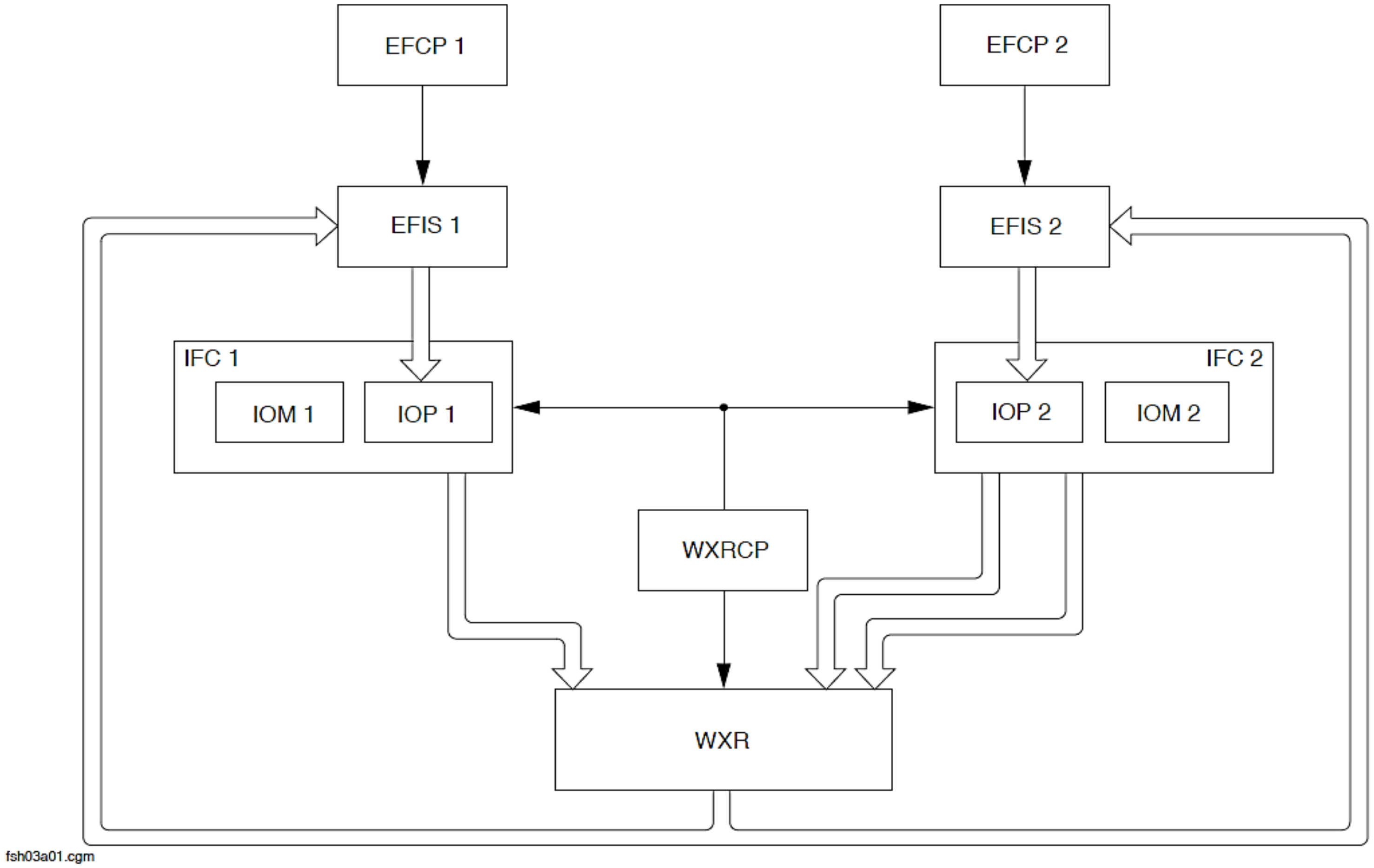


Figure - FDPS TCAS AND WXR CALCULATION AND CONCENTRATION

Refer Figure - FDPS GENERAL PURPOSE DATA BUS (GPDB) CONCENTRATION

The FDPS receives data from different systems and supplies it to other non-avionic systems through a General Purpose Data bus. It supplies the data as shown in the table that follows:

Parameter	Source
VHFCOM Frequency	VHFCOM1, VHFCOM2
ADF Frequency	ADF1, ADF2
Selected Heading	FGCP
Selected Altitude	FGCP
Standard Pressure Altitude	ADU1, ADU2
Barometric Corrected Altitude	ADU1, ADU2
Pitch	AHRS1, AHRS2
Magnetic Heading	AHRS1, AHRS2
Mach	ADU1, ADU2
Calibrated airspeed	ADU1, ADU2
VMO	ADU1, ADU2
True airspeed	ADU1, ADU2
Total Air Temperature	ADU1, ADU2
Altitude Rate	ADU1, ADU2
Static Air Temperature	ADU1, ADU2
Hydraulic pressure	Hydraulic Sensor
HP compress Exit Pres (P3.0)	FADEC1, FADEC2
Propeller speed	FADEC1, FADEC2
Fuel flow sensor	FADEC1, FADEC2
Vertical Speed	AHRS1, AHRS2
Roll	AHRS1, AHRS2

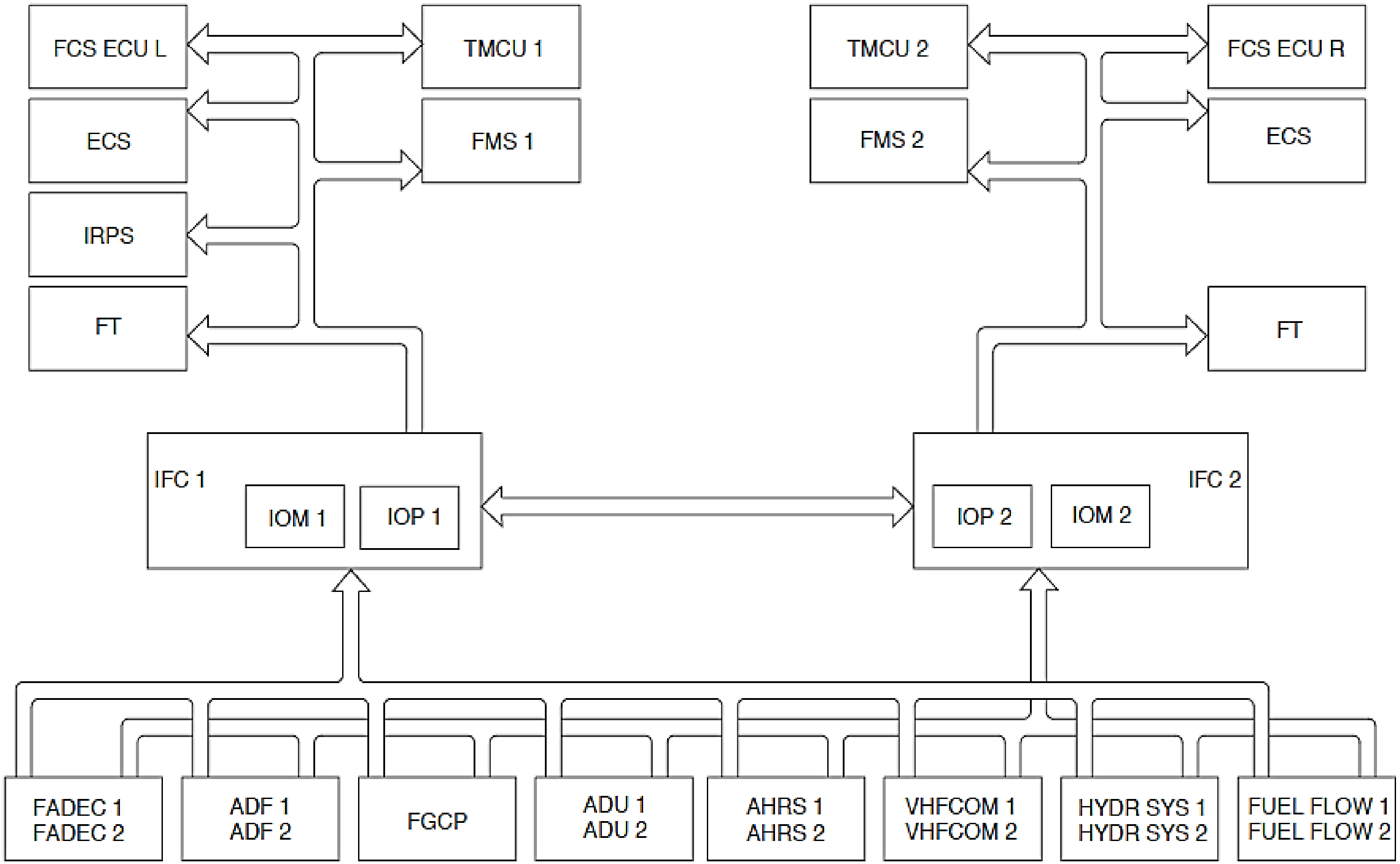


Figure - FDPS GENERAL PURPOSE DATA BUS (GPDB) CONCENTRATION

Refer Figure - FDPS DISPLAY UNIT CALCULATION AND CONCENTRATION

The FDPS receives a valid discrete from each EIS indication and sends it to the DUs.

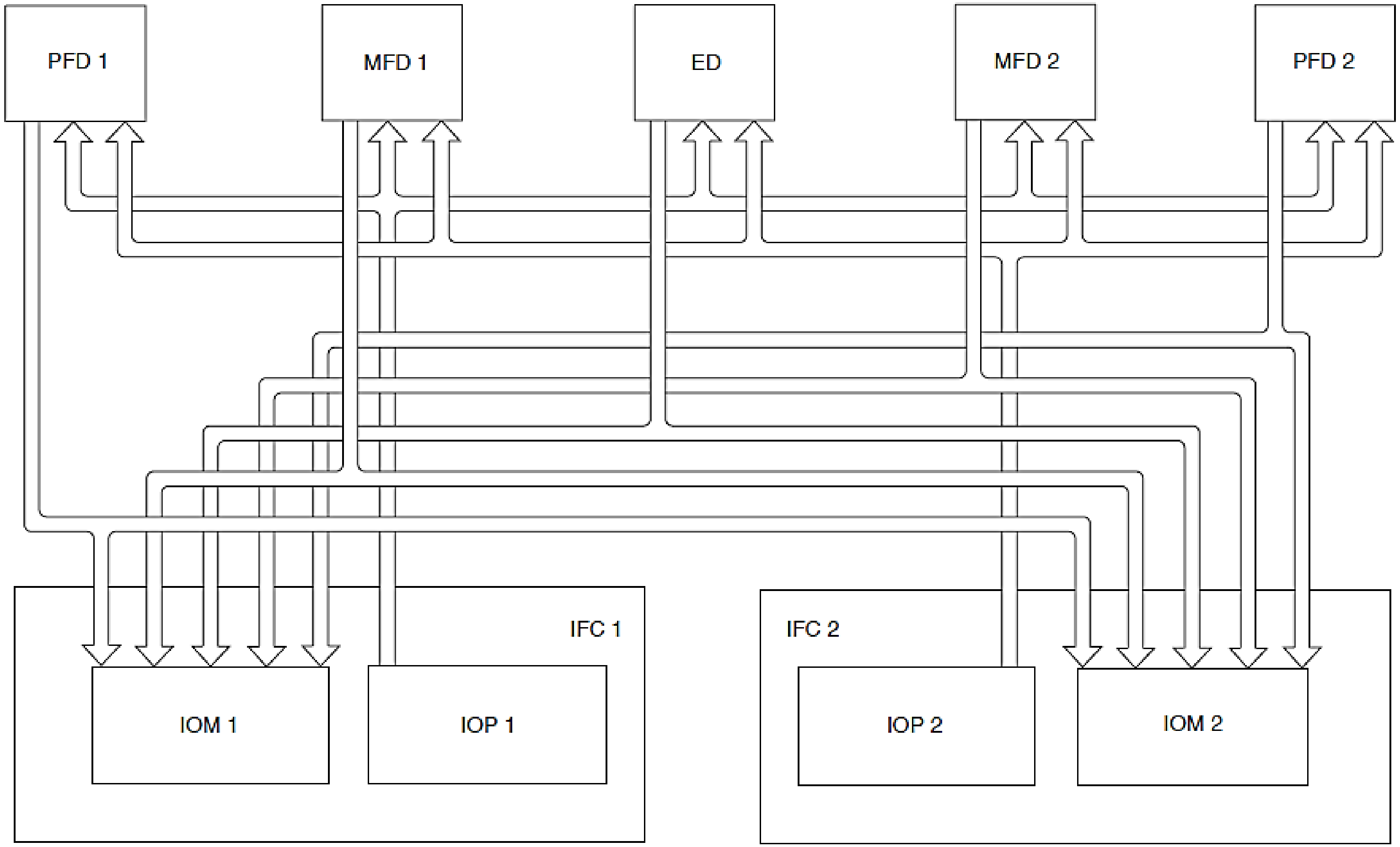


Figure - FDPS DISPLAY UNIT CALCULATION AND CONCENTRATION

Refer Figure - FDPS ICP CALCULATION

The FDPS1 calculates the data from Index Control Panel 1 (ICP1) and FDPS2 calculates data from ICP2. The ICPs supply the data to the FDPS's as shown in the table that follows:

Parameter	Source
Speed Bug Selector	ICP1, ICP2
Speed Bug Index Setting	ICP1, ICP2
Decision Height	ICP1, ICP2

The ICP1 barometric altitude setting is calculated by the ADC1 and ICP2 barometric altitude setting is calculated by the ADC2.

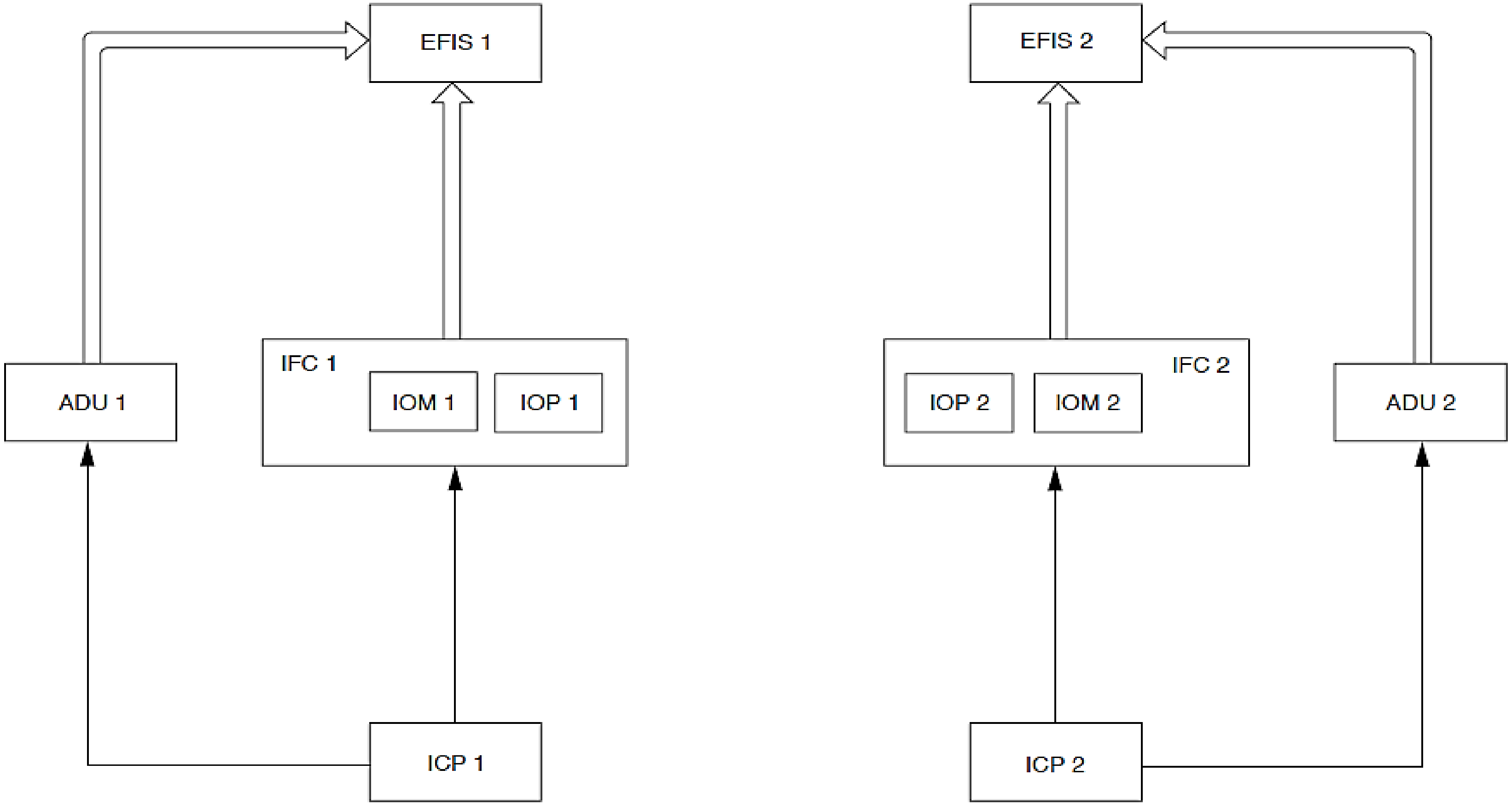


Figure - FDPS ICP CALCULATION

Refer Figure - FDPS FLIGHT GUIDANCE CONTROL PANEL (FGCP) CALCULATION

The FDPS calculates data from the FGCP and supplies it to the EIS, FGM and opposite IOP that follows:

Parameter	Source
Heading Selection	FGCP
Course Selection	FGCP
Attitude selection	FGCP
HSI SEL	FGCP
Navigation Source	FGCP

The IOP, when valid, calculates the parameters from the FGCP and transmits its result through IOP2 to the EIS.

If the data connection between IOP1 and the FGCP malfunctions, IOP1 also calculates parameters from the FGCP through IOP2.

If IOP1 malfunctions, the FDPS continues to operate with calculations from IOP2 and with parameters previously calculated by IOP1.

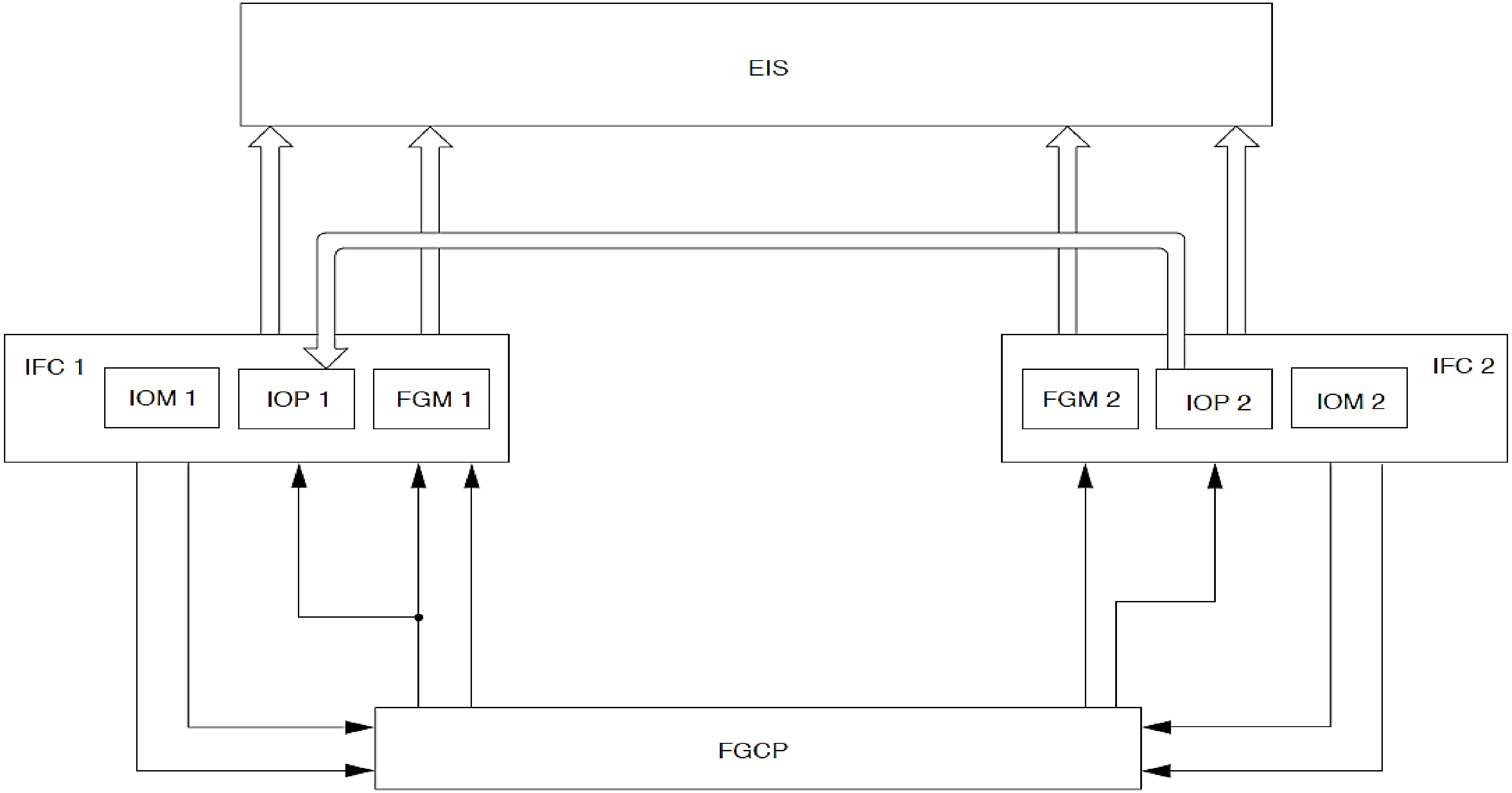


Figure - FDPS FLIGHT GUIDANCE CONTROL PANEL (FGCP) CALCULATION

Refer Figure - FDPS FGCP MONITORING

The FDPS1 calculates with IOP1 with data from FGCP and sends the result to FDPS2. The FDPS sends the data to the EIS. When the link malfunctions between IOP1 and the FGCP, the FDPS1 will calculate data from the FGCP through FDPS2.

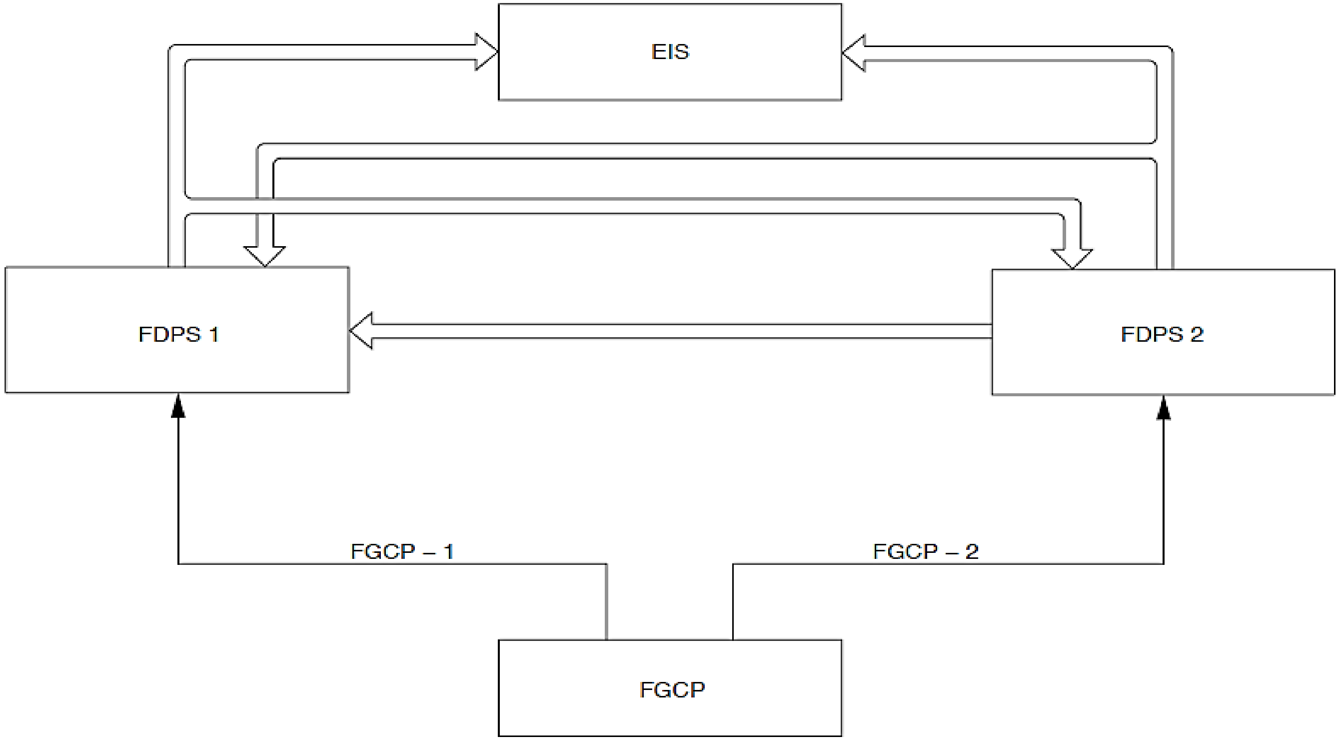


Figure - FDPS FGCP MONITORING

Refer Figure - FDPS ESID CONTROL PANEL (ESCP) CALCULATION

The FDPS1 calculates the data from the ESID Control Panel (ESCP). It supplies the data to the FDPS's as shown in the table that follows:

Parameter	Source
ED Brightness	ESCP
MFD1 Reversion	ESCP
MFD2 Reversion	ESCP
ATT/HDG Reversion	ESCP
ADU Reversion	ESCP

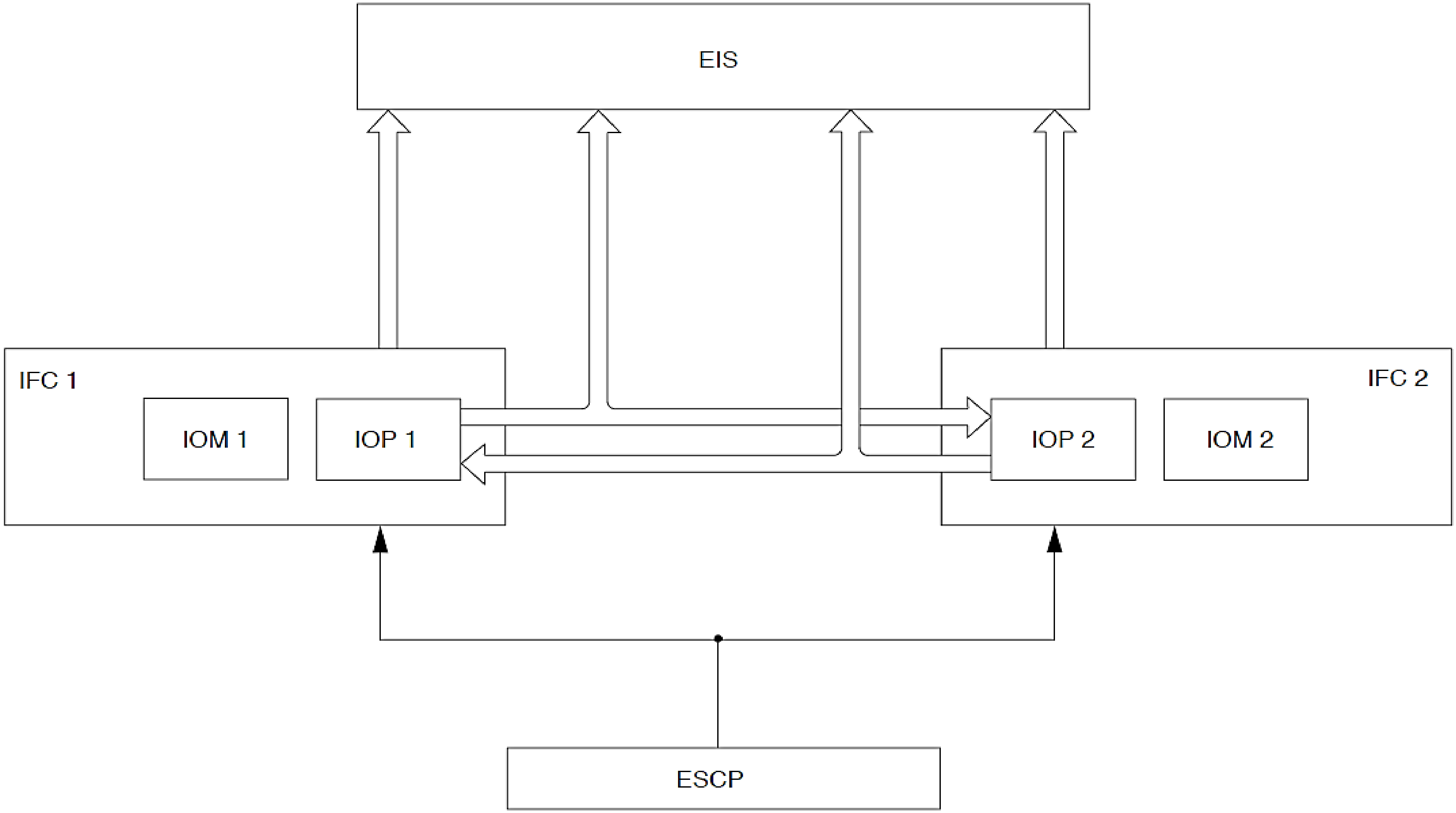


Figure - FDPS ESID CONTROL PANEL (ESCP) CALCULATION

Refer Figure - FDPS MLS/NAV TUNE SELECTION CALCULATION

The DME is tuned by the MLS or VHF NAV. The FDPS moves an external relay contact to select a navigation source, MLS or VHF NAV.

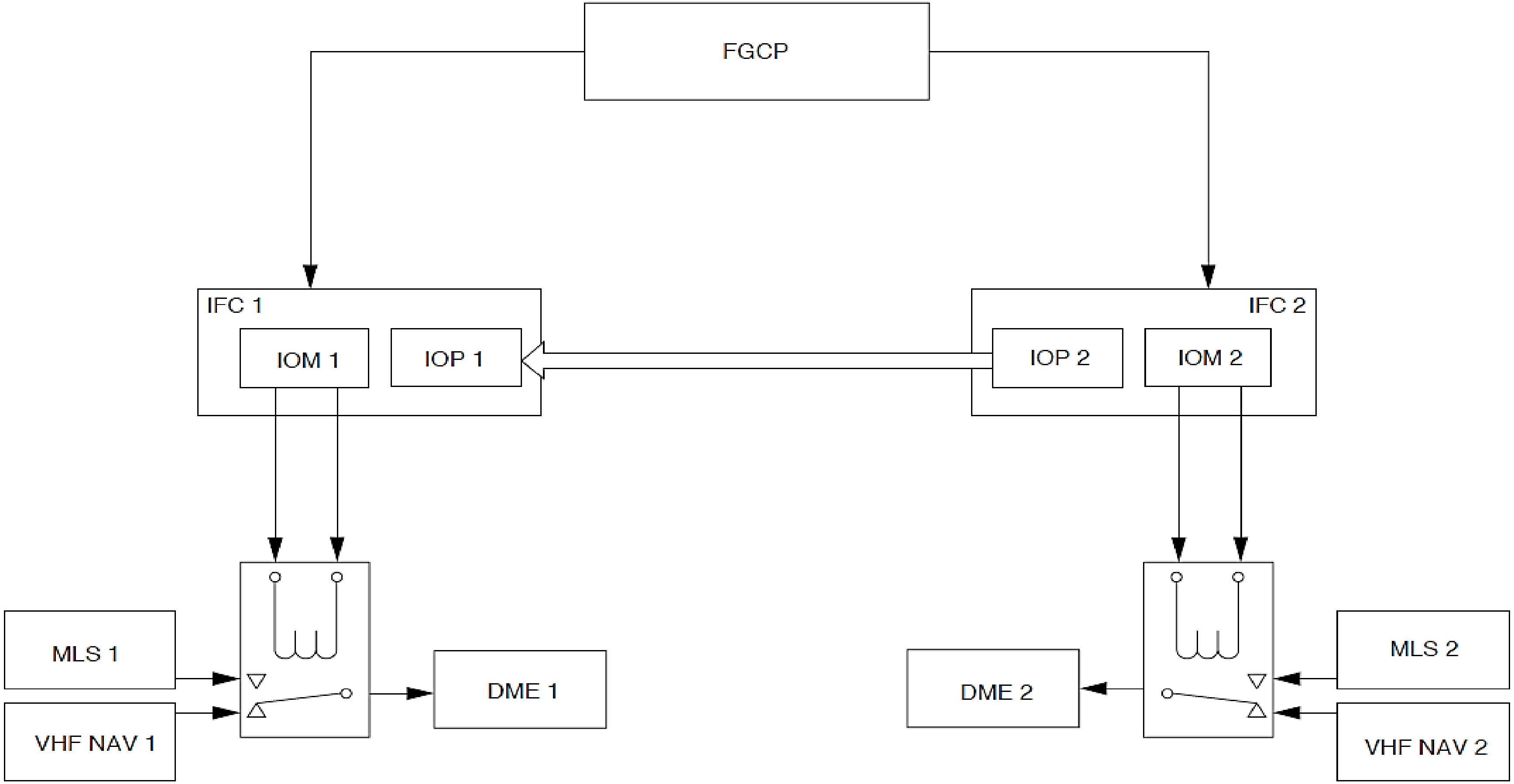


Figure - FDPS MLS/NAV TUNE SELECTION CALCULATION

Refer Figure - VOR/MLS AUDIO SELECTION CALCULATION

The ARCDUs control the VHF NAV or MLS audio level. The IOMS sends a discrete signal to control the Remote Control Audio Unit (RCAU) navigation source audio.

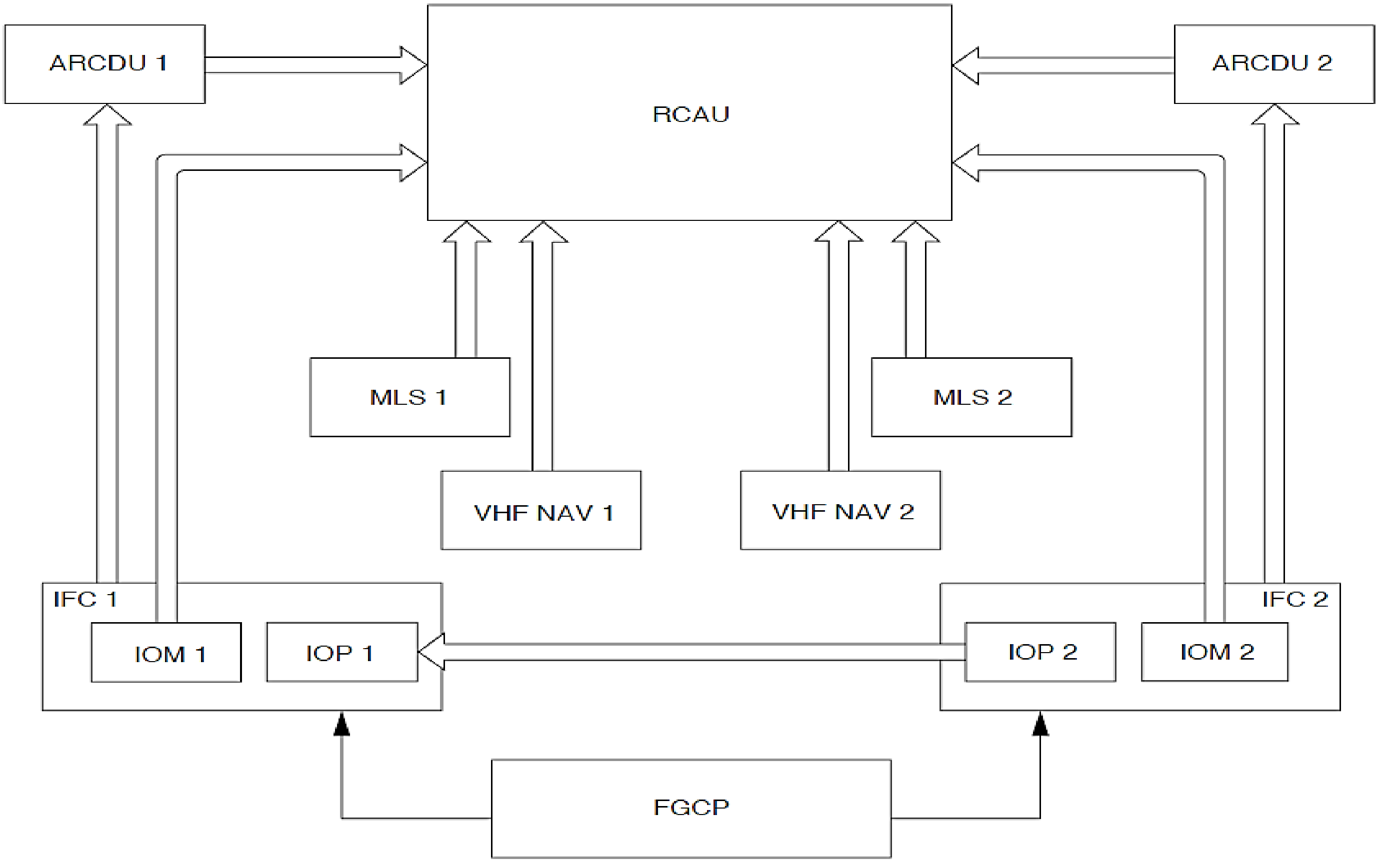


Figure - VOR/MLS AUDIO SELECTION CALCULATION

Refer Figure - FDPS ENGINES OPERATING/STOPPED CALCULATION

The FDPS uses data from the FADEC oil pressure discrete signal to change the Passenger Address Amplifier (PAA) gain. The FDPS supplies a discrete signal to the PAA when oil pressure from the two engines are less than their threshold. This discrete signal causes the PAA to reduce its PA gain by 6 dB when the two engine are not running.

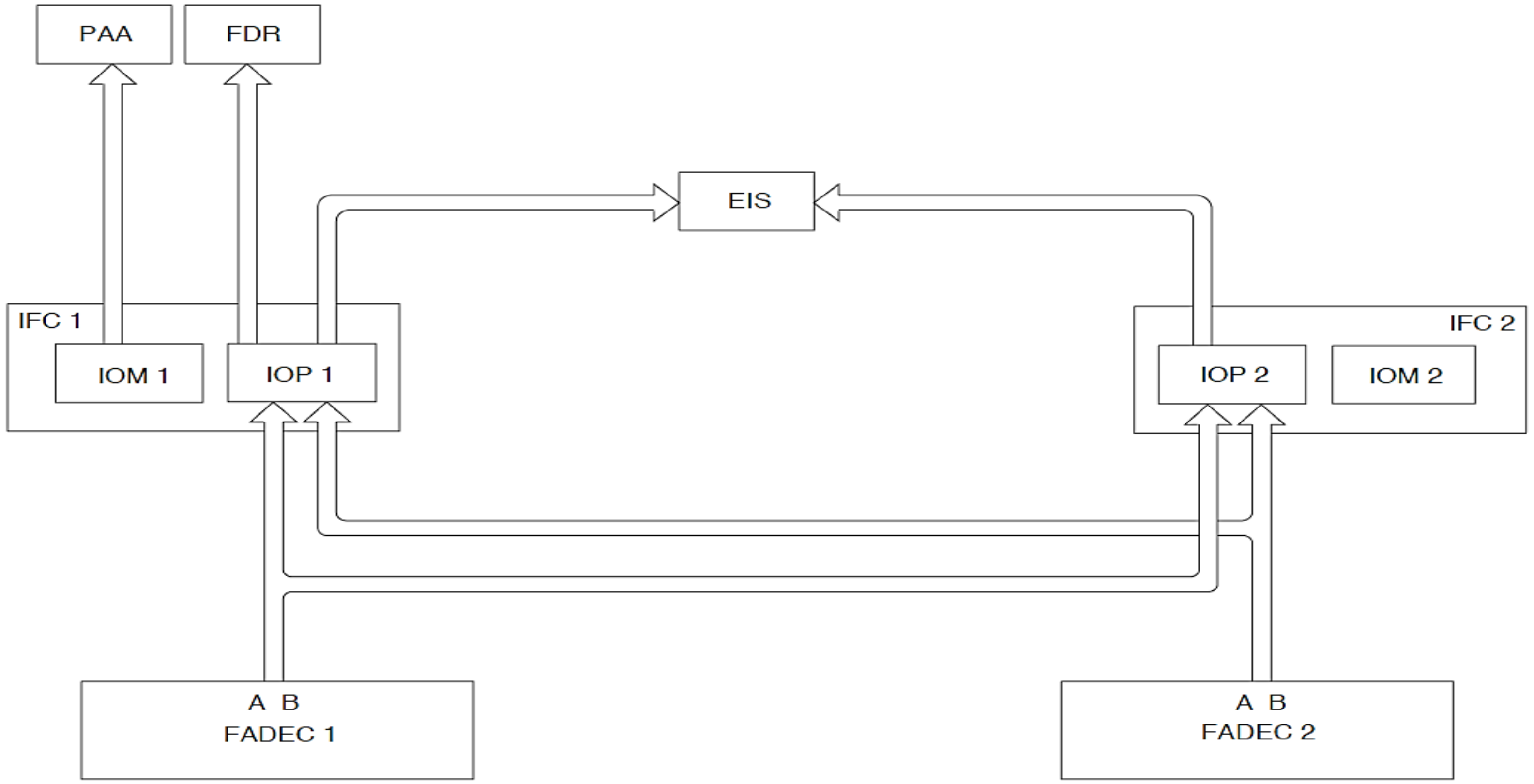


Figure - FDPS ENGINES OPERATING/STOPPED CALCULATION

Refer Figure - FDPS TCAS MANEUVER INHIBIT CALCULATION
Refer Figure - FDPS FLAP POSITION CALCULATION

The FDPS calculates data from the PSEU, PEC and FPIU to inhibit the climb and increase climb signals to the TCAS.

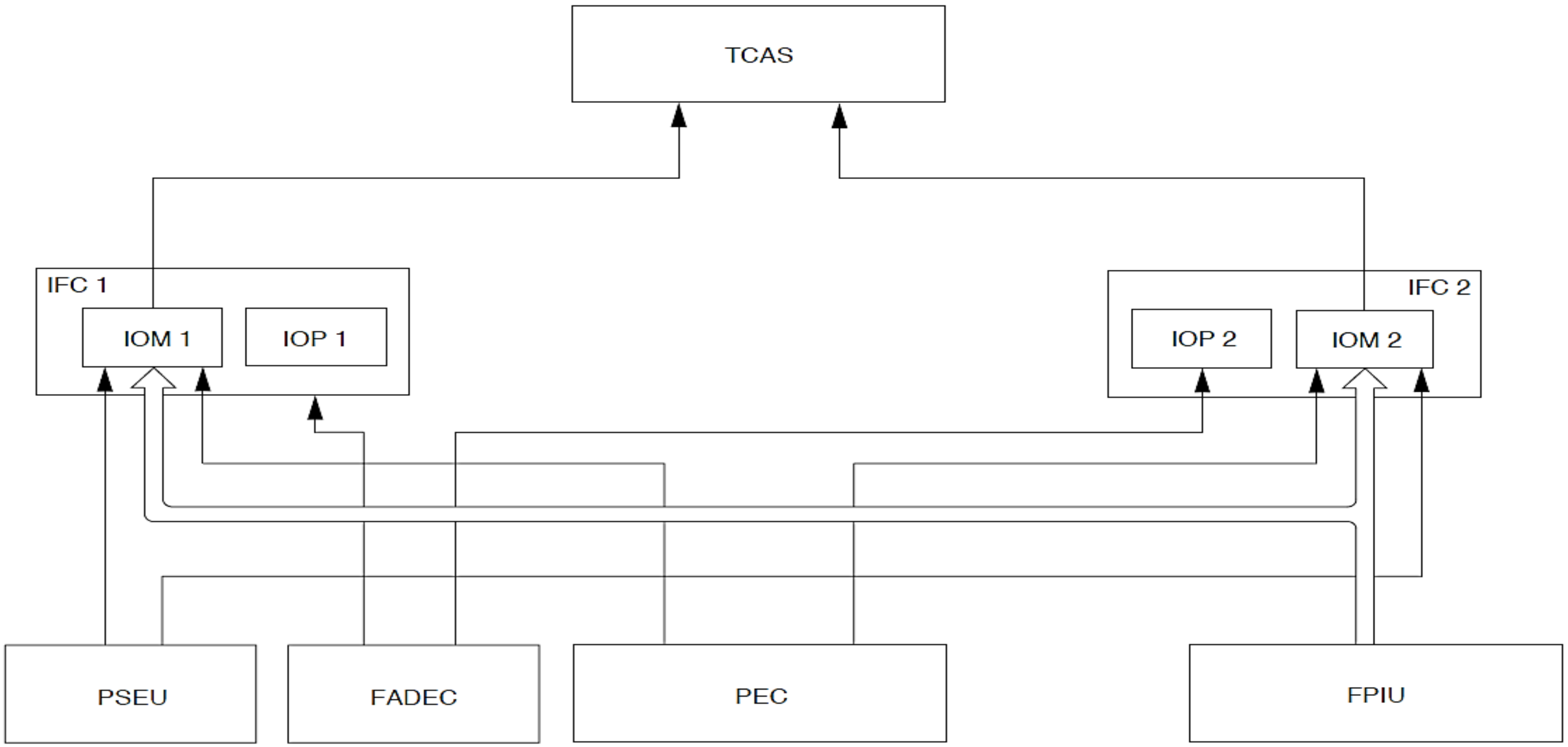


Figure - FDPS TCAS MANEUVER INHIBIT CALCULATION

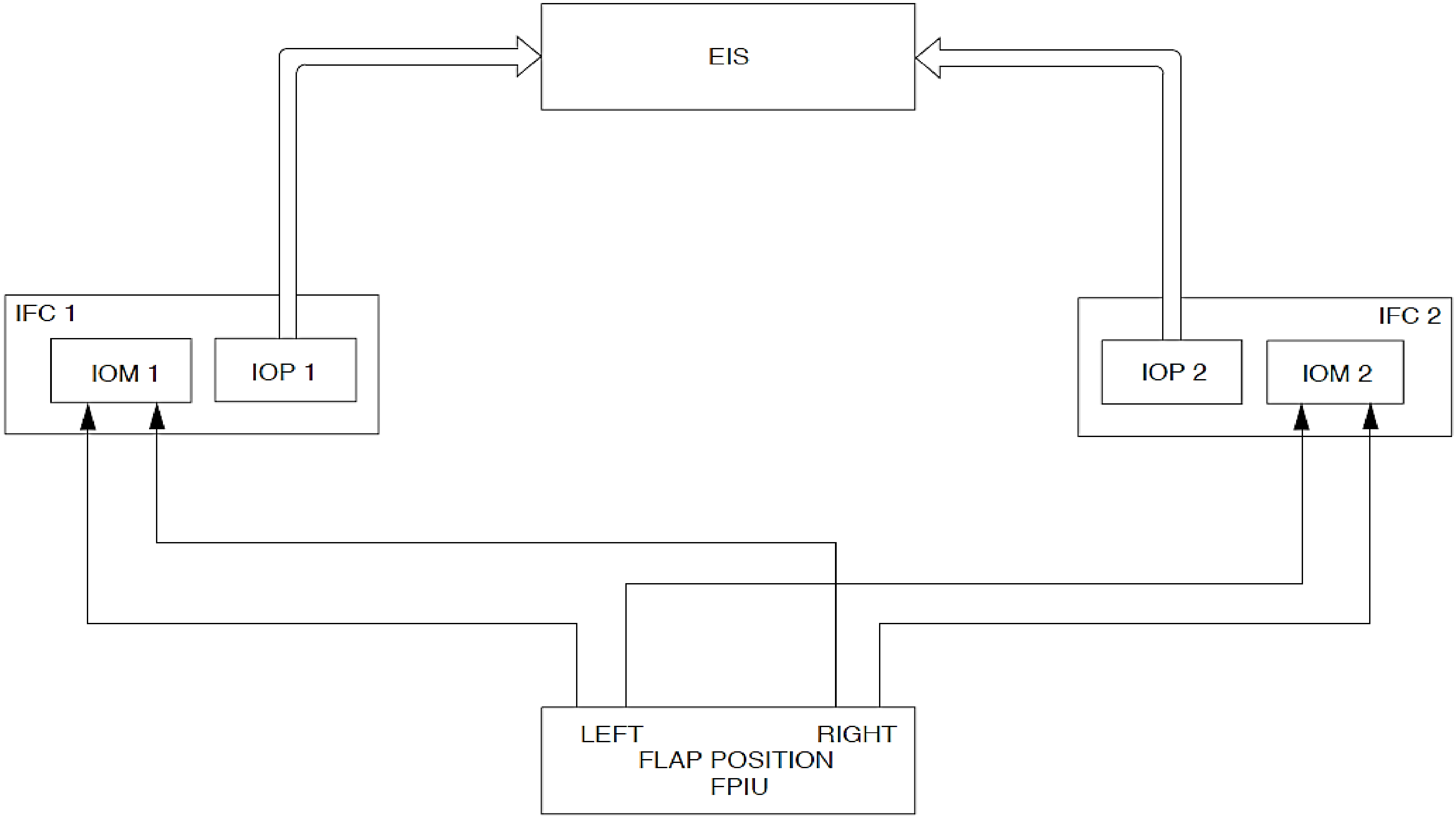


Figure - FDPS FLAP POSITION CALCULATION

Refer Figure - FDPS FUEL TANK TEMPERATURE CALCULATION

The FDPS changes the analog fuel flow signals from the fuel flow meter to digital data for transmission to the EIS through a data bus.

The range of the fuel flow signal is 80 to 3000pph.

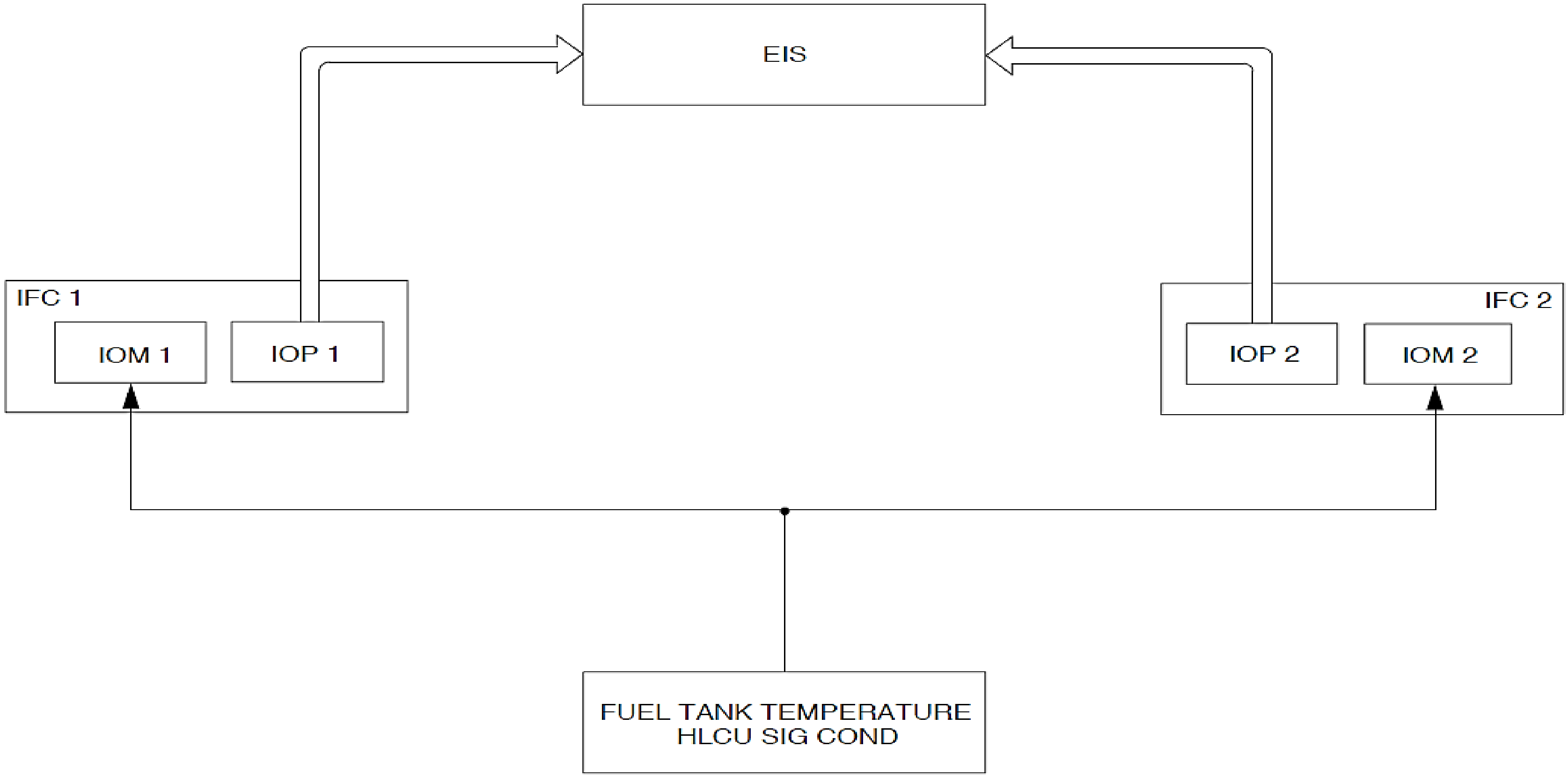


Figure - FDPS FUEL TANK TEMPERATURE CALCULATION

Refer Figure - FDPS FUEL INLET TEMPERATURE CALCULATION

The FDPS changes the analog fuel tank temperature signals from the HLCU signal conditioner to digital data for transmission to the EIS through a data bus.

The range of the fuel tank temperature signal is −70°C to +75°C.

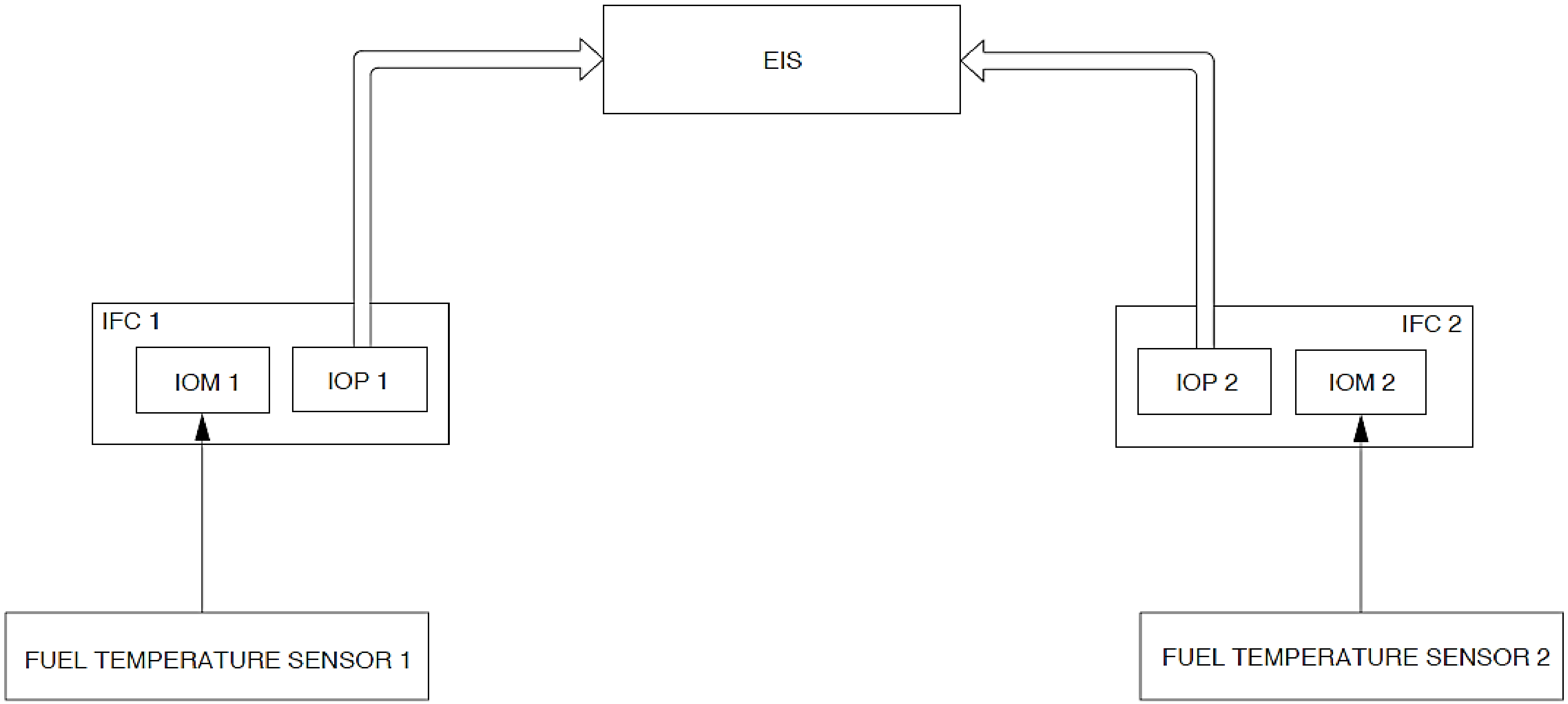


Figure - FDPS FUEL INLET TEMPERATURE CALCULATION

Refer Figure - FDPS FUEL INLET TEMPERATURE CALCULATION

The FDPS changes the analog fuel inlet temperature signals from the fuel temperature sensor to digital data for transmission to the EIS through a data bus.

The range of the fuel inlet temperature signal is −55°C to +121°C.

The FDPS changes the analog hydraulic pressure and quantity signals from the different hydraulic systems to digital data for transmission to the EIS through a data bus.

Refer Figure - FDPS HYDRAULIC PARAMETERS CALCULATION

The hydraulics pressure data from all hydraulic systems is supplied to the two IOMs. The hydraulics quantity data from system 1 and 3 are supplied to IOM1 and hydraulic quantity 2 to IOM2.

NOTE

Note: A FDPS1 malfunction causes hydraulic quantity indication 1 and 3 to also malfunction. If FDPS2 malfunctions, hydraulic quantity 2 indication also malfunctions.

The range of the hydraulic quantity signal is 0 to 4000PSI.

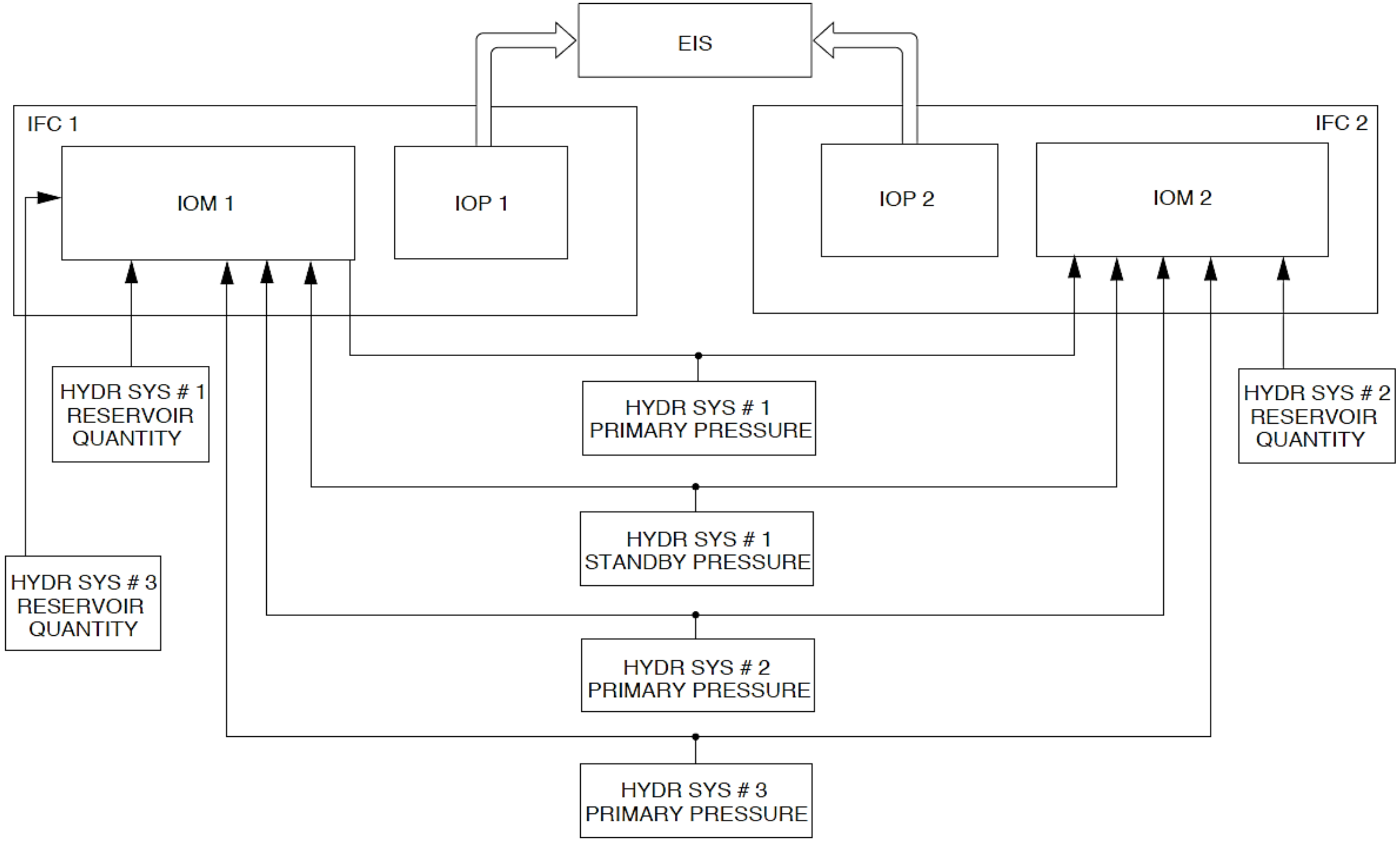


Figure - FDPS HYDRAULIC PARAMETERS CALCULATION

Refer Figure - FDPS MAIN OIL PRESSURE (MOP) CALCULATION

The FDPS changes the analog main oil pressure signals from the engine oil pressure sensors to digital data for transmission to the EIS through a data bus.

The MOP data is supplied to the two IOMs. MOP1 data is supplied through its related IOM1 to the EIS for indication. The MOP2 data is supplied through IOM2 to the EIS when the right main bus power is energized and by IOM1 when the right main bus is not energized.

NOTE

Note: A FDPS1 malfunction causes MOP1 to also malfunction.

NOTE

A FDPS2 malfunction causes MOP2 indication to also malfunction if the right main bus is energized.

The range of the main oil pressure signal is 0 to 260PSI.

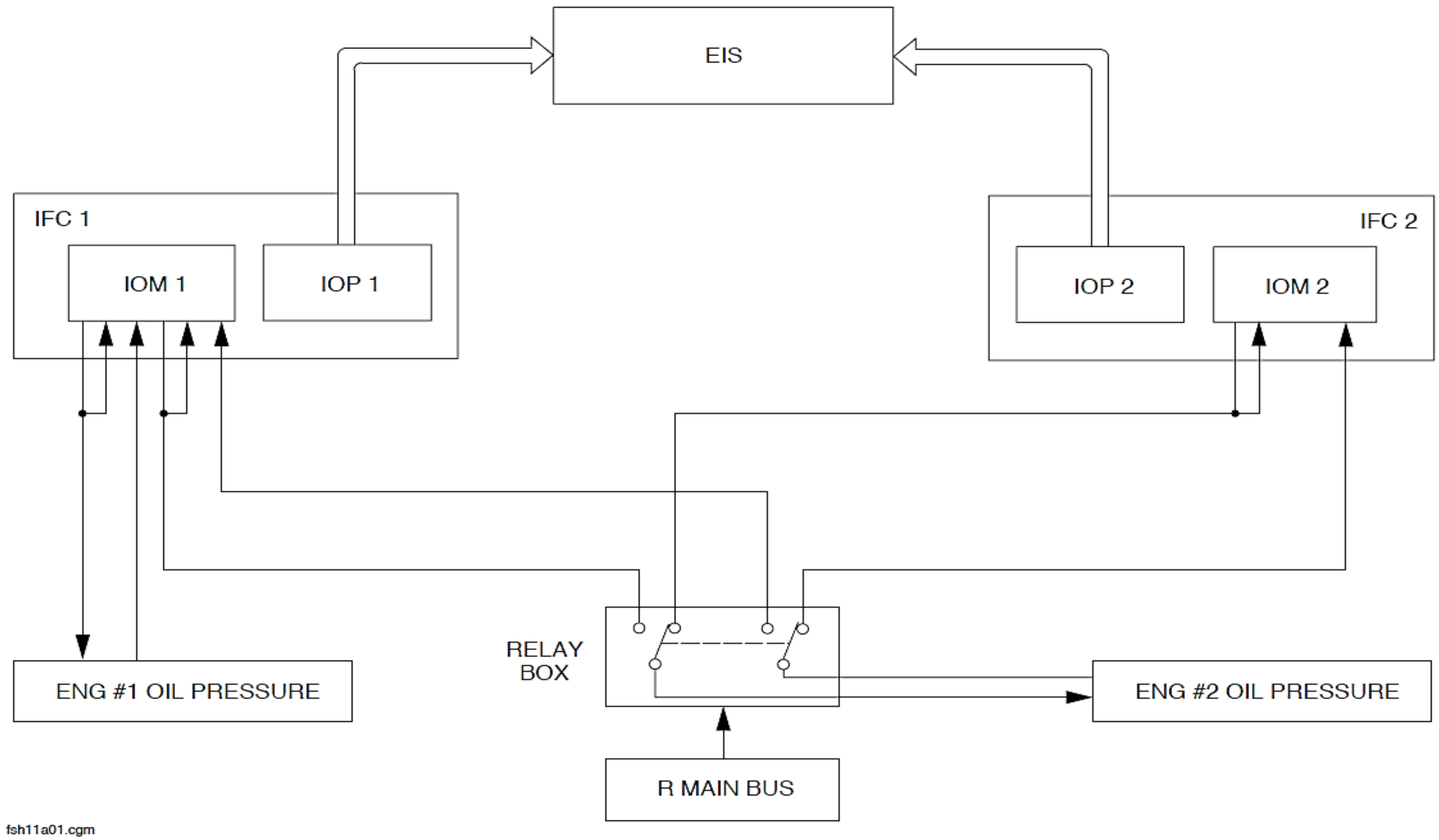


Figure - FDPS MAIN OIL PRESSURE (MOP) CALCULATION

Refer Figure - FDPS PARKING BRAKE HYDRAULIC PRESSURE CALCULATION

The FDPS changes the analog parking brake pressure signals from the parking brake hydraulic pressure transmitter to digital data for transmission to the EIS through a data bus.

The parking brake pressure data is supplied to the two IOMs and is supplied through IOM1 to the EIS for indication.

NOTE

Note: A FDPS1 malfunction causes the parking brake pressure indication to also malfunction.

The range of the parking brake pressure signal is 0 to 4000PSI.

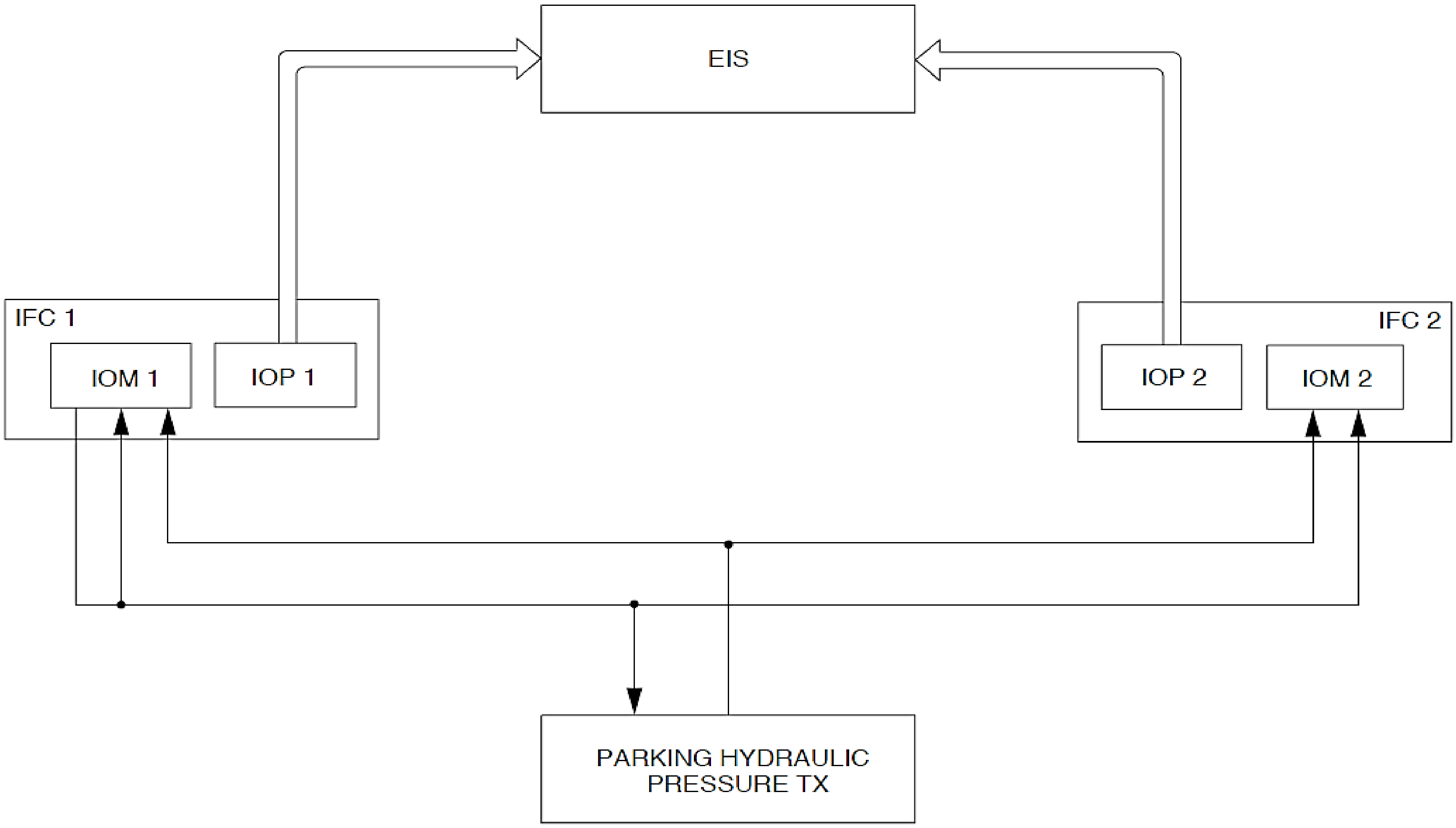


Figure - FDPS PARKING BRAKE HYDRAULIC PRESSURE CALCULATION

Refer Figure - FDPS FLAP POSITION CALCULATION

The FDPS changes the analog flap position signals from the FPIU to digital data for transmission to the EIS through a data bus.

The flap position data is supplied through the IOM1 from the right FPIU channel to the EIS for indication. The data changes to the left FPIU channel if the right malfunctions and to IOM2 if IOM1 malfunctions.

Refer Figure - FDPS AIRCRAFT CONFIGURATION MANAGEMENT

The Aircraft Configuration Module stores information. The ACM stores information from the systems that follow.

- TCAS
- ACARS
- SELCAL
- MLS
- FMS
- VHF COM
- HF
- SSM
- ADF
- DME
- ATC
- RA
- IRS
- Aircraft type
- Engine type
- Temperatures
- Quantities
- Fuel used
- Ground mapping
- Aircraft serial numbers
- AOA correction
- WTG outputs
- CRC LSB indications
- CRC MSB indication.

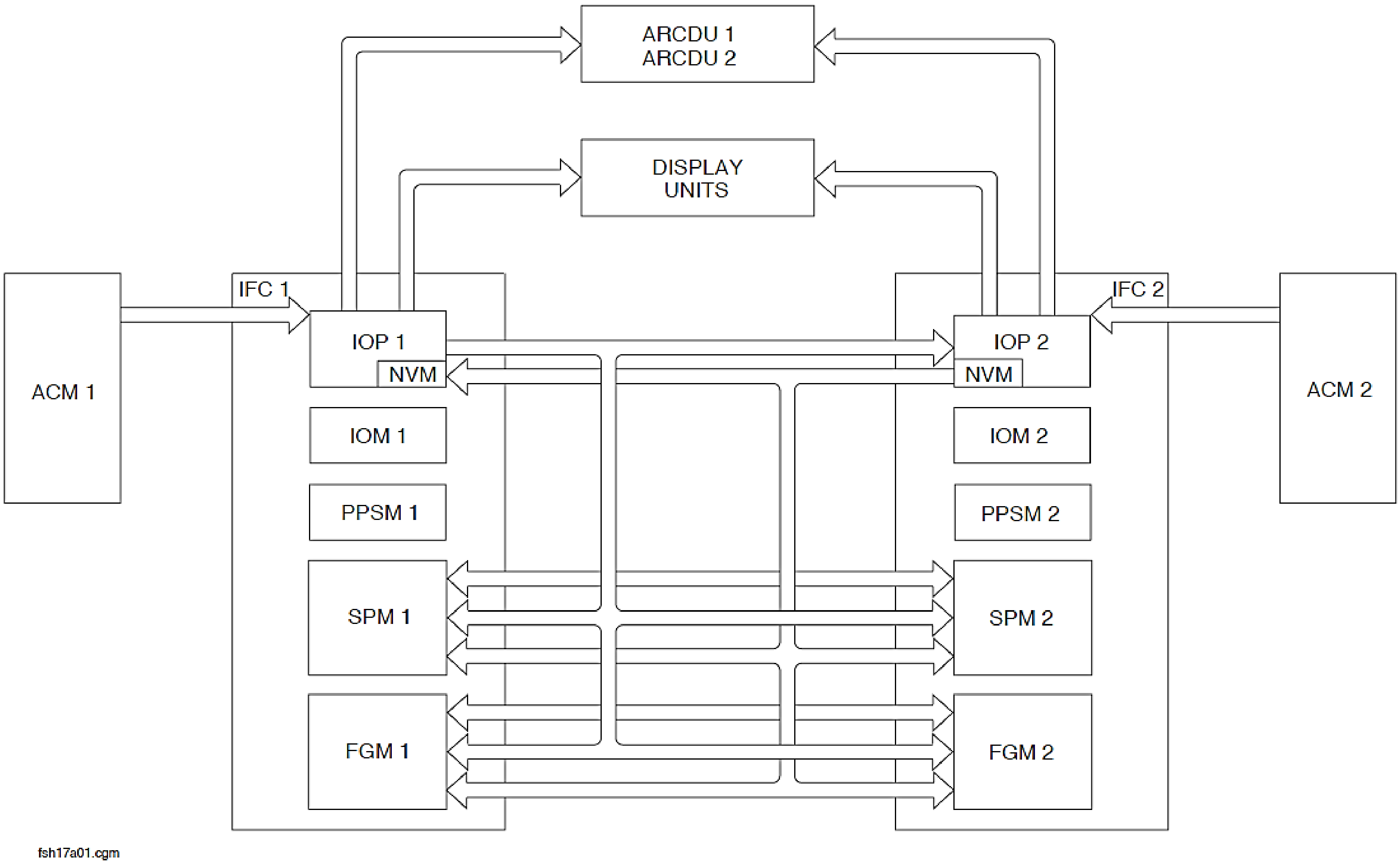


Figure - FDPS AIRCRAFT CONFIGURATION MANAGEMENT

Integrated Flight Cabinets (IFC1, IFC2)

Refer Figure - INTEGRATED FLIGHT CABINETS, LOCATOR

The Integrated Flight Cabinets (IFC 1, IFC 2) are Electromagnetic Containment (EMC) compartments that protect the components inside of it from High Intensity Radio Frequencies (HIRF).

It has two avionics cooling fans to remove air to cool the components inside of it.

The IFC have modules that are line replaceable. Each IFC contains the Line Replaceable Modules (LRM) that follow:

- Input/Output Processor (IOP 1, IOP 2)
- Input/Output Module (IOM 1, IOM 2)
- Aircraft Configuration Module (ACM)
- The Ground Proximity Warning System Converter (GPWSC) located in the Stall Protection Module (SPM 1, SPM 2)
- Prime Power Supply Module (PPSM 1, PPSM 2).

Each LRM is guided into the IFC using a guide rail and guide pins. The top and bottom guide pins at the Integrated Flight Cabinets (IFC 1, IFC 2) back panel makes sure that the module and the back panel come together precisely in the correct location.

The Line Replaceable Units (LRU) are held in place with front locking levers. This gives easy installation and removal of the Line Replaceable Unit (LRU) without any tools.

The IFC are installed in the avionics rack.

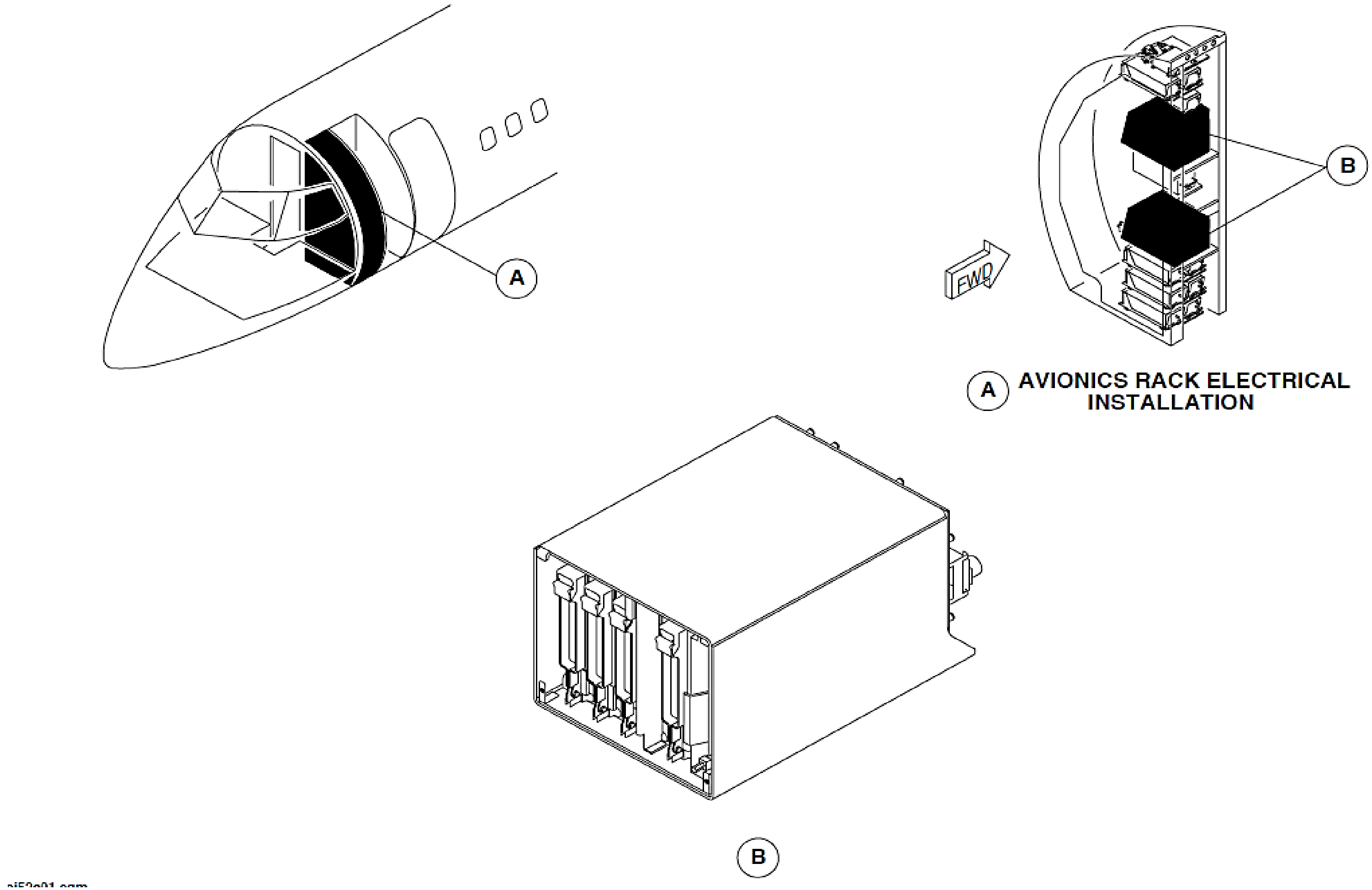


Figure - INTEGRATED FLIGHT CABINETS, LOCATOR

Input/Output Processor (IOP 1, IOP 2)

Refer Figure - INPUT/OUTPUT PROCESSORS, LOCATOR

The Input/Output Processors (IOP 1, IOP 2) receives, calculates, and then transmits data to other avionics systems.

Each FDPS has an IOP and IOM. The FDPS and the Centralized Diagnostic System (CDS) uses the same processor located on the Input/Output Processor (IOP1) module but they function independently.

Each IOP has two electronic boards a CPU Board and an ARINC Board.

The IOPs are Line Replaceable Modules (LRM) that are located in their related IFC.

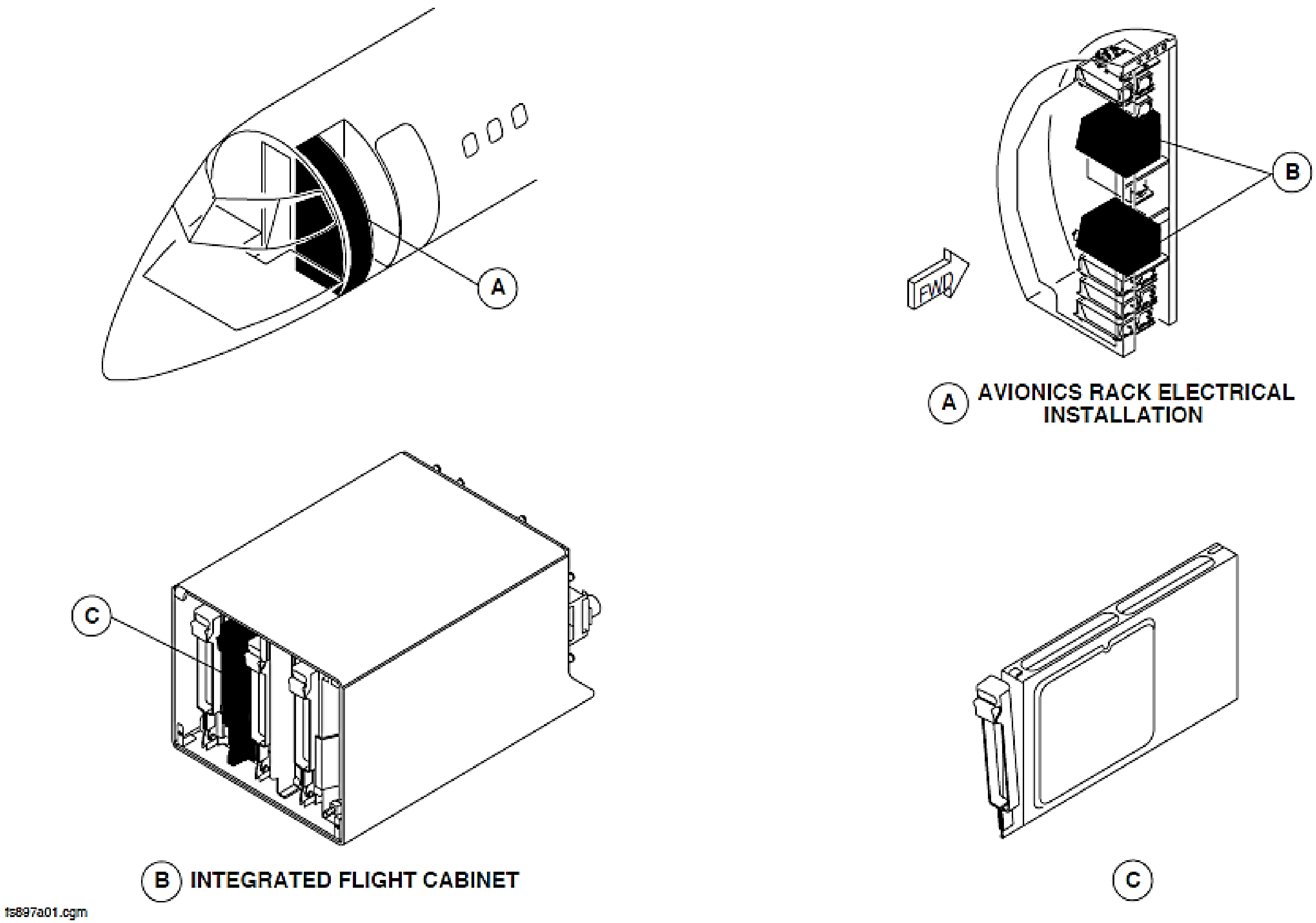


Figure - INPUT/OUTPUT PROCESSORS, LOCATOR

Input/Output Module (IOM 1, IOM 2)

Refer Figure - INPUT/OUTPUT MODULES, LOCATOR

The Input/Output Modules (IOM 1, IOM 2) receive different analog and discrete signals. It changes the inputs into an ARINC 429 format and transmits them to the IOPs for calculations.

The IOMs are Line Replaceable Modules (LRM) that are located in their related IFC.

Prime Power Supply Module (PPSM 1, PPSM 2)

The PPSM receives 28 Vdc from the main and essential busses.

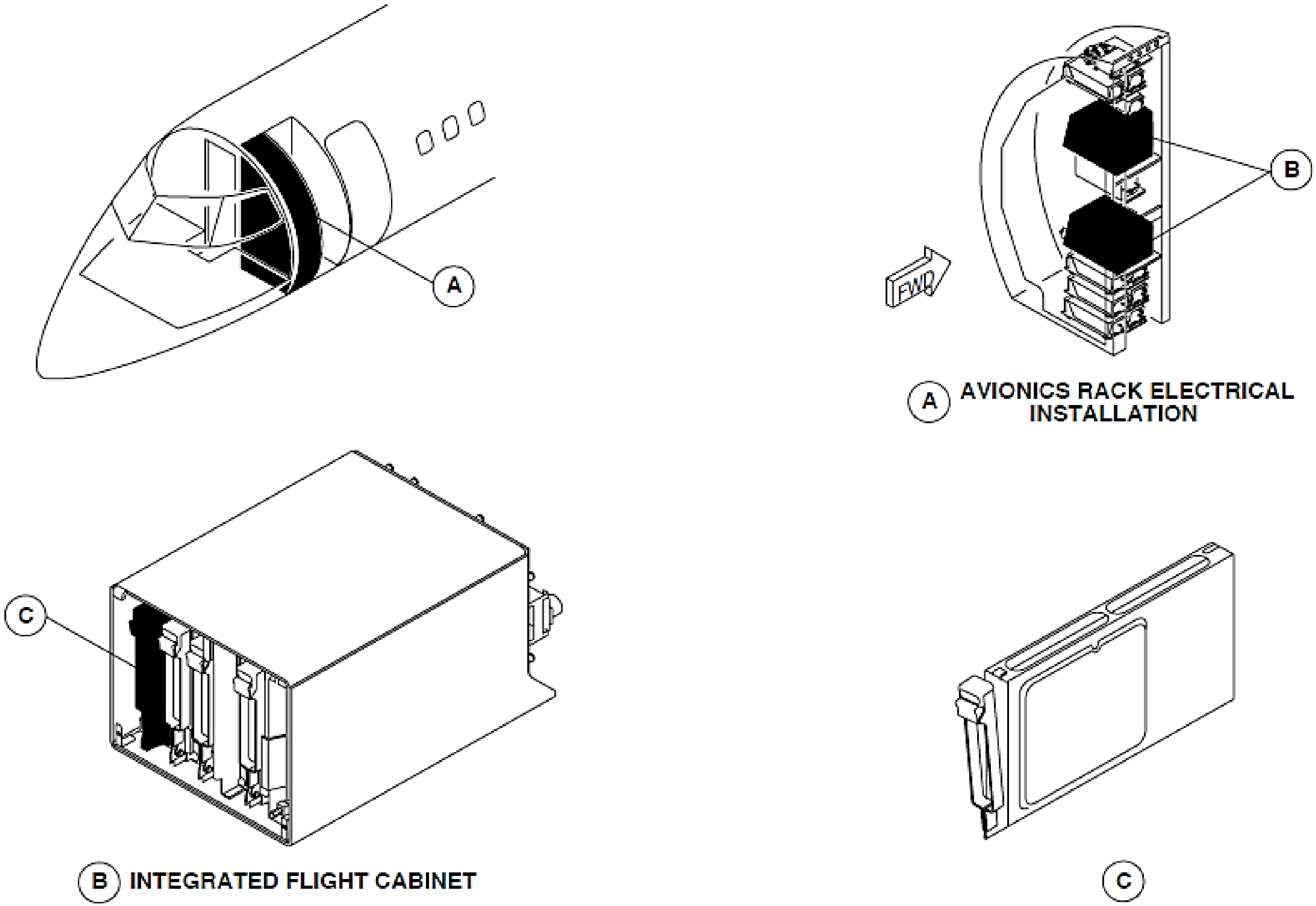


Figure - INPUT/OUTPUT MODULES, LOCATOR

CENTRAL WARNING SYSTEM

Introduction

The central warning system has different sub−systems to do the functions that follows:

- Show caution and warning lights
- Show advisory lights
- Take−off warning calculation.

General Description

Refer Figure - CENTRAL WARNING SYSTEM BLOCK DIAGRAM

The indicating and recording system has the sub−systems that follow:

- Caution and Warning Lights System
- Take−off Warning System.

Caution and Warning Lights System

The caution and warning light system is divided into the two parts that follow:

- Caution and warning lights
- Advisory lights.

Caution and Warning Lights: The caution and warning lights system shows system malfunctions and other conditions that require a corrective action.

Advisory Lights: The advisory lights show malfunctions and other conditions that require a corrective action and safe and normal system operation.

Take−off Warning System

Refer to Central Computer.

The take−off warning system supplies an aural warning when a take−off is attempted with the aircraft not in the correct take−off configuration.

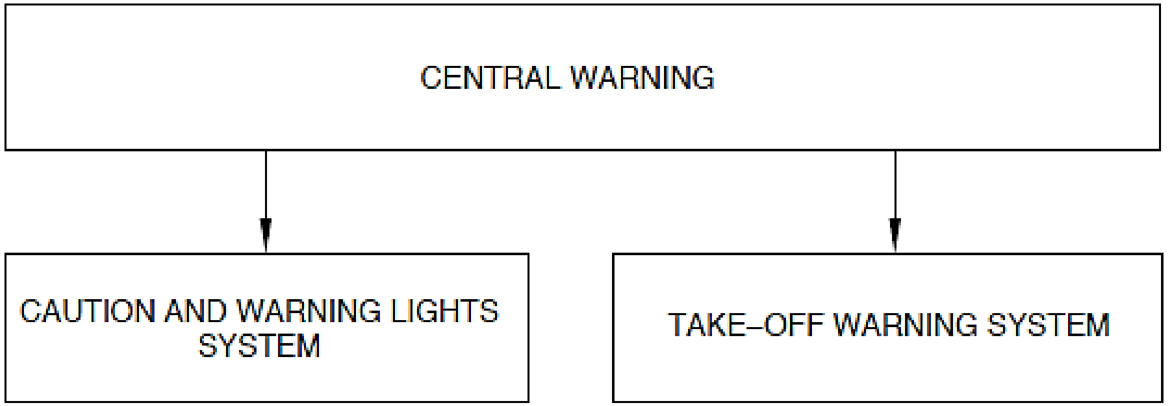


Figure - CENTRAL WARNING SYSTEM BLOCK DIAGRAM

CAUTION AND WARNING LIGHTS SYSTEM

Introduction

The caution and warning light system is divided into the two parts that follow:

- Caution and warning lights
- Advisory lights.

Caution and Warning Lights: The caution and warning lights system shows system malfunctions and other conditions that require a corrective action.

Advisory Lights: The advisory lights show malfunctions and other conditions that require a corrective action and safe and normal system operation.

General Description

Refer Figure - CAUTION AND WARNING LIGHTS SYSTEM BLOCK DIAGRAM

The caution and warning and advisory lights system has color attributes that follow:

- Red
- Amber or yellow
- Green, white, or blue.

The red warning lights show system malfunctions or the conditions that cause dangerous flight conditions. It shows a malfunction or condition that requires an immediate corrective action.

The amber or yellow caution lights show system malfunctions or conditions that do not immediately affect the aircraft flight capabilities. It shows that a future corrective action is possibly necessary.

The green lights show safe system operation and white or blue lights show normal operation.

The caution and warning panel receives input caution and warning signals from the different aircraft systems, and shows them on the caution and warning panel annunciator segments. The caution and warning panel is attached to the overhead panel frame.

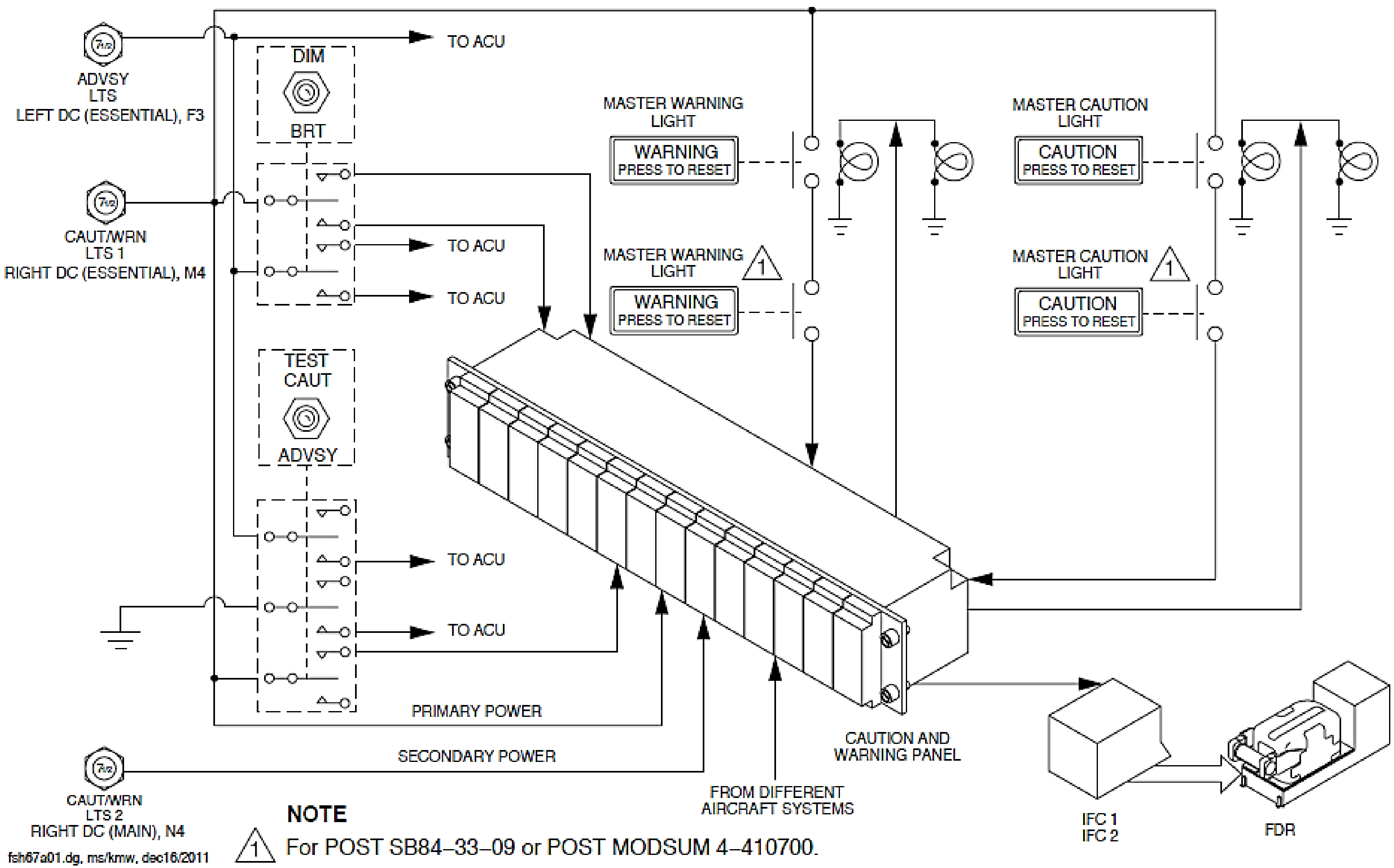


Figure - CAUTION AND WARNING LIGHTS SYSTEM BLOCK DIAGRAM

Figure - Pilot Glareshield Panel, Master Caution/Warning Indication
Figure - Co−pilot Glareshield Panel, Master Caution/Warning Indication

When the caution and warning panel senses a warning condition, it causes a related red warning annunciator to come on flashing with the pilot glare shield panel master warning switch and three chimes sounds. The master warning switch has a WARNING PRESS TO RESET label.

On aircraft with ModSum 4−410700 or SB84−33−09 incorporated, an additional master warning switch labeled WARNING PRESS TO RESET is installed on the co−pilot glare shield panel.

The caution and warning panel warning annunciator and the master warning switch indication flashes until the master warning switch is pushed. A switch selection causes the caution and warning panel warning indication to change to steady and the master warning indication to go out. When the condition is corrected, the caution and warning panel warning indication goes out.

NOTE

The No. 1 or No. 2 engine oil pressure red warning indication comes on steady. It will not flash.

When the caution and warning panel senses a caution condition, it causes a related yellow caution annunciator to come on with the copilot glare shield panel master caution switch and one chime sounds. The master caution switch has a CAUTION PRESS TO RESET label.

On aircraft with ModSum 4−410700 or SB84−33−09 incorporated, an additional master caution switch labeled CAUTION PRESS TO RESET is installed on the pilot glare shield panel.

The caution and warning panel caution annunciator and the master caution switch indication stays on until the master warning switch is pushed. A switch selection causes the master caution indication to go out. When the condition is corrected, the caution and warning panel caution indication goes out.

NOTE

The caution and warning panel FUELING ON caution light will not cause the master caution indicator to come on.

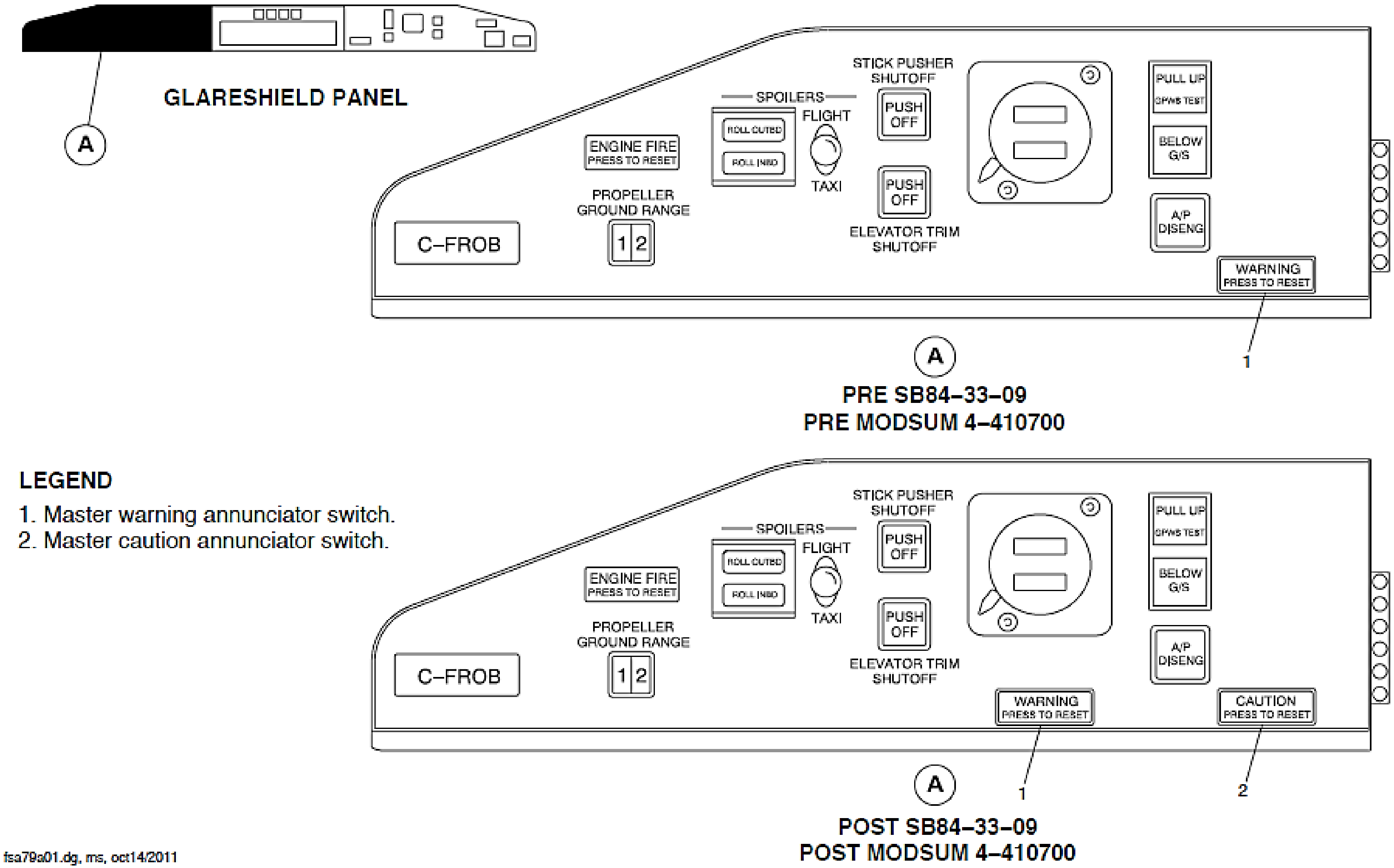


Figure - Pilot Glareshield Panel, Master Caution/Warning Indication

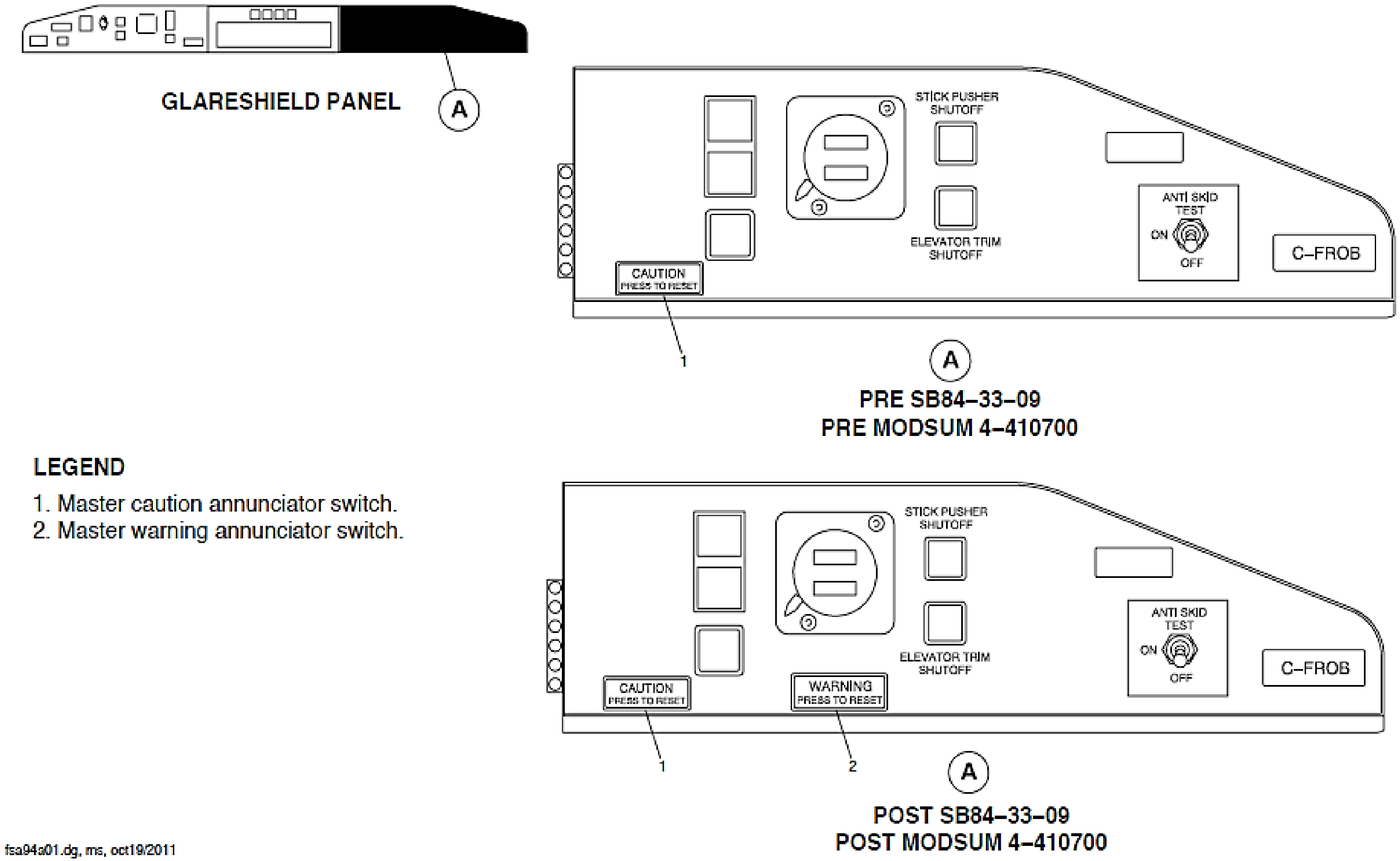


Figure - Co−pilot Glareshield Panel, Master Caution/Warning Indication

Refer Figure - ADVISORY LIGHTS SYSTEM BLOCK DIAGRAM

Advisory lights are located on the panels that follow:

- Overhead
- Glare shield
- Instrument
- Center console
- Pilot side console.

NOTE

An amber or yellow advisory indication does not cause the master caution light to come on and a red advisory indication does not cause the master warning light to come on.

An Advisory Lights Control Unit (ACU) is used to test and control the brightness of the advisory annunciators. It is controlled by the CAUTION/ADVSY LIGHTS toggle switches located on the passenger warning panel.

The signals from different aircraft systems are supplied to the related advisory annunciator through the ACU. A loss of electrical power to the ACU will not prevent individual indicators from coming on when it receives an advisory light signal. The advisory test, dim or bright selections will have no effect.

The caution and warning light system has the components that follow:

- Panel, Caution and Warning
- Unit, Advisory Control
- Switch, Master Warning
- Switch, Master Caution
- Switch, Test Caution/Advisory
- Switch, Dim/Bright.

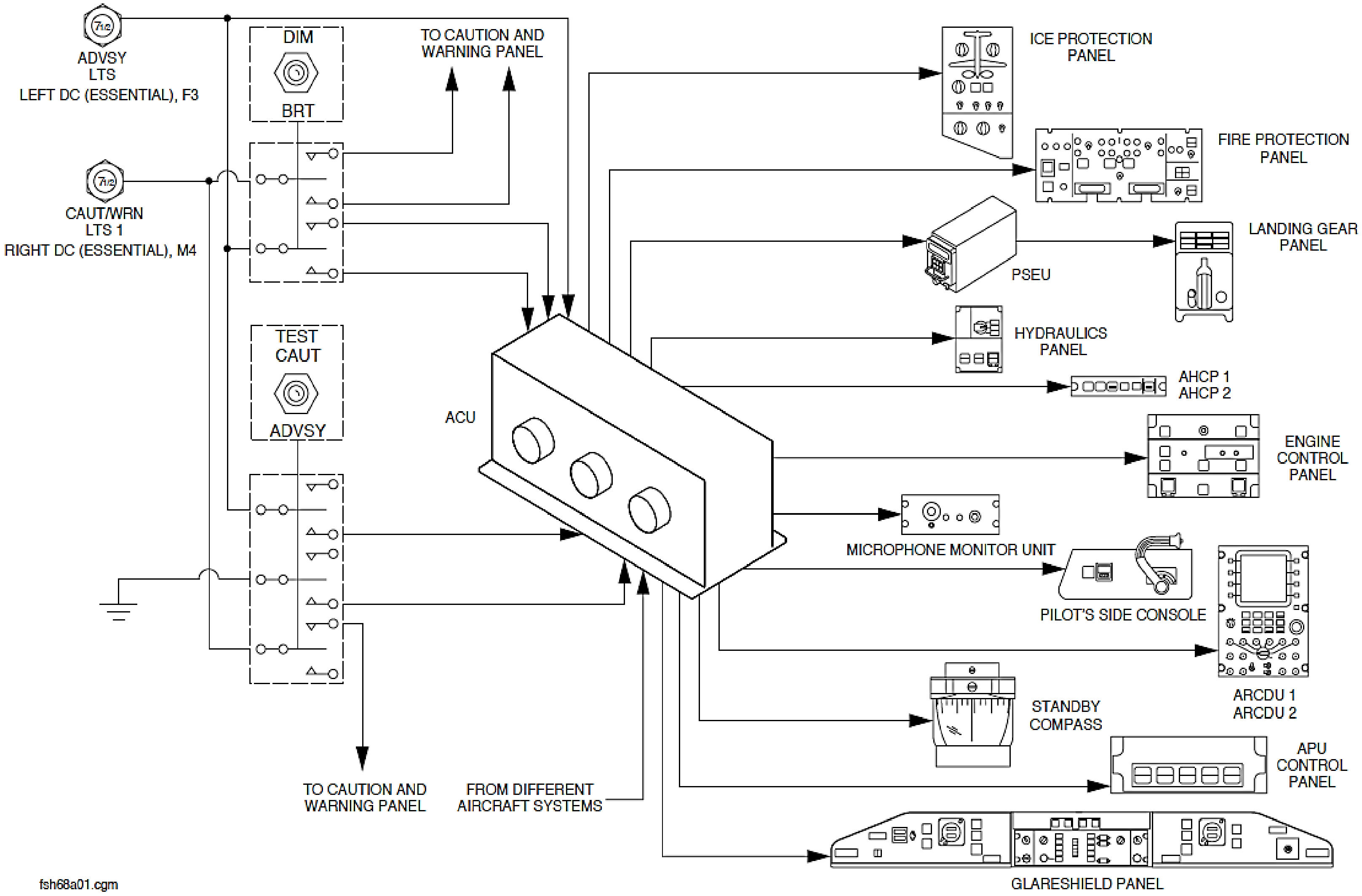


Figure - ADVISORY LIGHTS SYSTEM BLOCK DIAGRAM

Detailed Description

There are five different types of inputs to the caution and warning panel:

- Type I, supplied 28 V dc
- Type II, supplied ground
- Type III, 28 V dc removed
- Type IV, ground removed
- Type V, combination.

Type I: A malfunction condition is present if the input to the caution and warning panel is more than 10 V dc.

Type II: A condition is present if the impedance to ground potential is less than 50 Ohm.

Type III: A condition is present if the input to the caution and warning panel is less than 10 V dc.

Type IV: A condition is present if the impedance to ground potential is more than 2000 Ohm or if the total voltage across a switching element and a 50 Ohm resistor is more than 4.5 V dc.

Type V: A condition is present if the input is a combination of type I, II, III, or IV inputs.

There are five other control inputs from the caution and advisory lights toggle switches to the caution and warning panel:

- Master warning reset, 28 V dc
- Master caution reset, 28 V dc
- Test, 28 V dc
- Dim, 28 V dc
- Bright, 28 V dc.

A system malfunction or condition causes its related advisory annunciator to come on. The annunciator stays in view until the malfunction or condition is corrected.

There are two different types of inputs from the caution and advisory lights toggle switches to the caution and warning panel:

- Type I, supplied 28 V dc
- Type II, supplied ground.

Type I: A condition is present if the input to the ACU is more than 10 V dc.

Type II: A condition is present if the impedance to ground potential is less than 50 Ohm

There are three other control inputs to the ACU:

- Test, 28 V dc
- Dim, 28 V dc
- Bright, 28 V dc.

Refer Figure - Passenger Warning Panel

The warning, caution, and advisory annunciator lights are initially set to bright when the systems are energized. The DIM, BRT toggle switch located on the passenger warning panel is momentarily set to the DIM position to make the advisory and caution annunciators lights and compass light change to dim.

The DIM, BRT toggle switch located on the passenger warning panel is momentarily set to the BRT position to make the advisory and caution annunciators lights and compass light change back to bright.

The TEST CAUT/ADVSY toggle switch on the passenger warning panel is set to TEST CAUT position to test the caution and warning lights on the caution warning panel and the master caution and master warning indication.

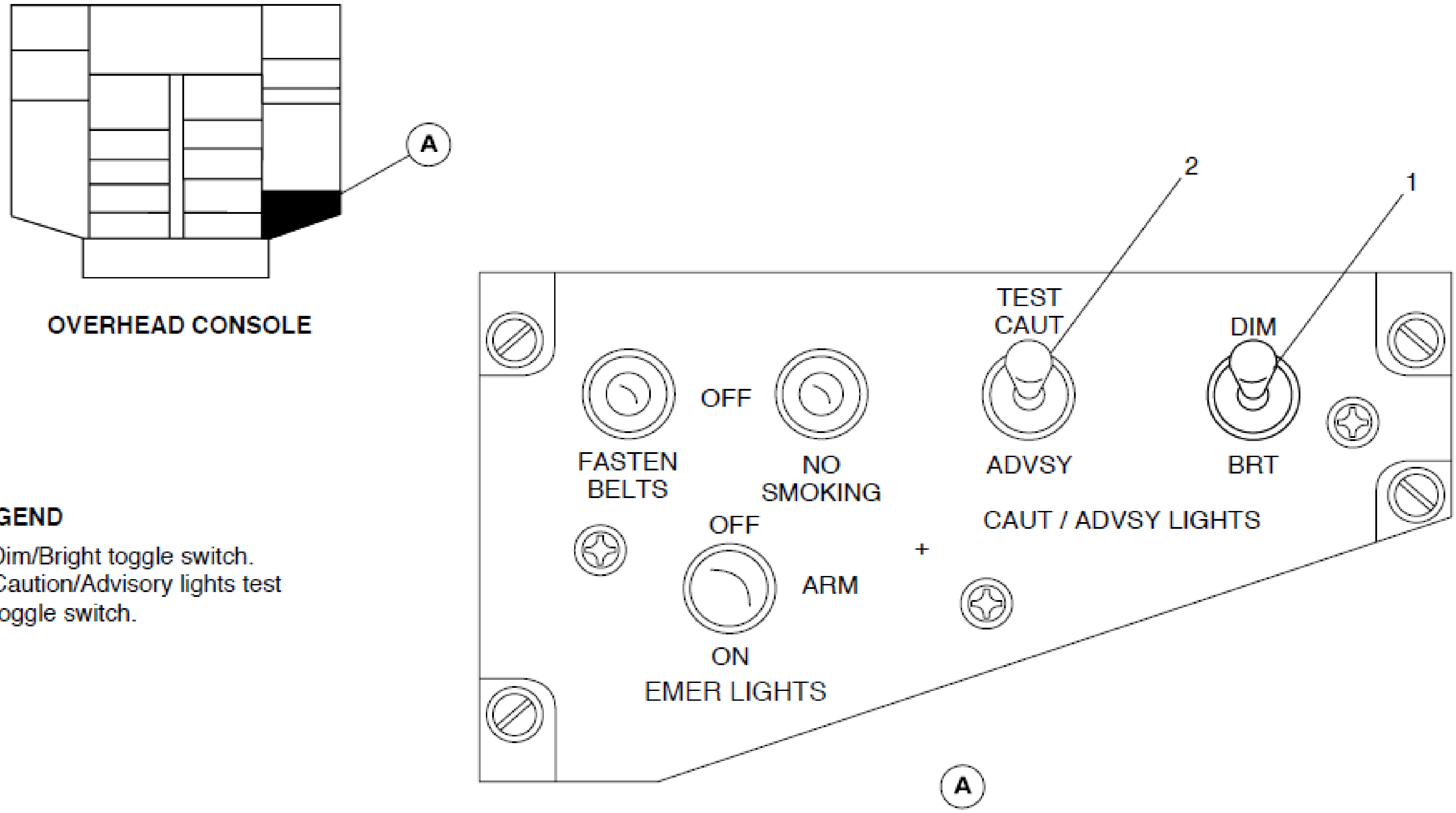
The caution and warning test gives the indications that follow:

- Warning lights come on bright and flashing
- Caution lights come on bright
- Master warning comes on flashing
- Master caution comes on flashing
- Three chimes sound.

The TEST CAUT/ADVSY toggle switch on the passenger warning panel is set to TEST ADVSY position to test the advisory annunciator lights.

The aircraft right essential and right main buses supply 28 V dc electrical power through two 7.5 A circuit breakers to the caution and warning panel. The circuit breaker that is connected to the right essential bus also supplies electrical power to the master warning and master caution switches and its related part of the DIM, BRT and TEST CAUT/ADVSY toggle switches. The circuit breakers are located in position M4 and N4 on the 28 right circuit breaker panels.

The aircraft left essential bus supplies 28 V dc electrical power through a 7.5 A circuit breaker to the ACU and its related part of the DIM, BRT and TEST CAUT/ADVSY toggle switches. The circuit breaker is located in position F3 on the 28 left circuit breaker panel.



LEGEND

1. Dim/Bright toggle switch.
2. Caution/Advisory lights test toggle switch.

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Figure - Passenger Warning Panel

Caution and Warning Annunciator Panel

Refer Figure - Figure - CAUTION AND WARNING **PANEL LOCATOR**

Caution and warning fault signals caused by important malfunctions or conditions are supplied to the caution and warning panel. These malfunctions or conditions are shown on the ninety six active annunciator segments on the panel. The panel is divided into 14 modules. Each module is further divided into seven segments, six yellow cautions and one red warning annunciator at the bottom. Each segment is labelled with an appropriate legend that is back lit with a LED. The legends are not readable until the LEDs come on.

Each module has a bezel and lens assembly that is keyed and marked with the module number. It is pulled removed from the module to get access to the LEDs.

The unit is attached to the overhead panel frame of the aircraft with four dzus fasteners.

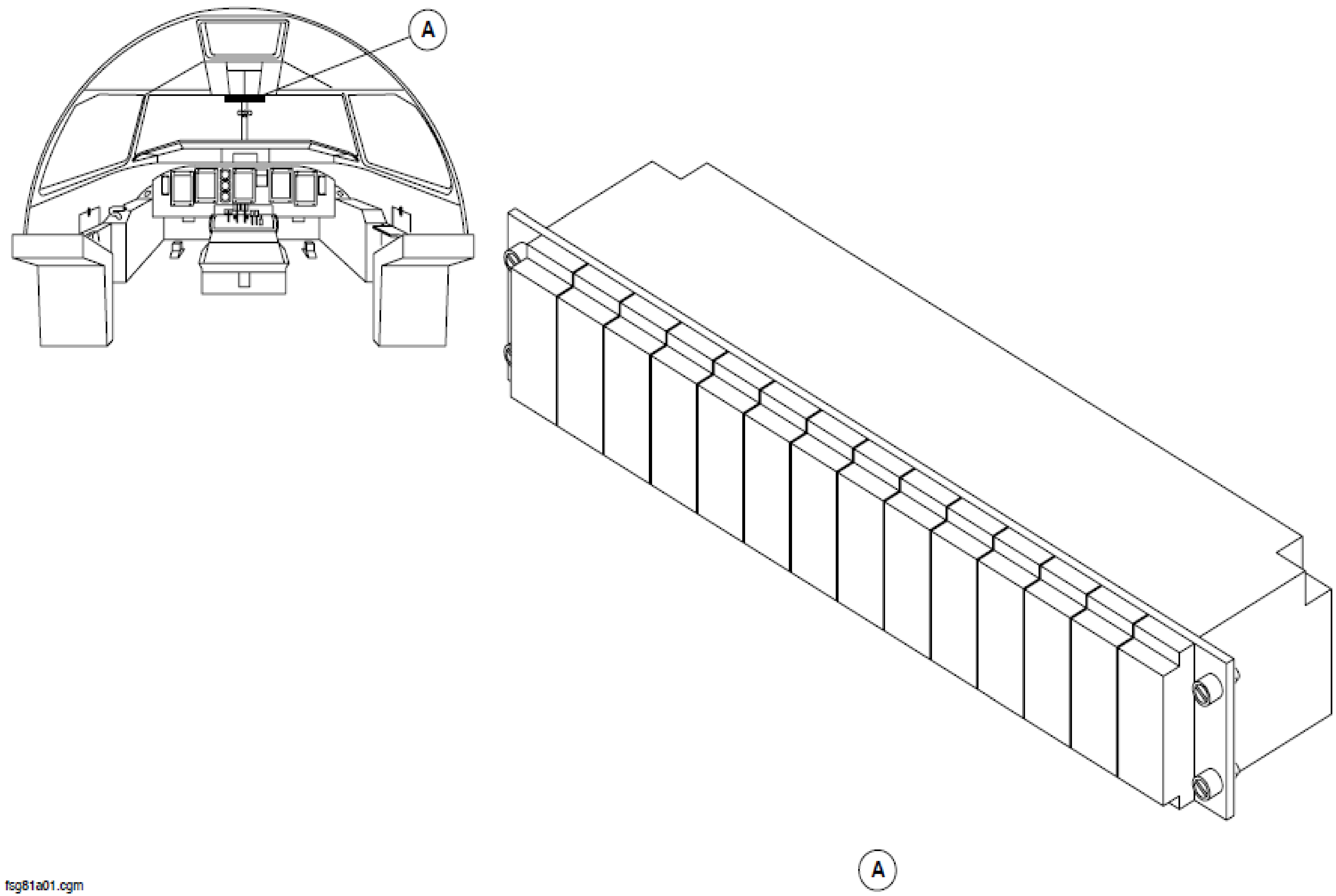

Figure - CAUTION AND WARNING PANEL LOCATOR

Advisory Lights Control Unit

Refer Figure - ADVISORY LIGHTS CONTROL UNIT LOCATOR

The Advisory Lights Control Unit (ACU) tests and controls the brightness of the advisory annunciators.

The advisory control unit receives 28 V dc input or ground input signals from different aircraft systems and supplies them to the related advisory lights on the aircraft.

The advisory control unit is attached to the bottom shelf of the right DC circuit breaker console with seven mounting screws.

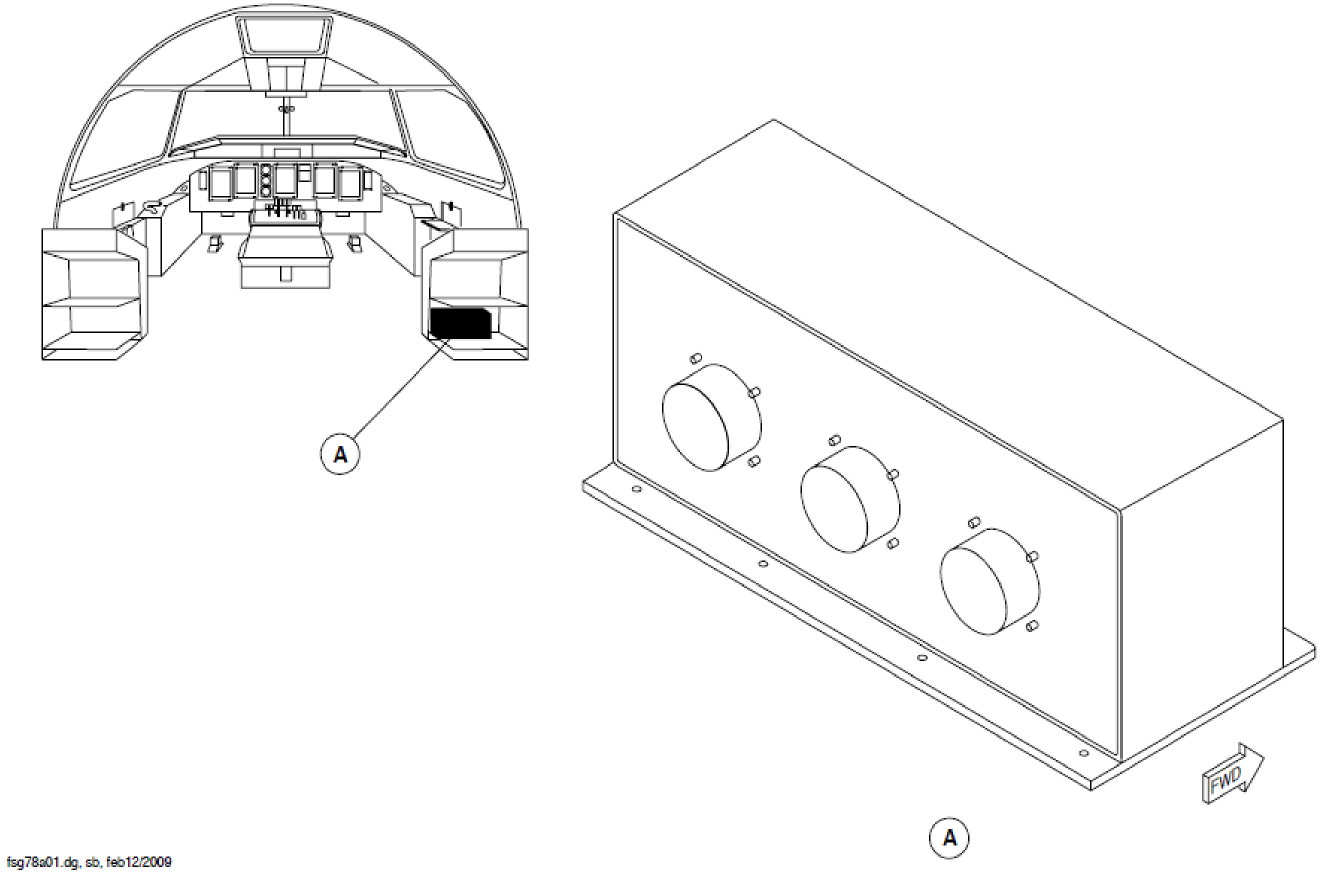

Figure - ADVISORY LIGHTS CONTROL UNIT LOCATOR

Master Warning Switch

Refer Figure - Master Warning Switch Locator

The master warning switch has a pushbutton reset switch and a WARNING PRESS TO RESET annunciator in red letters on a black background. It is back lit by a LED when a warning signal is received. The master warning switch is located on the pilot glare shield panel and is secured by two mounting screws.

On aircraft with ModSum 4−410700 or SB84−33−09 incorporated, a master warning switch is also installed on the co−pilot glare shield panel.

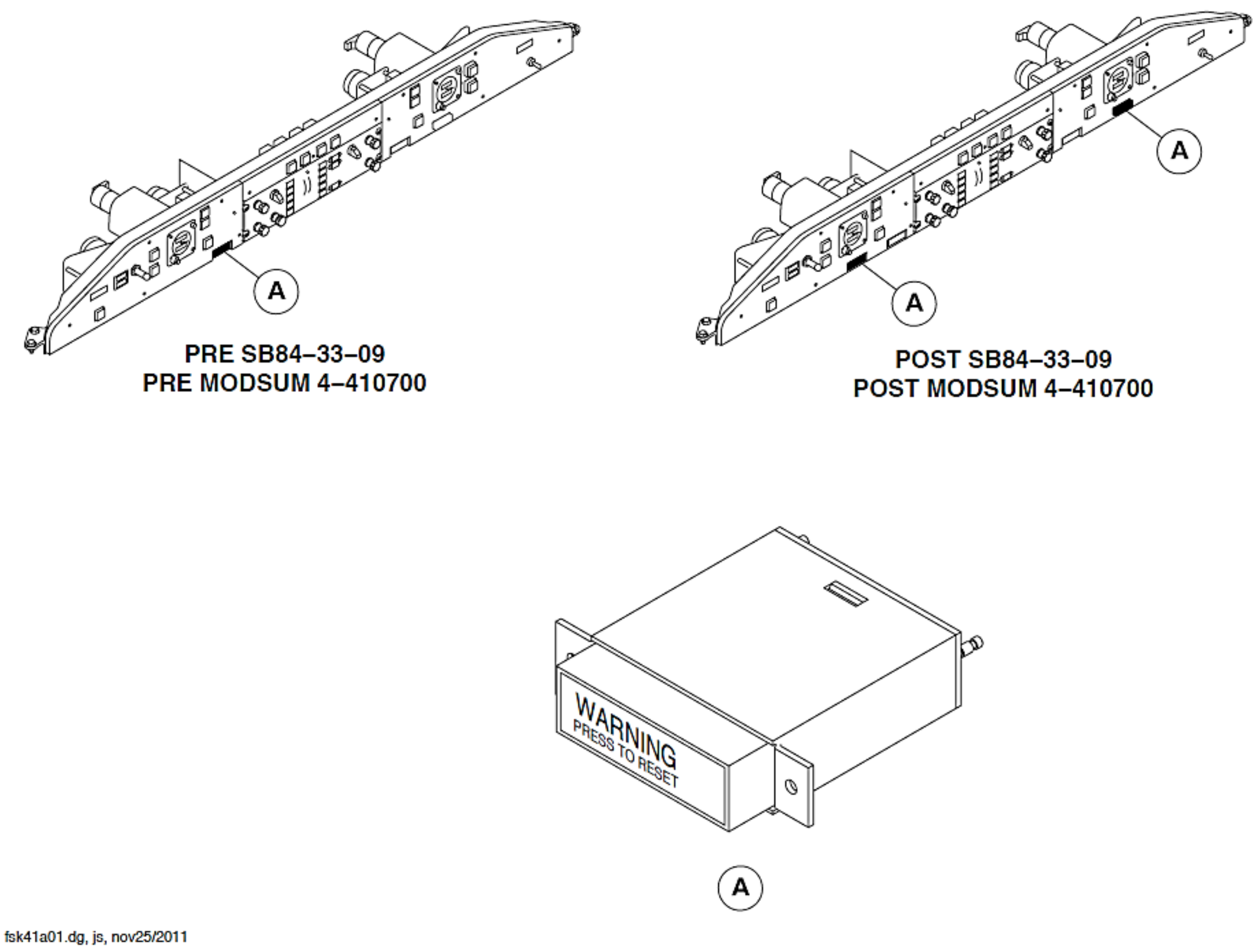

Figure - Master Warning Switch Locator

Master Caution Switch

Refer Figure - Master Caution Switch Locator

The master caution switch has a pushbutton reset switch and a CAUTION PRESS TO RESET annunciator in yellow letters on a black background. It is back lit by a LED when a caution signal is received. The master caution switch is located on the copilot glareshield panel and is secured by two mounting screws.

On aircraft with ModSum 4−410700 or SB84−33−09 incorporated, a master caution switch is also installed on the pilot glareshield panel.

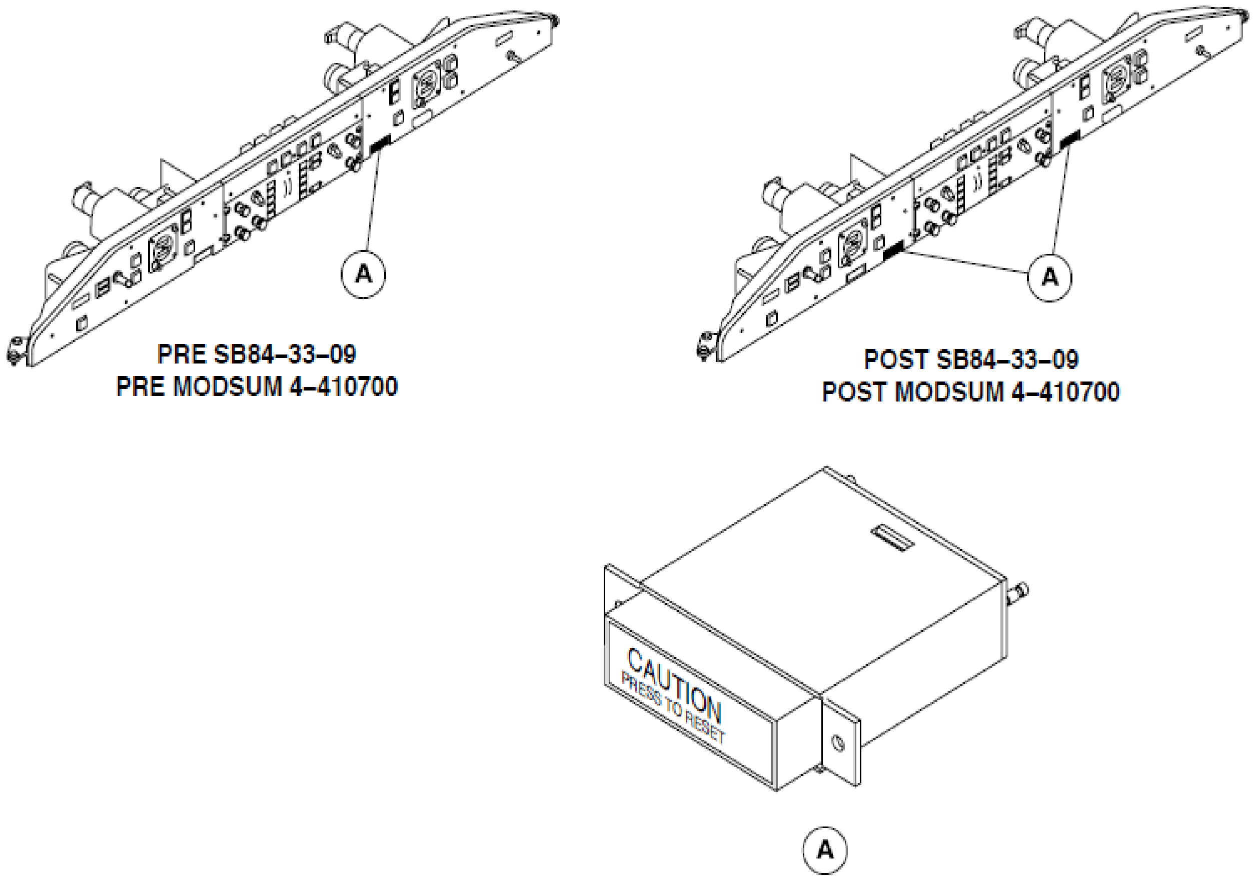

Figure - Master Caution Switch Locator

Dim/Bright Switch

Refer Figure - DIM/BRIGHT SWITCH LOCATOR

The warning, caution, and advisory annunciator lights are initially set to bright when the systems are energized. The DIM, BRT toggle switch is momentarily set to the DIM position to make the advisory and caution annunciators lights and compass light change to dim. It is momentarily set to the BRT position to make the advisory and caution annunciators lights and compass light change back to bright.

The dim/bright switch is a four pole, two position center–off toggle switch located on the right side of the passenger warning panel. It has a toggle lever that springs back to the center off position when released. The toggle lever is connected to a mechanism in the body of the switch that moves in the opposite direction as the toggle lever. The toggle switch has wire lug terminals for wiring connections.

The dim/bright switch is attached to the passenger warning panel assembly with a mounting nut and lock washer. The passenger warning panel assembly is attached to the overhead console with four dzus fasteners.

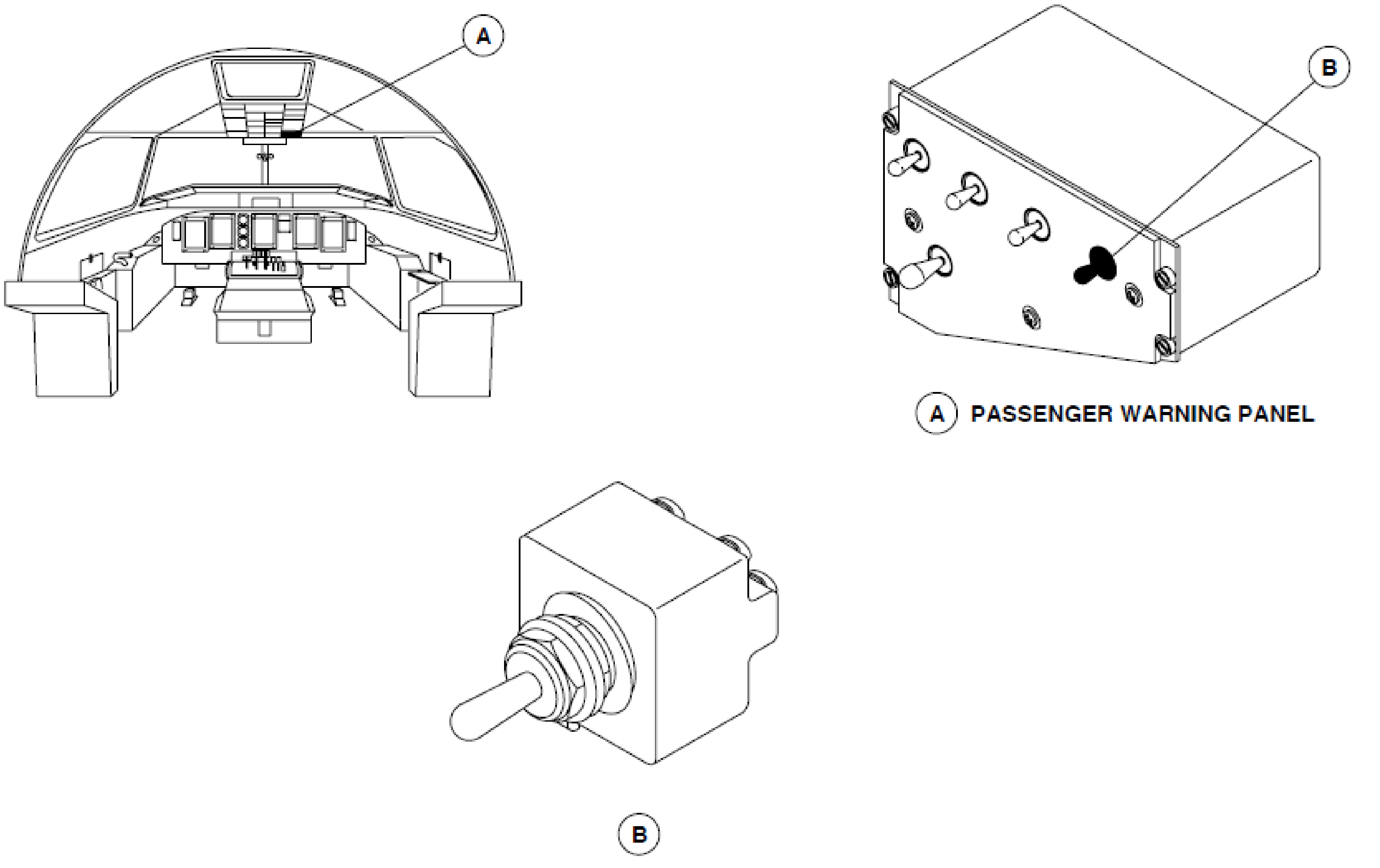

Figure - DIM/BRIGHT SWITCH LOCATOR

Caution/Advisory Test Switch

Refer Figure - CAUTION/ADVISORY TEST SWITCH LOCATOR

The caution/advisory test switch is set to TEST CAUT position to test the caution and warning lights on the caution warning panel and the master caution and master warning indication. It is set to TEST ADVSY position to test the advisory annunciator lights.

The caution/advisory test switch is a 6 pole, two position center−off toggle switch located on the right side of the passenger warning panel. It has a toggle lever that springs back to the center off position when released. The toggle lever is connected to a mechanism in the body of the switch that moves in the opposite direction as the toggle lever. The toggle switch has wire lug terminals for wiring connections.

The caution/advisory test switch is attached to the passenger warning panel assembly with a mounting nut and lock washer. The passenger warning panel assembly is attached to the overhead console with four dzus fasteners.

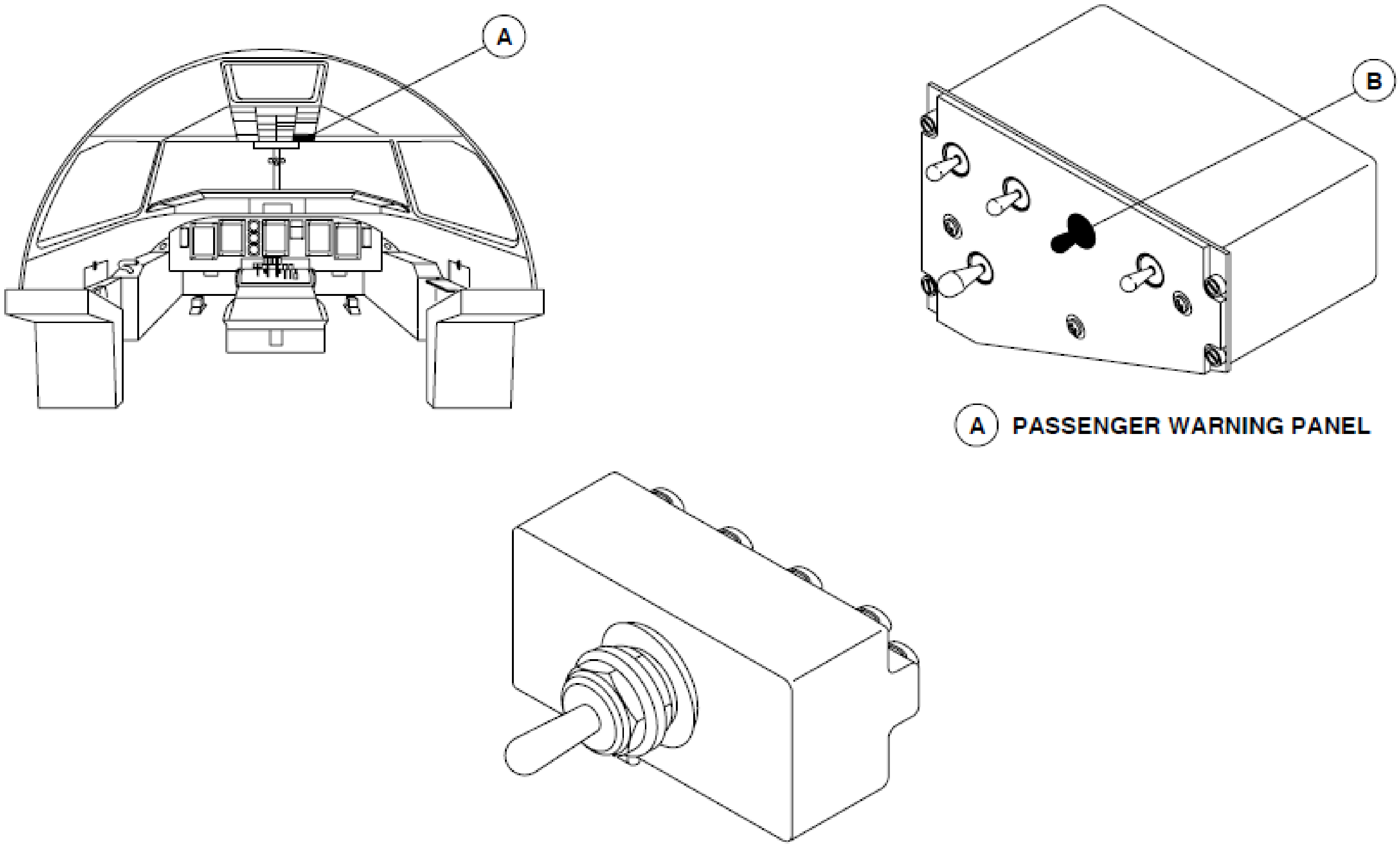

Figure - CAUTION/ADVISORY TEST SWITCH LOCATOR

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TAKE−OFF WARNING SYSTEM

Introduction

Refer to Central Computer.

The take−off warning system supplies an aural warning when a take−off is attempted with the aircraft not in the correct take−off configuration.

General Description

Refer Figure 31- 70 Flight Compartment Take−Off **Configuration**
Refer Figure - FDPS Incorrect Take−Off Configuration Warning

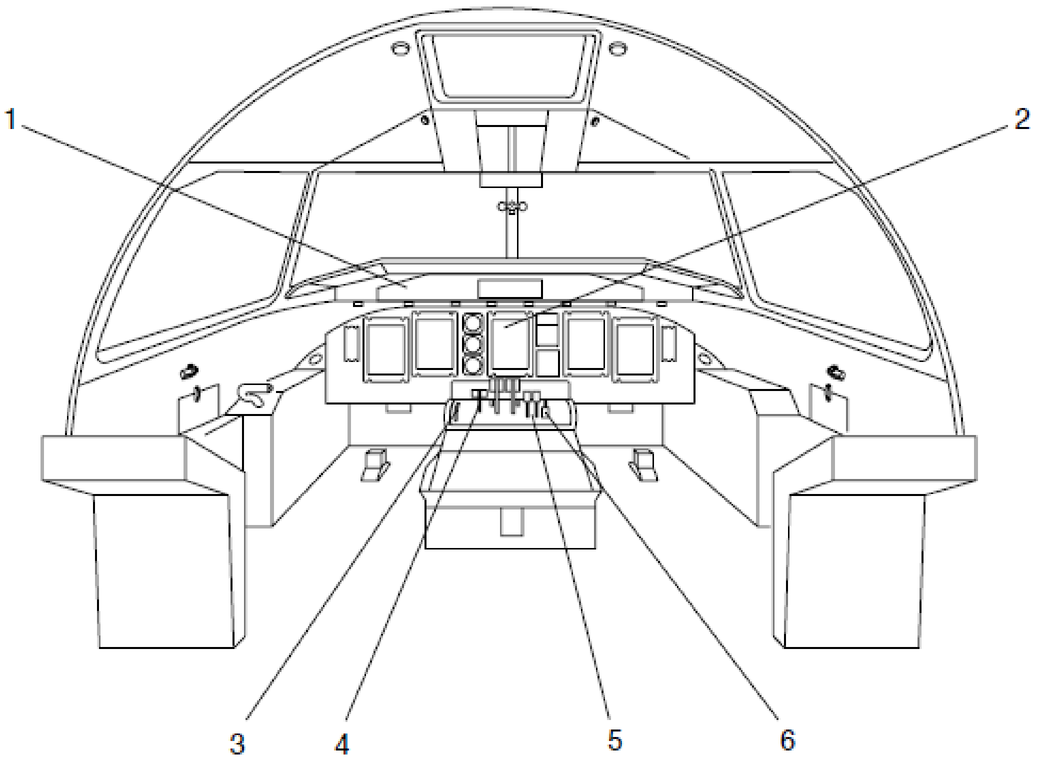
The Warning Tone Generator (WTG) sounds a 1000 Hz intermittent tone when the aircraft is not in a configuration which is safe for take−off. It functions when the two engine torque indications are more than 50% or the T/O WARN TEST Toggle switch on the pilot's side console is set to TEST and one or more of the parameters is sensed as follows:

- Flaps set to more than 20
- CLA 1 not set to 1020
- CLA 2 not set to 1020
- Parking brake set
- Inboard spoilers extended
- Outboard spoiler extended
- Left elevator trim out of range
- Right elevator trim out of range.

Detailed Description

The take-off warning system has the interfaces that follow:

Parameter	Source
T/O WARN MAINT	T/O WARN TEST Toggle switch
ENG1 TORQUE > 50%	FADEC
ENG2 TORQUE > 50%	FADEC
NGWOFW	PSEU
FLAPS ≤ 13.5°	FPIU
CLA1 ≠ MAX/1020	FADEC
CLA2 ≠ MAX/1020	FADEC
PARKING BRAKE ON	PARKING BRAKE LEVER
INBOARD SPOILERS EXTENDED	FCS ECU LEFT CHANNEL
LEFT ELEVATOR TRIM OUT OF RANGE	FCS ECU LEFT CHANNEL
OUTBOARD SPOILERS EXTENDED	FCS ECU RIGHT CHANNEL
RIGHT ELEVATOR TRIM OUT OF RANGE	FCS ECU RIGHT CHANNEL



LEGEND
1. Spoiler advisory lights.
2. Engine torque indication.
3. Elevator trim indicator.
4. Parking brake lever.
5. Condition levers.
6. Flap selector lever.

Figure 31- 2 Flight Compartment Take−Off Configuration

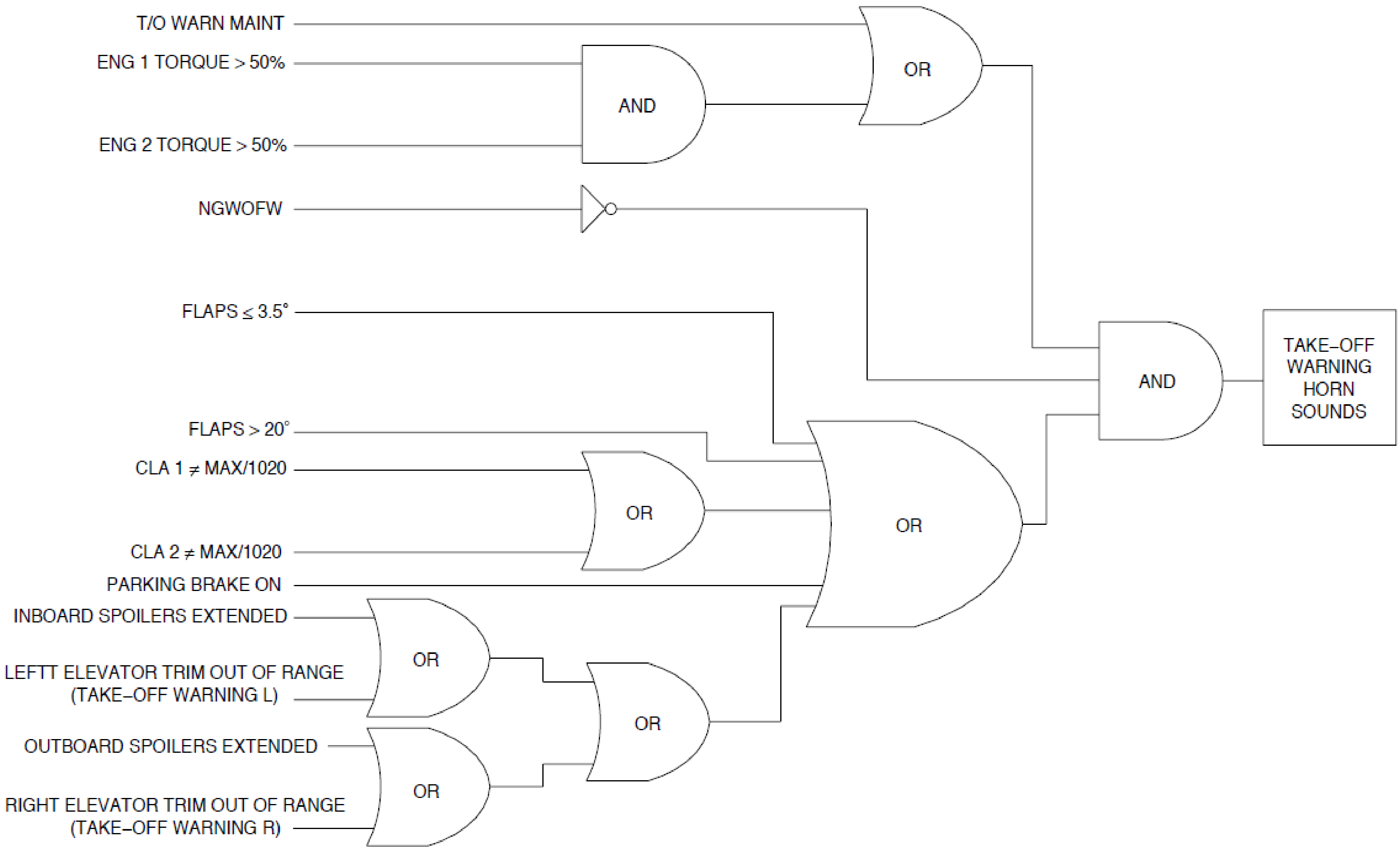


Figure - FDPS Incorrect Take−Off Configuration Warning

ELECTRONIC INSTRUMENTS SYSTEM

Introduction

The Electronic Instrument System (EIS) shows navigation, engine, and system parameters. It interfaces with other systems to calculate, make, and show their images. The Electronic Instrument System (EIS) also monitors its calculations to prevent misleading information from being shown.

General Description

Refer Figure - ELECTRONIC INSTRUMENT SYSTEM BLOCK DIAGRAM
Refer Figure - ELECTRONIC INSTRUMENT SYSTEM BLOCK DIAGRAM POWER

The electronic instruments system has five identical and interchangeable active matrix liquid crystal Display Units (DUs) with the same hardware and software part numbers. Different pin programming at each DU location gives the functions that follow:

- Pilot’s Primary Flight Display, PFD1
- Pilot’s Multi−Function Display, MFD1
- Engine Display, ED
- Copilot’s Primary Flight Display, PFD2
- Copilot’s Multi−Function Display, MFD2.

The Electronic Instrument System (EIS) is divided into the two subsystems that follow:

- Electronic Flight Instrument System (EFIS)
- Engine and System Integrated Display (ESID).

The Electronic Flight Instrument System (EFIS) is further subdivided into two systems, one for the pilot and one for the copilot. Each EFIS has the components that follow:

- Primary Flight Display (PFD)
- Multi−Function Display (MFD)
- Panels, EFIS Control (EFCP1, EFCP2)
- Panel, ESID Control (ESCP)
- Index Control Panels (ICP1, ICP2).

The Engine System Integrated Display (ESID) shares the MFDs with EFIS. It has the components that follow:

- Engine Display (ED)
- Panel, ESID Control (ESCP).

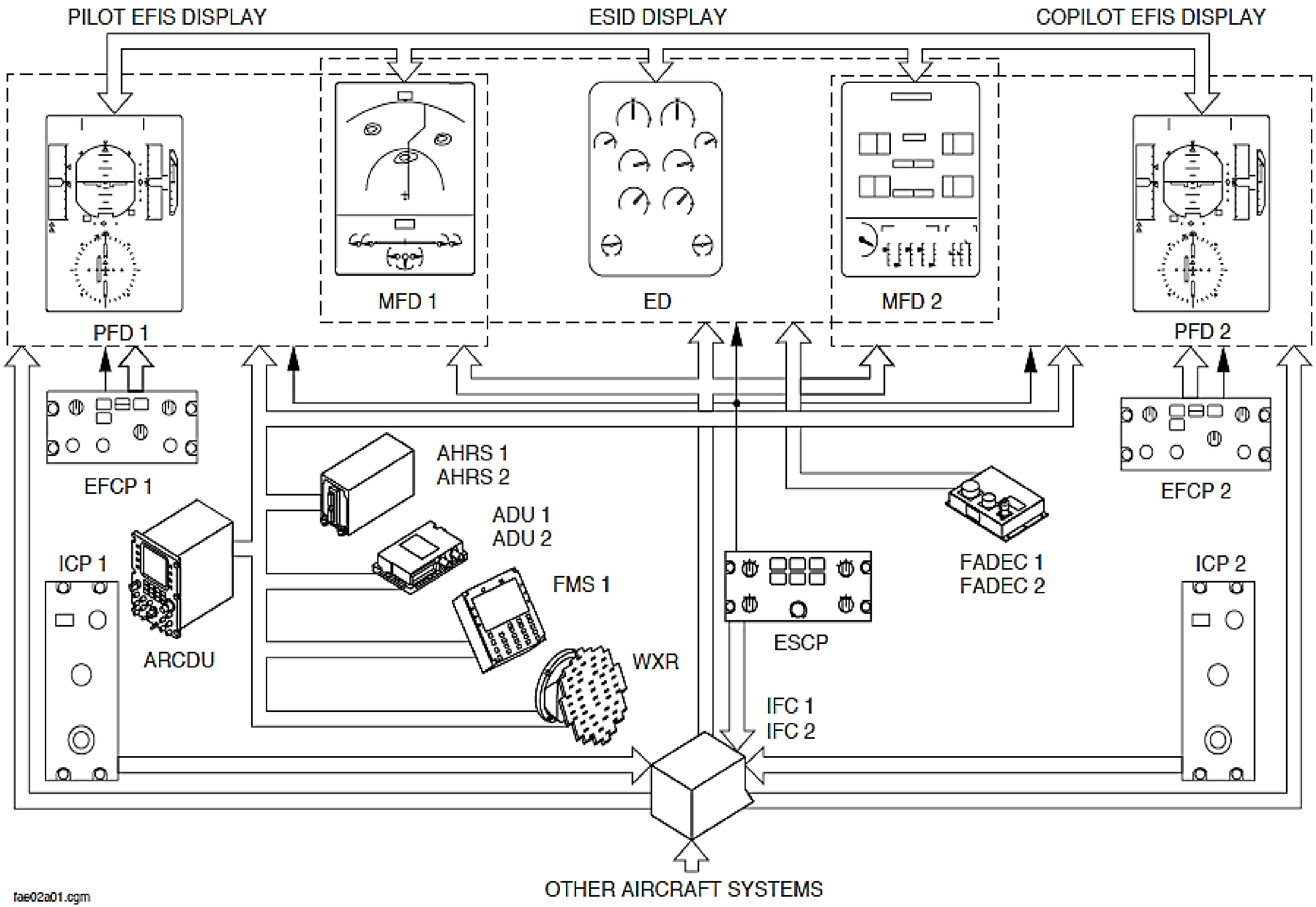


Figure - ELECTRONIC INSTRUMENT SYSTEM BLOCK DIAGRAM

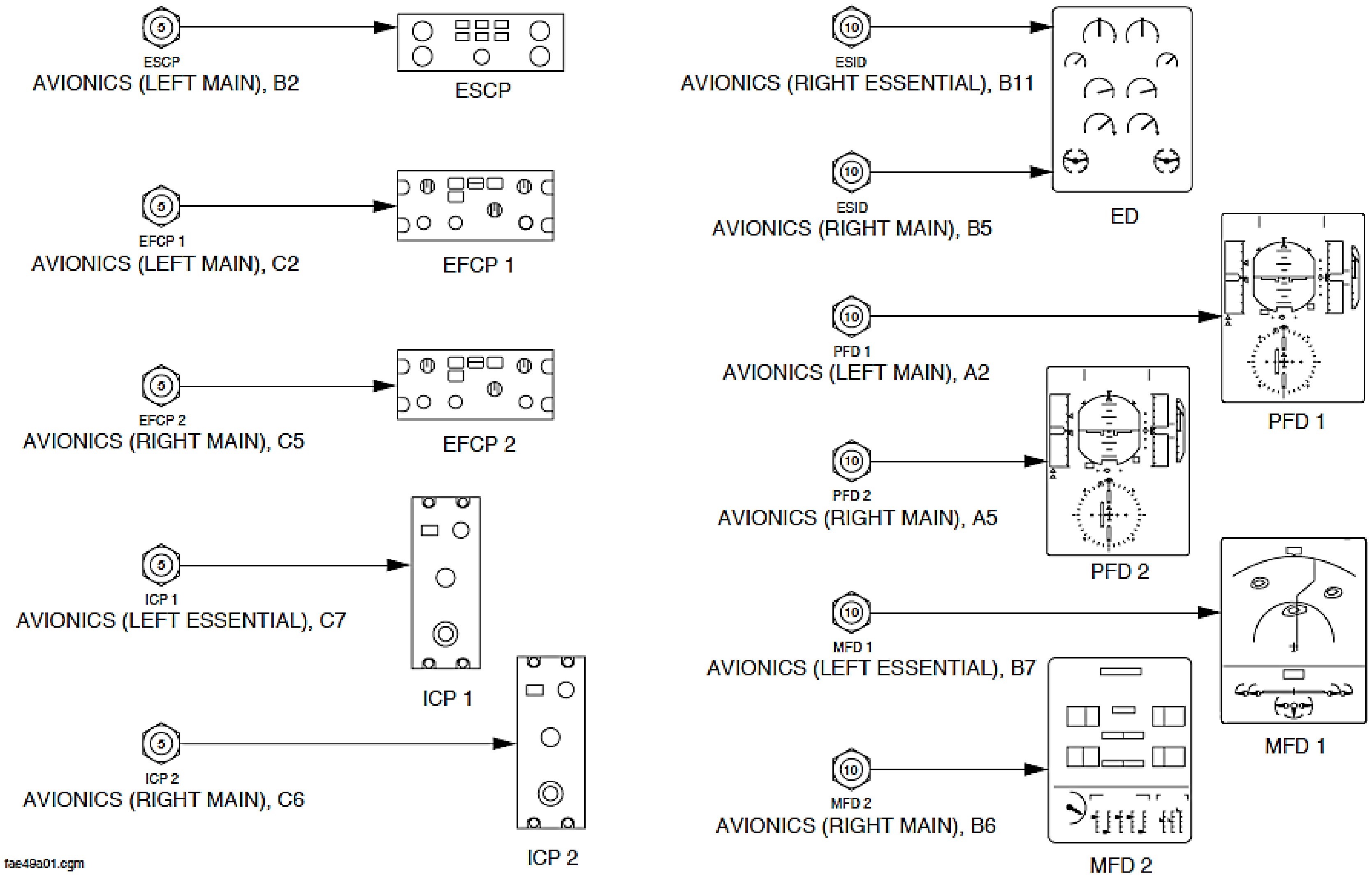


Figure - ELECTRONIC INSTRUMENT SYSTEM BLOCK DIAGRAM POWER

Detailed Description

The DUs function in the modes that follow:

- Power−On Self-Test (POST)
- Line operational
- Maintenance
- Teleloading.

Power−On Self-Test (POST): The Power−On Self-Test (POST) mode automatically continues for 2 seconds after power up to ensure correct operation before operational mode is started. The Power−On Self−Test (POST) does the checks that follow:

- Hardware program pin parity
- CPU/ASIC self-test
- RAM/EEPROM memory
- Input/Output (I/O) circuits and ports
- Hardware/Consistency.

During the self−test, the output data from DUs cannot be used. They show a green T during the Power−On Self−Test (POST) mode. The POST mode continues for 30 seconds.

During the POST mode, the DU checks the compatibility between the DU part number, hardware and software version. It will not start the operational mode if the checks are not correct.

The Display Units (DU) sense their internal temperature. They automatically turn an internal heater on while the temperature is too low to make sure the correct display brightness. It shows a WARMING UP message in white in the center of the otherwise blank screen.

The normal display replaces the message as soon as the internal temperature reaches the minimum operational requirement.

The WARMING UP message is shown after the Power−On Self−Test (POST) mode.

From −55 to −25 °C (−67 to −13 °F), the display can be of less than the nominal quality but shows the correct data. These images are available in less than 15 minutes after the POWER ON.

Nominal performance is available in less than 20 minutes of POWER ON.

From −25 °C (−13 °F) and above, the display can be of less than the nominal quality but shows the correct data. These images are available in less than 5 minutes after the POWER ON.

Nominal performance is available in less than 15 minutes of POWER ON.

Line Operational: Each DU calculates the position, size and the value of all parameters that is shown.

The DUs makes images similar to existing conventional symbology. These images have line strokes and raster to display critical flight data and guidance commands. Raster is a scan pattern in which an area is scanned from side to side in lines from top to bottom.

A circuit independent of channel that makes the images does a monitoring calculation to make sure that incorrect parameters are not shown. The DUs shows warnings when it senses a malfunction by continuous monitoring.

The indication brightness is automatically and manually controlled.

The DU brightness changes automatically with the ambient lighting conditions in the flight compartment. A light detector located on the front face of each Display Unit (DU) provides an ambient light input for automatic brightness control.

The manual brightness controls enable the pilot and copilot to change the brightness of each DU from a minimum to a maximum level. The brightness level is stays the same after a power interruption.

The manual brightness control can change the automatic control level in normal and failure conditions.

Maintenance Mode: The Built in Test Equipment (BITE) uses the Central Diagnostic System (CDS) to give the condition of the component. It saves faults in a Non-Volatile Memory (NVM) for reporting to line and shop maintenance.

A short or medium power interruption causes the DUs to go off and then start an initialization of the hardware. It does not start the Power−On Self-Test (POST) mode or show a malfunction indication. The DUs operates in the line operational mode after the power interruption. (The display flickers).

A long power interruption while the aircraft is on the ground causes a POST that continues for 30 seconds.

A long power interruption while the aircraft is airborne starts a short hardware and RAM initialization that continues for 2 seconds.

When a DU senses an internal malfunction, it will stay in the POST mode for the duration of the malfunction.

The BITE allows aircraft maintenance personnel to:

- Do fault isolation and return to service testing after completing maintenance actions
- Access failure reports from last or previous flight legs
- Get the avionics status report
- Get the part number of an LRU.

The BITE mode monitors the condition of the component as follows:

- Power−On Self-Test (POST)
- Continuous monitoring.

Power−On Self-Test (POST): The POST checks the condition of the component at power−up or after a long power interruption more than 200 milliseconds.

Refer Figure - ELECTRONIC INSTRUMENT SYSTEM DISPLAY MONITORING

Continuous Monitoring: The continuous monitoring checks the status of the components in flight. It records faults in a Non-Volatile Memory (NVM) for later troubleshooting using the Central Diagnostic System (CDS).

The DUs monitor the critical indication parameters that follow:

- Pitch
- Roll
- Altitude
- Airspeed
- Heading
- Torque
- ITT
- NH
- NP.

A mismatch condition between inputs from critical parameters is monitored by the Flight Data Processing System (FDPS). When a difference is sensed by the FDPS, it causes the PFDs to show a mismatch message.

Display monitoring makes sure that the symbols are shown correctly. A feedback bus is used to compare the shown indication with the direct input.

The monitoring is done by one or two DUs as follows:

- The PFD is monitored by its adjacent MFD (if not set to PFD) and opposite PFD. If the opposite PFD is de–energized, it uses the opposite MFD.
- The ED is monitored by its adjacent MFDs. If one MFD is set to PFD, it uses the other MFD.

The threshold for PFD monitoring is shown in the table that follows:

PARAMETER	THRESHOLD
Roll	4 degrees
Pitch	4 degrees
Heading	6 degrees
Indicated Air Speed	10 knots
Corrected Barometric Altitude	100 feet

The threshold for ED monitoring is shown in the table that follows:

PARAMETER	THRESHOLD
Torque	5 percent
NH	5 percent
NP	50 rpm
ITT	50 °C

When a difference between the shown indication and the direct input is more than the threshold, the EIS shows the messages that follow:

- CHECK PFD on PFDs and MFDs
- CHECK ED on ED and MFDs.

An incorrect response from a switch selection is immediately sensed by the pilot but a malfunction causes set parameters to change to dashes.

Teleloading: The operational software serial data is supplied to the DUs.

NOTE

Note: The DU must be removed from the aircraft to receive a teleload.

The DUs do a software configuration check before the teleloading mode starts. It makes sure that the software version to be teleloaded is the same as the equipment part number.

The teleload time is less than 5 minutes for each DU. the DUs show a green L during the Teleloading mode.

The Display Unit (DU) restarts after a complete and successful teleload. It initiates a complete Power−On Self-Test (POST) within 10 seconds.

The DU starts a POST less than 10 seconds after the teleload mode is complete.

There are programming pins located on the DU connectors to supply data to the DUs that follows:

- Aircraft type
- Location of the DU on the aircraft
- Selection of the DU
- Shop maintenance
- Teleloading.

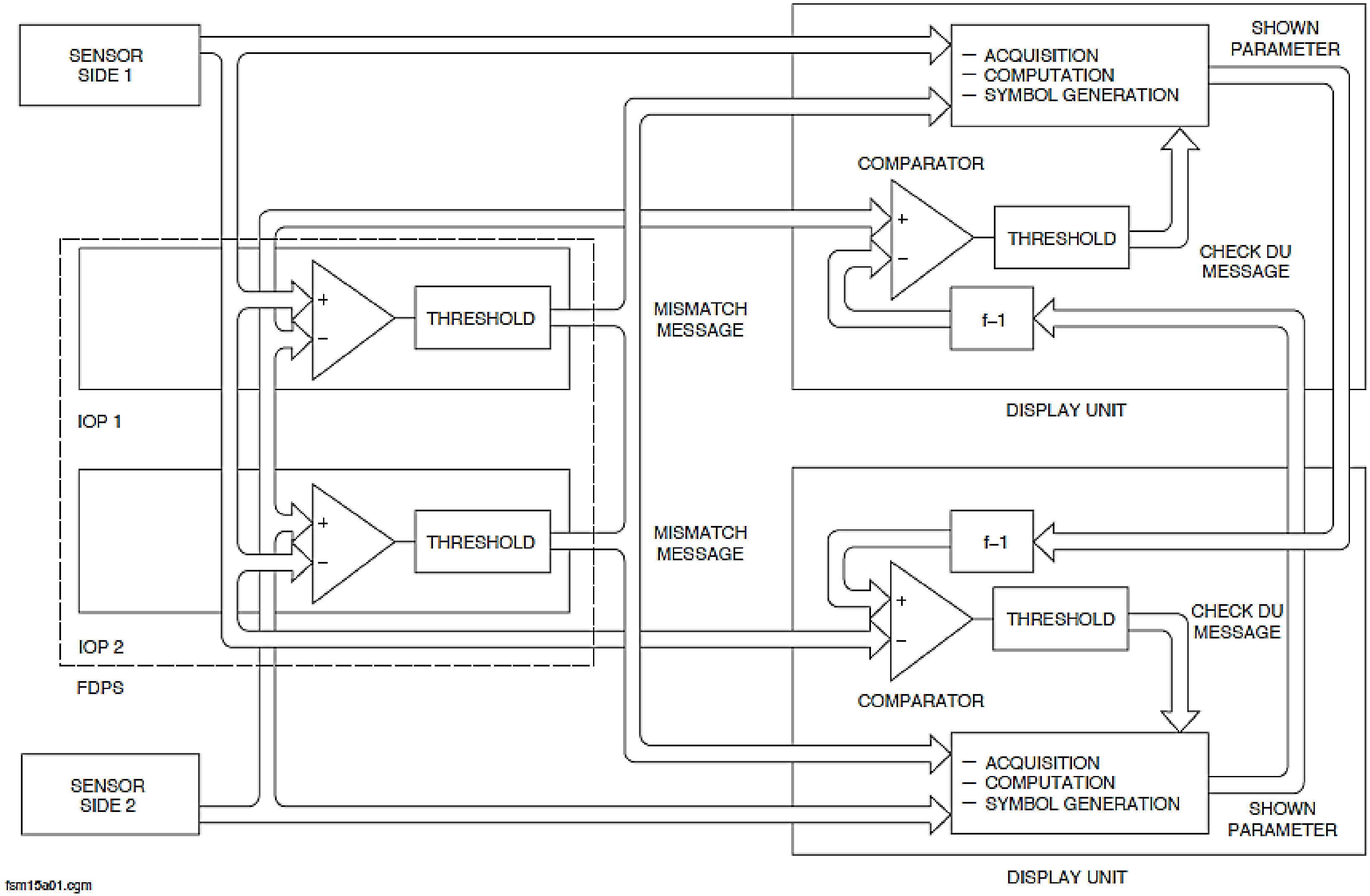


Figure - ELECTRONIC INSTRUMENT SYSTEM DISPLAY MONITORING

Refer Figure - ELECTRONIC INSTRUMENT SYSTEM INDICATION SELECTION

When the pilot is flying the aircraft, the MFD1 reversion selector is set to the NAV position and the MFD2 reversion selector is set to the SYS position on the ESCP.

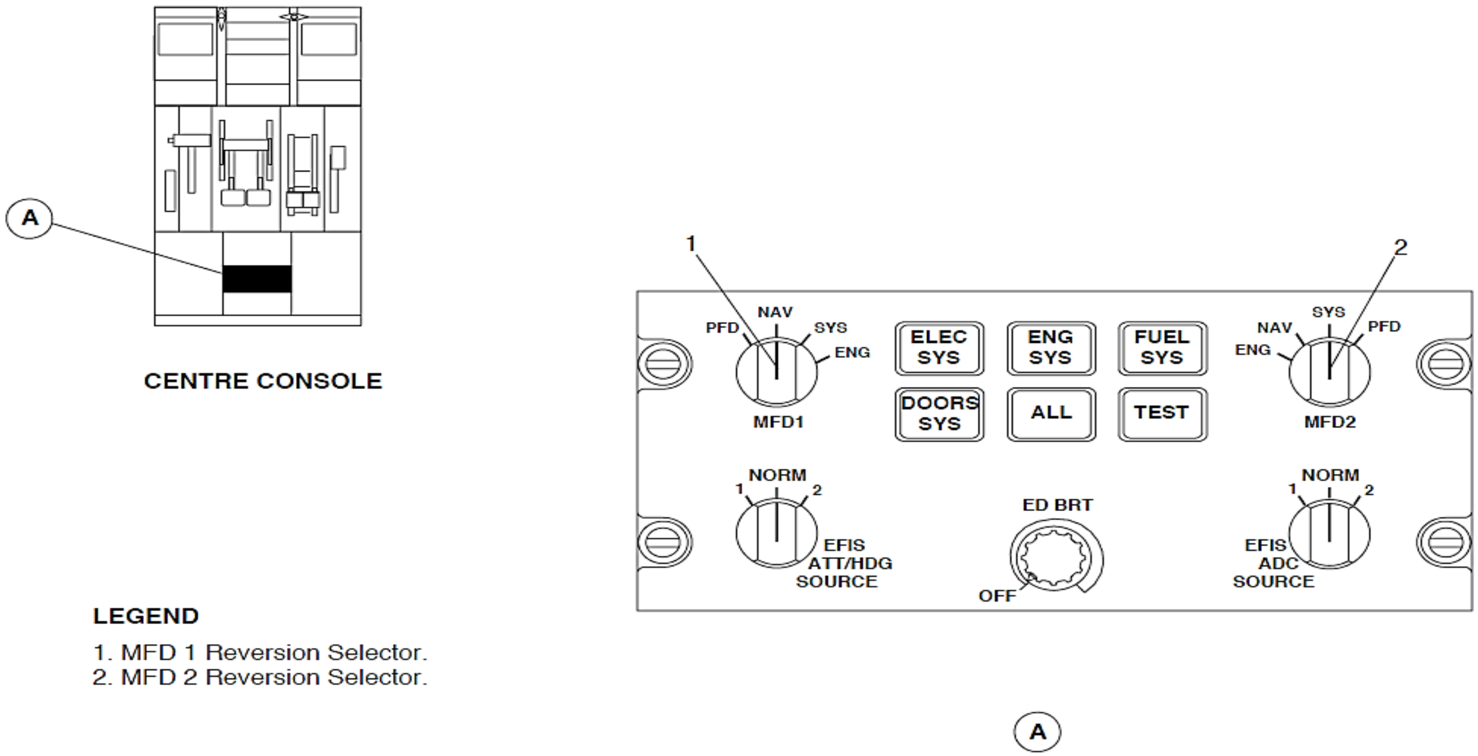


Figure - ELECTRONIC INSTRUMENT SYSTEM INDICATION SELECTION

Figure 31- 76 ELECTRONIC INSTRUMENT SYSTEM INDICATIONS PAGE **1**

The DUs show the Electronic Instruments System (EIS) data that follows:

- Primary Flight Display (PFD1)
- Primary Flight Display (PFD2)
- Engine Display (ED).
- Multi–Function Displays (MFD1)
- Multi–Function Displays (MFD2).

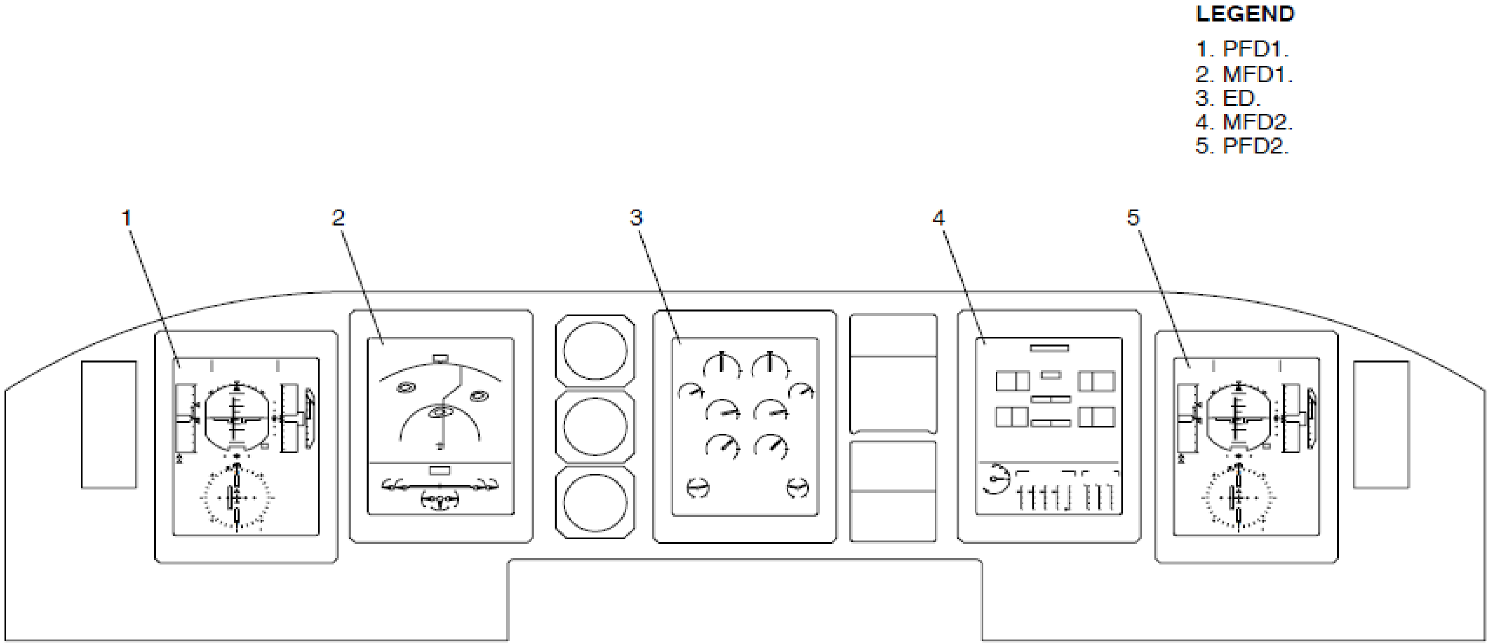


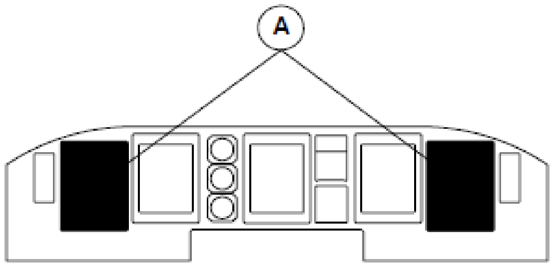
Figure 31- 3 ELECTRONIC INSTRUMENT SYSTEM INDICATIONS PAGE 1

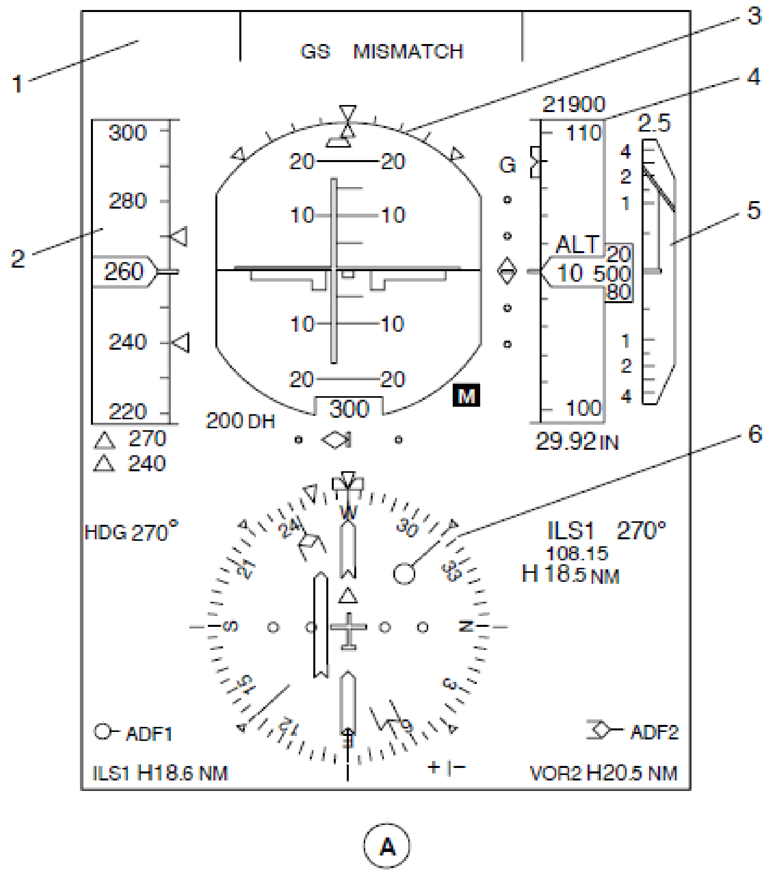
Refer Figure - ELECTRONIC INSTRUMENT SYSTEM PFD INDICATIONS

Primary Flight Displays (PFD1, PFD2): The Primary Flight Displays (PFD) are the outer displays of the Electronic Instrument System (EIS). Each display shows the primary navigation information that follows:

- Flight Mode Annunciator (FMA)
- Indicated Air Speed (IAS)
- Altimeter (ALT)
- Electronic Attitude Direction Indicator (EADI)
- Electronic Horizontal Situation Indicator (EHSI)
- Inertial Vertical Speed Indicator (IVSI)
- Traffic Collision Avoidance System II (TCAS II).

The MFDs are located inboard of the PFDs. Their indication is divided into parts when they are not showing a Primary Flight Display (PFD) or Engine Display (ED) reversionary indication.





LEGEND

1. Flight Mode Annunciator (FMA).
2. Air Speed Indicator (IAS).
3. Electronic Attitude Direction Indicator (EADI).
4. Altimeter (ALT).
5. Inertial Vertical Speed Indicator (IVSI), TCAS.
6. Electronic Horizontal Situation Indicator (EHSI).

Figure - ELECTRONIC INSTRUMENT SYSTEM PFD INDICATIONS

Refer Figure - ELECTRONIC INSTRUMENT SYSTEM MFD1 INDICATION, MAP FORMAT

The MFD1 is usually set to show the Electronic Flight Instrument System (EFIS) navigation page, map format at the top part of the indication and the Engine and System Integrated Display (ESID) permanent system data area at the bottom. Its permanent system data area usually shows the Powered Flight Control Surface Position (PFCS) parameters.

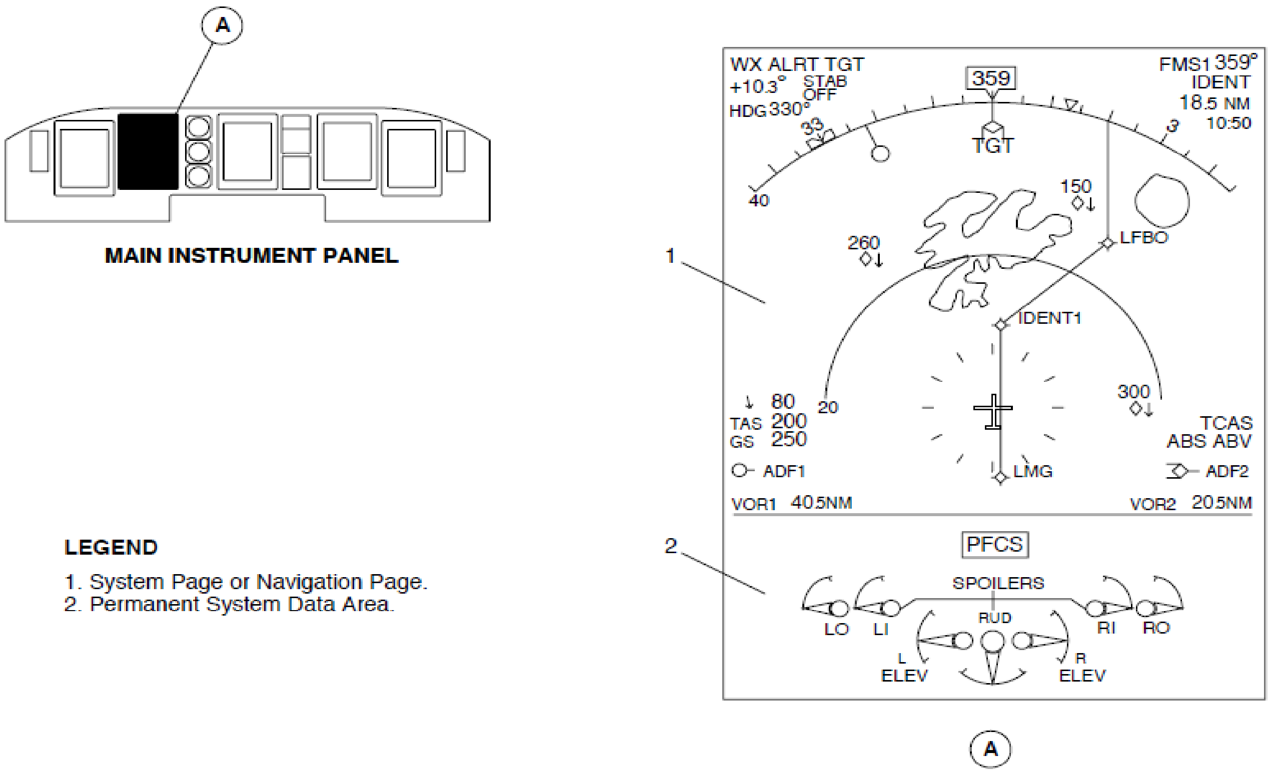


Figure - ELECTRONIC INSTRUMENT SYSTEM MFD1 INDICATION, MAP FORMAT

Refer Figure - ELECTRONIC INSTRUMENT SYSTEM MFD1 INDICATION, ARC FORMAT

The navigation page shows also shows the arc format.

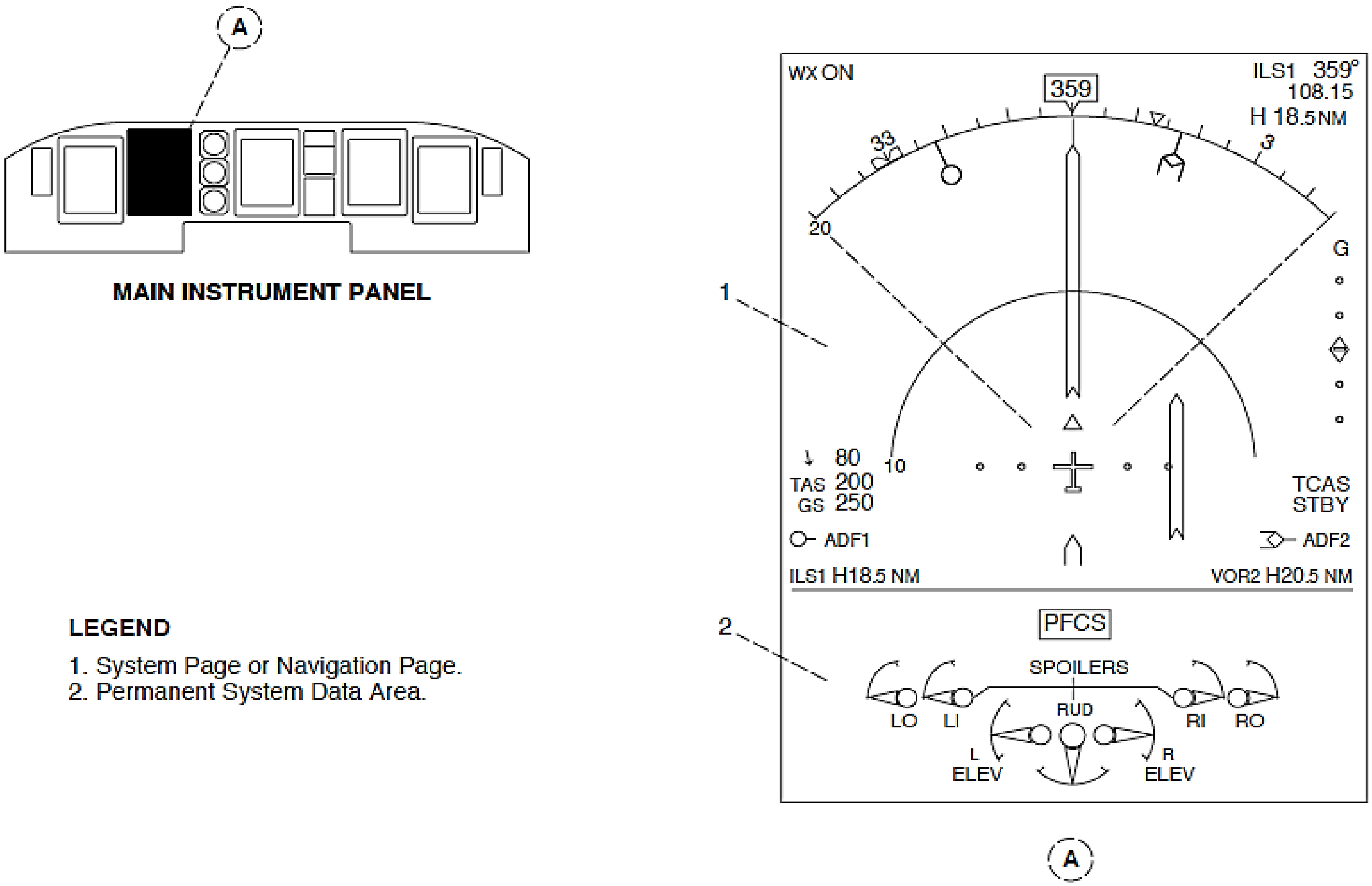


Figure - ELECTRONIC INSTRUMENT SYSTEM MFD1 INDICATION, ARC FORMAT

Refer Figure - ELECTRONIC INSTRUMENT SYSTEM MFD1 INDICATION, PLAN FORMAT

The navigation page shows also shows the plan format.

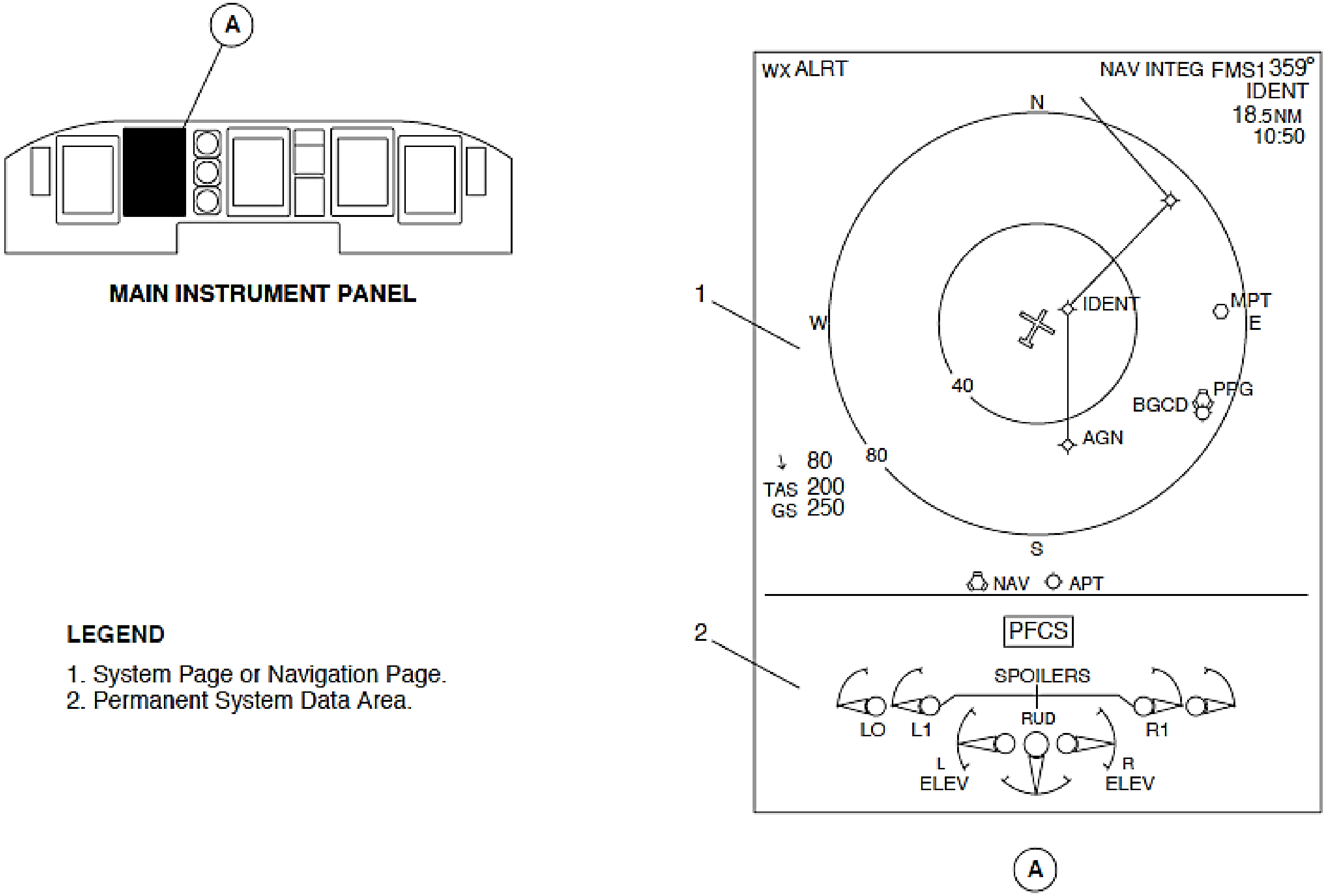


Figure - ELECTRONIC INSTRUMENT SYSTEM MFD1 INDICATION, PLAN FORMAT

Refer Figure - ELECTRONIC INSTRUMENT SYSTEM MFD2 INDICATIONS

The MFD2 is usually set to show the Engine and System Integrated Display (ESID) electrical system page at the top part of the indication and permanent system data area at the bottom. Its permanent system data area usually shows the parameters that follow:

- Flap position
- Hydraulic pressure
- Hydraulic quantity.

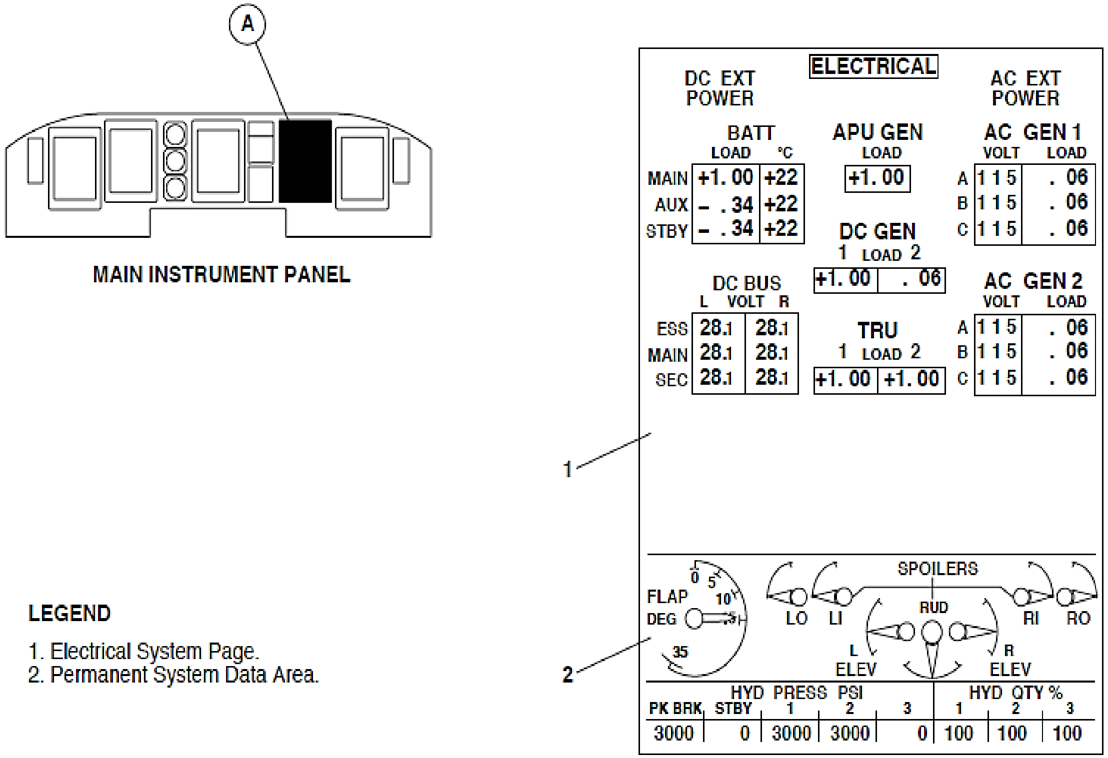


Figure - ELECTRONIC INSTRUMENT SYSTEM MFD2 INDICATIONS

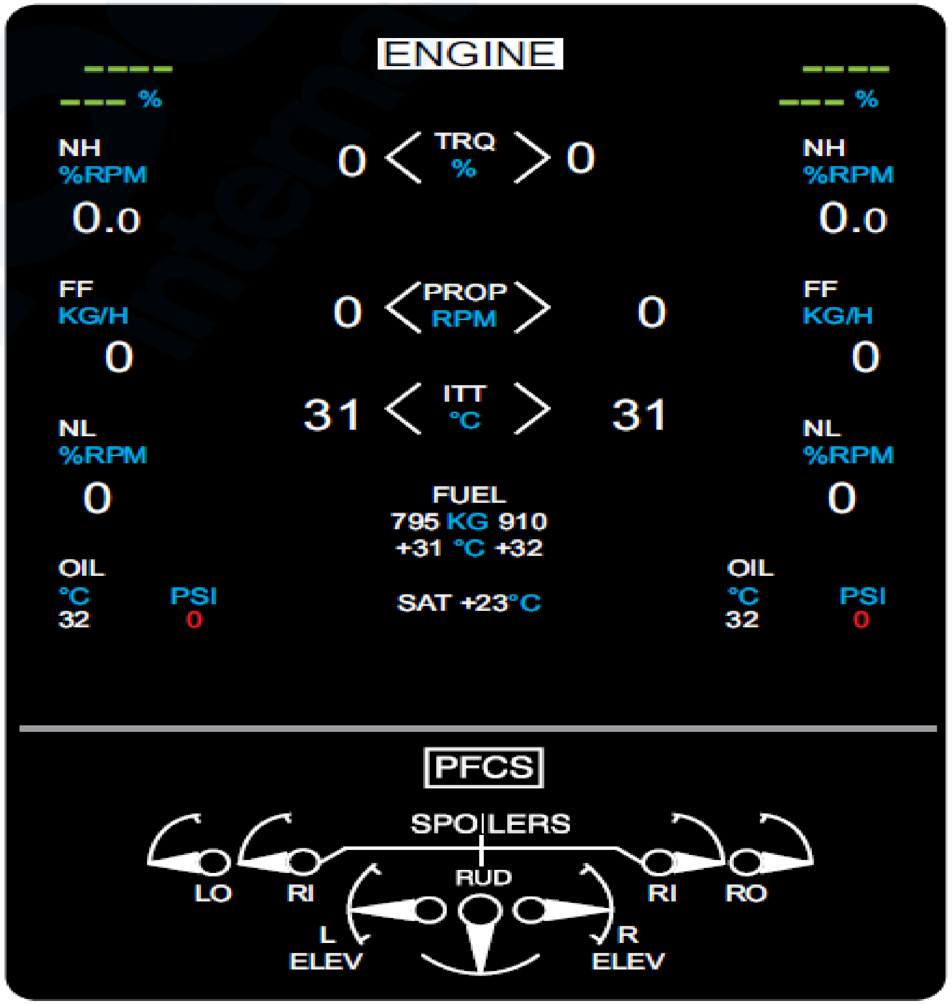


Figure - ELECTRONIC INSTRUMENT SYSTEM ENGINE SYSTEM PAGE

Refer Figure - ELECTRONIC INSTRUMENT SYSTEM ENGINE SYSTEM PAGE
Refer Figure - ELECTRONIC INSTRUMENT SYSTEM DOORS SYSTEM PAGE
Refer Figure - ELECTRONIC INSTRUMENT SYSTEM FUEL SYSTEM PAGE

The Engine and System Integrated Display (ESID) system shows the additional system pages that follows:

- Engine, or
- Fuel, or
- Doors.

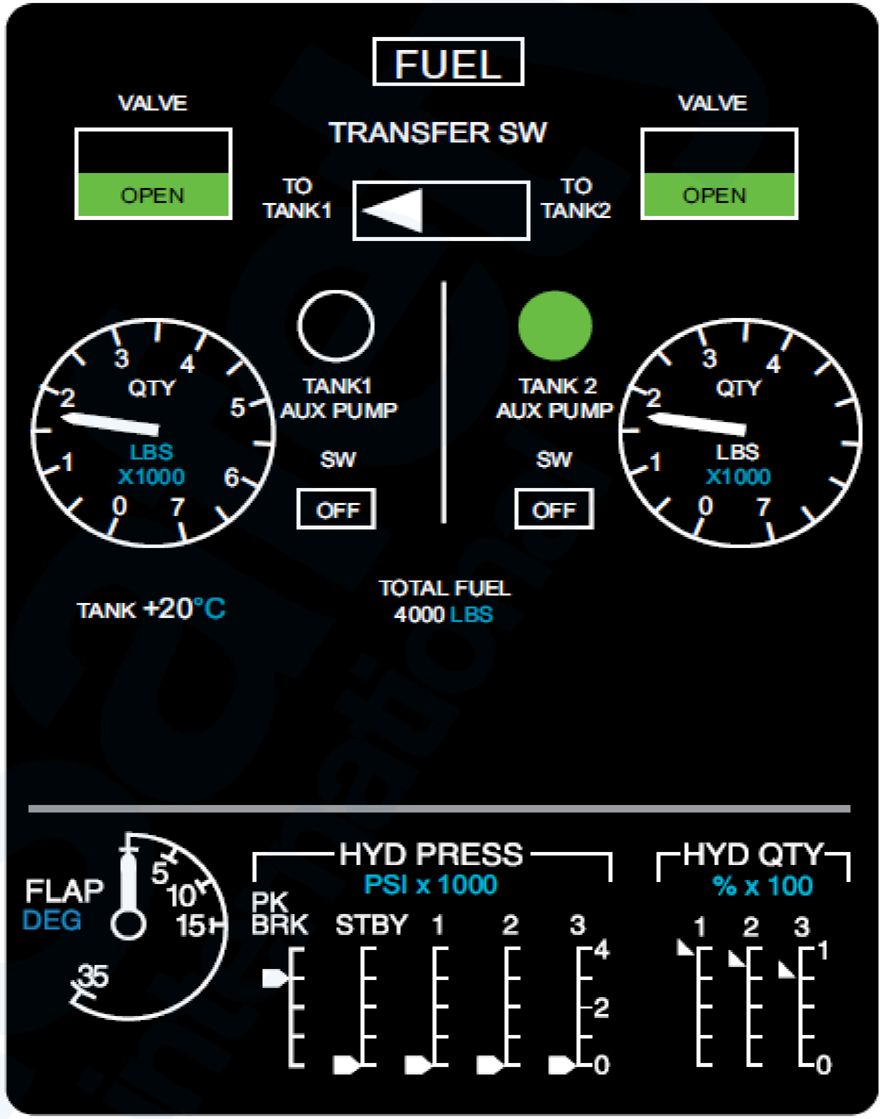


Figure - ELECTRONIC INSTRUMENT SYSTEM FUEL SYSTEM PAGE

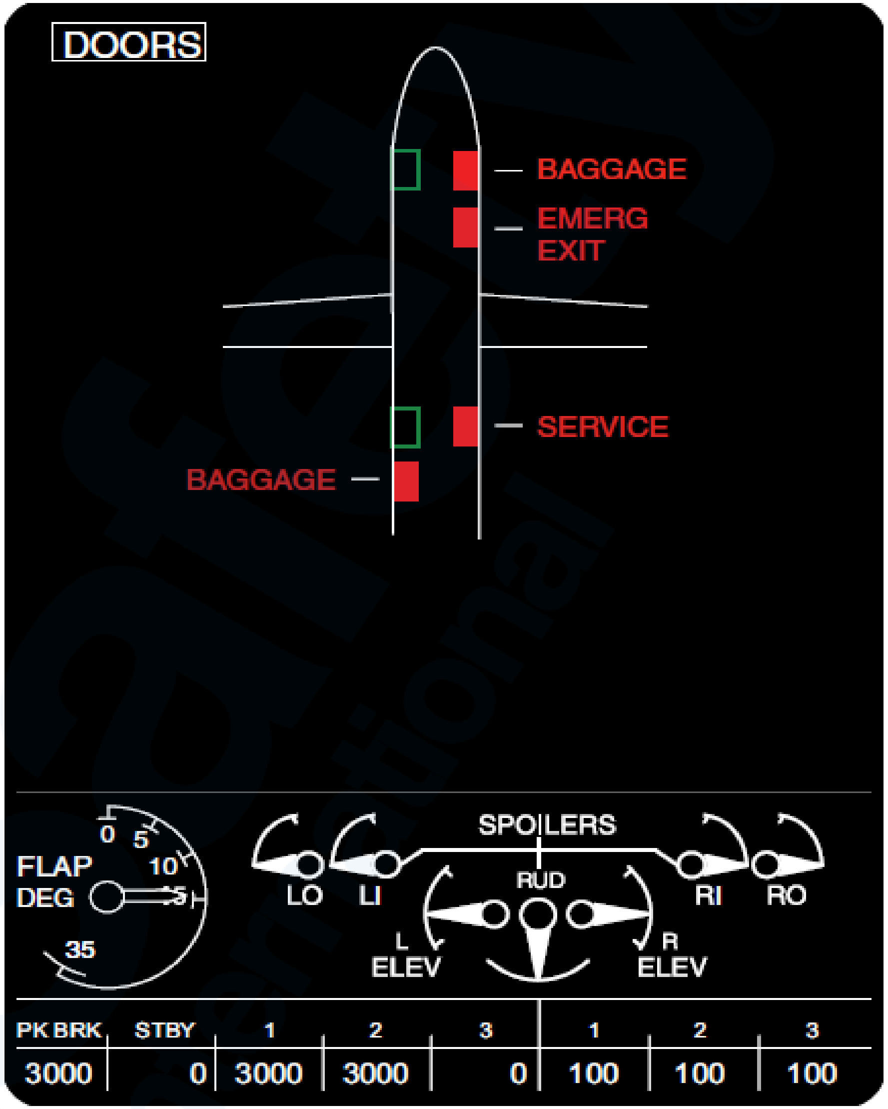


Figure - ELECTRONIC INSTRUMENT SYSTEM DOORS SYSTEM PAGE

Refer Figure - Engine Display (ED)

The Engine Display (ED) is located at the center of the instrument panel. It shows primary engine and aircraft systems data that follows:

- Torque
- NH High Pressure Compressor Rotational Speed
- NP Propeller Rotational Speed
- ITT Inter−Turbine Temperature
- Fuel Flow
- NL Low Pressure Compressor Rotational Speed
- Engine rating mode
- Oil Pressure
- Oil Temperature
- Tank Fuel Quantity
- Fuel inlet temperature
- Advisory messages
- Static Air Temperature.
- System Page (Mono Mode).

Three different types of advisory messages are shown in white letters at the bottom part of the Engine Display (ED). An IFC, powerplant, or display related message is shown at a different location while the condition is present.

The messages are shown in order of decreasing importance. The most important display message is shown at the bottom right part of the ED. The DUs cause the ED to show the advisory messages that follow:

- DU BAD CONF
- ED MON FAIL
- PFD1,PFD2 MON FAIL

- HOT PFD1, MFD1, ED, MFD2 or PFD2
- HOT DISPLAYS

- PFD1, MFD1, MFD2 or PFD2 LINK FAIL.

DU BAD CONF Message: The DU BAD CONF message comes into view when a bad or incorrect aircraft configuration is sensed by a display unit.

NOTE

Note: The DU BAD CONF message comes into view after a POST while the aircraft is on the ground.

ED MON FAIL Message: The ED MON FAIL message comes into view when the ED critical parameters are not monitored another display.

PFD1, PFD2 MON FAIL Message: The PFD1 or PFD2 MON FAIL message comes into view when the PFD critical parameters are not monitored another display unit.

HOT PFD1, MFD1, ED, MFD2 or PFD2 Message: The HOT display unit message comes into view when one display unit senses an overheat condition.

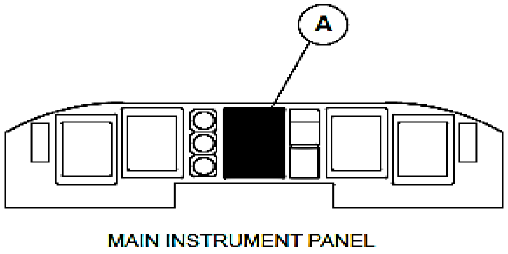
NOTE

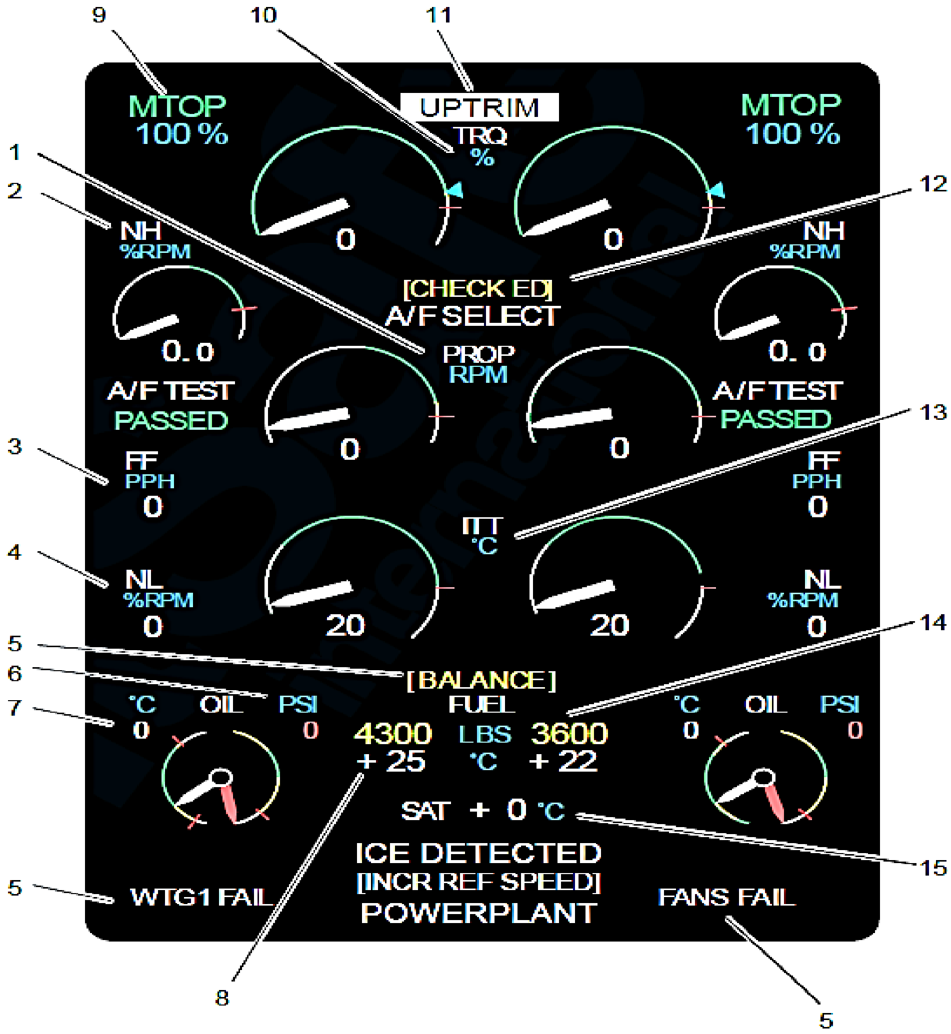
Note: The related display unit is automatically dimmed to decrease its temperature.

HOT DISPLAYS Message: The HOT DISPLAYS message comes into view when two or more display units sense an overheat condition. PFD1, MFD1, MFD2 or PFD2 LINK FAIL Message: The PFD1, MFD1, MFD2 or PFD2 LINK FAIL message comes into view to tell the pilots that the ED will not show an advisory message from the related display unit.

The Electronic Instruments System (EIS) has the control panels that follow to control the DU indications:

- EFIS Control Panels (EFCP1)
- EFIS Control Panels (EFCP2)
- ESID Control Panel (ESCP).
- Index Control Panel (ICP1)
- Index Control Panel (ICP2).





LEGEND

1. NP Propeller Rotational Speed Indicator.
2. NH High Pressure Compressor Rotational Speed Indicator.
3. Fuel Flow Indicator.
4. NL Low Pressure Compressor Rotational Speed Indicator.
5. Advisory Messages.
6. Oil Pressure Indication.
7. Oil Temperature Indication.
8. Fuel Inlet Temperature Indication.
9. Engine Rating Mode Indication.
10. Torque Indication.
11. Powerplant Messages.
12. Check DU Message.
13. ITT Inter-Turbine Temperature.
14. Tank Fuel Quantity Indication.
15. Static Air Temperature Indication.

Figure - Engine Display (ED)

Refer Figure - ELECTRONIC INSTRUMENT SYSTEM EFCP PAGE 1

EFIS Control Panels (EFCP1, EFCP2): On aircraft with the MFD navigation source selection independent of the PFD navigation source selection (823CH00015): The EFCPs have the controls that follow:

- Bearing 1 selector knob
- Bearing 2 selector knob
- TCAS pushbutton switch
- WX/TERR pushbutton switch
- DATA pushbutton switch
- FORMAT pushbutton switch
- RANGE selector knob
- PFD OFF, BRT knob
- MFD OFF, BRT knob
- WX/TERR BRT knob.

Bearing 1, 2 Selector Knob: The bearing 1 or 2 selector knob is turned to make an AUX, VOR, ADF, or FMS bearing data selection for indication on the EFIS PFD.

TCAS pushbutton Switch: The TCAS pushbutton switch is pushed to change the TCAS indication from automatic (pop−up) to on. It is pushed again to show the automatic indication. The automatic selection, only shows Traffic Advisory (TA) and Resolution Advisory (RA) indications. The on selection shows all TCAS traffic indications.

NOTE

Note: A TCAS pushbutton switch selection has no effect when the navigation page is not in view.

WX/TERR Pushbutton Switch: The WX/TERR pushbutton switch is pushed to change the weather radar indication from on to off.

It is pushed again to show the EGPWS terrain indication. A third WX/TERR pushbutton switch selection causes the MFD to show the weather radar indication again.

If the navigation page is set to show a weather radar indication (ARC or MAP format) and an EGPWS terrain alert is sensed, the terrain indication automatically comes into view. If the navigation page is set to the PLAN format, an EGPWS terrain alert causes it to change the ARC format. An EGPWS terrain alert always causes the weather radar range indication to change to 10 NM.

NOTE

Note: A WX/TERR pushbutton switch selection has no effect when the navigation page is not in view.

DATA Pushbutton Switch: The DATA pushbutton switch is pushed to show FMS navigation aids and airports when the navigation page is set to MAP or PLAN format. The first selection causes the navigation page to show the nearest navigation aids. It is pushed again to shown the nearest airports. The third selection causes the navigation page to show the nearest navigation aids and airports. The fourth selection causes the data indications to go out of view.

NOTE

Note: A DATA pushbutton switch selection has no effect when the navigation page is set to show an ARC VOR/ILS or ARC MLS format.

FORMAT Pushbutton Switch: The FORMAT pushbutton switch is pushed to change to the navigation page indication from a map (FMS) to arc (VOR/ILS) indication. It is pushed again to show the map indication.

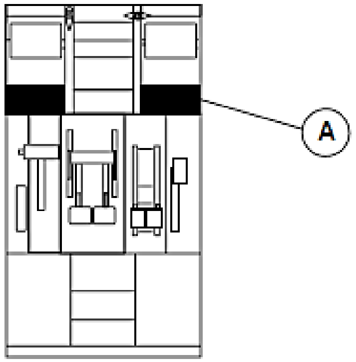
If the FORMAT pushbutton switch is pushed for more than one second, the plan format is shown. It is pushed again to show the map indication.

RANGE selector knob: The RANGE pushbutton switch is pushed to change the weather radar range indications on the navigation page.

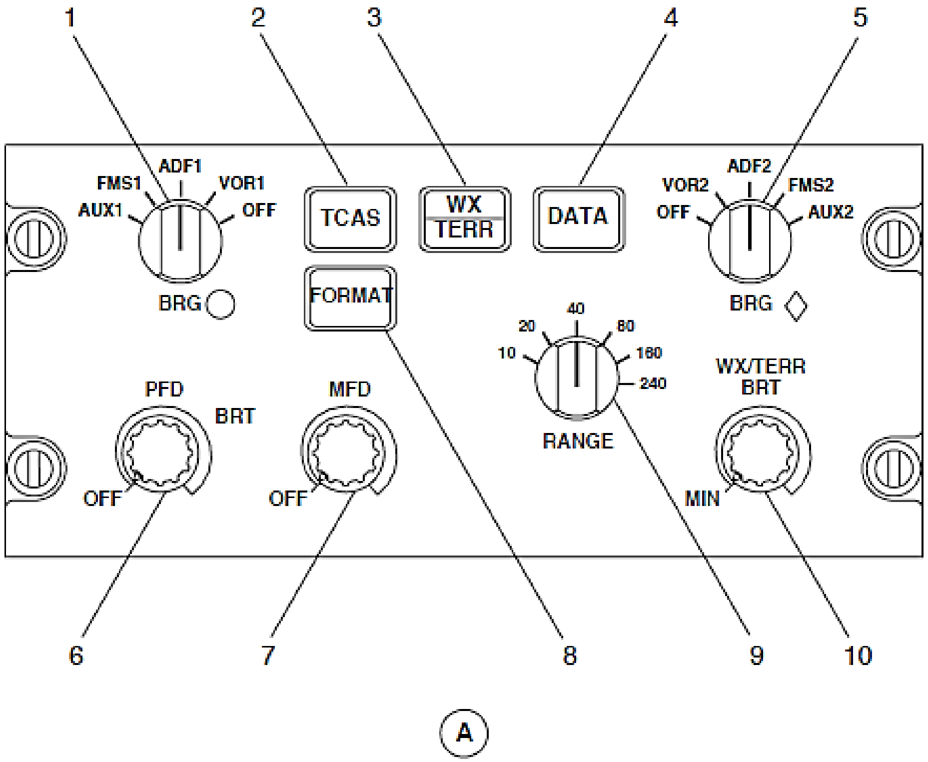
PFD OFF, BRT Knob: The PFD OFF, BRT Knob is turned to energize or de−energize the PFD and to control its brightness.

MFD OFF, BRT Knob: The MFD OFF, BRT Knob is turned to energize or de−energize the MFD and to control its brightness.

WX/TERR BRT knob: The WX/TERR BRT knob is turned to adjust the weather radar image brightness.



CENTRE CONSOLE



LEGEND

1. BRG 1 Selector Knob.
2. TCAS Pushbutton Switch.
3. WX/TERR Pushbutton Switch.
4. DATA Pushbutton Switch.
5. BRG 2 Selector Knob.
6. PFD OFF, BRT Knob .
7. MFD OFF, BRT Knob.
8. FORMAT Pushbutton Switch.
9. RANGE Selector Knob.
10. WX/TERR BRT Knob.

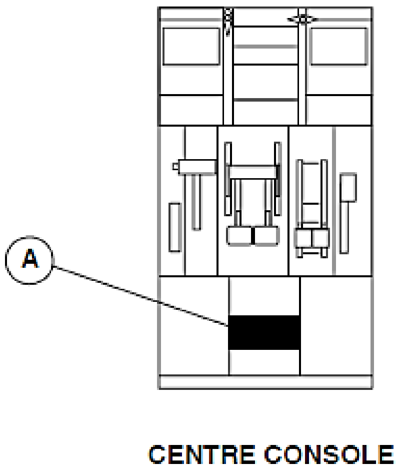
Figure - ELECTRONIC INSTRUMENT SYSTEM EFCP PAGE 1

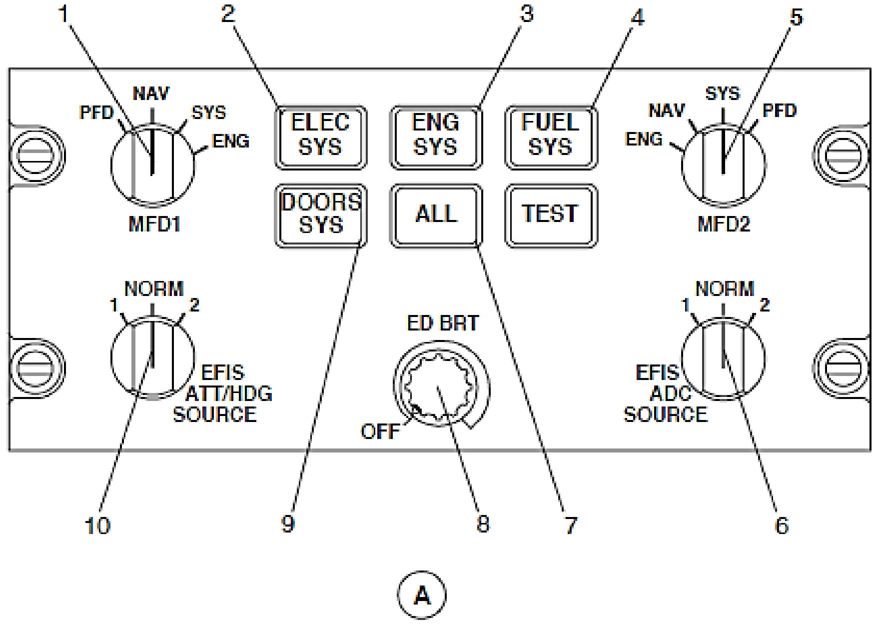
Refer Figure - ELECTRONIC INSTRUMENT SYSTEM ESCP PAGE 2

ESID Control Panels (ESCP): To control the Engine and System Integrated Display ESID) indications and some EFIS indications, the ESCP has the controls that follow:

- MFD1 and MFD2 Reversion Switches
- ELEC SYS Pushbutton Switch
- ENG SYS Pushbutton Switch
- FUEL SYS Pushbutton Switch
- DOORS SYS Pushbutton Switch
- ALL Pushbutton Switch
- EFIS ATT/HDG SOURCE reversion selector
- EFIS ADC SOURCE Reversion Selector
- ED OFF, BRT Knob.

MFD1 and MFD2 Reversion Switches: The MFD1 and MFD2 reversion switches are used to make indication selections on the ESID DUs.





LEGEND

1. MFD 1 Reversion Switches.
2. ELEC SYS Pushbutton Switch.
3. ENG SYS Pushbutton Switch.
4. FUEL SYS Pushbutton Switch.
5. MFD 2 Reversion Switches.
6. EFIS ADC Source Reversion Selector.
7. ALL Pushbutton Switch.
8. ED OFF, BRT Knob.
9. Door System Pushbutton Switch.
10. EFIS ATT/HDG Source Reversion Selector.

Figure - ELECTRONIC INSTRUMENT SYSTEM ESCP PAGE 2

Refer Figure - ELECTRONIC INSTRUMENT SYSTEM ESCP, ELECTRICAL SYSTEM PAGE SELECTION
Refer Figure - Multi-Function Display 2(MFD 2) – Electrical System Page

ELEC SYS Pushbutton Switch: The ELEC SYS pushbutton switch is pushed to show the electrical system page on the ESID DUs.

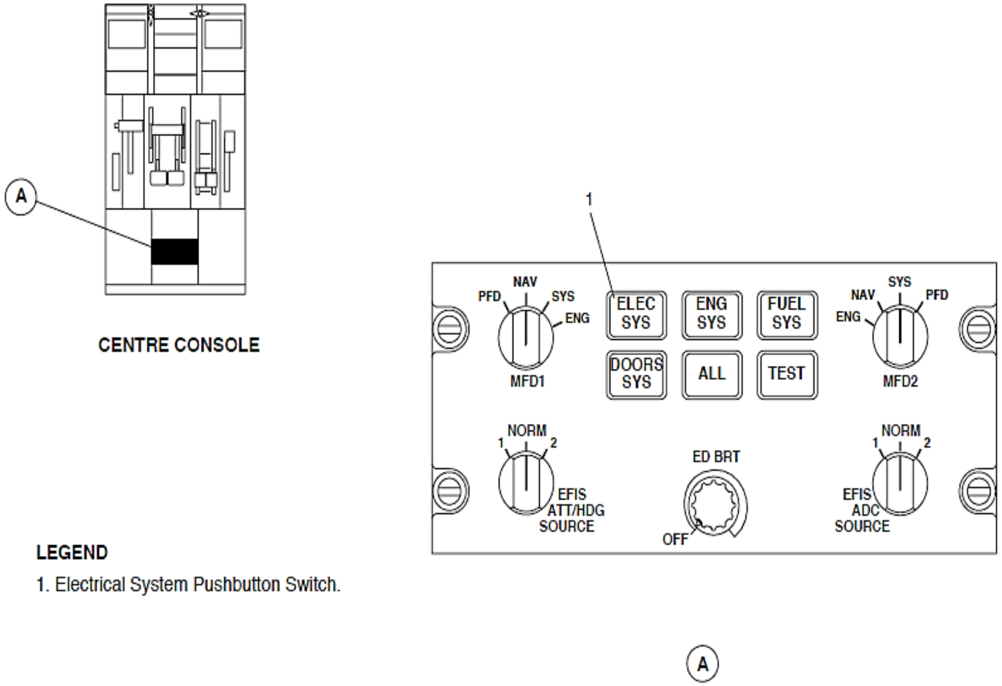


Figure - ELECTRONIC INSTRUMENT SYSTEM ESCP, ELECTRICAL SYSTEM PAGE SELECTION

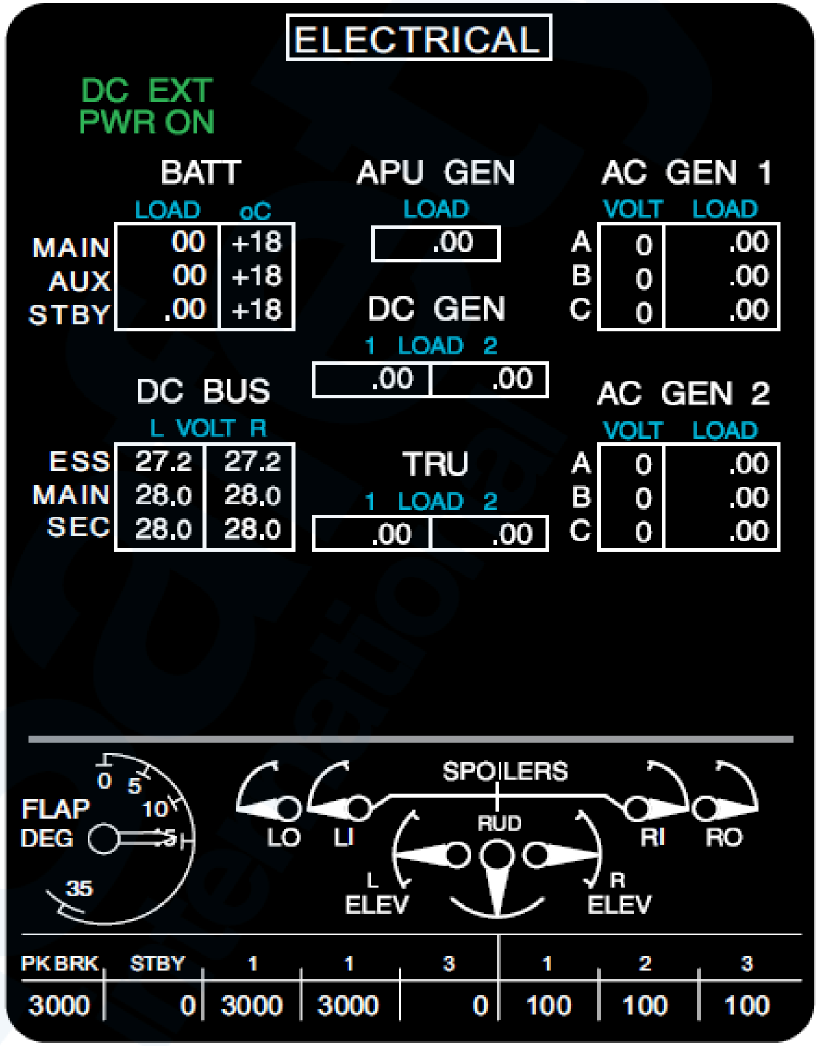


Figure - Multi-Function Display 2(MFD 2) – Electrical System Page

Refer Figure - ELECTRONIC INSTRUMENT SYSTEM ESCP, ENGINE SYSTEM PAGE SELECTION
Refer Figure - ENGINE SYSTEM PAGE

ENG SYS Pushbutton Switch: The ENG SYS pushbutton switch is pushed to show the engine system page on the ESID DUs.

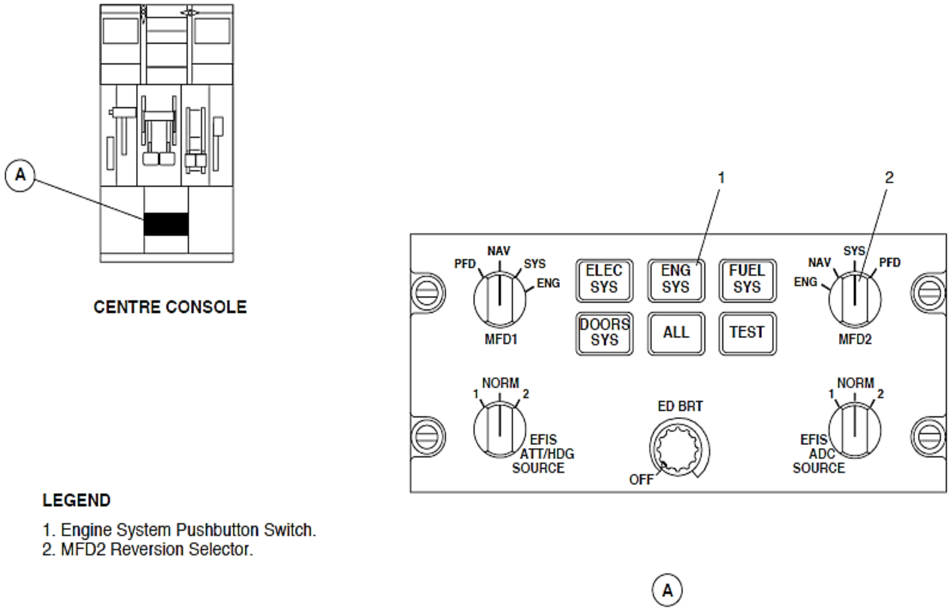


Figure - ELECTRONIC INSTRUMENT SYSTEM ESCP, ENGINE SYSTEM PAGE SELECTION

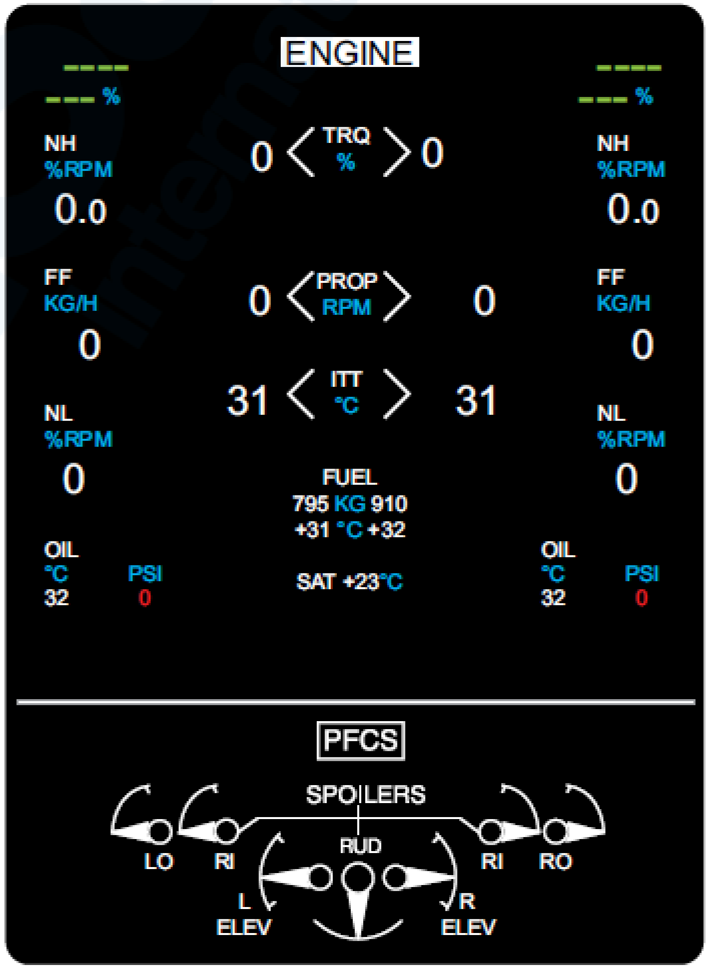


Figure - ENGINE SYSTEM PAGE

Refer Figure - ELECTRONIC INSTRUMENT SYSTEM ESCP, FUEL SYSTEM PAGE SELECTION
Refer Figure - FUEL SYSTEM PAGE

FUEL SYS Pushbutton Switch: The FUEL SYS pushbutton switch is pushed to show the fuel system page on the ESID DUs.

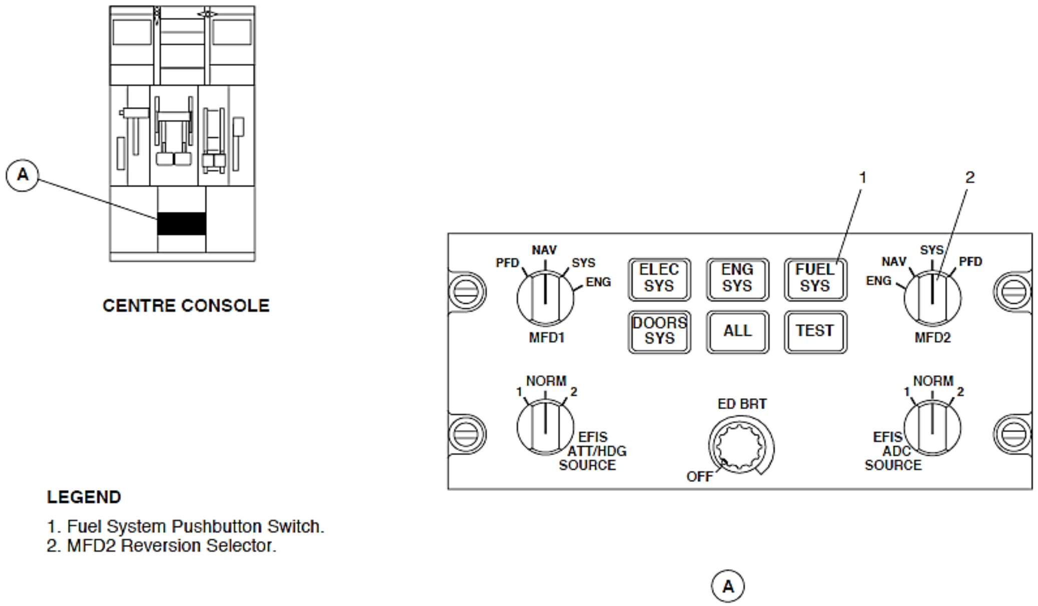


Figure - ELECTRONIC INSTRUMENT SYSTEM ESCP, FUEL SYSTEM PAGE SELECTION

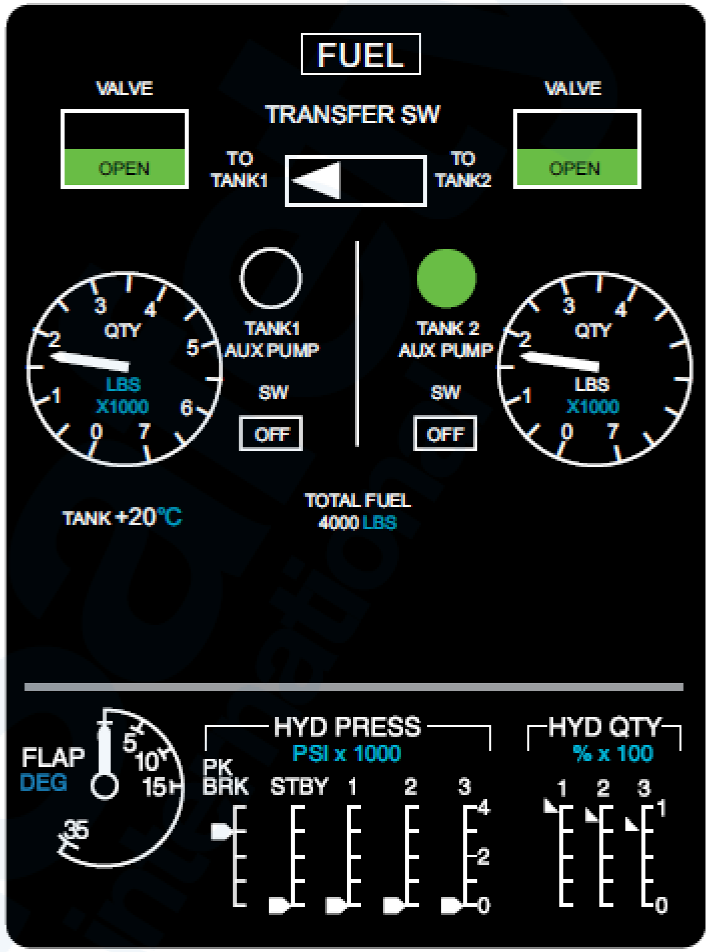


Figure - FUEL SYSTEM PAGE

Refer Figure - ELECTRONIC INSTRUMENT SYSTEM ESCP, DOORS SYSTEM PAGE SELECTION
Refer Figure - DOOR SYSTEM PAGE

DOORS SYS Pushbutton Switch: The DOORS SYS pushbutton switch is pushed to show the doors system page on the ESID DUs.

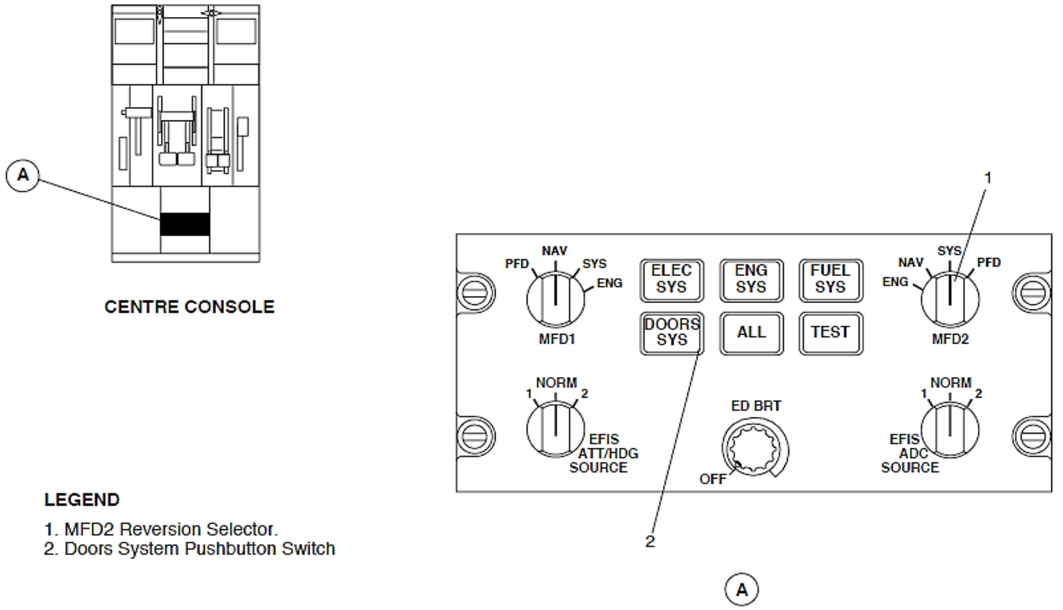


Figure - ELECTRONIC INSTRUMENT SYSTEM ESCP, DOORS SYSTEM PAGE SELECTION

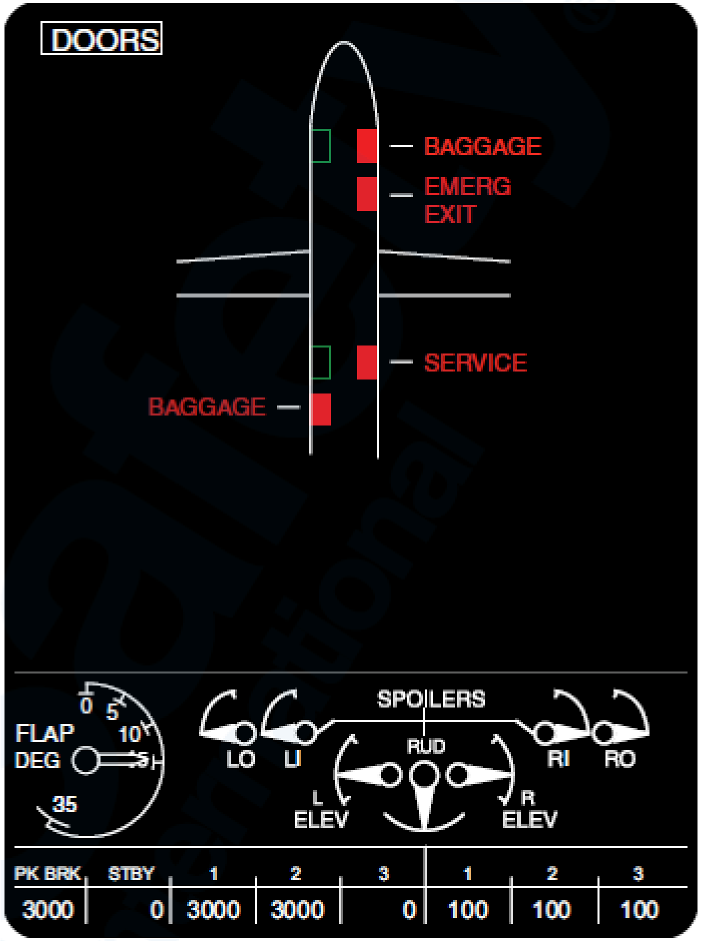


Figure - DOOR SYSTEM PAGE

ALL Pushbutton Switch: The ALL SYS pushbutton switch is pushed to sequence through all the system pages on the ESID DUs.

EFIS ATT/HDG SOURCE Reversion Selector: The EFIS ATT/HDG SOURCE reversion selector is used to make an Attitude and Heading Reference Unit (AHRU 1, AHRU2) indication selection on the EFIS DUs.

EFIS ADC SOURCE Reversion Selector: The EFIS ADC SOURCE reversion selector is used to make Attitude and Heading Reference Units, ADU 1 or ADU 2 indication selections on the EFIS DUs.

ED OFF, BRT Knob: The ED OFF, BRT Knob is turned to energize or de−energize the ED and to control its brightness.

Figure - Multi-Function Display 1 (MFD1) – Composite Indication

The MFD1 permanent system data area shows the Powered Flight Control Surface Position (PFCS) parameters and the MFD 2 permanent system data area shows flap position, hydraulic pressure, and hydraulic quantity indications.

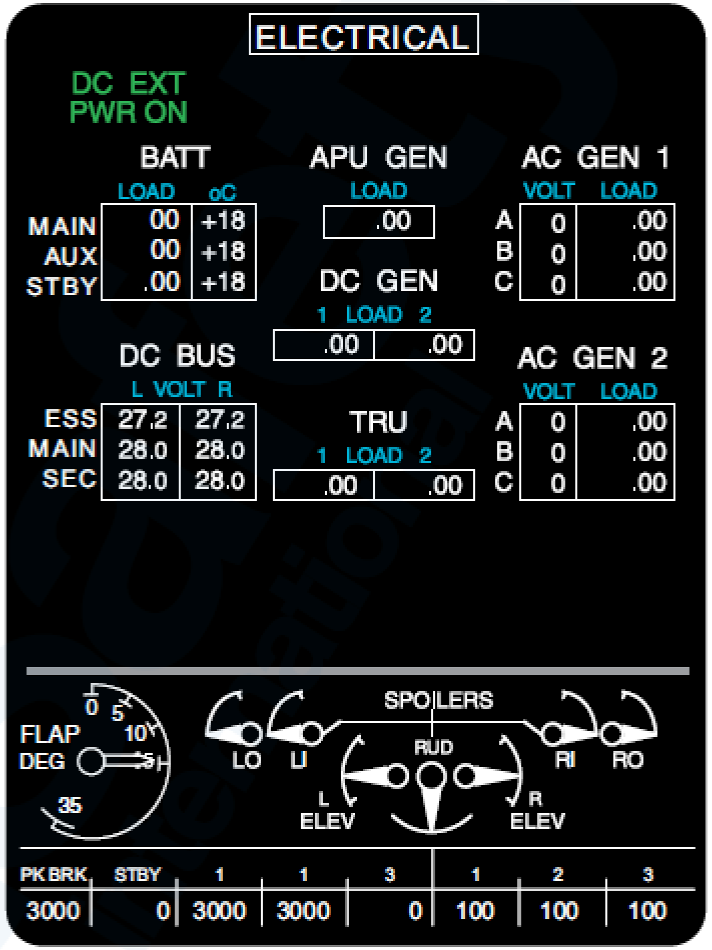
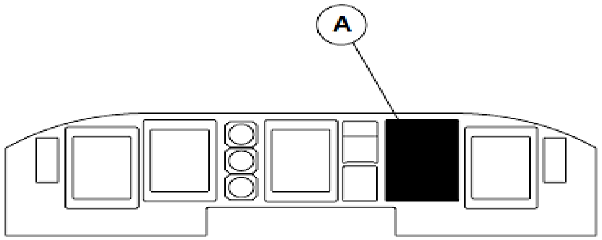


Figure - Multi-Function Display 1 (MFD1) – Composite Indication

Refer Figure - ELECTRONIC INSTRUMENT SYSTEM PERMANENT SYSTEM DATA AREA, COMPOSITE VIEW

When a MFD malfunctions, its permanent system data moves to the other MFD and the permanent system data indication changes to a composite indication to show the two permanent system data together.

When a PFD or ED malfunctions, the MFD1 or MFD2 reversion switch located on the ESCP is used to manually move the indications from the defective DU to a MFD. As part of this reversionary mode, the permanent system data moves to the other MFD and changes to a composite indication to show the two MFD1 and MFD2 permanent system data together.



MAIN INSTRUMENT PANEL

LEGEND

1. System Page.
2. Permanent System Data Area.

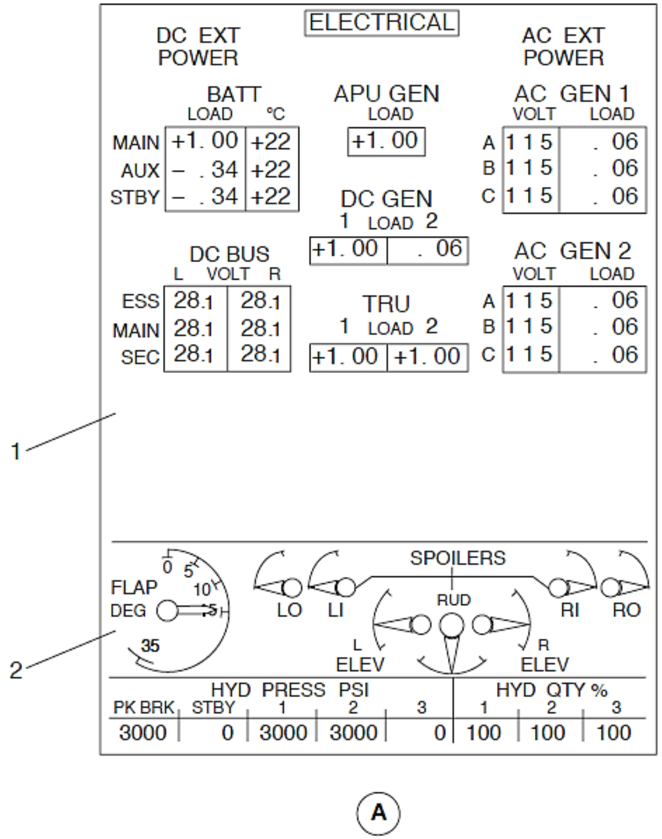


Figure - ELECTRONIC INSTRUMENT SYSTEM PERMANENT SYSTEM DATA AREA, COMPOSITE VIEW

Figure - ELECTRONIC INSTRUMENT SYSTEM ED MALFUNCTION

There is one automatic DU reversionary mode. When the Engine Display (ED) malfunctions while airborne, the ED images automatically move to the MFD1 and the MFD2 shows a composite permanent system data indication. This automatic reversionary mode occurs only if the MFD1 is not set to show the PFD images.

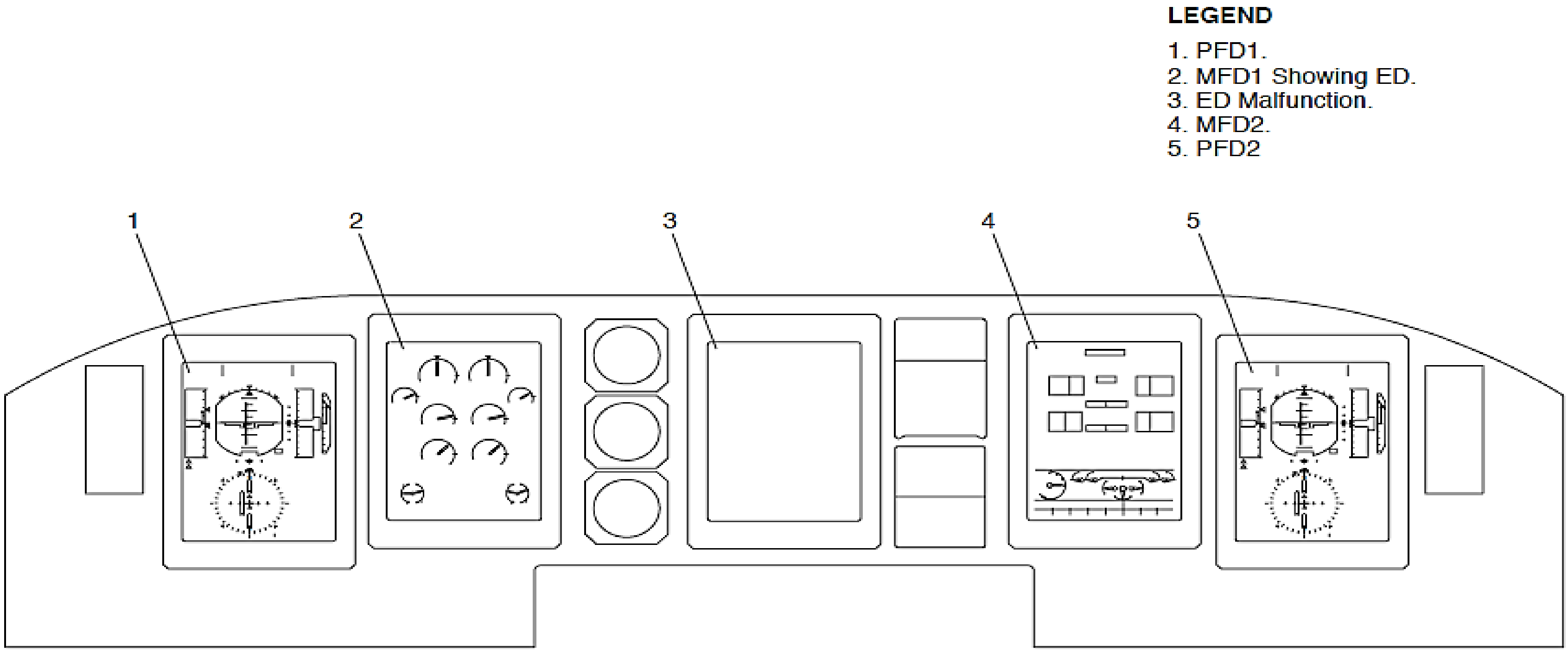


Figure - ELECTRONIC INSTRUMENT SYSTEM ED MALFUNCTION

Refer Figure - ELECTRONIC INSTRUMENT SYSTEM ESCP, ED MFD1 REVERSION SELECTION

After the automatic reversion, the MFD1 reversion switch is manually set to the ENG position to make the DU indications agree with the switch selection.

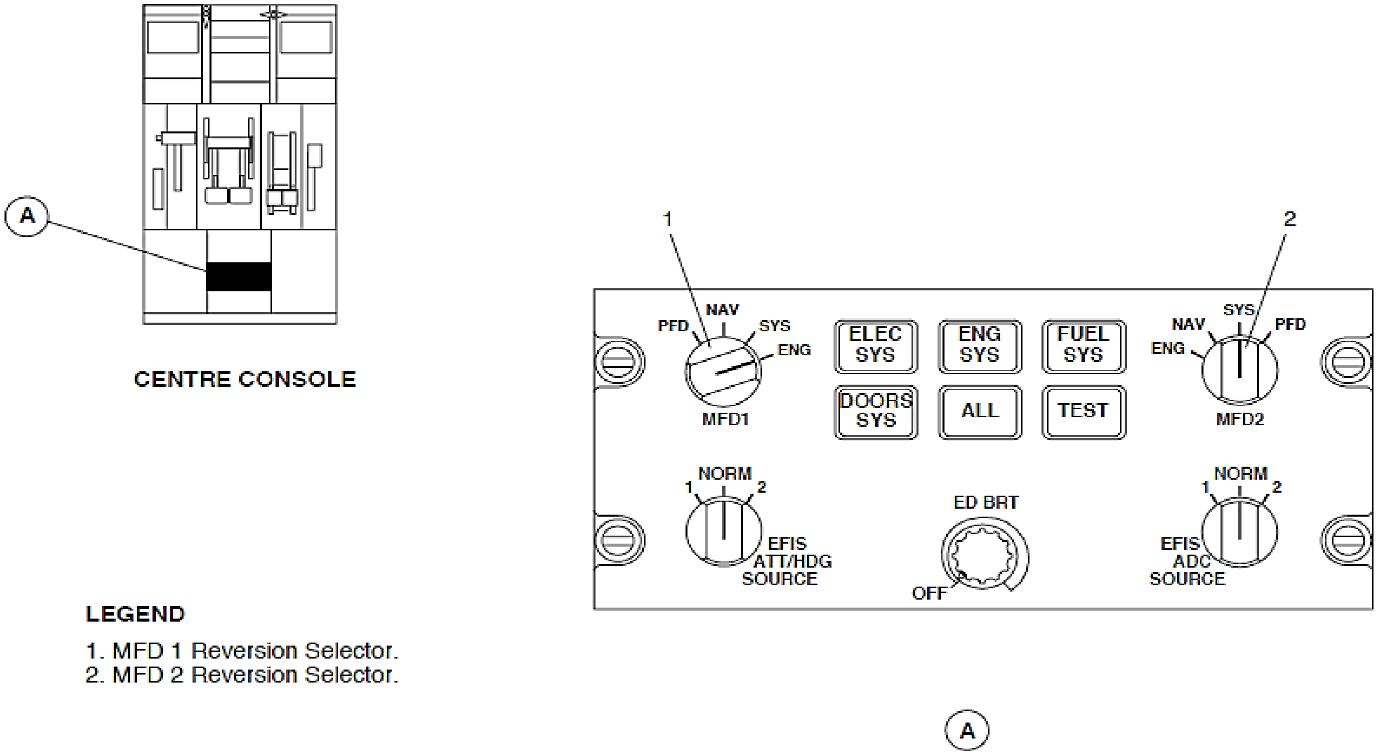


Figure - ELECTRONIC INSTRUMENT SYSTEM ESCP, ED MFD1 REVERSION SELECTION

Refer Figure - ELECTRONIC INSTRUMENT SYSTEM PFD1 AND ED MALFUNCTION

If the Engine Display (ED) and Primary Flight Display (PFD1) malfunction at the same time while the TAS is more than 50 kts and the MFD 1 reversion selector is not set to PFD, the ED images automatically move to the MFD.

The MFD2 shows a composite permanent system data indication.

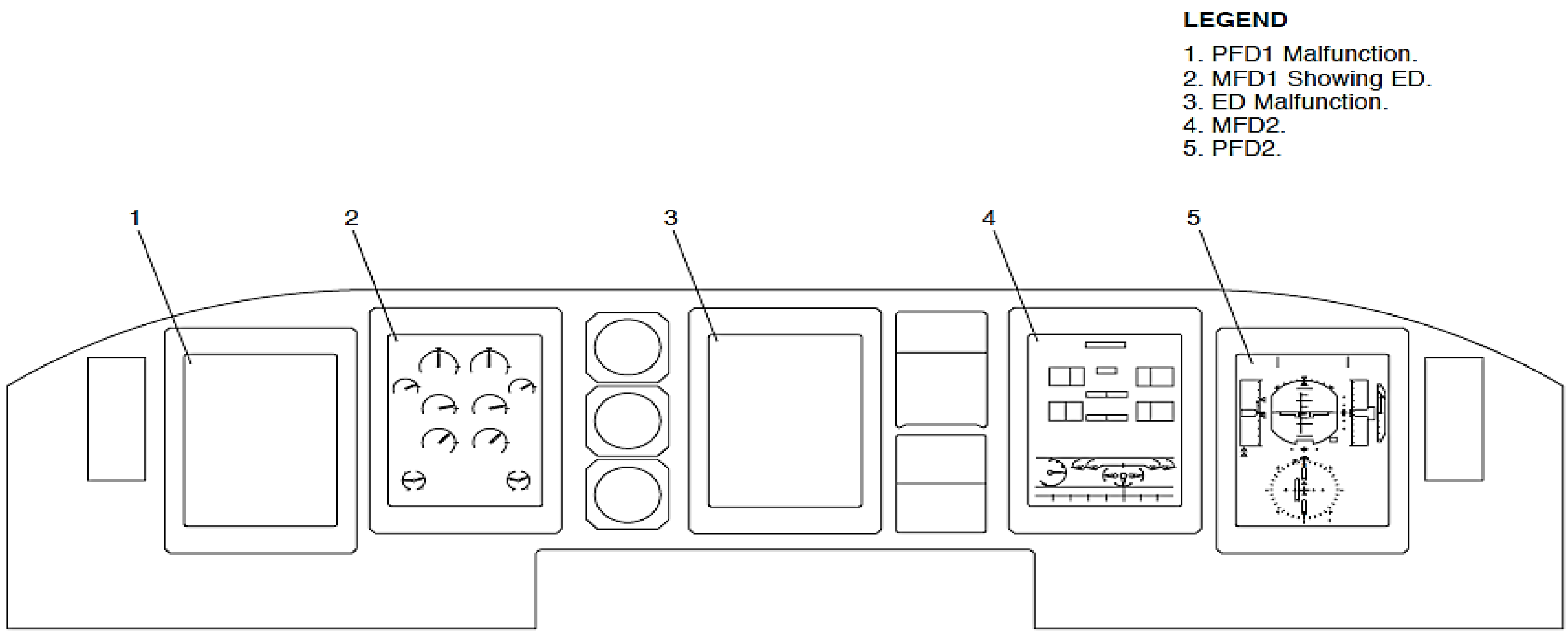
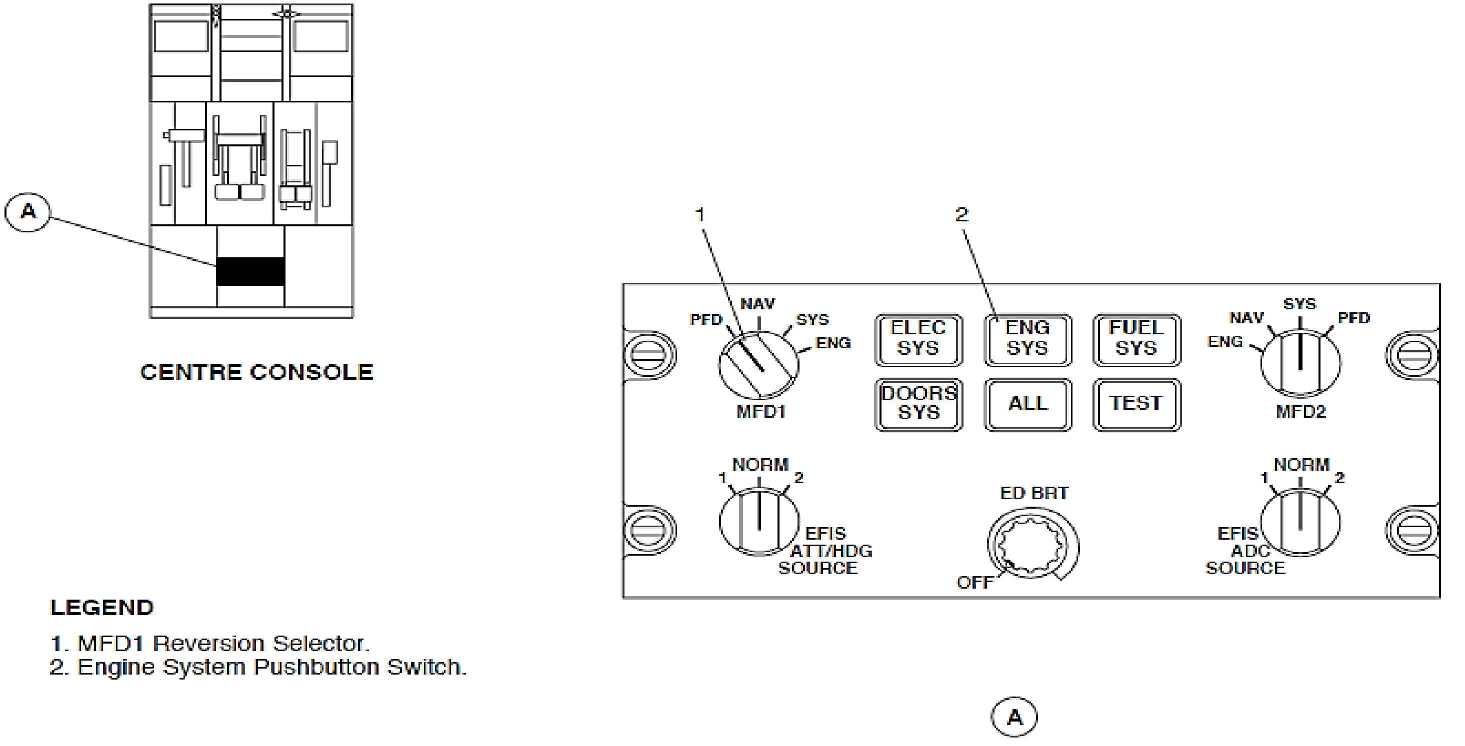


Figure - ELECTRONIC INSTRUMENT SYSTEM PFD1 AND ED MALFUNCTION

Refer Figure - ELECTRONIC INSTRUMENT SYSTEM ESCP, PFD MFD1 REVERSION AND ENGINE SYSTEM PAGE SELECTIONS

The MFD1 reversion switch is set to PFD and the ENG SYS pushbutton switch is then pushed.



LEGEND

1. MFD1 Reversion Selector.
2. Engine System Pushbutton Switch.

Figure - ELECTRONIC INSTRUMENT SYSTEM ESCP, PFD MFD1 REVERSION AND ENGINE SYSTEM PAGE SELECTIONS

Refer Figure - ELECTRONIC INSTRUMENT SYSTEM, PFD1 AND ED REVERSIONARY INDICATION

The PFD indications are manually moved from the defective PFD1 to its related MFD and the other MFD will show the engine system page with a composite permanent system data indication.

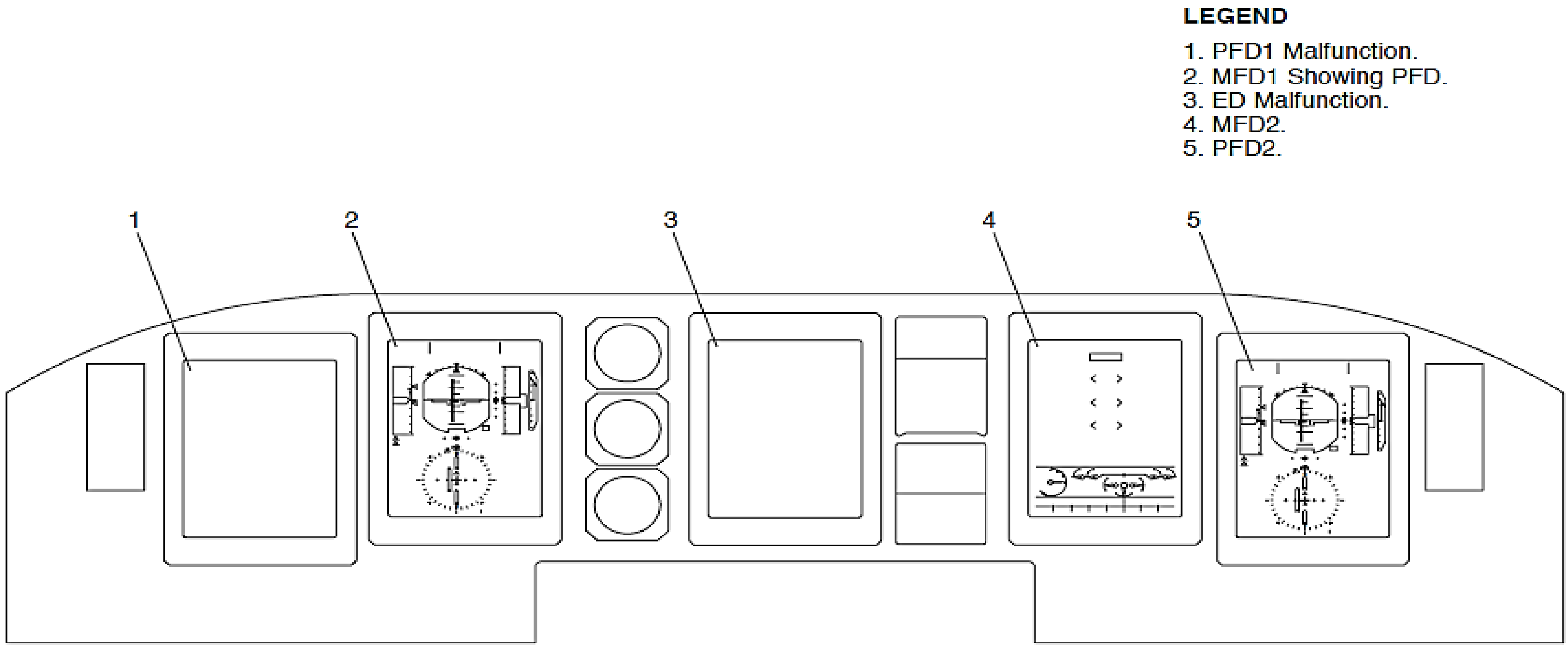


Figure - ELECTRONIC INSTRUMENT SYSTEM, PFD1 AND ED REVERSIONARY INDICATION

Refer Figure - ELECTRONIC INSTRUMENT SYSTEM MFD1 AND MFD2 MALFUNCTION

If the MFD1and MFD2 malfunction at the same time, the navigation page, system page, and the permanent system data indication is not in view.

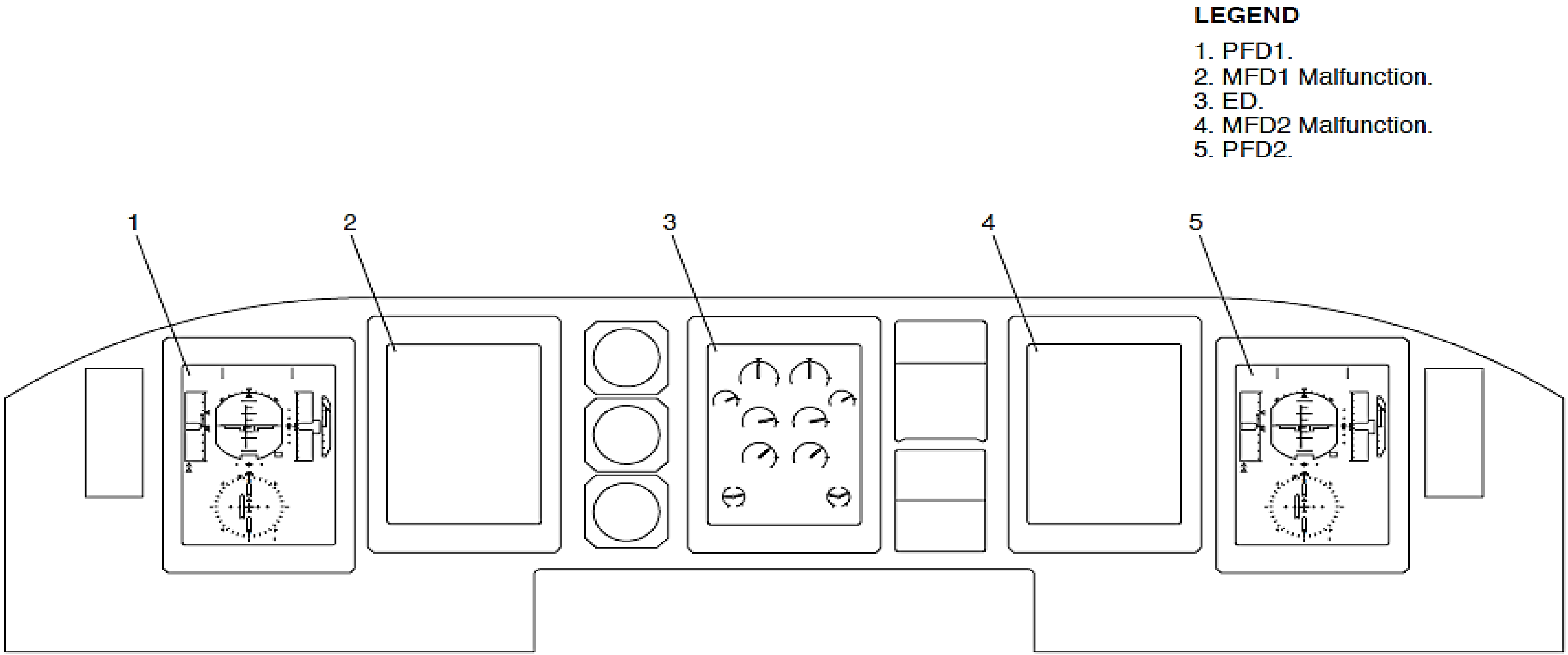
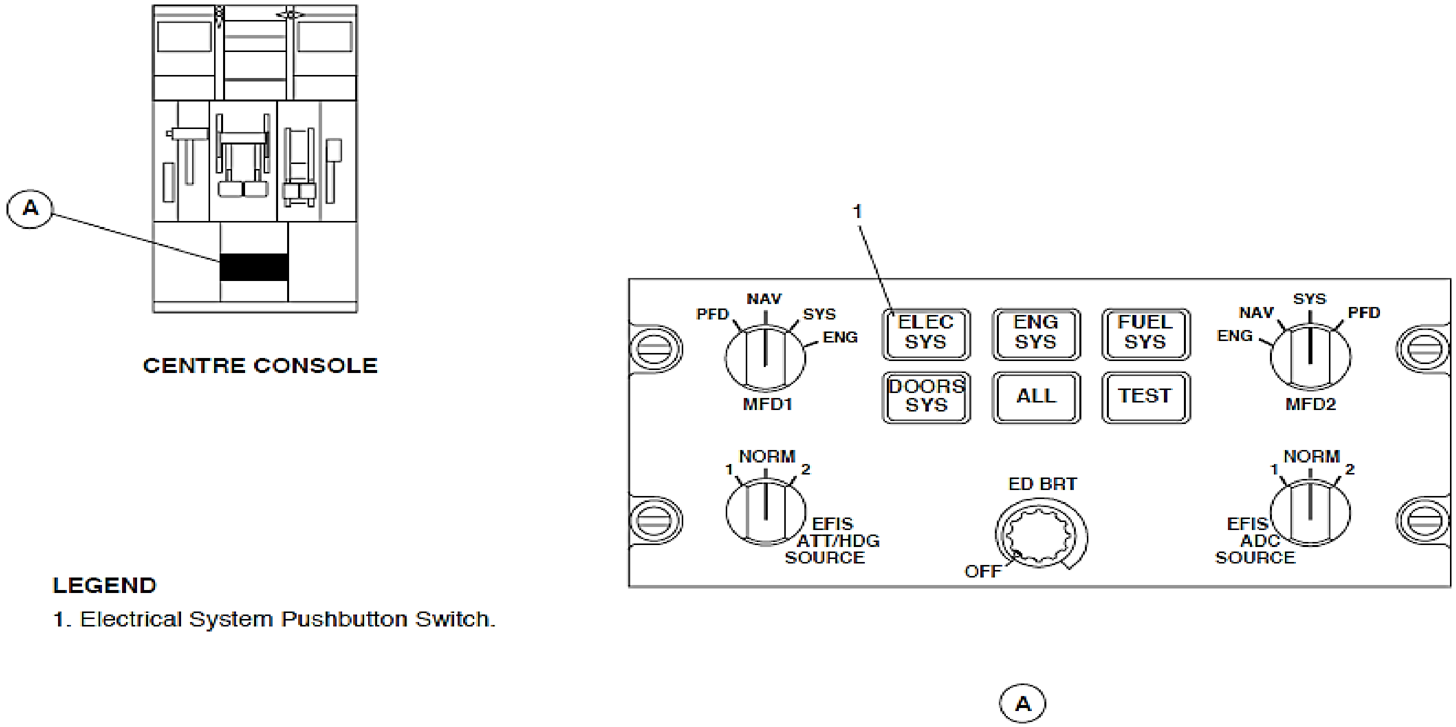


Figure - ELECTRONIC INSTRUMENT SYSTEM MFD1 AND MFD2 MALFUNCTION

Refer Figure - ELECTRONIC INSTRUMENT SYSTEM ESCP, MONO ELECTRICAL SYSTEM PAGE SELECTION

The Engine and System Integrated Display (ESID) system has a mono mode to let the pilots view the system page with a composite permanent system data indication when the two MFDs malfunction or are set to show PFD, ED, or navigation page indications.



LEGEND
1. Electrical System Pushbutton Switch.

Figure - ELECTRONIC INSTRUMENT SYSTEM ESCP, MONO ELECTRICAL SYSTEM PAGE SELECTION

Refer Figure - ELECTRONIC INSTRUMENT SYSTEM ESCP, MONO ELECTRICAL SYSTEM PAGE INDICATION

A system pushbutton switch is pushed to show its applicable system page with a composite permanent system data indication on the Engine Display (ED).

The system page is shown while the pushbutton switch is pushed. The ED images are shown again when the push button switch is released.

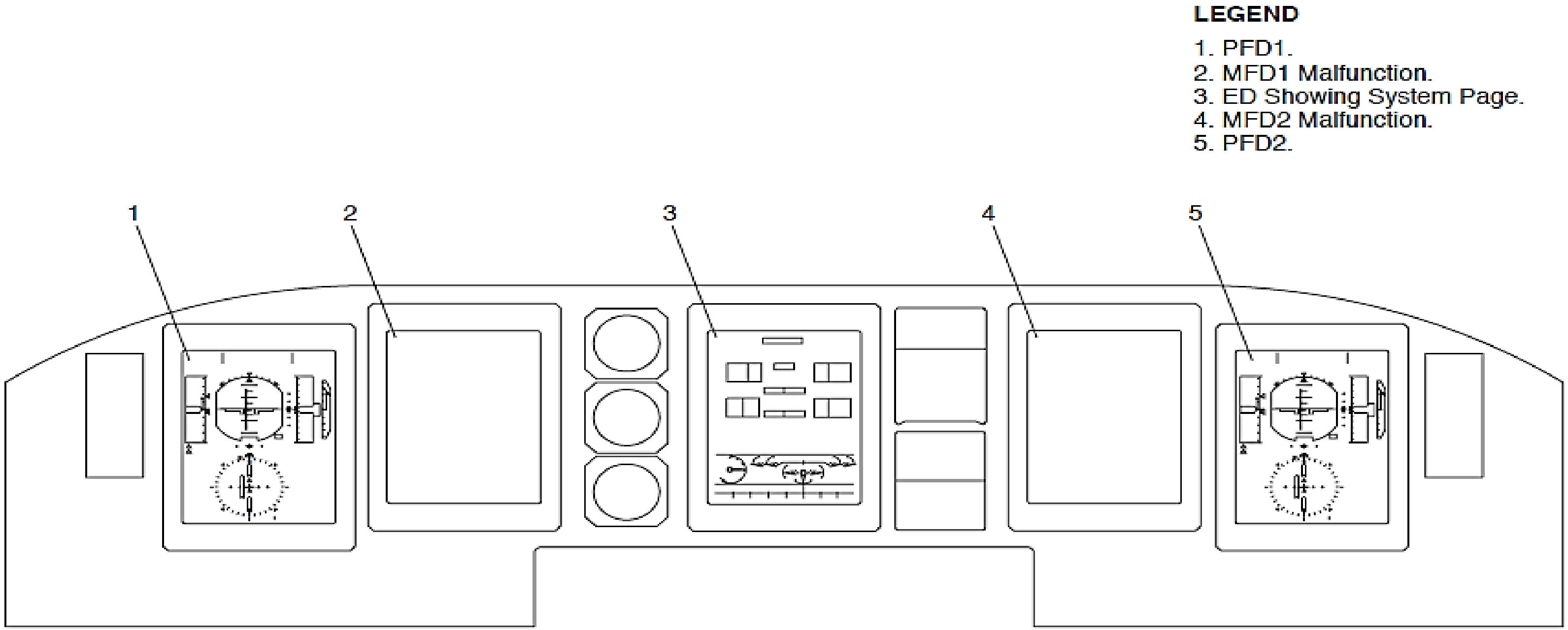


Figure - ELECTRONIC INSTRUMENT SYSTEM ESCP, MONO ELECTRICAL SYSTEM PAGE INDICATION

Figure - ELECTRONIC INSTRUMENT SYSTEM ICP

Index Control Panels (ICP1, ICP2): On aircraft with the alternate ICPs (823CH00013): The ICPs have the controls that follow:

- SPEED BUG SEL pushbutton switch
- SPEED BUG rotary selector
- BARO SET knob/PUSH TO STD switch
- DH/MDA rotary switch
- DH/MDA knob.

SPEED BUG SEL Pushbutton Switch: The SPEED BUG SEL pushbutton switch located on the ICP is pushed to set the speed bugs that follow:

- V1, take off decision speed
- VR, rotation speed
- V2, take off safety speed
- No. 1 (Solid)
- No. 2 (Outline).

The SPEED BUG SEL pushbutton switch is pushed to set speed bugs V1, VR, and V2 when the aircraft is on the ground and speed bugs No. 1 and No. 2 when the aircraft is airborne.

The SPEED BUG SEL pushbutton switch is pushed to change speed bug V1 or No. 1. It is pushed again to set V1 or No. 1 and to change VR or No. 2. The SPEED BUG SEL pushbutton switch is pushed again to set V2. Each speed bug is also set five seconds after the last SPEED BUG SEL pushbutton switch or SPEED BUG rotary selection.

SPEED BUG rotary selector: The SPEED BUG rotary selector is turned to change the speed bug value.

BARO SET Knob/PUSH TO STD Switch: The knob is turned to supply the related Air Data Unit (ADU1, ADU2) with a barometric correction value and it is pushed to supply a standard barometric correction value.

DH/MDA rotary switch: The DH/MDA rotary switch is set to DH to let the DH/MDA knob change the DH value. It is set to MDA to let the DH/MDA knob change the MDA value.

DH/MDA knob: The DH/MDA knob is turned to set a DH altitude for Decision Height (DH) calculations or MDA for Minimum Descent Altitude calculations.

The Electronic Instrument System (EIS) units are energized by the main 28 V dc generation system. The electrical power source for each unit is shown in the table that follows:

PART	LEFT MAIN	LEFT ESS	RIGHT MAIN	RIGHT ESS
PFD1	X			
MFD1		X		
ED			X	X
MFD2			X	
PFD2			X	
EFCP1	X			
ICP1		X		
ESCP	X			
EFCP 2			X	
ICP2			X	

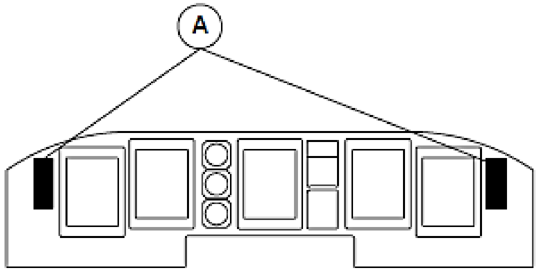
Each Display Unit (DU) dissipates 170 W of power when the display heater is on and the brightness is set to a maximum level. It is 110 W when the heater turns off. The DUs dissipates 90 W of power when its heater is off and the brightness is set to a maximum level with low ambient flight compartment illumination.

The left main bus supplies electrical power through a 10 A circuit breaker and two 5 A circuit breakers to the PFD, EFCP, and ESCP. The circuit breakers are located in position A2, C2, and B2 on the avionics circuit breaker panel.

The left essential bus supplies electrical power through a 10 A circuit breaker and a 5 A circuit breaker to the MFD1 and ICP1. The circuit breakers are located in position B7 and C7 on the avionics circuit breaker panel.

The right main bus supplies electrical power through three 10 A circuit breakers and two 5 A circuit breakers to the ED, MFD2, PFD2, EFCP2, and ICP2.The circuit breakers are located in position A5, B5, B6, C5, and C6 on the avionics circuit breaker panel.

The right essential bus also supplies electrical power through a 10A circuit breaker to the ED. The circuit breaker is located in position B11 on the avionics circuit breaker panel.



MAIN INSTRUMENT PANEL

LEGEND

1. Speed Bug Index SEL Pushbutton Switch.
2. Speed Bug Rotary Selector.
3. BARO SET Knob / PUSH TO STD Switch.
4. DH Knob.
5. DH/MDA Switch.

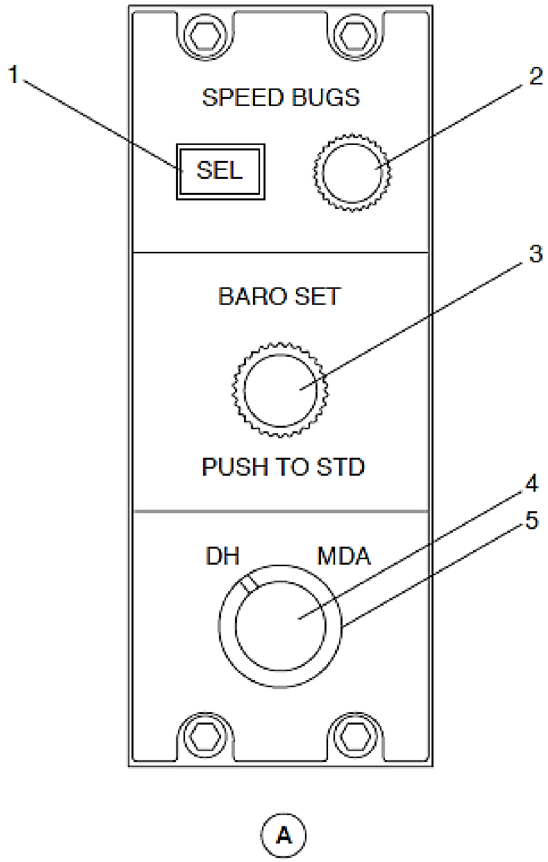


Figure - ELECTRONIC INSTRUMENT SYSTEM ICP

Display Units (DU)

Figure - DISPLAY UNIT LOCATOR

Each display unit has a 6 by 8 in. (152 by 203 mm) active matrix liquid crystal display with a resolution of 167 Dots Per Inch (DPI). It uses the color rules shown in the table that follows:

COLOUR	INDICATION
Red	Warning
Yellow	Caution
White	Advisory, labels, scales
Green	Active modes, passed test
Cyan	Selectable parameters
Magenta	FMS
Brown	Attitude sphere
Blue	Attitude sphere, ESID units
Grey	Indicator background
Black	Aircraft symbol

The DU shows messages with the attributes that follows:

- Flashing
- Reverse video
- Brackets.

Flashing: Flashing is used when a new message that requires pilot attention comes into view. It is time limited to 5 seconds or stays in view until the pilot action is completed. The flashing frequency is 1 Hz.

Reverse Video: Reverse video is used to show a change in the operating mode that was not pilot set. It is time limited to 5 seconds.

Brackets: The message in the brackets are pilot instructions.

The dynamic data of the images are received and updated on the screen at a refresh frequency of 20 Hz. The stroke and raster images are updated at a nominal 60 Hz rate.

The DU has the sub−assemblies that follow:

- Interface Connection Module (ICM)
- Digital Processing Module (DPM)
- Backlighting Module (BLM)
- LCD Assembly Module (LAM)
- Power Supply Module (PSM)
- Housing Assembly Module (HAM).

The DUs are attached to the instrument panel with four mounting screws. They are electrical bonded to the instrument panel through wires in the rear connector and by surface contact between the DU chassis and the instrument panel mounting frame.

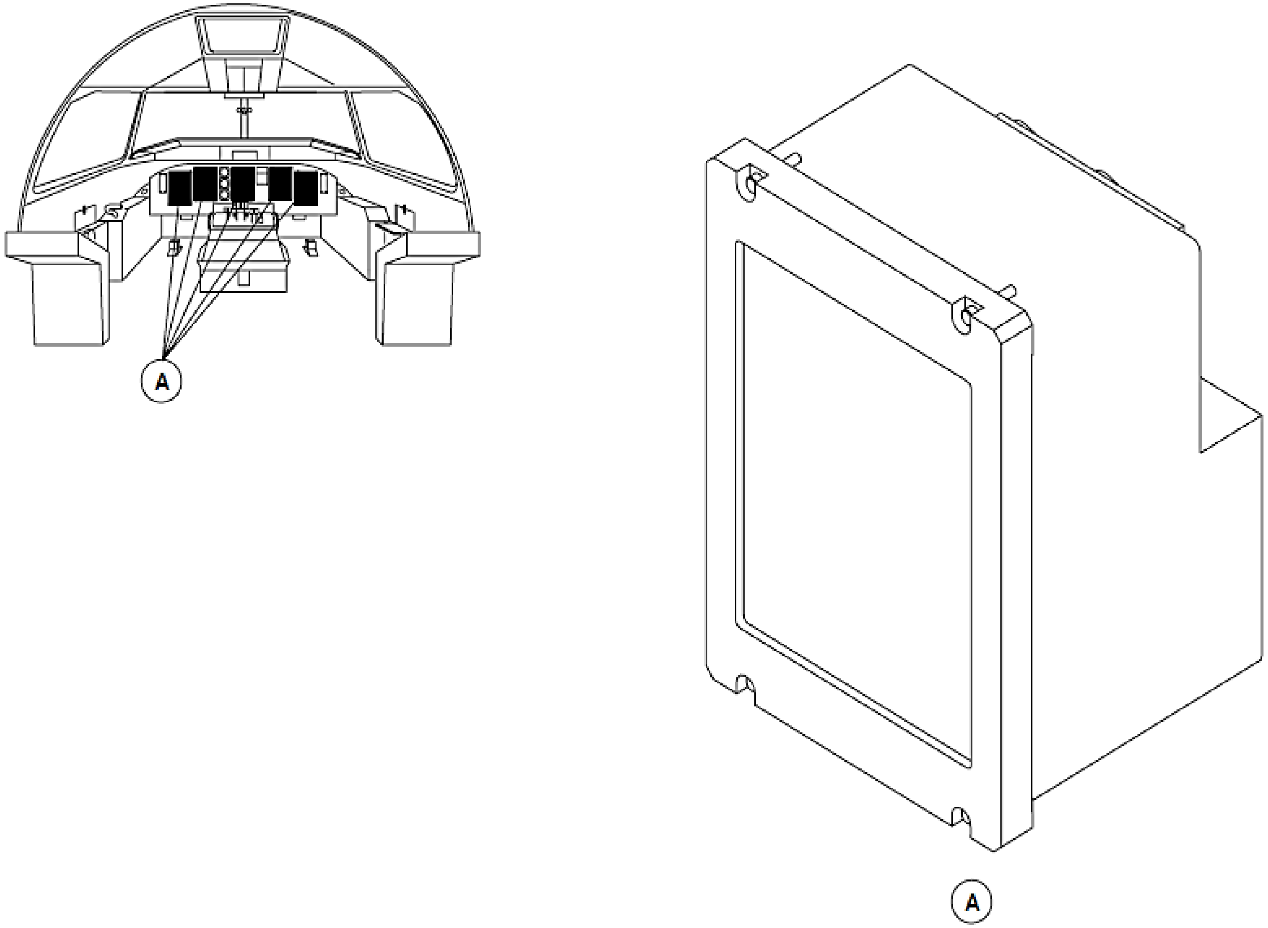


Figure - DISPLAY UNIT LOCATOR

EFIS Control Panels (EFCP)

Refer Figure - EFIS CONTROL PANEL (EFCP) LOCATOR

To control some related EFIS PFD and navigation page indications, the EFCPs have the controls that follow:

- Bearing 1 selector knob
- Bearing 2 selector knob
- TCAS pushbutton switch
- WX/TERR pushbutton switch
- DATA pushbutton switch
- FORMAT pushbutton switch
- RANGE selector knob
- PFD OFF, BRT knob
- MFD OFF, BRT knob
- WX/TERR BRT knob.

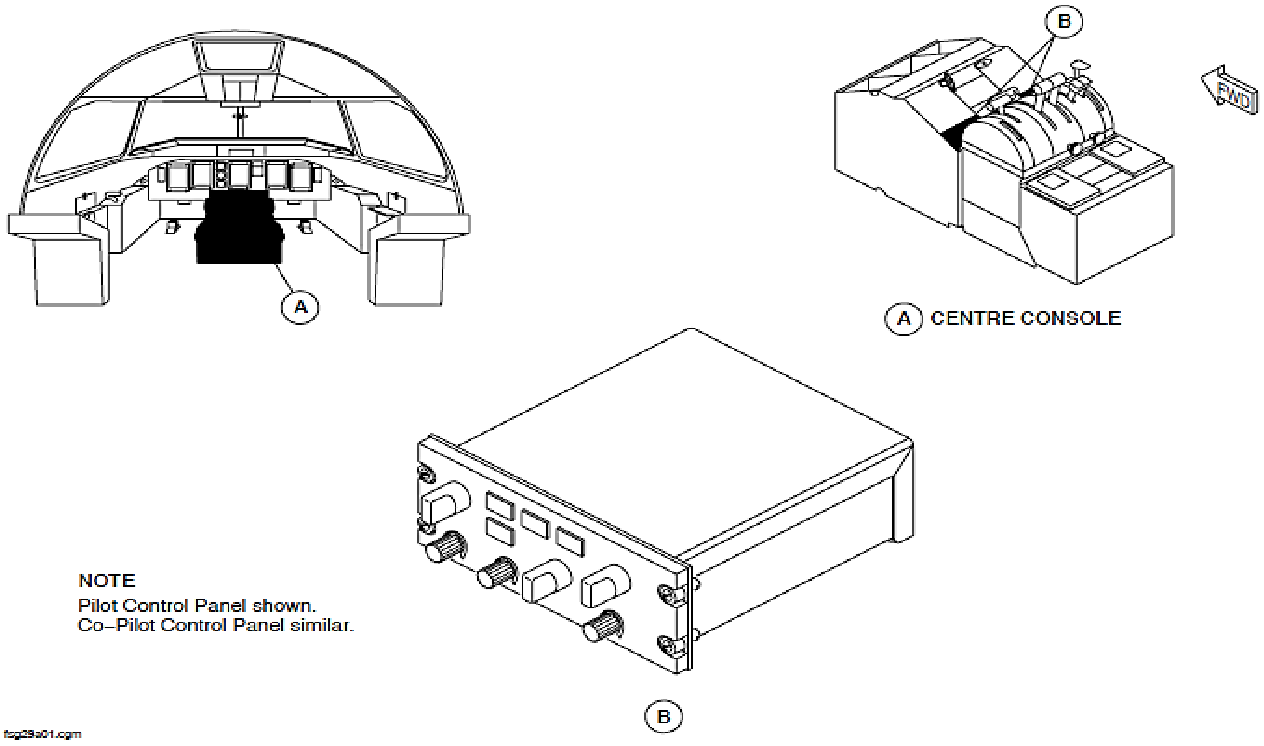


Figure - EFIS CONTROL PANEL (EFCP) LOCATOR

Refer Figure - CONTROL PANEL BLOCK DIAGRAM

The EFCP has the sub−assemblies that follow:

- Front face module
- Interface board
- Outer box.

The Interface board has the functions that follow:

- Power supply
- Parallel/serial converter
- ARINC emitter/receiver
- EMI, lightning protection.

The EFIS selection and the type of interface signal is shown in the table that follows:

SELECTION	SIGNAL TYPE
Bearing 1 selector knob	ARINC 429
Bearing 2 selector knob	ARINC 429
FULL/ARC mode pushbutton switch	ARINC 429
WX mode pushbutton switch	ARINC 429
FMS MAP mode pushbutton switch	ARINC 429
TCAS pushbutton switch	ARINC 429
RANGE selector knob	ARINC 429
PFD OFF, BRT knob	Discrete, ARINC 429
MFD OFF, BRT knob	Discrete, ARINC 429
WX/TERR BRT knob	ARINC 429

NOTE

Note: The EFCP PFD and MFD OFF, BRT knobs supply a discrete signal to the DUs for ON/OFF control and ARINC 429 data for brightness control.

The EFCPs weigh 2.2 lb. (1.0 kg). The EFCP is 2.625 in. (66.68 mm) wide, 5.75 in. (146 mm) high and 5.9 in. (150 mm) deep. The EFCPs are attached to the center console with four DZUS fasteners. The fasteners and a bonding wire directly connect to the chassis to make a ground continuity connection between control panel chassis and the aircraft structure.

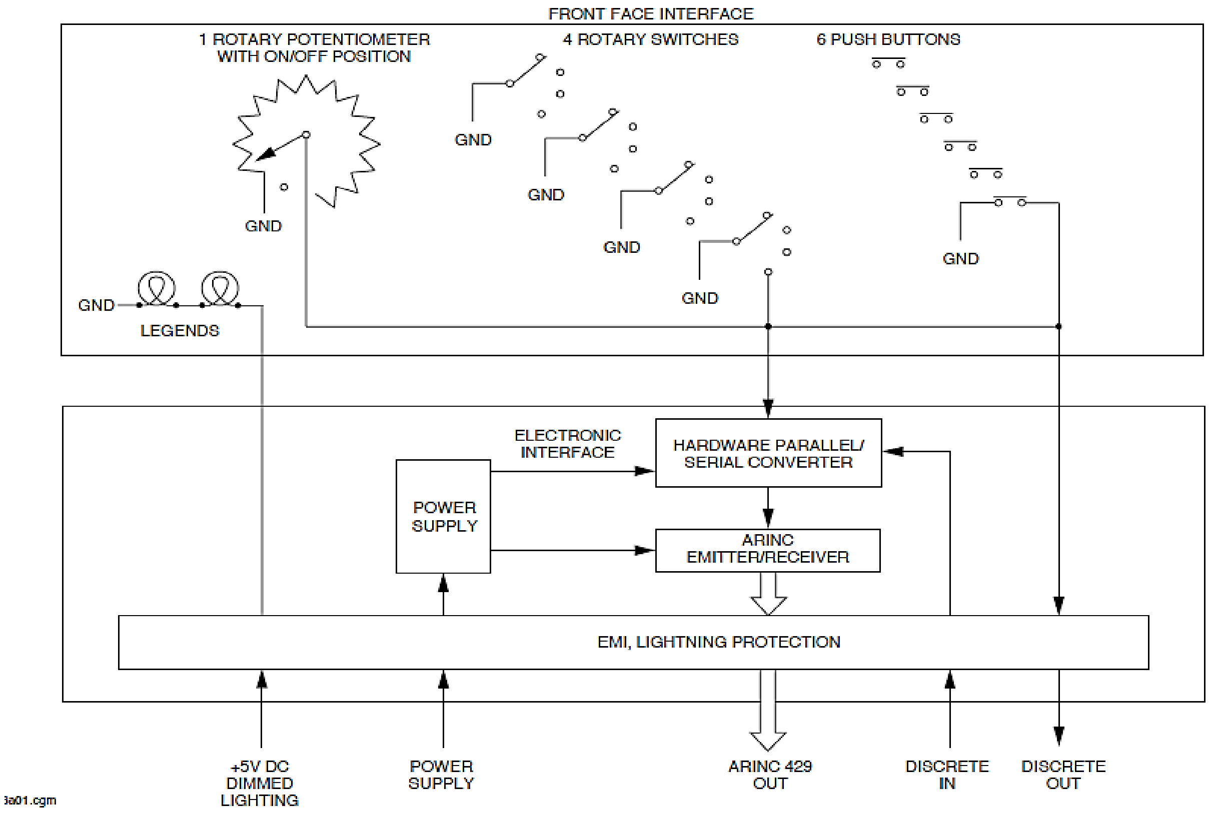


Figure - CONTROL PANEL BLOCK DIAGRAM

Refer Figure - ESID CONTROL PANEL (ESCP) LOCATOR

ESID Control Panel (ESCP)

To control some related EFIS PFD and navigation page indications, the ESCP has the controls that follow:

- MFD1 and MFD2 Reversion Switches
- ELEC SYS Pushbutton Switch
- ENG SYS Pushbutton Switch
- FUEL SYS Pushbutton Switch
- DOORS SYS Pushbutton Switch
- ALL Pushbutton Switch
- EFIS ATT/HDG SOURCE reversion selector
- EFIS ADC SOURCE Reversion Selector
- ED OFF, BRT Knob.

The ESCP has the sub−assemblies that follow:

- Front face module
- Interface board
- Outer box.

The Interface board has the functions that follow:

- Power supply
- Parallel/serial converter
- ARINC emitter/receiver
- EMI, lightning protection.

The ESID selection and the type of interface signal is shown in the table that follows:

SELECTION	SIGNAL TYPE
MFD1 and MFD2 Reversion Switches	ARINC 429
ELEC SYS Pushbutton Switch	ARINC 429
ENG SYS Pushbutton Switch	ARINC 429
FUEL SYS Pushbutton Switch	ARINC 429
DOORS SYS Pushbutton Switch	ARINC 429
ALL Pushbutton Switch	Discrete
EFIS ATT/HDG SOURCE reversion selector	Discrete
EFIS ADC SOURCE Reversion Selector	Discrete
ED OFF, BRT Knob	Discrete, ARINC 429

NOTE

Note: ED OFF, BRT knob supplies a discrete signal to the DU for ON/OFF control and ARINC 429 data for brightness control.

The ESIS Control Panel (ESCP) is attached to the center console with four DZUS fasteners. The fasteners and a bonding wire directly connect to the chassis to make a ground continuity connection between control panel chassis and the aircraft structure.

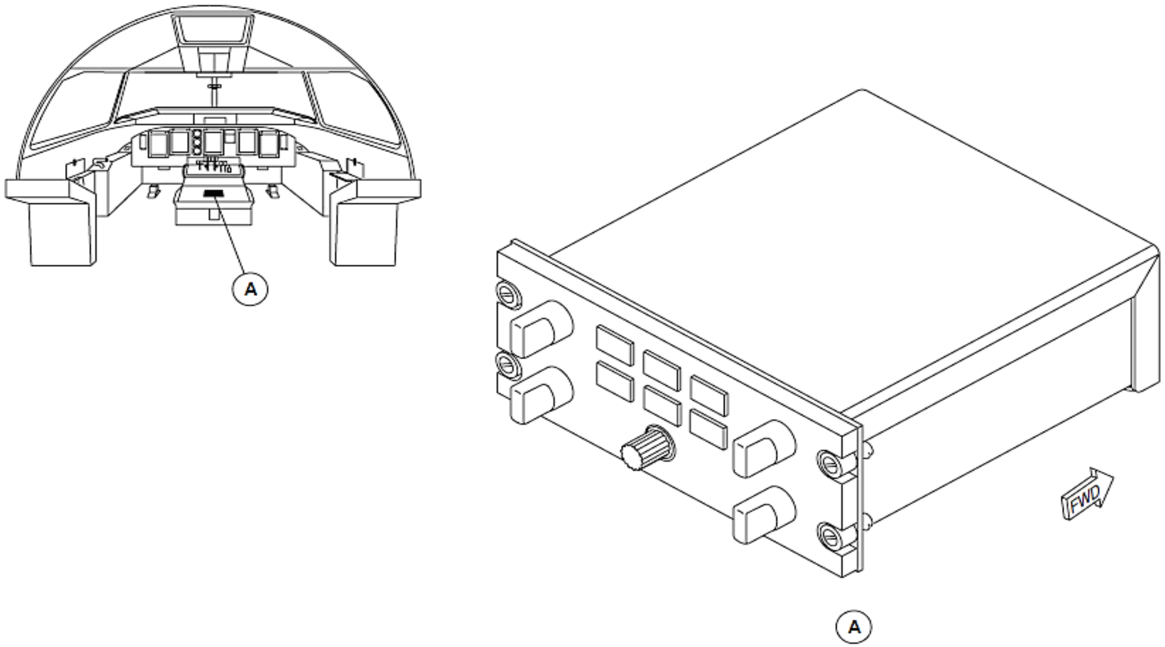


Figure - ESID CONTROL PANEL (ESCP) LOCATOR

Index Control Panels (ICP)

Figure - INDEX CONTROL PANEL (ICP) LOCATOR

To control some related EFIS PFD indications, the ICPs have the controls that follow:

- SPEED BUG index switch
- SPEED BUG rotary selector
- BARO SET knob
- DH knob.

The ICP has the sub–assemblies that follow:

- Front face module
- Interface board
- Outer box.

The Interface board has the functions that follow:

- Power supply
- Parallel/serial converter
- ARINC emitter/receiver
- EMI, lightning protection.

Each ICP supplies data to its related IOP and ADU through ARINC 429 data buses.

The ICPs are attached to the instrument panel with four mounting screws. The fasteners and a bonding wire directly connect to the chassis to make a ground continuity connection between control panel chassis and the aircraft structure.

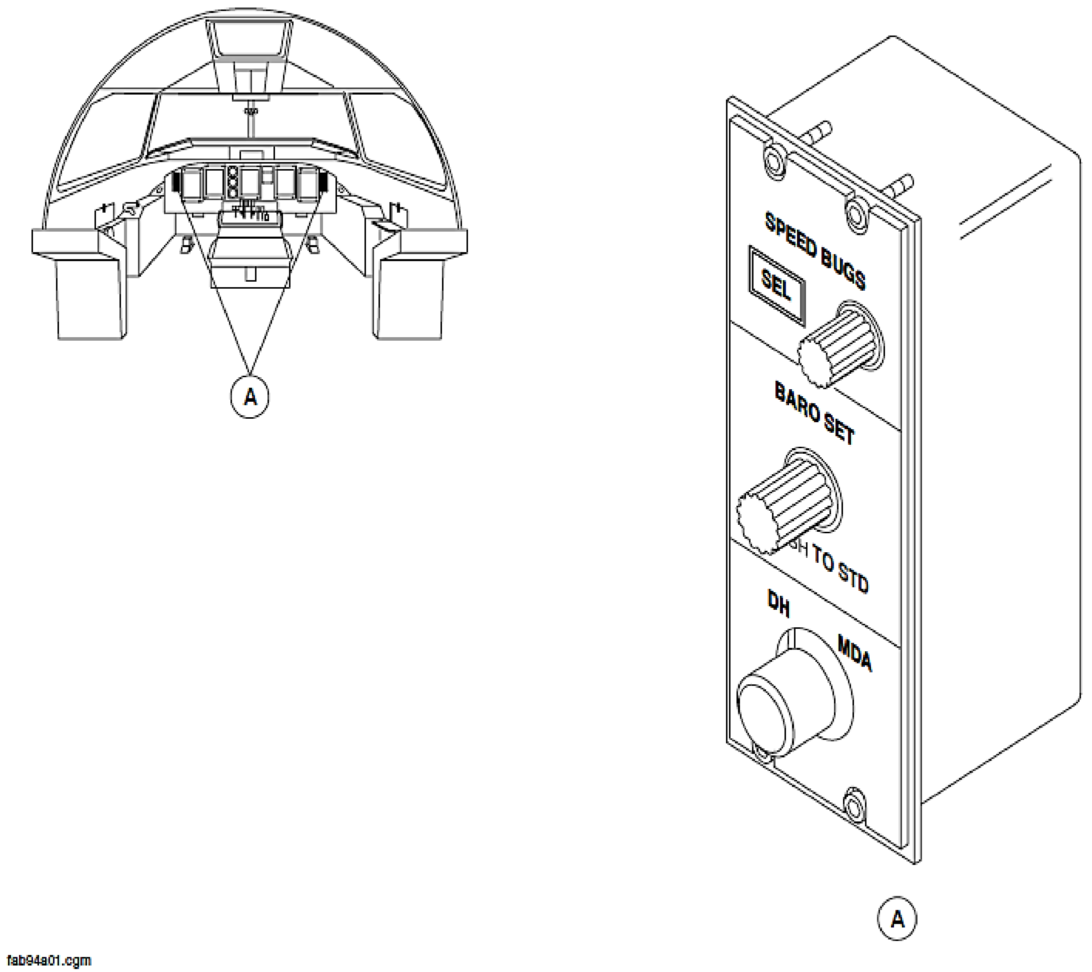


Figure - INDEX CONTROL PANEL (ICP) LOCATOR

Refer Figure - Indication of Navigation Data

The Air Data Units (ADU1 and ADU2) and Attitude and Heading Reference Units (AHRS1 and AHRS2) supply data through different ARINC 429 buses to all the Display Units (DUs) of EFIS (PFD1, PFD2, MFD1, and MFD2).

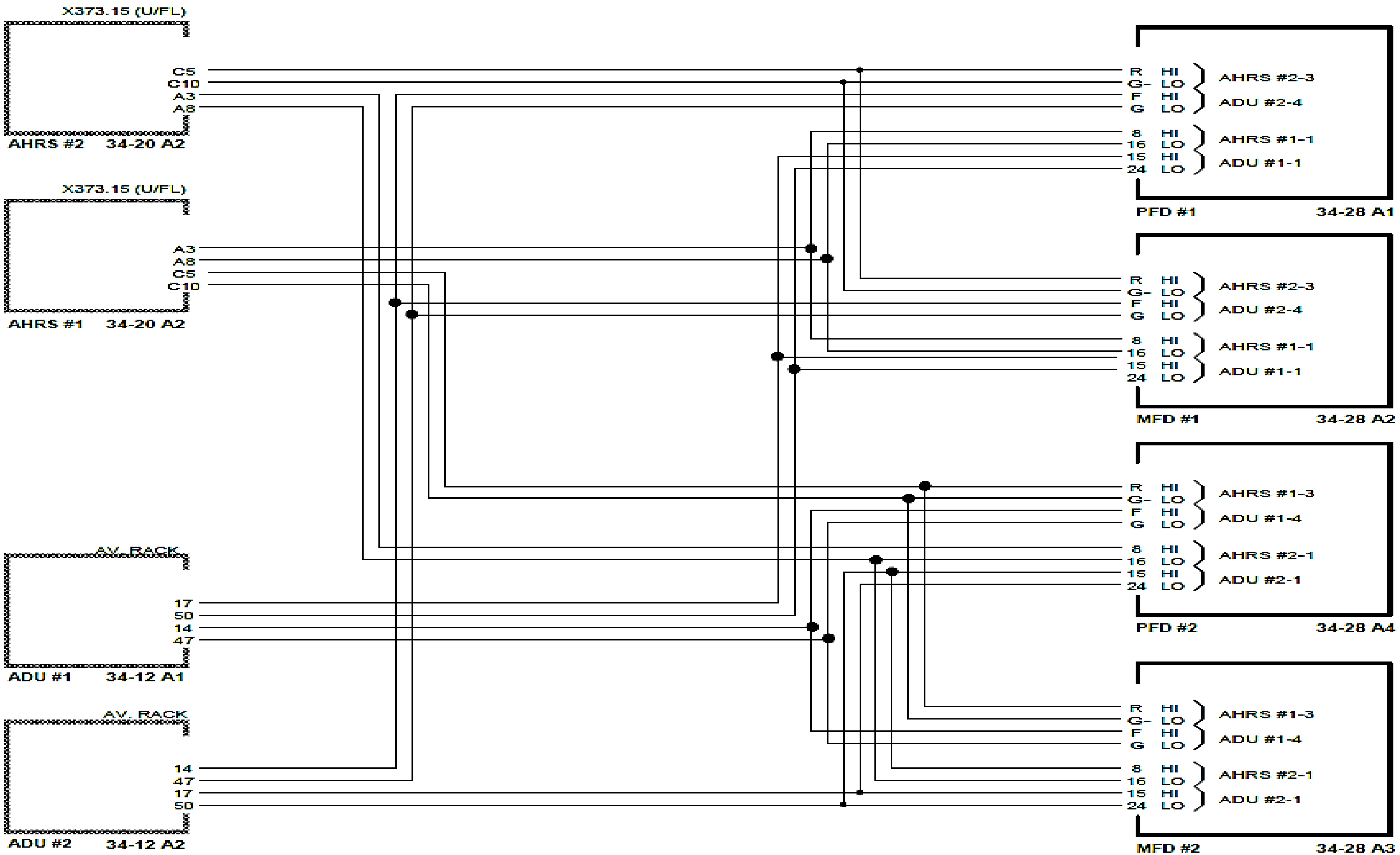


Figure - Indication of Navigation Data

Refer Figure - EIS Reversion

Three [ATT REV] and [ADU REV] discrete signals are supplied from the ESID control panel (ESCP) to tell the DUs to show the No.1 or No. 2 parameter.

The MFD1 and MFD2 reversions are done the same way.

NOTE

Two [ADU REV] discrete signals are also supplied to the AHRU1 and AHRU2 to make them switch their input from ADU1 or ADU2.

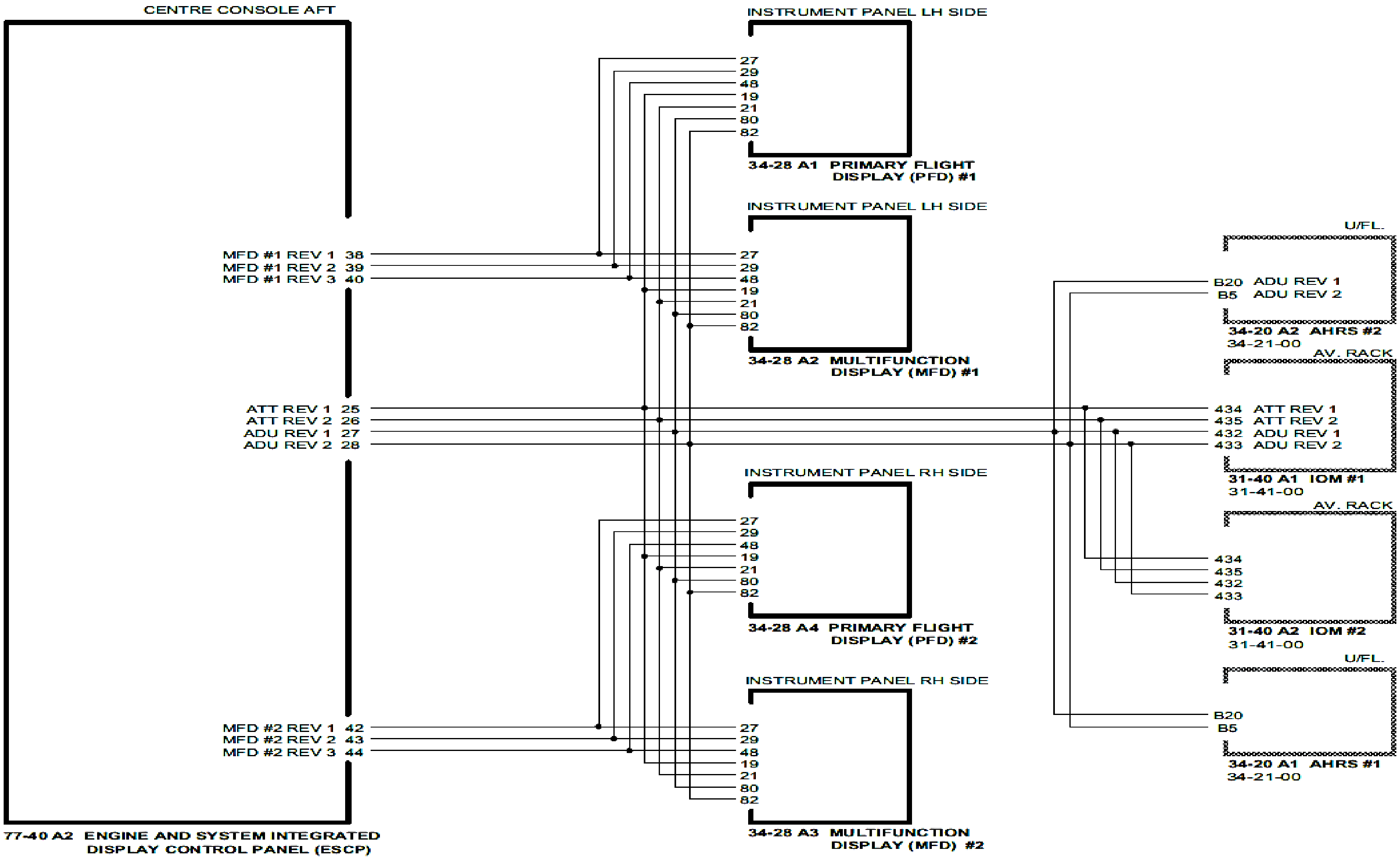


Figure - EIS Reversion

Refer Figure - Power Supply

28 VDC is supplied directly to the DUs.

The DUs supply indication data through different ARINC 429 feedback buses to other display units to make sure that the important parameters are correctly drawn.

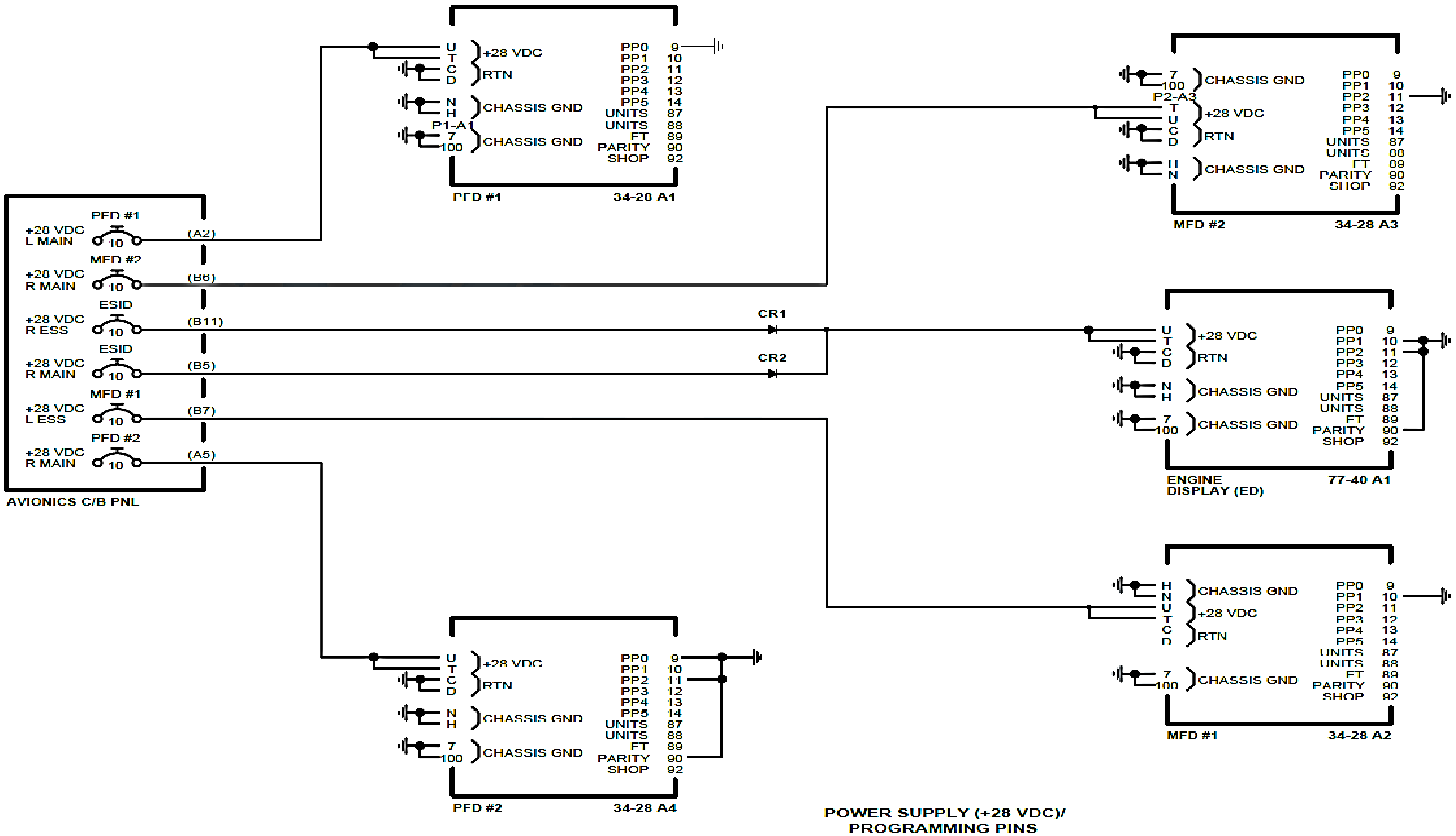


Figure - Power Supply

Refer Figure - ON/OFF Control

The EFIS control panels (EFCP1 and EFCP2) and ESID control panel (ESCP) supply a [ON/ OFF] discrete signal to the DU to energize and de-energize it.

The EFCP1 and EFCP2 also supply control data through ARINC 429 buses directly to the DUs.

The ESID supplies control data through the two input/output processors (IOP1 and IOP2) of the flight data processing system (FDPS) to the DUs.

The ESID supplies an [ALL] discrete signal to the DU to sequence it to the next system page. This is used if the ARINC 429 bus malfunctions.

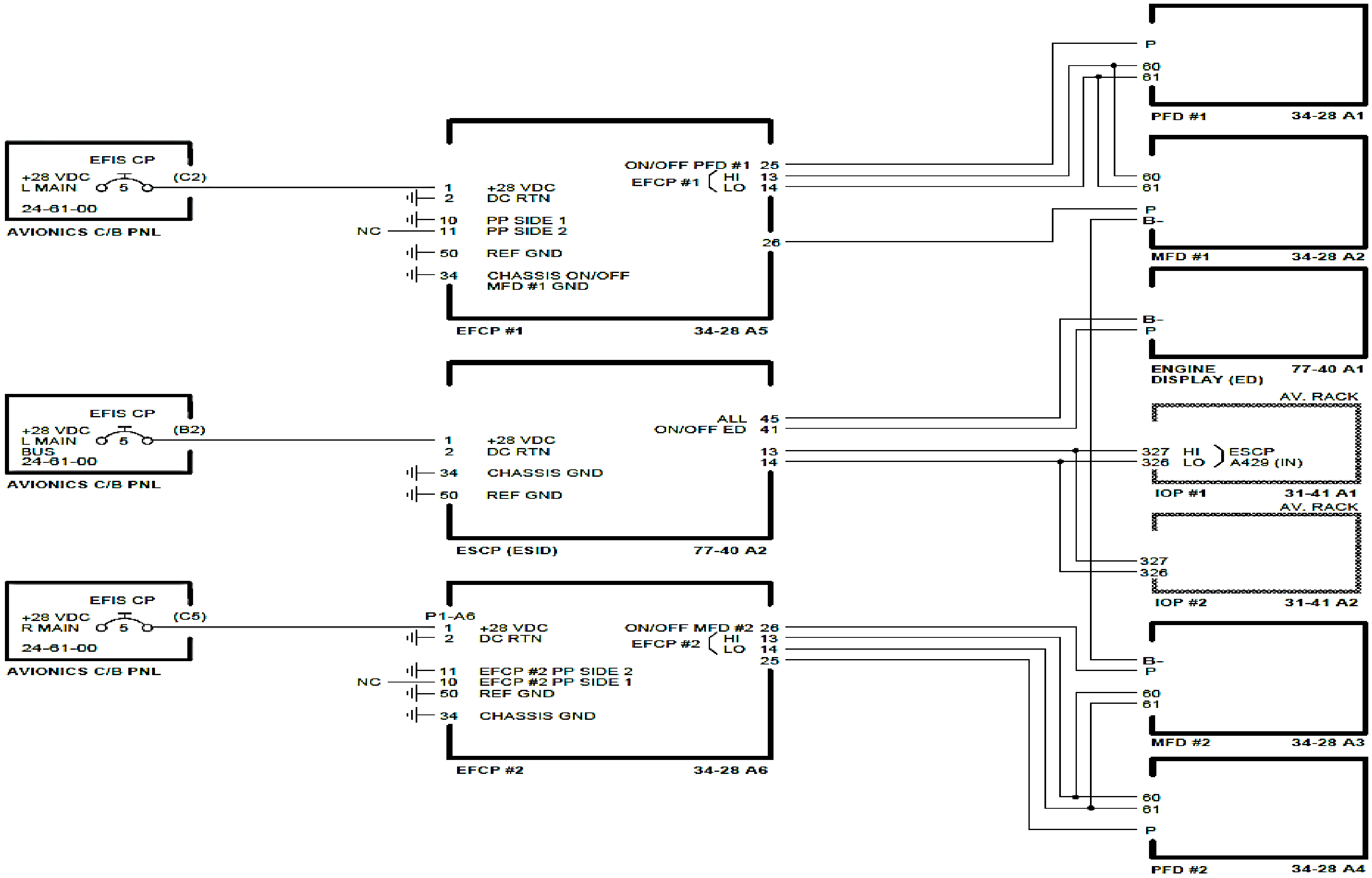


Figure - ON/OFF Control

Refer Figure - Continuous Monitoring

The DUs supply [HEALTHY] discretes through the input/output module 1 (IOM) to the central diagnostic system (CDS) in input/output processor 1 (IOP1).

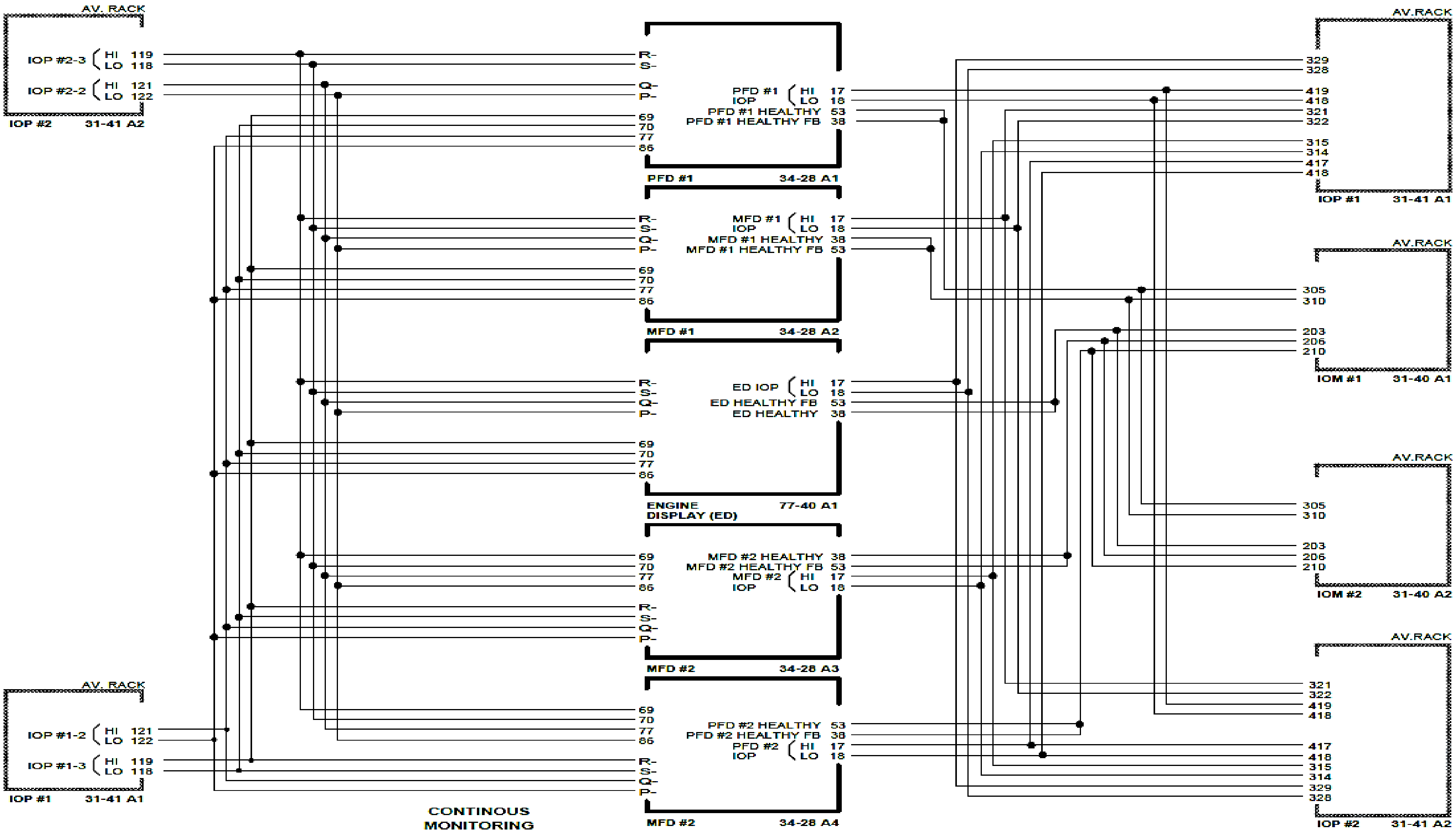


Figure - Continuous Monitoring